Accelerating Variance Reduction: Theory Notes

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1 Proposition 5.9 as a chain of inequalities: exact one-step identity, tightness ratios, optimal (γ, Δ), trajectory-dependent version

This note takes the proof of Proposition 5.9 of [6] (linear convergence of SAGA [7]) and splits it into an *exact* one-step identity followed by six inequalities (I1)–(I6). Each inequality receives a tightness ratio in $[0, 1]$ (in $(0, 1]$ for all of them except r_{var}) that is computable from the state of the algorithm, so that the experiments of hypothesis (B3) can measure, on ℓ_2-regularised logistic regression, which inequality is responsible for the gap between the worst-case contraction factor $1 - \min\{1/(3n),\, 3\mu/(16R^2)\}$ and the contraction actually observed. We then prove that no choice of the two constants (γ, Δ) of the proof can improve the certified one-step rate $1 - \rho_{\mathrm{bound}}$ by more than a factor 3 (a factor $4/3$ when $\kappa \geq 9n/16$), the optimal constants being exactly those of [17], so that a larger gap must come from the inequalities themselves; we show by an elementary argument, valid for every n and every constant step, that the exponent 1 of κ in the complexity of SAGA cannot be removed by a better analysis; and we derive the trajectory-dependent version of the proposition that hypothesis (B2) builds on.

1.1 Setting, the SAGA iteration and Proposition 5.9

Assumption 1.1 (finite sum; notation of the book)**.** *$F(\theta) = \frac{1}{n}\sum_{i=1}^{n} f_i(\theta)$ on $\mathbb{R}^d$, where each f_i is convex, differentiable and R^2-smooth, i.e. (Definition 5.3 of the book, Lipschitz-gradient form) $\|f_i'(\theta) - f_i'(\eta)\| \leq R^2 \|\theta - \eta\|$ for all $\theta, \eta \in \mathbb{R}^d$; and F is μ-strongly convex, $\mu > 0$ (Definition 5.2 of the book: $F(\eta) \geq F(\theta) + F'(\theta)^\top(\eta - \theta) + \frac{\mu}{2}\|\eta - \theta\|^2$ for all θ, η). We write $\kappa = R^2/\mu$.*

Lemma 1.2 (consequences of the assumptions)**.** *Under Assumption 1.1:*

(a) F has a unique minimiser η_*, characterised by $F'(\eta_*) = 0$, i.e. $\sum_{i=1}^{n} f_i'(\eta_*) = 0$;

(b) for all θ, $\langle F'(\theta), \theta - \eta_* \rangle \geq \mu \|\theta - \eta_*\|^2$;

(c) F is R^2-smooth and $\mu \le R^2$, i.e. $\kappa \ge 1$.

Proof. (a) is Proposition 5.2 and Exercise 5.6 of the book: a strongly convex function is coercive and strictly convex, hence has exactly one minimiser, and for a differentiable convex function the minimisers are exactly the stationary points. (b) Write the strong convexity inequality at the pairs $(\theta,\eta) = (\theta,\eta_*)$ and (η_*,θ) and add: $0 \ge (F'(\theta) - F'(\eta_*))^\top(\eta_* - \theta) + \mu\,\|\theta-\eta_*\|^2$; conclude with $F'(\eta_*) = 0$. (c) $\|F'(\theta) - F'(\eta)\| \le \frac{1}{n}\sum_i \|f_i'(\theta) - f_i'(\eta)\| \le R^2\,\|\theta-\eta\|$ by the triangle inequality. For $\theta \ne \eta_*$, (b) and Cauchy–Schwarz give $\mu\,\|\theta-\eta_*\|^2 \le \|F'(\theta)\|\,\|\theta-\eta_*\| \le R^2\,\|\theta-\eta_*\|^2$. □

SAGA. Let $i(1), i(2), \dots$ be i.i.d. uniform on $\{1,\dots,n\}$, let $\mathcal{F}_t = \sigma(i(1),\dots,i(t))$ ($\mathcal{F}_0$ trivial) and $\mathbb{E}_{i(t)}[\,\cdot\,] := \mathbb{E}[\,\cdot \mid \mathcal{F}_{t-1}]$. Given $\theta_0 \in \mathbb{R}^d$ and stored gradients $z_1^{(0)},\dots,z_n^{(0)} \in \mathbb{R}^d$, the iteration of [6, Sec. 5.4.4] is

$$\begin{aligned} \theta_t &= \theta_{t-1} - \gamma\,\omega_t, \qquad \omega_t := f_{i(t)}'(\theta_{t-1}) + \frac{1}{n}\sum_{j=1}^n z_j^{(t-1)} - z_{i(t)}^{(t-1)}, \\ z_{i(t)}^{(t)} &= f_{i(t)}'(\theta_{t-1}), \qquad z_j^{(t)} = z_j^{(t-1)} \ \ (j \ne i(t)). \end{aligned} \tag{1}$$

We write $\bar z_{t-1} := \frac{1}{n}\sum_j z_j^{(t-1)}$. The variables θ_t, $z^{(t)}$ are $\mathcal{F}_t$-measurable, and $i(t)$ is independent of $\mathcal{F}_{t-1}$ and uniform, so that for any $\mathcal{F}_{t-1}$-measurable vectors $(v_i)_{i\le n}$, $\mathbb{E}_{i(t)}[v_{i(t)}] = \frac{1}{n}\sum_i v_i$. The Lyapunov function of the book is, for a constant $\Delta > 0$,

$$V_t := \|\theta_t - \eta_*\|^2 + \Delta \sum_{i=1}^n \left\| z_i^{(t)} - f_i'(\eta_*) \right\|^2, \qquad \text{with } \gamma = \frac{1}{4R^2},\ \Delta = 3\gamma^2 = \frac{3}{16R^4} \text{ in the book.} \tag{2}$$

Proposition 1.3 (Proposition 5.9 of [6], convergence of SAGA)**.** *Under Assumption 1.1, if $z_i^{(0)} = f_i'(\theta_0)$ for all i and $\gamma = \frac{1}{4R^2}$, then for all $t \ge 0$,*

$$\mathbb{E}\,\|\theta_t - \eta_*\|^2 \le \left(1 - \min\left\{\frac{1}{3n}, \frac{3\mu}{16R^2}\right\}\right)^t \left(1 + \frac{n}{4}\right) \|\theta_0 - \eta_*\|^2 . \tag{3}$$

The book's proof (pp. 147–149) is reproduced in Section 1.3 as the exact identity of Section 1.2 followed by the six inequalities (I1)–(I6).

1.2 The exact one-step identity

Fix $t \ge 1$ and write $(\theta, z) = (\theta_{t-1}, z^{(t-1)})$ for the state before the t-th draw; everything below is $\mathcal{F}_{t-1}$-measurable. Define

$$D_{t-1} := \|\theta - \eta_*\|^2, \qquad M_{t-1} := \sum_{i=1}^n \left\|z_i - f_i'(\eta_*)\right\|^2, \qquad G_{t-1} := (\theta - \eta_*)^\top F'(\theta),$$

$$S_{t-1} := \sum_{i=1}^n \left\|f_i'(\theta) - f_i'(\eta_*)\right\|^2, \qquad a_i := f_i'(\theta) - f_i'(\eta_*), \qquad b_i := f_i'(\eta_*) - z_i + \bar z_{t-1},$$

so that $\omega_t = a_{i(t)} + b_{i(t)}$ (indeed $a_i + b_i = f_i'(\theta) - z_i + \bar z_{t-1}$), and

$$\Omega_{t-1} := \mathbb{E}_{i(t)}\,\|\omega_t\|^2 = \frac{1}{n}\sum_{i=1}^n \|a_i + b_i\|^2, \qquad C_{t-1} := \frac{1}{n}\sum_{i=1}^n \langle a_i, b_i\rangle,$$

$$B_{t-1} := \mathbb{E}_{i(t)}\left\|b_{i(t)}\right\|^2 = \frac{1}{n}\sum_{i=1}^n \left\|f_i'(\eta_*) - z_i + \frac{1}{n}\sum_{j=1}^n z_j\right\|^2.$$

Thus $V_{t-1} = D_{t-1} + \Delta M_{t-1}$. We drop the index $t-1$ when no confusion is possible.

Lemma 1.4 (exact one-step identities). *For every $t \geq 1$, almost surely:*

(a) $\mathbb{E}_{i(t)}[\omega_t] = F'(\theta_{t-1})$ *and*

$$\mathbb{E}_{i(t)}\big[D_t\big] = D_{t-1} - 2\gamma\, G_{t-1} + \gamma^2\, \Omega_{t-1}; \tag{4}$$

(b)

$$\mathbb{E}_{i(t)}\big[M_t\big] = \Big(1 - \frac{1}{n}\Big) M_{t-1} + \frac{1}{n} S_{t-1}; \tag{5}$$

(c)

$$B_{t-1} = \frac{1}{n} M_{t-1} - \|\bar{z}_{t-1}\|^2; \tag{6}$$

(d) $\Omega_{t-1} = \frac{1}{n} S_{t-1} + 2C_{t-1} + B_{t-1}$, *and* $\Omega_{t-1} = \|F'(\theta_{t-1})\|^2 + \mathbb{E}_{i(t)} \|\omega_t - F'(\theta_{t-1})\|^2$;

(e) $\mathbb{E}_{i(t)} \big\|a_{i(t)}\big\|^2 = S_{t-1}/n$, $\mathbb{E}_{i(t)}[b_{i(t)}] = 0$ *and* $|C_{t-1}| \leq \sqrt{(S_{t-1}/n)\, B_{t-1}}$.

Proof. (a) Since θ_{t-1} and $z^{(t-1)}$ are $\mathcal{F}_{t-1}$-measurable and $i(t)$ is uniform and independent of $\mathcal{F}_{t-1}$, $\mathbb{E}_{i(t)}[f'_{i(t)}(\theta_{t-1})] = \frac{1}{n}\sum_i f'_i(\theta_{t-1}) = F'(\theta_{t-1})$ and $\mathbb{E}_{i(t)}[z^{(t-1)}_{i(t)}] = \bar{z}_{t-1}$, hence $\mathbb{E}_{i(t)}[\omega_t] = F'(\theta_{t-1}) + \bar{z}_{t-1} - \bar{z}_{t-1} = F'(\theta_{t-1})$. Expanding the square in $\theta_t - \eta_* = (\theta_{t-1} - \eta_*) - \gamma\omega_t$, $D_t = D_{t-1} - 2\gamma\,\langle \theta_{t-1} - \eta_*, \omega_t\rangle + \gamma^2 \|\omega_t\|^2$; taking $\mathbb{E}_{i(t)}$ and pulling the $\mathcal{F}_{t-1}$-measurable vector $\theta_{t-1} - \eta_*$ out of the inner product gives (4). (b) Only the coordinate $i(t)$ of the memory changes, so $M_t = M_{t-1} - \Big\|z^{(t-1)}_{i(t)} - f'_{i(t)}(\eta_*)\Big\|^2 + \Big\|f'_{i(t)}(\theta_{t-1}) - f'_{i(t)}(\eta_*)\Big\|^2$, and $\mathbb{E}_{i(t)}$ of the last two terms equals M_{t-1}/n and S_{t-1}/n. (c) Let $Z := f'_{i(t)}(\eta_*) - z^{(t-1)}_{i(t)}$. Then $\mathbb{E}_{i(t)}[Z] = \frac{1}{n}\sum_i f'_i(\eta_*) - \bar{z}_{t-1} = -\bar{z}_{t-1}$ by Lemma 1.2(a), so $b_{i(t)} = Z - \mathbb{E}_{i(t)}[Z]$ and $B_{t-1} = \mathbb{E}_{i(t)} \big\|Z - \mathbb{E}_{i(t)}Z\big\|^2 = \mathbb{E}_{i(t)} \|Z\|^2 - \big\|\mathbb{E}_{i(t)}Z\big\|^2 = M_{t-1}/n - \|\bar{z}_{t-1}\|^2$. (d) Expand $\|a_i + b_i\|^2 = \|a_i\|^2 + 2\,\langle a_i, b_i\rangle + \|b_i\|^2$ and average over i. The second identity is $\mathbb{E}_{i(t)} \|\omega_t\|^2 = \big\|\mathbb{E}_{i(t)}\omega_t\big\|^2 + \mathbb{E}_{i(t)} \big\|\omega_t - \mathbb{E}_{i(t)}\omega_t\big\|^2$ with (a). (e) The first two claims are the computations in (a) and (c). For the third, Cauchy–Schwarz in $\mathbb{R}^d$ and then in $\mathbb{R}^n$: $|C| \leq \frac{1}{n}\sum_i \|a_i\|\,\|b_i\| \leq \big(\frac{1}{n}\sum_i \|a_i\|^2\big)^{1/2}\big(\frac{1}{n}\sum_i \|b_i\|^2\big)^{1/2}$. □

Proposition 1.5 (exact conditional contraction). *For every $t \geq 1$ and every $\gamma, \Delta > 0$, almost surely,*

$$\mathbb{E}_{i(t)}[V_t] = D_{t-1} - 2\gamma G_{t-1} + \gamma^2 \Omega_{t-1} + \Delta\Big[\Big(1 - \frac{1}{n}\Big) M_{t-1} + \frac{1}{n} S_{t-1}\Big], \tag{7}$$

so that, on $\{V_{t-1} > 0\}$, the exact conditional contraction factor is

$$\rho_{\text{exact}}(t) := \frac{\mathbb{E}_{i(t)}[V_t]}{V_{t-1}} = 1 - \frac{2\gamma G_{t-1} - \gamma^2 \Omega_{t-1} + \Delta\,(M_{t-1} - S_{t-1})/n}{D_{t-1} + \Delta M_{t-1}}. \tag{8}$$

Proof. Add (4) and Δ times (5), then subtract $V_{t-1} = D_{t-1} + \Delta M_{t-1}$. □

Remark 1.6 (reading the identity). The numerator of (8) is the one-step *progress*: $2\gamma G - \gamma^2\Omega$ is the progress of the parameter (as for gradient descent, with the second moment $\Omega = \|F'(\theta)\|^2 + \mathrm{Var}_{i(t)}(\omega_t)$ of the direction in place of $\|F'(\theta)\|^2$), and $\Delta(M - S)/n$ is the progress of the memory: one stored error $\|z_i - f'_i(\eta_*)\|^2$ (of average size M/n) is replaced by the current error $\|f'_i(\theta) - f'_i(\eta_*)\|^2$ (of average size S/n). For admissible constants (Proposition 1.14 below, in particular the textbook ones), the expected progress is bounded below by $(1 - \rho_{\text{bound}})V_{t-1} > 0$ whenever $V_{t-1} > 0$, by the chain (I1)–(I5) of Section 1.3: the memory progress $\Delta(M - S)/n$ alone can be negative (when

$S > M$), but it is then compensated by the parameter term. Only the realised one-step change $V_t - V_{t-1}$ can be positive. In the experiments, a negative numerator in (8) with the textbook constants would therefore signal a bug, not a feature. Note that $V_{t-1} = 0$ iff $\theta_{t-1} = \eta_*$ and $z_i = f_i'(\eta_*)$ for all i; then $\omega_t = 0$ and the state is absorbing, so ρ_{exact} is only needed on $\{V_{t-1} > 0\}$.

1.3 The proof of Proposition 5.9 as a chain of six inequalities

We now follow the book's proof line by line; the only steps that are not identities are the six inequalities below (named as in the project description).

(I1) Square-norm inequality. For $a, b \in \mathbb{R}^d$, $\|a+b\|^2 = 2\|a\|^2 + 2\|b\|^2 - \|a-b\|^2 \le 2\|a\|^2 + 2\|b\|^2$. Applied to $\omega_t = a_{i(t)} + b_{i(t)}$ and averaged with Lemma 1.4(e):

$$\Omega_{t-1} \;\le\; \frac{2}{n} S_{t-1} + 2B_{t-1}, \qquad \text{with} \quad \frac{2}{n} S_{t-1} + 2B_{t-1} - \Omega_{t-1} = \frac{1}{n}\sum_{i=1}^{n} \|a_i - b_i\|^2 . \tag{I1}$$

(I2) Co-coercivity (Proposition 5.4 of the book, eq. (5.31)). Proposition 5.4 states that for a convex L-smooth function f and all θ, η, $\frac{1}{L}\|f'(\theta) - f'(\eta)\|^2 \le (f'(\theta) - f'(\eta))^\top(\theta - \eta)$. Applied to each f_i with $L = R^2$, $\eta = \eta_*$, summed over i, divided by n, and using $\frac{1}{n}\sum_i f_i'(\eta_*) = F'(\eta_*) = 0$:

$$\frac{1}{n} S_{t-1} = \frac{1}{n}\sum_{i=1}^{n} \big\|f_i'(\theta_{t-1}) - f_i'(\eta_*)\big\|^2 \;\le\; \frac{R^2}{n}\sum_{i=1}^{n} \big(f_i'(\theta_{t-1}) - f_i'(\eta_*)\big)^\top(\theta_{t-1} - \eta_*) = R^2\, G_{t-1}. \tag{I2}$$

This inequality is used *twice*: once in the recursion for D (through (I1)) and once in the recursion for M (through (5)).

(I3) Variance $\le$ second moment. By Lemma 1.4(c) and $\|\bar z_{t-1}\|^2 \ge 0$:

$$B_{t-1} = \mathbb{E}_{i(t)} \big\|Z - \mathbb{E}_{i(t)} Z\big\|^2 \;\le\; \mathbb{E}_{i(t)} \|Z\|^2 = \frac{1}{n} M_{t-1}, \qquad Z = f_{i(t)}'(\eta_*) - z_{i(t)}^{(t-1)}. \tag{I3}$$

Step 1 of the book (parameter recursion). Inserting (I1), then (I2), then (I3) into (4):

$$\mathbb{E}_{i(t)}[D_t] \le D_{t-1} - 2\gamma G_{t-1} + 2\gamma^2\Big(\frac{1}{n} S_{t-1} + B_{t-1}\Big) \le D_{t-1} - 2\gamma(1 - \gamma R^2)\, G_{t-1} + \frac{2\gamma^2}{n} M_{t-1}. \tag{9}$$

Step 2 of the book (memory recursion). Inserting (I2) into (5):

$$\mathbb{E}_{i(t)}[M_t] \le \Big(1 - \frac{1}{n}\Big) M_{t-1} + R^2\, G_{t-1}. \tag{10}$$

Combination. Adding (9) and Δ times (10), for any $\gamma, \Delta > 0$,

$$\mathbb{E}_{i(t)}[V_t] \le D_{t-1} - 2\gamma\Big(1 - \gamma R^2 - \frac{R^2\Delta}{2\gamma}\Big) G_{t-1} + \Big[\frac{2\gamma^2}{n\Delta} + 1 - \frac{1}{n}\Big]\Delta M_{t-1}. \tag{11}$$

With $\Delta = 3\gamma^2$ and $\gamma = \frac{1}{4R^2}$: $1 - \gamma R^2 - \frac{R^2\Delta}{2\gamma} = 1 - \frac{1}{4} - \frac{3}{8} = \frac{3}{8}$ and $\frac{2\gamma^2}{n\Delta} = \frac{2}{3n}$, so the memory coefficient is $1 - \frac{1}{n} + \frac{2}{3n} = 1 - \frac{1}{3n}$ and

$$\mathbb{E}_{i(t)}[V_t] \le D_{t-1} - \frac{3\gamma}{4}\, G_{t-1} + \Big(1 - \frac{1}{3n}\Big)\Delta M_{t-1}. \tag{12}$$

(I4) Strong convexity. By Lemma 1.2(b),

$$G_{t-1} = (\theta_{t-1} - \eta_*)^\top F'(\theta_{t-1}) \;\geq\; \mu\, D_{t-1}. \tag{I4}$$

Since the coefficient $\frac{3\gamma}{4}$ of G_{t-1} in (12) is positive, $-\frac{3\gamma}{4}G_{t-1} \leq -\frac{3\gamma\mu}{4}D_{t-1} = -\frac{3\mu}{16R^2}D_{t-1}$, hence

$$\mathbb{E}_{i(t)}[V_t] \leq \Big(1 - \frac{3\mu}{16R^2}\Big) D_{t-1} + \Big(1 - \frac{1}{3n}\Big)\Delta M_{t-1}. \tag{13}$$

(I5) The min step. For $\alpha, \beta \geq 0$ and $x, y \geq 0$, $\alpha x + \beta y \leq \max\{\alpha, \beta\}(x+y)$. With $\alpha = 1 - \frac{3\mu}{16R^2} \in (0,1)$ (recall $\mu \leq R^2$) and $\beta = 1 - \frac{1}{3n} \in (0,1)$:

$$\mathbb{E}_{i(t)}[V_t] \;\leq\; \Big(1 - \min\Big\{\frac{1}{3n}, \frac{3\mu}{16R^2}\Big\}\Big) V_{t-1} =: \rho_{\text{bound}}\, V_{t-1}. \tag{I5}$$

Taking expectations and iterating (tower property; V_0 is deterministic): $\mathbb{E}[V_t] \leq \rho_{\text{bound}}^t V_0$, and $\mathbb{E}\,\|\theta_t - \eta_*\|^2 \leq \mathbb{E}[V_t]$ because $\Delta M_t \geq 0$.

(I6) Initialisation (Lipschitz continuity of each f_i'). With $z_i^{(0)} = f_i'(\theta_0)$ one has $M_0 = S_0$ and

$$M_0 = \sum_{i=1}^n \big\| f_i'(\theta_0) - f_i'(\eta_*) \big\|^2 \;\leq\; n\, R^4\, \|\theta_0 - \eta_*\|^2, \tag{I6}$$

so that $V_0 = D_0 + \frac{3}{16R^4} M_0 \leq \big(1 + \frac{3n}{16}\big) D_0 \leq \big(1 + \frac{n}{4}\big) D_0$, which is (3). (The last step, $1 + \frac{3n}{16} \leq 1 + \frac{n}{4}$, is a cosmetic loss by a factor at most $4/3$ on the constant, not on the rate.) □

Remark 1.7 (the constant R^2 for logistic regression). For $f_i(\theta) = \ell(y_i x_i^\top \theta) + \frac{\lambda}{2}\|\theta\|^2$ with $\ell(u) = \log(1 + e^{-u})$, $y_i \in \{-1, 1\}$, one has $\ell''(u) = \sigma(u)\sigma(-u) \in (0, \frac{1}{4}]$ where $\sigma(u) = 1/(1+e^{-u})$, so $f_i''(\theta) = \ell''(y_i x_i^\top \theta)\, x_i x_i^\top + \lambda I \preceq \big(\frac{\|x_i\|^2}{4} + \lambda\big) I$. Hence f_i is L_i-smooth with $L_i = \|x_i\|^2/4 + \lambda$, and the smallest constant valid for all i is

$$R^2 := \max_i \Big(\frac{\|x_i\|^2}{4} + \lambda \Big).$$

With features bounded by R_x in ℓ_2-norm, the book's constant $R^2 := R_x^2$ ("features bounded by R in ℓ_2-norm") is therefore a valid smoothness constant whenever $\lambda \leq \frac{3}{4}R_x^2$, but a loose one: it ignores the factor $\frac{1}{4}$ of the logistic second derivative (and λ), so that it exceeds $\max_i L_i$ by a factor close to 4 when $\lambda \ll R_x^2$. Since every constant of the proof is expressed through R^2 ($\gamma = 1/(4R^2)$, $\Delta = 3/(16R^4)$, the rate $3\mu/(16R^2)$), using the loose value would divide the step and the guaranteed rate by about 4; the experiments must use $R^2 = \max_i(\|x_i\|^2/4 + \lambda)$. The global strong convexity constant is $\mu = \lambda$ (the logistic part is convex), typically much smaller than the curvature actually met along the trajectory, see Section 1.6.

1.4 Tightness ratios and attribution of the gap

Definition 1.8 (tightness ratios). *For a state $(\theta, z) = (\theta_{t-1}, z^{(t-1)})$ with $\theta \neq \eta_*$, define, with the quantities of Section 1.2,*

$$r_{\mathrm{sq}} := \frac{\Omega}{\frac{2}{n}S + 2B} \quad \textit{(I1)}, \qquad r_{\mathrm{coco}} := \frac{S/n}{R^2 G} = \frac{L_{\mathrm{eff}}}{R^2}, \quad L_{\mathrm{eff}} := \frac{S}{nG} \quad \textit{(I2)},$$

$$r_{\mathrm{var}} := \frac{B}{M/n} = 1 - \frac{n\,\|\bar{z}\|^2}{M} \quad \textit{(I3)}, \qquad r_{\mathrm{sc}} := \frac{\mu D}{G} = \frac{\mu}{\mu_{\mathrm{eff}}}, \quad \mu_{\mathrm{eff}} := \frac{G}{D} \quad \textit{(I4)},$$

$$r_{\min} := \frac{\big(1 - \frac{3\mu}{16R^2}\big)D + \big(1 - \frac{1}{3n}\big)\Delta M}{\big(1 - \min\{\frac{1}{3n}, \frac{3\mu}{16R^2}\}\big)(D + \Delta M)} \quad \textit{(I5)}, \qquad r_{\mathrm{init}} := \frac{\sum_i \|f_i'(\theta_0) - f_i'(\eta_*)\|^2}{nR^4\,\|\theta_0 - \eta_*\|^2} \quad \textit{(I6)},$$

with the convention $r_{\mathrm{var}} := 1$ if $M = 0$ (then (I3) is the equality $0 \leq 0$; no convention is needed for r_{sq}, since $S > 0$ when $\theta \neq \eta_*$, Lemma 1.9(i)). Each ratio is the smaller side divided by the larger side of the corresponding inequality of Section 1.3 (for (I4), $G \geq \mu D$, this is $\mu D/G$); $r_{\mathrm{sq}}, r_{\mathrm{coco}}, r_{\mathrm{var}}, r_{\mathrm{sc}}$ depend on the state only (not on γ, Δ), $r_{\min}$ also depends on (Δ, n, μ, R^2).

Lemma 1.9 (ranges and degenerate cases). *Let $\theta \neq \eta_*$. Then:*

(i) $G > 0$, $S > 0$ and $0 < \mu \leq \mu_{\mathrm{eff}} \leq L_{\mathrm{eff}} \leq R^2$; hence $r_{\mathrm{sc}} \in (0, 1]$ and $r_{\mathrm{coco}} \in (0, 1]$.

(ii) $0 \leq r_{\mathrm{var}} \leq 1$; for $M > 0$, $r_{\mathrm{var}} = 0$ iff the memory errors $z_i - f_i'(\eta_*)$ are the same (nonzero) vector for all i; if $M = 0$ then $B = 0$ and $r_{\mathrm{var}} = 1$ by convention.

(iii) $S/n + B > 0$, $\Omega > 0$ and $r_{\mathrm{sq}} \in (0, 1]$; moreover

$$r_{\mathrm{sq}} = \frac{1}{2} + \frac{C}{S/n + B}, \qquad 1 - r_{\mathrm{sq}} = \frac{\frac{1}{n}\sum_i \|a_i - b_i\|^2}{2(S/n + B)},$$

so $r_{\mathrm{sq}} = 1$ iff $a_i = b_i$ for all i, and $r_{\mathrm{sq}} = \frac{1}{2}$ iff $C = 0$ (gradient changes a_i and memory errors b_i uncorrelated under $i(t)$).

(iv) $r_{\min} \in (0, 1]$, with $r_{\min} = 1$ iff $\kappa = 9n/16$ (the two coefficients coincide) or $M = 0$ and $\kappa \geq 9n/16$ (the Lyapunov mass sits on the parameter component, which is then the slower one). Since $D > 0$, the mass never sits on the memory component alone.

(v) If $\theta_0 \neq \eta_*$ then $r_{\mathrm{init}} \in (0, 1]$.

Proof. (i) $G \geq \mu D > 0$ by (I4), which is $\mu_{\mathrm{eff}} \geq \mu$. By Jensen, $\|F'(\theta)\|^2 = \big\|\frac{1}{n}\sum_i a_i\big\|^2 \leq \frac{1}{n}\sum_i \|a_i\|^2 = S/n$, and $F'(\theta) \neq 0$ by Lemma 1.2(a), so $S > 0$. By Cauchy–Schwarz $G \leq \|F'(\theta)\|\sqrt{D}$, hence $S/n \geq \|F'(\theta)\|^2 \geq G^2/D$, i.e. $L_{\mathrm{eff}} = S/(nG) \geq G/D = \mu_{\mathrm{eff}}$. Finally $L_{\mathrm{eff}} \leq R^2$ is (I2). (ii) By (6), $0 \leq B \leq M/n$, and $B = 0$ iff $Z = f_{i(t)}'(\eta_*) - z_{i(t)}^{(t-1)}$ is almost surely equal to its mean, i.e. $f_i'(\eta_*) - z_i$ does not depend on i. If $M = 0$ then $z_i = f_i'(\eta_*)$ for all i and $B = 0$. (iii) $S/n + B \geq S/n > 0$ by (i). The two formulas follow from Lemma 1.4(d) and from the identity in (I1); $r_{\mathrm{sq}} \leq 1$ is (I1) and $r_{\mathrm{sq}} \geq 0$ is $\Omega \geq 0$ (equivalently $|C| \leq \sqrt{(S/n)B} \leq \frac{1}{2}(S/n + B)$ by Lemma 1.4(e)). Suppose $\Omega = 0$; then $a_i + b_i = f_i'(\theta) - z_i + \bar{z} = 0$ for all i; averaging over i gives $F'(\theta) = 0$, so $\theta = \eta_*$, which is excluded. Hence $\Omega > 0$ and $r_{\mathrm{sq}} > 0$. (iv) The numerator is $\alpha D + \beta \Delta M$ with $\alpha = 1 - \frac{3\mu}{16R^2}$ and $\beta = 1 - \frac{1}{3n}$ in $(0, 1)$, and the denominator is $\max\{\alpha, \beta\}(D + \Delta M)$; the ratio is in $(0, 1]$, with equality iff $\alpha = \beta$ (i.e. $16R^2 = 9n\mu$, $\kappa = 9n/16$) or the component with the strictly smaller coefficient has zero mass. Since $D > 0$ (because $\theta \neq \eta_*$), the latter can only be the memory component: $M = 0$ and $\beta \leq \alpha$, i.e. $\frac{1}{3n} \geq \frac{3\mu}{16R^2}$, i.e. $\kappa \geq 9n/16$. (v) $S_0 > 0$ by (i) and $S_0 \leq nR^4 D_0$ is (I6). □

Proposition 1.10 (the exact contraction in terms of the ratios). *Let $\theta_{t-1} \neq \eta_*$ and $M_{t-1} > 0$, and let $\gamma, \Delta > 0$. Then*

$$\mathbb{E}_{i(t)}[V_t] = c_D\, D_{t-1} + c_M\, \Delta M_{t-1}, \qquad \begin{aligned} c_D &= 1 - 2\gamma\,\mu_{\text{eff}}\Big(1 - \gamma\, r_{\text{sq}} L_{\text{eff}} - \frac{L_{\text{eff}}\Delta}{2\gamma}\Big), \\ c_M &= 1 - \frac{1}{n}\Big(1 - \frac{2\gamma^2}{\Delta}\, r_{\text{sq}}\, r_{\text{var}}\Big), \end{aligned} \tag{14}$$

where $\mu_{\text{eff}}, L_{\text{eff}}, r_{\text{sq}}, r_{\text{var}}$ are evaluated at the state $(\theta_{t-1}, z^{(t-1)})$. For the textbook constants $\gamma = \frac{1}{4R^2}$, $\Delta = 3\gamma^2$,

$$c_D = 1 - \frac{\mu_{\text{eff}}}{2R^2}\Big(1 - \frac{L_{\text{eff}}}{R^2}\Big(\frac{r_{\text{sq}}}{4} + \frac{3}{8}\Big)\Big), \qquad c_M = 1 - \frac{1}{n}\Big(1 - \frac{2}{3}\, r_{\text{sq}}\, r_{\text{var}}\Big). \tag{15}$$

Substituting the worst-case values $r_{\text{sq}} = r_{\text{var}} = 1$, $L_{\text{eff}} = R^2$, $\mu_{\text{eff}} = \mu$ gives exactly the two coefficients $1 - \frac{3\mu}{16R^2}$ and $1 - \frac{1}{3n}$ of (13).

Proof. By definition of the ratios, $S/n = L_{\text{eff}} G = L_{\text{eff}}\mu_{\text{eff}} D$, $B = r_{\text{var}} M/n$ and $\Omega = r_{\text{sq}}(2S/n + 2B) = 2r_{\text{sq}}(L_{\text{eff}}\mu_{\text{eff}} D + r_{\text{var}} M/n)$. Inserting into (7):

$$\mathbb{E}_{i(t)}[V_t] = D\big[1 - 2\gamma\mu_{\text{eff}} + 2\gamma^2 r_{\text{sq}} L_{\text{eff}}\mu_{\text{eff}} + \Delta L_{\text{eff}}\mu_{\text{eff}}\big] + \Delta M\Big[1 - \frac{1}{n} + \frac{2\gamma^2}{n\Delta} r_{\text{sq}} r_{\text{var}}\Big],$$

and factoring $2\gamma\mu_{\text{eff}}$ in the first bracket gives c_D. With $\gamma = 1/(4R^2)$, $\Delta = 3\gamma^2$: $2\gamma\mu_{\text{eff}} = \mu_{\text{eff}}/(2R^2)$, $\gamma L_{\text{eff}} = L_{\text{eff}}/(4R^2)$, $L_{\text{eff}}\Delta/(2\gamma) = 3\gamma L_{\text{eff}}/2 = 3L_{\text{eff}}/(8R^2)$ and $2\gamma^2/\Delta = 2/3$. □

Proposition 1.10 says that the book's proof *is* the substitution of worst-case values into an exact formula: the two coefficients of (13) are c_D and c_M at $(r_{\text{sq}}, L_{\text{eff}}, \mu_{\text{eff}}, r_{\text{var}}) = (1, R^2, \mu, 1)$, and (I5) replaces the weighted mean $(c_D D + c_M \Delta M)/V$ by $\max\{c_D, c_M\}$. This gives a canonical way of attributing the gap $\rho_{\text{bound}} - \rho_{\text{exact}}$ to the individual inequalities: undo them one at a time.

Proposition 1.11 (attribution of the gap). *Let $\theta_{t-1} \neq \eta_*$, $M_{t-1} > 0$, $\gamma, \Delta > 0$ with $\alpha := 1 - \gamma R^2 - \frac{R^2\Delta}{2\gamma} > 0$, and write D, M, G, S, B, Ω, V for the state quantities. Define the coefficient functions*

$$c_D(r, L, m) := 1 - 2\gamma m\Big(1 - \gamma r L - \frac{L\Delta}{2\gamma}\Big), \qquad c_M(r, v) := 1 - \frac{1}{n}\Big(1 - \frac{2\gamma^2}{\Delta} r v\Big),$$

and

$$\rho_{\text{bound}}(\gamma, \Delta) := \max\{c_D(1, R^2, \mu),\, c_M(1, 1)\} = 1 - \min\Big\{2\gamma\mu\alpha,\ \frac{1}{n}\Big(1 - \frac{2\gamma^2}{\Delta}\Big)\Big\},$$

which is the worst-case coefficient of Proposition 1.14 below (it is < 1 iff $2\gamma^2 < \Delta$) and equals the ρ_{bound} of (I5) for the textbook constants; in the rest of this proposition we write ρ_{bound} for $\rho_{\text{bound}}(\gamma, \Delta)$.

Order A (I4 $\to$ I2 $\to$ I3 $\to$ I1 $\to$ I5; the min is kept until the end).

$$
\begin{aligned}
\rho_0^A &:= \max\{c_D(1, R^2, \mu),\, c_M(1,1)\} = \rho_{\text{bound}},\\
\rho_1^A &:= \max\{c_D(1, R^2, \mu_{\text{eff}}),\, c_M(1,1)\} = \max\Big\{1 - 2\gamma\alpha\frac{G}{D},\; 1 - \frac{1}{n} + \frac{2\gamma^2}{n\Delta}\Big\},\\
\rho_2^A &:= \max\{c_D(1, L_{\text{eff}}, \mu_{\text{eff}}),\, c_M(1,1)\} = \max\Big\{1 - 2\gamma\frac{G}{D} + (2\gamma^2+\Delta)\frac{S}{nD},\; 1 - \frac{1}{n} + \frac{2\gamma^2}{n\Delta}\Big\},\\
\rho_3^A &:= \max\{c_D(1, L_{\text{eff}}, \mu_{\text{eff}}),\, c_M(1, r_{\text{var}})\} = \max\Big\{1 - 2\gamma\frac{G}{D} + (2\gamma^2+\Delta)\frac{S}{nD},\; 1 - \frac{1}{n} + \frac{2\gamma^2 B}{\Delta M}\Big\},\\
\rho_4^A &:= \max\{c_D(r_{\text{sq}}, L_{\text{eff}}, \mu_{\text{eff}}),\, c_M(r_{\text{sq}}, r_{\text{var}})\}\\
&\quad = \max\Big\{1 - 2\gamma\frac{G}{D} + \frac{2\gamma^2 r_{\text{sq}} S}{nD} + \frac{\Delta S}{nD},\; 1 - \frac{1}{n} + \frac{2\gamma^2 r_{\text{sq}} B}{\Delta M}\Big\},\\
\rho_5^A &:= \frac{c_D(r_{\text{sq}}, L_{\text{eff}}, \mu_{\text{eff}})\, D + c_M(r_{\text{sq}}, r_{\text{var}})\, \Delta M}{D + \Delta M} = \rho_{\text{exact}}(t),
\end{aligned}
$$

with $r_{\text{sq}} = \Omega/(2S/n + 2B)$.

Order B (I5 $\to$ I4 $\to$ I2 $\to$ I3 $\to$ I1; reverse proof order, the min is undone first).

$$
\begin{aligned}
\rho_0^B &:= \rho_{\text{bound}},\\
\rho_1^B &:= \frac{(1 - 2\gamma\mu\alpha)D + \big(1 - \frac{1}{n} + \frac{2\gamma^2}{n\Delta}\big)\Delta M}{V} \quad \big(= r_{\min}\,\rho_{\text{bound}} \textit{ for the textbook constants}\big),\\
\rho_2^B &:= \frac{D - 2\gamma\alpha G + \big(1 - \frac{1}{n} + \frac{2\gamma^2}{n\Delta}\big)\Delta M}{V},\\
\rho_3^B &:= \frac{D - 2\gamma G + (2\gamma^2+\Delta)\frac{S}{n} + \frac{2\gamma^2}{n} M + \big(1 - \frac{1}{n}\big)\Delta M}{V},\\
\rho_4^B &:= \frac{D - 2\gamma G + (2\gamma^2+\Delta)\frac{S}{n} + 2\gamma^2 B + \big(1 - \frac{1}{n}\big)\Delta M}{V},\\
\rho_5^B &:= \frac{D - 2\gamma G + \gamma^2\Omega + \Delta\frac{S}{n} + \big(1 - \frac{1}{n}\big)\Delta M}{V} = \rho_{\text{exact}}(t).
\end{aligned}
$$

Then both sequences are nonincreasing:

$$
\rho_{\text{bound}} = \rho_0^A \ge \rho_1^A \ge \rho_2^A \ge \rho_3^A \ge \rho_4^A \ge \rho_5^A = \rho_{\text{exact}}(t), \qquad \rho_{\text{bound}} = \rho_0^B \ge \rho_1^B \ge \cdots \ge \rho_5^B = \rho_{\text{exact}}(t),
$$

and consequently the nonnegative increments $\delta_k := \rho_{k-1} - \rho_k$ *(additive attribution) and, when* $\rho_{\text{exact}}(t) > 0$, $\lambda_k := \log(\rho_{k-1}/\rho_k)$ *(attribution in nats per iteration) satisfy*

$$
\sum_{k=1}^{5} \delta_k = \rho_{\text{bound}} - \rho_{\text{exact}}(t), \qquad \sum_{k=1}^{5} \lambda_k = \log\frac{1}{\rho_{\text{exact}}(t)} - \log\frac{1}{\rho_{\text{bound}}}, \qquad \prod_{k=1}^{5} \frac{\rho_k}{\rho_{k-1}} = \frac{\rho_{\text{exact}}(t)}{\rho_{\text{bound}}}.
$$

In order A the increments are attributed to (I4), (I2), (I3), (I1), (I5) in this order; in order B to (I5), (I4), (I2), (I3), (I1).

Proof. The closed forms follow from $\mu_{\text{eff}} = G/D$, $L_{\text{eff}}\mu_{\text{eff}} = S/(nD)$, $r_{\text{var}}M/n = B$ and $r_{\text{sq}}(2S/n + 2B) = \Omega$, exactly as in the proof of Proposition 1.10; e.g. $c_D(1, L_{\text{eff}}, \mu_{\text{eff}}) = 1 - 2\gamma G/D + 2\gamma^2 L_{\text{eff}}\mu_{\text{eff}} + \Delta L_{\text{eff}}\mu_{\text{eff}} = 1 - 2\gamma G/D + (2\gamma^2+\Delta)S/(nD)$, and $c_D(r_{\text{sq}}, L_{\text{eff}}, \mu_{\text{eff}})D + c_M(r_{\text{sq}}, r_{\text{var}})\Delta M$ equals the right-hand side of (7).

Monotonicity. For $r \le 1$ and $L \le R^2$ the bracket in c_D satisfies $1-\gamma r L - \frac{L\Delta}{2\gamma} \ge 1-\gamma R^2 - \frac{R^2\Delta}{2\gamma} = \alpha > 0$, so $m \mapsto c_D(r, L, m)$ is nonincreasing; moreover c_D is nondecreasing in L and in r, and c_M is nondecreasing in r and v. Lemma 1.9 gives $\mu \le \mu_{\text{eff}}$, $L_{\text{eff}} \le R^2$, $r_{\text{var}} \le 1$, $r_{\text{sq}} \le 1$, so each substitution lowers (weakly) the coefficient it touches; max and the weighted mean $(x, y) \mapsto (xD + y\Delta M)/V$ are monotone in both arguments; and finally $\max\{x, y\} \ge (xD + y\Delta M)/V$ since $D, \Delta M \ge 0$. This proves both chains; the telescoping identities are immediate. □

Remark 1.12 (reading the two orders). The attribution is path dependent, as for any decomposition of a difference of a nonlinear function. Order B (undo from the end) attributes to each inequality its full effect on the weighted contraction; order A keeps the max structure, so that undoing (I4) and (I2), which touch c_D alone, has *no* effect as long as the memory coefficient c_M is the larger one (regime $\kappa < 9n/16$ for the textbook constants), and undoing (I3), which touches c_M alone, has no effect as long as c_D is the larger one. Undoing (I1) is different: r_{sq} enters both coefficients (through $2\gamma^2 r_{\text{sq}} L_{\text{eff}} \mu_{\text{eff}}$ in c_D and $2\gamma^2 r_{\text{sq}} r_{\text{var}}/\Delta$ in c_M), so the step $\rho_3^A \to \rho_4^A$ lowers both of them and has an effect in both regimes, except in the degenerate cases $r_{\text{sq}} = 1$, or $r_{\text{var}} = 0$ with c_M the larger coefficient. Reporting both orders therefore shows, in addition to the individual contributions, which component (D or M) is the bottleneck of the bound. Since ρ is close to 1, $\lambda_k \approx \delta_k$; per effective pass the natural unit is $n\lambda_k$.

Remark 1.13 ((I1) throws away the control variate). Suppose the memory is fresh, $z_i = f_i'(\theta)$ for all i (e.g. at $t = 1$ with the book's initialisation, or approximately when the iterates move slowly). Then $\omega_t = F'(\theta)$ is deterministic, $\Omega = \|F'(\theta)\|^2$, $M = S$, $B = S/n - \|F'(\theta)\|^2$ and

$$r_{\text{sq}} = \frac{\|F'(\theta)\|^2}{4S/n - 2\|F'(\theta)\|^2} \le \frac{1}{2}, \qquad \text{with } r_{\text{sq}} \ll \frac{1}{2} \text{ when } \frac{S}{n} \gg \left\|F'(\theta)\right\|^2,$$

i.e. when the per-sample gradient changes a_i are diverse. In general, if index i was last drawn at time $u_i \le t-1$, then $z_i = z_i^{(u_i)} = f_i'(\theta_{u_i - 1})$ by (1); writing $s_i := u_i - 1 \in \{0, \dots, t-2\}$, and $s_i := 0$ for the indices not yet drawn (with the book's initialisation $z_i^{(0)} = f_i'(\theta_0)$), one has $z_i = f_i'(\theta_{s_i})$ in all cases, and

$$C = \left\langle F'(\theta), \bar{z} \right\rangle - \frac{1}{n}\sum_{i=1}^{n} \left\langle a_i(\theta), a_i(\theta_{s_i}) \right\rangle, \qquad a_i(\theta') := f_i'(\theta') - f_i'(\eta_*) :$$

the correlation between the current gradient change and the stored one is exactly what the control variate $-z_{i(t)} + \bar{z}$ exploits, and (I1) discards it. One therefore expects $r_{\text{sq}} \le \frac{1}{2}$ along most of the trajectory: a factor 2 in the γ^2 terms, which the experiments can confirm.

1.5 The best constants of the proof: general (γ, Δ)

The inequalities (I1)–(I3) and the identities do not depend on γ or Δ, so (11) holds for all $\gamma, \Delta > 0$. We now optimise the two constants.

Proposition 1.14 (general step and weight). *Let $\gamma, \Delta > 0$ satisfy*

$$\alpha(\gamma, \Delta) := 1 - \gamma R^2 - \frac{R^2\Delta}{2\gamma} > 0 \qquad \textit{and} \qquad \frac{2\gamma^2}{\Delta} < 1.$$

Then for all $t \ge 1$, almost surely,

$$\mathbb{E}_{i(t)}[V_t] \le \Big(1 - \min\Big\{2\gamma\mu\,\alpha(\gamma, \Delta),\ \frac{1}{n}\Big(1 - \frac{2\gamma^2}{\Delta}\Big)\Big\}\Big) V_{t-1}, \tag{16}$$

and both arguments of the min lie in $(0, \frac{1}{2}]$ and $(0, \frac{1}{n})$ respectively.

Proof. In (11) the coefficient $2\gamma\alpha$ of G_{t-1} is positive, so (I4) gives $-2\gamma\alpha G_{t-1} \le -2\gamma\mu\alpha D_{t-1}$ and $\mathbb{E}_{i(t)}[V_t] \le (1-2\gamma\mu\alpha)D_{t-1} + (1-\frac{1}{n}(1-\frac{2\gamma^2}{\Delta}))\Delta M_{t-1}$. Since $\alpha > 0$ forces $\gamma R^2 < 1$, we have $2\gamma\mu\alpha \le 2\gamma\mu(1-\gamma R^2) \le 2x(1-x) \le \frac{1}{2}$ with $x = \gamma R^2$ (using $\mu \le R^2$), and $0 < \frac{1}{n}(1-\frac{2\gamma^2}{\Delta}) < \frac{1}{n}$. Both coefficients are thus in $[\frac{1}{2}, 1)$ and $(1-\frac{1}{n}, 1)$, and (I5) concludes. □

Corollary 1.15 (parametrisation $\Delta = c\gamma^2$)**.** *Write $\Delta = c\gamma^2$ with $c > 2$. Then $\alpha = 1 - \gamma R^2(1+\frac{c}{2})$, the condition $\alpha > 0$ reads $\gamma < \frac{2}{R^2(2+c)}$, and (16) becomes*

$$\begin{aligned} &\mathbb{E}_{i(t)}[V_t] \le \big(1 - \min\{\tau_D(\gamma, c), \tau_M(c)\}\big)V_{t-1}, \\ &\tau_D(\gamma, c) := 2\gamma\mu\Big(1 - \gamma R^2\Big(1 + \frac{c}{2}\Big)\Big), \qquad \tau_M(c) := \frac{1 - 2/c}{n}. \end{aligned} \tag{17}$$

For fixed c, $\gamma \mapsto \tau_D(\gamma, c)$ is a concave parabola, maximal at

$$\gamma^\star(c) = \frac{1}{R^2(2+c)}, \qquad \tau_D(\gamma^\star(c), c) = \frac{\mu}{R^2(2+c)} = \frac{1}{\kappa(2+c)},$$

while τ_M does not depend on γ. Hence the best rate of this proof for a given c is

$$\tau(c) := \min\Big\{\frac{1}{\kappa(2+c)},\ \frac{c-2}{cn}\Big\}.$$

Proof. $\alpha = 1 - \gamma R^2 - \frac{R^2 c\gamma^2}{2\gamma} = 1 - \gamma R^2(1+\frac{c}{2})$ and $\frac{2\gamma^2}{\Delta} = \frac{2}{c}$. Then $\partial_\gamma \tau_D = 2\mu(1 - 2\gamma R^2(1+\frac{c}{2}))$ vanishes at $\gamma^\star = \frac{1}{2R^2(1+c/2)} = \frac{1}{R^2(2+c)}$, where $1 - \gamma^\star R^2(1+\frac{c}{2}) = \frac{1}{2}$. For fixed c the min of a function of γ and a constant is maximised by maximising the function. □

Proposition 1.16 (optimal constants of the proof structure)**.** *Let $s := n/\kappa = n\mu/R^2$. The maximum over $c > 2$ and admissible γ of the rate $\min\{\tau_D, \tau_M\}$ in (17) is attained at*

$$\begin{aligned} c^\star &= \frac{s + \sqrt{s^2+16}}{2}, \qquad \Delta^\star = c^\star(\gamma^\star)^2, \\ \gamma^\star &= \frac{1}{R^2(2+c^\star)} = \frac{2}{R^2\big(4 + s + \sqrt{s^2+16}\big)} = \frac{2}{4R^2 + n\mu + \sqrt{n^2\mu^2 + 16R^4}}, \end{aligned}$$

and equals

$$\tau^\star = \frac{1}{\kappa(2+c^\star)} = \frac{2}{4\kappa + n + \sqrt{n^2 + 16\kappa^2}}, \qquad \frac{1}{n+4\kappa} \ \le\ \tau^\star \ \le\ \frac{4 - 2\sqrt{2}}{n+4\kappa} \approx \frac{1.172}{n+4\kappa}.$$

In particular $\tau^\star \to \frac{1}{4\kappa}$ with $\gamma^\star \to \frac{1}{4R^2}$, $c^\star \to 2$ when $n/\kappa \to 0$, and $\tau^\star \to \frac{1}{n}$ with $\gamma^\star \sim \frac{1}{n\mu}$, $c^\star \sim n/\kappa$ when $n/\kappa \to \infty$.

Proof. By Corollary 1.15 we must maximise $\tau(c) = \min\{\frac{1}{\kappa(2+c)}, \frac{c-2}{cn}\}$ over $c > 2$. The first argument is decreasing in c, the second is increasing from 0 (at $c = 2$) to $\frac{1}{n}$; the maximum of the min is therefore attained where they cross: $cn = \kappa(c+2)(c-2)$, i.e. $\kappa c^2 - nc - 4\kappa = 0$, whose positive root is $c^\star = \frac{n + \sqrt{n^2+16\kappa^2}}{2\kappa} = \frac{s + \sqrt{s^2+16}}{2}$; it satisfies $c^\star > 2$ for $s > 0$ because $\sqrt{s^2+16} > 4 - s$. Then $2 + c^\star = \frac{4 + s + \sqrt{s^2+16}}{2}$, which gives $\tau^\star$ and $\gamma^\star$; the last expression for $\gamma^\star$ follows from $R^2 s = n\mu$ and $R^2\sqrt{s^2+16} = \sqrt{n^2\mu^2 + 16R^4}$. For the bounds, $\sqrt{n^2+16\kappa^2} \le n + 4\kappa$ gives $\tau^\star \ge \frac{2}{2(n+4\kappa)}$, and $\sqrt{n^2+16\kappa^2} \ge \frac{n+4\kappa}{\sqrt{2}}$ (since $(a+b)^2 \le 2(a^2+b^2)$) gives $\tau^\star \le \frac{2}{(1+1/\sqrt{2})(n+4\kappa)} = \frac{4-2\sqrt{2}}{n+4\kappa}$. The limits are read off the formulas ($c^\star = 2 + \frac{s}{2} + O(s^2)$ as $s \to 0$; $c^\star = s + \frac{4}{s} + O(s^{-3})$ as $s \to \infty$). □

These are the constants of Hofmann et al. Proposition 1.16 is not new: $\tau^\star$ and $\gamma^\star$ are exactly the optimal rate and step of [17, Theorem 1 and Corollary 1] for SAGA, i.e. $q=1$ in their notation, with $L=R^2$. Their Theorem 1 states, for a "uniform q-memorization algorithm" (SAGA is the case $q=1$ of their Definition 1) and any step $\gamma=a/(4L)$ with $a<1$, a geometric rate $1-\rho(\gamma)$ with $\rho(\gamma)=\frac{q}{n}\cdot\frac{1-a}{1-a/2}$ if $\gamma\geq\gamma^\star(K)$ and $\rho(\gamma)=\mu\gamma$ otherwise, where $\gamma^\star(K):=a^\star(K)/(4L)$, $a^\star(K):=\frac{2K}{1+K+\sqrt{1+K^2}}$ and $K:=\frac{4qL}{n\mu}$; their Corollary 1 states that ρ is maximised at $\gamma=\gamma^\star(K)$, with $\rho^\star(K)=\frac{q}{n}\big[\frac{2}{1+K+\sqrt{1+K^2}}\big]$. With $q=1$ and $K=4/s$,

$$a^\star(K)=\frac{8}{4+s+\sqrt{s^2+16}},\qquad \gamma^\star(K)=\frac{a^\star(K)}{4R^2}=\frac{2}{R^2(4+s+\sqrt{s^2+16})}=\gamma^\star,$$
$$\rho^\star(K)=\frac{1}{n}\cdot\frac{2}{1+\frac{4}{s}+\sqrt{1+\frac{16}{s^2}}}=\frac{2s}{n(4+s+\sqrt{s^2+16})}=\frac{2}{\kappa(4+s+\sqrt{s^2+16})}=\tau^\star,$$

and moreover $c^\star=4/a^\star(K)-2$. Their Corollary 3 ("choosing $\gamma=\frac{a}{4L}$ with $a<1$ yields $\rho=\min\{\frac{1-a}{1-\frac12 a}\frac{q}{n},\frac{a}{4}\frac{\mu}{L}\}$; in particular, for $\gamma=\frac{1}{5L}$, $\rho=\min\{\frac13\frac{q}{n},\frac15\frac{\mu}{L}\}$") is Corollary 1.15 at $\gamma=\gamma^\star(c)$: with $a=4/(2+c)$ one has $\frac{1-a}{1-a/2}=1-\frac{2}{c}$ and $\frac{a}{4}\frac{\mu}{L}=\frac{1}{\kappa(2+c)}$, i.e. $\tau(c)$, and $\gamma=\frac{1}{5L}$ is $c=3$. The coincidence is structural. Their Lyapunov function $\mathcal{L}_\sigma(w,H)=\|w-w^*\|^2+S\sigma\bar{H}$ with $S=\gamma n/(Lq)$ has the two-term form (2) with $\Delta=\gamma\sigma/(Lq)$ (their H_i are upper bounds on $\|\alpha_i-f_i'(w^*)\|^2$ obtained from the Bregman divergences of f_i at the stored points, their eqs. (10)–(12)), and their Lemma 1 uses the same square-norm inequality (I1) ("we sacrifice tightness and choose $\epsilon=1$"). The difference is that they apply strong convexity and co-coercivity through function values (their eqs. (5), (7)–(8)) instead of through G, and that they require the resulting coefficient of $f(w)-f(w^*)$ to be nonnegative and drop it (their eq. (31): $\gamma\leq\frac{1-\sigma}{2L}$); this caps their parameter rate at $\mu\gamma$ ("we will have $\rho\leq\gamma\mu$ based on the $\|w-w^*\|^2$ part", proof of their Lemma 3) and restricts their constants to $\Delta\leq\gamma(1-2\gamma R^2)/R^2$, i.e. to $\gamma\leq\gamma^\star(c)$ in the parametrisation $\Delta=c\gamma^2$. On the curve $\gamma=\gamma^\star(c)$, where the bracket $1-\gamma R^2(1+\frac{c}{2})$ equals $\frac12$, the two one-step bounds coincide, and the joint maximiser lies on this curve: this is why the optima agree. Off the curve the bounds differ: for $\gamma<\gamma^\star(c)$ the book's argument certifies the parameter rate $2\gamma\mu(1-\gamma R^2(1+\frac{c}{2}))>\mu\gamma$, and the textbook constants $\gamma=\frac{1}{4R^2}$, $c=3$ (their $a=1$, excluded in their Theorem 1) are admissible in Proposition 1.14 but not in their Lemma 3.

Comparison with the textbook choice. The book takes $c=3$ and $\gamma=\frac{1}{4R^2}$, i.e. $\tau_{\text{book}}=\min\{\frac{1}{3n},\frac{3}{16\kappa}\}$. (For $c=3$ the best step would be $\gamma^\star(3)=\frac{1}{5R^2}$, with $\tau_D=\frac{1}{5\kappa}$ instead of $\frac{3}{16\kappa}$: the step $\frac{1}{4R^2}$ is slightly too large for $c=3$, by 6.7% on the rate; this is the choice of Corollary 3 of [17] quoted above.) The ratio $Q(s):=\tau^\star/\tau_{\text{book}}$ between the best rate this proof structure can certify and the textbook rate is, with $s=n/\kappa$,

$$Q(s)=\begin{cases}\dfrac{32}{3\big(4+s+\sqrt{s^2+16}\big)}, & s\leq\frac{16}{9}\quad(\text{i.e. }\kappa\geq\frac{9n}{16},\ \text{parameter term is the bottleneck}),\\[2ex] \dfrac{6s}{4+s+\sqrt{s^2+16}}, & s\geq\frac{16}{9}\quad(\text{memory term is the bottleneck}),\end{cases}$$

which decreases from $Q(0)=\frac43$ to $Q(\frac{16}{9})\approx1.050$ and then increases to $Q(\infty)=3$; numerically $Q(0.5)\approx1.25$, $Q(1)\approx1.17$, $Q(5)\approx1.95$, $Q(20)\approx2.70$, $Q(100)\approx2.94$. Thus:

- in the ill-conditioned regime $\kappa\geq\frac{9n}{16}$ (the regime of hypothesis (A4), $\kappa\gg n$), the textbook constants are within a factor $\frac43$ of the best this proof can give, and the optimal step is $\gamma^\star=\frac{\tau^\star\kappa}{R^2}\in[\frac{0.197}{R^2},\frac{1}{4R^2}]$;

- in the big-data regime $n \gg \kappa$ the textbook rate $\frac{1}{3n}$ can be improved to at most $\frac{1}{n}$ (factor 3), at the price of the much smaller step $\gamma^\star \approx \frac{1}{n\mu}$ (this is the step $\frac{1}{2(\mu n+L)}$ of [7] up to a factor 2).

In both regimes the certified complexity is $\Theta\big((n+\kappa)\log(1/\varepsilon)\big)$: by Proposition 1.16 the best iteration complexity of the proof structure is $(n+4\kappa)\log(V_0/\varepsilon)$ up to a factor 1.172, and the textbook's is $\max\{3n, \frac{16}{3}\kappa\}\log(V_0/\varepsilon)$.

Proposition 1.17 (SAGA with a constant step is not accelerated: a lower bound valid for every n)**.** *Let $n \geq 1$, $0 < \mu < R^2$, $d = 2$, and $f_i(\theta) = \frac{1}{2}(\mu\theta_1^2 + R^2\theta_2^2)$ for all $i \leq n$. Assumption 1.1 holds with these constants, $F = f_1$, $F'(\theta) = Q\theta$ with $Q = \operatorname{diag}(\mu, R^2)$, and $\eta_* = 0$. Run (1) with any constant step $\gamma > 0$ (which may depend on μ) and any deterministic initial memory $z^{(0)}$. Then for all $t \geq 0$,*

$$\mathbb{E}[\theta_t] = (I-\gamma Q)^t\theta_0 \qquad \text{and} \qquad \mathbb{E}\,\|\theta_t-\eta_*\|^2 \;\geq\; \|\mathbb{E}[\theta_t]-\eta_*\|^2 \;\geq\; \Big(\frac{\kappa-1}{\kappa+1}\Big)^{2t}\min\{\theta_{0,1}^2,\theta_{0,2}^2\}. \quad (18)$$

In particular, for $\theta_0 = (r,r)$ with $r \neq 0$ and $\varepsilon < \frac{1}{2}$, reaching $\mathbb{E}\,\|\theta_t-\eta_*\|^2 \leq \varepsilon\,\|\theta_0-\eta_*\|^2$ requires $t \geq \frac{\kappa-1}{4}\log\frac{1}{2\varepsilon}$, whatever the step.

Proof. By Lemma 1.4(a), $\mathbb{E}[\omega_t \mid \mathcal{F}_{t-1}] = F'(\theta_{t-1}) = Q\theta_{t-1}$ whatever the content of the memory, so $\mathbb{E}[\theta_t \mid \mathcal{F}_{t-1}] = (I-\gamma Q)\theta_{t-1}$; taking expectations (the map is linear) and iterating from the deterministic θ_0 gives the first identity. Next, $\mathbb{E}\,\|X\|^2 = \|\mathbb{E}X\|^2 + \mathbb{E}\,\|X-\mathbb{E}X\|^2 \geq \|\mathbb{E}X\|^2$ with $X = \theta_t$. Finally, Q is diagonal, so $\big\|(I-\gamma Q)^t\theta_0\big\|^2 = (1-\gamma\mu)^{2t}\theta_{0,1}^2 + (1-\gamma R^2)^{2t}\theta_{0,2}^2 \geq \max\{|1-\gamma\mu|, \big|1-\gamma R^2\big|\}^{2t}\min\{\theta_{0,1}^2,\theta_{0,2}^2\}$, and $\max\{|1-\gamma\mu|, \big|1-\gamma R^2\big|\} \geq \frac{\kappa-1}{\kappa+1}$ for every $\gamma > 0$: if $\gamma \leq \frac{2}{\mu+R^2}$ then $1-\gamma\mu \geq 1-\frac{2\mu}{\mu+R^2} = \frac{R^2-\mu}{R^2+\mu}$, and if $\gamma \geq \frac{2}{\mu+R^2}$ then $\gamma R^2 - 1 \geq \frac{2R^2}{\mu+R^2} - 1 = \frac{R^2-\mu}{R^2+\mu}$. For the last claim, $\min\{\theta_{0,1}^2,\theta_{0,2}^2\} = r^2 = \frac{1}{2}\,\|\theta_0-\eta_*\|^2$, so the target forces $(\frac{\kappa-1}{\kappa+1})^{2t} \leq 2\varepsilon$, i.e. $t \geq \log\frac{1}{2\varepsilon}\big/\big(2\log\frac{\kappa+1}{\kappa-1}\big) \geq \frac{\kappa-1}{4}\log\frac{1}{2\varepsilon}$, using $\log\frac{\kappa+1}{\kappa-1} = \log(1+\frac{2}{\kappa-1}) \leq \frac{2}{\kappa-1}$. □

The argument uses only that the SAGA direction is unbiased for $F'(\theta_{t-1})$ and that F' is affine: on any finite sum with a quadratic F (least squares, for instance), the *mean* SAGA iterate is the gradient-descent iterate with the same step, and the variance $\mathbb{E}\,\|\theta_t - \mathbb{E}\theta_t\|^2$ only adds to the error. Hence no analysis of SAGA with a constant step can certify, uniformly over Assumption 1.1, a per-iteration contraction of $\mathbb{E}\,\|\theta_t-\eta_*\|^2$ better than $(\frac{\kappa-1}{\kappa+1})^2 \geq 1-\frac{4}{\kappa+1}$. For $n = 1$ the memory is useless and (1) *is* gradient descent; the proposition shows that the obstruction persists for every n.

Remark 1.18 (conclusion: the gap to practice is not a matter of constants). Within the proof structure of Proposition 5.9 (Lyapunov function $D + \Delta M$, inequalities (I1)–(I5)), optimising (γ, Δ) changes the certified one-step rate by a factor between 1.05 and 3, and by at most 4/3 when $\kappa \geq 9n/16$. On the hard instances of hypothesis (A4) ($\kappa \gg n$), the rate per effective pass certified by any constants is at most $n\tau^\star \leq \frac{1.172\,n}{4\kappa} \ll 1$. Whenever SAGA is observed to contract V_t much faster than ρ_{bound}^t, the discrepancy must therefore be located in the state-dependent ratios $r_{\text{sq}}, r_{\text{coco}}, r_{\text{var}}, r_{\text{sc}}$ (inequalities (I1)–(I4)) and in $r_{\min}$, which is exactly what the chains of Proposition 1.11 measure. Two caveats delimit the claim. (i) Other analyses give constants of the same order, not better ones. [17] give exactly $\tau^\star$ and $\gamma^\star$ (see the paragraph following Proposition 1.16); [7] (Theorem 1, Corollary 1 and the "adaptivity to strong convexity" comment at the end of the proof of Theorem 1) prove, with a Lyapunov function containing the Bregman divergences $f_i(\phi_i) - f_i(x^*) - \langle f_i'(x^*), \phi_i - x^*\rangle$ and assuming every f_i to be μ-strongly convex, the rate $1-\frac{\mu}{2(\mu n+L)}$ for $\gamma = \frac{1}{2(\mu n+L)}$ and the rate $1-\min\{\frac{1}{4n}, \frac{\mu}{3L}\}$ for the μ-free step $\gamma = \frac{1}{3L}$; when strong convexity only holds on average (our Assumption 1.1) they state the rate $1-\frac{\mu}{6(\mu n+L)}$ for $\gamma = \frac{1}{3(\mu n+L)}$; the LMI-based analysis of [18] (Corollary 3, eq. (42)) certifies $1-\min\{\frac{1}{3n}, \frac{\mu}{10L}\}$ for $\gamma = \frac{1}{3L}$ under

assumptions implied by Assumption 1.1: each f_i convex and L-smooth ($f_i \in \mathcal{F}(0,L)$ in their notation) and F in their class $\mathcal{S}(m,L)$ of functions with sector-bounded gradients, which contains the m-strongly convex L-smooth functions but, as they stress, also non-convex ones. All of these have the form $1-\min\{\Theta(\frac{1}{n}),\Theta(\frac{1}{\kappa})\}$. (ii) The exponent 1 of κ is intrinsic to SAGA with a constant step, not to its analysis. Proposition 1.17 shows that on an instance of Assumption 1.1 with n identical quadratic components, for every n and every constant step (whether or not it depends on μ), $\mathbb{E}\,\|\theta_t-\eta_*\|^2$ decays no faster than $(1-\frac{2}{\kappa+1})^{2t}$, so that at least $\frac{\kappa-1}{4}\log\frac{1}{2\varepsilon}$ iterations are needed, against $O((n+\sqrt{n\kappa})\log\frac{1}{\varepsilon})$ for accelerated methods (hypotheses (A2)–(A4)). For $\kappa \gg n$ the certified complexity $\Theta(\kappa\log\frac{1}{\varepsilon})$ is therefore optimal for SAGA up to a constant factor: between the textbook bound, which guarantees the target after $\frac{16\kappa}{3}\log\frac{1+n/4}{\varepsilon}$ iterations, and the lower bound, the factor on the leading term $\kappa\log\frac{1}{\varepsilon}$ is $\frac{64}{3}\approx 21$. For oblivious but non-constant step schedules the same conclusion follows from [4, Theorem 2]: "Suppose that only L the smoothness parameter is known, i.e. $\mathcal{I}(\cdot)=\{L\}$. If the iteration complexity of a given oblivious, possibly stochastic, p-CLI optimization algorithm is $\tilde{O}(\kappa(f)^\alpha\ln(1/\epsilon))$, then $\alpha\geq 1$"; SAGA (with SAG, SVRG and SDCA) is listed there (Section 2.2) as a stationary stochastic p-CLI, and the book's step $\gamma=\frac{1}{4R^2}$, or any schedule depending on R^2 only, is oblivious in their sense (coefficients depending on the side-information $\{L\}$ and on time only). When μ is also known, their Theorem 1 only gives the lower bound $\tilde{\Omega}(\sqrt{\kappa}\ln(1/\epsilon))$ for oblivious schedules, which is what acceleration achieves; this does not contradict Proposition 1.17, which concerns constant steps. Improving κ to $\sqrt{n\kappa}$ therefore requires changing the algorithm (acceleration, hypotheses (A2)–(A3)), consistently with the lower bound of hypothesis (A1). The computer-aided approach of [31] automates exactly this kind of search over Lyapunov functions and constants, with tightness guarantees for the chosen potential class.

1.6 Trajectory-dependent version (hook for hypothesis (B2))

Strong convexity (I4) and co-coercivity (I2) are only used at the current point θ_{t-1}. Define, as functions of $\theta\neq\eta_*$ (they do not depend on the memory),

$$\mu_{\mathrm{eff}}(\theta):=\frac{(\theta-\eta_*)^\top F'(\theta)}{\|\theta-\eta_*\|^2},\qquad L_{\mathrm{eff}}(\theta):=\frac{\frac{1}{n}\sum_{i=1}^n\|f_i'(\theta)-f_i'(\eta_*)\|^2}{(\theta-\eta_*)^\top F'(\theta)},$$

and $\mu_{\mathrm{eff}}(t-1):=\mu_{\mathrm{eff}}(\theta_{t-1})$, $L_{\mathrm{eff}}(t-1):=L_{\mathrm{eff}}(\theta_{t-1})$. By Lemma 1.9(i), $\mu\leq\mu_{\mathrm{eff}}(\theta)\leq L_{\mathrm{eff}}(\theta)\leq R^2$.

Proposition 1.19 (a-posteriori rates)**.** *Let $\gamma=\frac{1}{4R^2}$ and $\Delta=3\gamma^2$. For every $t\geq 1$, almost surely on $\{\theta_{t-1}\neq\eta_*\}$:*

$$\mathbb{E}[V_t\mid\mathcal{F}_{t-1}]\leq\Big(1-\min\Big\{\frac{1}{3n},\ \frac{3\,\mu_{\mathrm{eff}}(t-1)}{16R^2}\Big\}\Big)V_{t-1},\tag{19}$$

$$\mathbb{E}[V_t\mid\mathcal{F}_{t-1}]\leq\Big(1-\min\Big\{\frac{1}{3n},\ \frac{\mu_{\mathrm{eff}}(t-1)}{2R^2}\Big(1-\frac{5\,L_{\mathrm{eff}}(t-1)}{8R^2}\Big)\Big\}\Big)V_{t-1},\tag{20}$$

and the coefficient in (20) satisfies $\frac{3\mu_{\mathrm{eff}}(t-1)}{16R^2}\leq\frac{\mu_{\mathrm{eff}}(t-1)}{2R^2}\big(1-\frac{5L_{\mathrm{eff}}(t-1)}{8R^2}\big)\leq\frac{\mu_{\mathrm{eff}}(t-1)}{2R^2}\leq\frac{1}{2}$, so that both contraction factors lie in $[\frac{1}{2},1)$. More generally, for γ,Δ as in Proposition 1.14,

$$\mathbb{E}[V_t\mid\mathcal{F}_{t-1}]\leq\Big(1-\min\Big\{2\gamma\,\mu_{\mathrm{eff}}(t-1)\Big(1-\gamma L_{\mathrm{eff}}(t-1)-\frac{L_{\mathrm{eff}}(t-1)\Delta}{2\gamma}\Big),\ \frac{1}{n}\Big(1-\frac{2\gamma^2}{\Delta}\Big)\Big\}\Big)V_{t-1}.\tag{21}$$

On $\{\theta_{t-1}=\eta_*\}$ one has $\mathbb{E}[V_t\mid\mathcal{F}_{t-1}]\leq\big(1-\frac{1}{n}+\frac{2\gamma^2}{n\Delta}\big)V_{t-1}$, so that (19)–(21) hold there with any value assigned to $\mu_{\mathrm{eff}}(t-1)$ and $L_{\mathrm{eff}}(t-1)$.

Proof. Apply (I1) and (I3) to (4) and keep (I2) as the identity $S/n = L_{\mathrm{eff}}(t-1)\,G$: with $G = \mu_{\mathrm{eff}}(t-1)D$,

$$\mathbb{E}_{i(t)}[D_t] \le D - 2\gamma G + 2\gamma^2 L_{\mathrm{eff}} G + \frac{2\gamma^2}{n} M, \qquad \mathbb{E}_{i(t)}[M_t] = \Big(1 - \frac{1}{n}\Big) M + L_{\mathrm{eff}} G,$$

hence $\mathbb{E}_{i(t)}[V_t] \le \big[1 - 2\gamma\mu_{\mathrm{eff}}\,\beta\big] D + \big[1 - \frac{1}{n}(1 - \frac{2\gamma^2}{\Delta})\big]\Delta M$ with $\beta := 1 - \gamma L_{\mathrm{eff}} - \frac{L_{\mathrm{eff}}\Delta}{2\gamma}$ (this is (14) with $r_{\mathrm{sq}} = r_{\mathrm{var}} = 1$, and it follows from it since c_D, c_M are nondecreasing in $r_{\mathrm{sq}}, r_{\mathrm{var}}$). Since $0 < L_{\mathrm{eff}} \le R^2$, we have $\alpha(\gamma,\Delta) \le \beta \le 1$, so $2\gamma\mu_{\mathrm{eff}}\beta > 0$; and since $\mu_{\mathrm{eff}} \le L_{\mathrm{eff}}$ (Lemma 1.9(i)) and $\gamma L_{\mathrm{eff}} \le \gamma R^2 < 1$, we have $2\gamma\mu_{\mathrm{eff}}\beta \le 2\gamma L_{\mathrm{eff}}(1 - \gamma L_{\mathrm{eff}}) \le \frac{1}{2}$. Hence the first coefficient lies in $[\frac{1}{2}, 1)$ and the second in $(1 - \frac{1}{n}, 1)$, and the step (I5) gives (21). For the textbook constants, $2\gamma\mu_{\mathrm{eff}} = \frac{\mu_{\mathrm{eff}}}{2R^2}$, $\gamma L_{\mathrm{eff}} + \frac{L_{\mathrm{eff}}\Delta}{2\gamma} = \frac{5L_{\mathrm{eff}}}{8R^2}$ and $\frac{1}{n}(1 - \frac{2\gamma^2}{\Delta}) = \frac{1}{3n}$, which is (20); replacing $L_{\mathrm{eff}}(t-1)$ by its upper bound R^2 (the bracket is decreasing in L_{eff}) gives $\frac{\mu_{\mathrm{eff}}}{2R^2} \cdot \frac{3}{8} = \frac{3\mu_{\mathrm{eff}}}{16R^2}$, which is (19); the chain of inequalities between the coefficients follows from $\mu \le \mu_{\mathrm{eff}} \le R^2$ and $0 \le L_{\mathrm{eff}} \le R^2$. On $\{\theta_{t-1} = \eta_*\}$: $S = G = D = 0$, $\omega_t = b_{i(t)}$, so $\mathbb{E}_{i(t)}[D_t] = \gamma^2 B \le \gamma^2 M/n$ by (I3) and $\mathbb{E}_{i(t)}[M_t] = (1 - \frac{1}{n})M$; thus $\mathbb{E}_{i(t)}[V_t] \le (1 - \frac{1}{n} + \frac{\gamma^2}{n\Delta})\Delta M \le (1 - \frac{1}{n} + \frac{2\gamma^2}{n\Delta})V_{t-1}$. □

Remark 1.20 (the step is fixed in advance). One may not replace R^2 by $L_{\mathrm{eff}}(t-1)$ inside γ: the step was chosen before the trajectory was observed, and $\Delta = 3\gamma^2$ with it. With $\gamma = \frac{1}{4R^2}$ fixed, a smaller effective smoothness $L_{\mathrm{eff}} < R^2$ enters only through the bracket $1 - \frac{5L_{\mathrm{eff}}}{8R^2} \in [\frac{3}{8}, 1)$: the gain from (I2) being loose is at most a factor $\frac{8}{3}$ on the parameter rate. By contrast $\mu_{\mathrm{eff}}(t-1)$ enters linearly and may exceed μ by orders of magnitude (for logistic regression, $\mu = \lambda$ while μ_{eff} is of the order of the curvature of the data term), which is why hypothesis (B2) focuses on μ_{eff}. A step adapted to L_{eff} would be a different algorithm, and Δ would have to follow γ; see also the discussion of step-size selection in [14, Sec. 3.A].

A posteriori versus a priori. The factors in Proposition 1.19 are random and $\mathcal{F}_{t-1}$-measurable: writing $\rho_{\mathrm{eff}}(t)$ for any of them, we have $\mathbb{E}[V_t \mid \mathcal{F}_{t-1}] \le \rho_{\mathrm{eff}}(t) V_{t-1}$, but *not* $\mathbb{E}[V_T] \le \mathbb{E}\big[\prod_{t \le T} \rho_{\mathrm{eff}}(t)\big] V_0$ in general: $\rho_{\mathrm{eff}}(t)$ and V_{t-1} are correlated, and $\rho_{\mathrm{eff}}(t)$ is not $\mathcal{F}_{t-2}$-measurable, so the conditional expectations cannot be iterated. The experiments can still report the realised products $\prod_{t \le T} \rho_{\mathrm{eff}}(t)$ (and $\prod_t \rho_{\mathrm{exact}}(t)$) next to the realised V_T/V_0; the difference is a martingale-type fluctuation. To obtain a rigorous a-priori statement one needs a deterministic bound on the effective constants over a region.

Proposition 1.21 (a-priori rate on a region)**.** *Let $\Theta \subset \mathbb{R}^d$ and*

$$\mu_\Theta := \inf_{\theta \in \Theta \setminus \{\eta_*\}} \mu_{\mathrm{eff}}(\theta) \;\ge \mu, \qquad L_\Theta := \sup_{\theta \in \Theta \setminus \{\eta_*\}} L_{\mathrm{eff}}(\theta) \;\le R^2,$$
$$\rho_\Theta := 1 - \min\Big\{\frac{1}{3n},\; \frac{\mu_\Theta}{2R^2}\Big(1 - \frac{5L_\Theta}{8R^2}\Big)\Big\} \;\le\; 1 - \min\Big\{\frac{1}{3n}, \frac{3\mu_\Theta}{16R^2}\Big\},$$

and let $\tau := \inf\{t \ge 0 : \theta_t \notin \Theta\}$ (with $\inf \emptyset = +\infty$). For $\gamma = \frac{1}{4R^2}$, $\Delta = 3\gamma^2$ and every $T \ge 0$,

$$\mathbb{E}\big[V_T\, \mathbf{1}_{\{\tau > T\}}\big] \le \rho_\Theta^T\, V_0. \tag{22}$$

In particular, if $\theta_t \in \Theta$ for all t almost surely, then $\mathbb{E}[V_T] \le \rho_\Theta^T V_0$ and $\mathbb{E}\,\|\theta_T - \eta_*\|^2 \le \rho_\Theta^T(1 + \frac{n}{4})\,\|\theta_0 - \eta_*\|^2$ for the book's initialisation.

Proof. Since $\theta_0, \dots, \theta_t$ are $\mathcal{F}_t$-measurable, $\{\tau > t\} \in \mathcal{F}_t$, and τ is a stopping time. Let $W_t := \rho_\Theta^{-(t\wedge\tau)} V_{t\wedge\tau}$. On $\{\tau \le t\}$, $W_{t+1} = W_t$. On $\{\tau > t\}$, $\theta_t \in \Theta$, so Proposition 1.19 (inequality (20) at time $t+1$, whose coefficient is nondecreasing in μ_{eff} and nonincreasing in L_{eff}, together with the case $\theta_t = \eta_*$) gives $\mathbb{E}[V_{t+1} \mid \mathcal{F}_t] \le \rho_\Theta V_t$, hence $\mathbb{E}[W_{t+1}\mathbf{1}_{\{\tau>t\}} \mid \mathcal{F}_t] = \mathbf{1}_{\{\tau>t\}}\rho_\Theta^{-(t+1)}\mathbb{E}[V_{t+1} \mid \mathcal{F}_t] \le \mathbf{1}_{\{\tau>t\}}\rho_\Theta^{-t} V_t = \mathbf{1}_{\{\tau>t\}} W_t$. Thus (W_t) is a supermartingale (all variables are integrable since $i(1), \dots, i(T)$ take finitely many values), $\mathbb{E}[W_T] \le W_0 = V_0$, and $W_T \ge \mathbf{1}_{\{\tau>T\}}\rho_\Theta^{-T} V_T$ gives (22). The last claim uses $\tau = \infty$ a.s., $D_T \le V_T$ and (I6). □

For ℓ_2-regularised logistic regression the infimum μ_Θ over a ball can be bounded below explicitly, thanks to the self-concordance-type property of the logistic loss that underlies the adaptivity results of [5] (Assumption (A2) there: along any segment, $|\varphi'''| \le R\,\|\theta_1 - \theta_2\|\,\varphi''$).

Proposition 1.22 (local strong convexity of logistic regression along segments)**.** *Let $f_i(\theta) = \ell(y_i x_i^\top \theta) + \frac{\lambda}{2}\|\theta\|^2$ as in Remark 1.7, $\lambda > 0$, and $R_x := \max_i \|x_i\|$. Then for every $\theta \ne \eta_*$,*

$$\mu_{\text{eff}}(\theta) \;\ge\; \psi\big(R_x\,\|\theta - \eta_*\|\,\big)\,\lambda_{\min}\big(F''(\eta_*)\big), \qquad \psi(x) := \frac{1 - e^{-x}}{x}\ (\textit{decreasing},\ \psi(0^+) = 1),$$

where $F''(\eta_*) = \lambda I + \frac{1}{n}\sum_i \ell''(y_i x_i^\top \eta_*) x_i x_i^\top \succeq \lambda I$. Consequently, on the ball $\Theta_r = \{\theta : \|\theta - \eta_*\| \le r\}$, $\mu_{\Theta_r} \ge \psi(R_x r)\lambda_{\min}(F''(\eta_*))$, and $\mu_{\Theta_r} \ge \frac{1}{2}\lambda_{\min}(F''(\eta_*))$ as soon as $R_x r \le \frac{3}{2}$ (since $\psi(3/2) \approx 0.518$).

Proof. Let $u = \theta - \eta_*$ and $\varphi(s) := F(\eta_* + su)$, $s \in [0,1]$. With $m_i(s) = y_i x_i^\top(\eta_* + su)$ and $y_i^2 = 1$,

$$\varphi''(s) = \lambda\,\|u\|^2 + \frac{1}{n}\sum_{i=1}^n \ell''(m_i(s))(x_i^\top u)^2 > 0, \qquad \varphi'''(s) = \frac{1}{n}\sum_{i=1}^n \ell'''(m_i(s))\,y_i (x_i^\top u)^3.$$

Since $\ell'''(u) = \ell''(u)(1 - 2\sigma(u))$, $|\ell'''| \le \ell''$, hence

$$\big|\varphi'''(s)\big| \le \frac{1}{n}\sum_i \ell''(m_i(s))\,\Big|x_i^\top u\Big|^3 \le R_x\,\|u\| \cdot \frac{1}{n}\sum_i \ell''(m_i(s))(x_i^\top u)^2 \le R_x\,\|u\|\ \varphi''(s).$$

Therefore $(\log\varphi'')'(s) \ge -R_x\,\|u\|$ and $\varphi''(s) \ge \varphi''(0) e^{-R_x\|u\|s}$ for $s \in [0,1]$. Now

$$G = \big\langle u, F'(\eta_* + u) - F'(\eta_*)\big\rangle = \int_0^1 \big\langle u, F''(\eta_* + su)\,u\big\rangle\,ds = \int_0^1 \varphi''(s)\,ds \;\ge\; \varphi''(0)\,\psi(R_x\,\|u\|),$$

and $\varphi''(0) = u^\top F''(\eta_*) u \ge \lambda_{\min}(F''(\eta_*))\,\|u\|^2$. Divide by $D = \|u\|^2$. The function ψ is decreasing on $(0,\infty)$ (its derivative is $\frac{(1+x)e^{-x} - 1}{x^2} \le 0$ because $e^x \ge 1 + x$), which gives the statement on balls. □

Combining Propositions 1.21 and 1.22: as long as the SAGA iterates stay within distance $\frac{3}{2R_x}$ of η_*, the Lyapunov function contracts in expectation at the rate $1 - \min\{\frac{1}{3n}, \frac{3\lambda_{\min}(F''(\eta_*))}{32R^2}\}$ instead of $1 - \min\{\frac{1}{3n}, \frac{3\lambda}{16R^2}\}$, i.e. the global constant λ is replaced by half the smallest eigenvalue of the Hessian at the optimum, which is the quantity to which averaged SGD is shown to adapt in [5]. The experiments of (B2) measure $\mu_{\text{eff}}(t)$ directly and compare it with λ, with $\lambda_{\min}(F''(\eta_*))$ and with the bound of Proposition 1.22.

1.7 What to measure

At a measurement time (typically once per effective pass, i.e. every n iterations, so that the $O(nd)$ cost of a full gradient pass does not dominate), the state is $(\theta, z) = (\theta_{t-1}, z^{(t-1)})$. The code needs η_* (computed beforehand to a precision far below the smallest V_t that will be reported, e.g. by a second-order method or by SAGA itself run to machine precision), the stored vectors $g_i^* := f_i'(\eta_*)$ ($n \times d$, computed once), and, at each measurement, one pass computing $g_i := f_i'(\theta)$ for all i. For logistic regression, $f_i'(\theta) = -y_i\sigma(-y_i x_i^\top \theta)x_i + \lambda\theta$, $R^2 = \max_i(\|x_i\|^2/4 + \lambda)$, $\mu = \lambda$, $R_x = \max_i \|x_i\|$.

Ineq.	Ratio	Closed form	State quantities	Code
(I1)	r_{sq}	$\Omega/(2S/n + 2B) = \frac{1}{2} + C/(S/n + B)$	Ω, S, B (or C)	$a_i = g_i - g_i^*$, $b_i = g_i^* - z_i + \bar{z}$, $\Omega = \frac{1}{n}\sum_i \|a_i + b_i\|^2$, $C = \frac{1}{n}\sum_i \langle a_i, b_i\rangle$
(I2)	$r_{\mathrm{coco}} = L_{\mathrm{eff}}/R^2$	$(S/n)/(R^2 G)$	S, G	$S = \sum_i \|a_i\|^2$, $F'(\theta) = \frac{1}{n}\sum_i g_i$, $G = (\theta - \eta_*)^\top F'(\theta)$
(I3)	r_{var}	$B/(M/n) = 1 - n\|\bar{z}\|^2/M$	$B, M, \bar{z}$	$M = \sum_i \|z_i - g_i^*\|^2$; compute $B = \frac{1}{n}\sum_i \|b_i\|^2$ directly (avoids cancellation)
(I4)	$r_{\mathrm{sc}} = \mu/\mu_{\mathrm{eff}}$	$\mu D/G$	D, G	$D = \|\theta - \eta_*\|^2$; also report $\mu_{\mathrm{eff}} = G/D$ (B2)
(I5)	$r_{\min}$	Definition 1.8	D, M	needs Δ, n, μ, R^2; equals $\rho_1^B/\rho_{\mathrm{bound}}$
(I6)	r_{init}	$S_0/(nR^4 D_0)$	S_0, D_0	at $t = 0$ only
exact	$\rho_{\mathrm{exact}}(t)$	(8)	D, M, G, S, Ω	compare with the realised V_t/V_{t-1} and with ρ_{bound}
chains	ρ_k^A, ρ_k^B	Proposition 1.11	all of the above	report δ_k, λ_k and $n\lambda_k$ (per pass) for both orders
(B2)	$\rho_{\mathrm{eff}}(t)$	(19), (20)	$\mu_{\mathrm{eff}}, L_{\mathrm{eff}}$	compare $\mu_{\mathrm{eff}}(t)$ with λ, with $\lambda_{\min}(F''(\eta_*))$ and with Proposition 1.22

Practical remarks. (a) The four ratios of (I1)–(I4) are independent of (γ, Δ); the same measured state can therefore be re-used to evaluate the chains of Proposition 1.11 for any (γ, Δ), e.g. the textbook constants and the optimal ones of Proposition 1.16. (b) Stop measuring when D or M reaches the precision of η_* (the ratios become numerical noise). (c) Expect $r_{\mathrm{sq}} \lesssim \frac{1}{2}$ (Remark 1.13), r_{var} close to 1 once the memory errors have no common direction, $r_{\mathrm{coco}} = L_{\mathrm{eff}}/R^2$ well below 1 (the per-sample curvature along $\theta - \eta_*$ is typically much smaller than the maximal one), and $r_{\mathrm{sc}} = \lambda/\mu_{\mathrm{eff}}$ very small when λ is small compared with the curvature of the data term. With (γ, Δ) fixed to the textbook values, (15) shows that r_{sq}, r_{var} and r_{coco} each change the corresponding certified rate by a bounded factor only: the parameter rate $1 - c_D = \frac{\mu_{\mathrm{eff}}}{2R^2}\big(1 - r_{\mathrm{coco}}(\frac{r_{\mathrm{sq}}}{4} + \frac{3}{8})\big)$ varies by at most a factor $\frac{8}{3}$ with r_{coco} and by at most $\frac{5}{3}$ with r_{sq} (the bracket lies in $[\frac{3}{8}, 1)$, Remark 1.20), and the memory rate $1 - c_M = \frac{1}{n}(1 - \frac{2}{3}r_{\mathrm{sq}}r_{\mathrm{var}})$ varies by at most a factor 3 with $r_{\mathrm{sq}}r_{\mathrm{var}}$. Only r_{sc} (through μ_{eff}, which is unbounded relative to μ) and $r_{\min}$ (when the Lyapunov mass sits on the faster component: for $\kappa \geq 9n/16$ and $\Delta M \gg D$, $(1 - \rho_1^B)/(1 - \rho_{\mathrm{bound}}) \approx \frac{1/(3n)}{3/(16\kappa)} = \frac{16\kappa}{9n}$, e.g. ≈ 1778 for $n = 10$, $\kappa = 10^4$) can explain a gap larger than a constant factor.

2 Effective curvature and effective smoothness along the trajectory (hypothesis B2)

The proof of Proposition 5.9 of [6] (convergence of SAGA, transcribed in the B3 note) uses exactly two problem constants. The per-sample smoothness constant R^2 enters through the co-coercivity

inequality (equation (5.31) of the book),

$$\frac{1}{n}\sum_{i=1}^{n}\left\|f_i'(\theta_{t-1})-f_i'(\eta_*)\right\|^2 \le R^2\,F'(\theta_{t-1})^\top(\theta_{t-1}-\eta_*),$$

and the strong-convexity constant μ enters through $(\theta_{t-1}-\eta_*)^\top F'(\theta_{t-1}) \ge \mu\,\|\theta_{t-1}-\eta_*\|^2$. Both inequalities are applied at the *current iterate* θ_{t-1}, always with the same second point η_*. Hypothesis (B2) of the project is that the slack of these two inequalities along the trajectory actually followed by the algorithm explains a large part of the gap between the rate $1-\min\{1/(3n), 3\mu/(16R^2)\}$ and the observed contraction. This note defines the two trajectory-dependent constants that make the two inequalities equalities, proves their basic properties, makes them explicit for ℓ_2-regularized logistic regression, analyses their dependence on the direction of the error $\theta_t-\eta_*$ (which differs between gradient descent and SAGA, and which is constrained by an exact invariance property: a trajectory started in the span of the features never leaves it, Lemma 2.16), and finally explains how they enter the SAGA bound and how to measure the fraction of the theory–practice gap they explain.

2.1 Setting and definitions

Assumption 2.1 (standing assumptions of this note)**.** *(i) Each $f_i:\mathbb{R}^d\to\mathbb{R}$ is convex, differentiable and R^2-smooth, i.e. f_i' is R^2-Lipschitz-continuous.*

(ii) $F=\frac{1}{n}\sum_{i=1}^n f_i$ is μ-strongly convex with $\mu>0$; its (unique) minimizer is η_*, so that $F'(\eta_*)=0$ (Proposition 5.2 of [6]). We denote by $L\le R^2$ the smoothness constant of F (F' is L-Lipschitz-continuous; $L\le R^2$ because an average of R^2-Lipschitz maps is R^2-Lipschitz), and by $\kappa=R^2/\mu$ the condition number used in the book.

(iii) Whenever second derivatives appear (Lemma 2.5, Proposition 2.8, Section 2.3 and after), each f_i is twice continuously differentiable.

Throughout, for $\theta\neq\eta_*$ we write

$$v=\theta-\eta_*,\qquad u=v/\|v\|,$$

and we use repeatedly that $F'(\theta)=F'(\theta)-F'(\eta_*)=\frac{1}{n}\sum_i[f_i'(\theta)-f_i'(\eta_*)]$.

Definition 2.2 (effective strong convexity and effective co-coercivity constants)**.** *For $\theta\neq\eta_*$, define*

$$\mu_{\mathrm{eff}}(\theta) \;=\; \frac{(\theta-\eta_*)^\top F'(\theta)}{\|\theta-\eta_*\|^2}, \tag{23}$$

$$L_{\mathrm{eff}}(\theta) \;=\; \frac{\frac{1}{n}\sum_{i=1}^n\|f_i'(\theta)-f_i'(\eta_*)\|^2}{(\theta-\eta_*)^\top F'(\theta)}. \tag{24}$$

Along a trajectory $(\theta_t)_{t\ge0}$ we abbreviate $\mu_{\mathrm{eff}}(t)=\mu_{\mathrm{eff}}(\theta_t)$ and $L_{\mathrm{eff}}(t)=L_{\mathrm{eff}}(\theta_t)$.

Lemma 2.4 below shows that $(\theta-\eta_*)^\top F'(\theta)\ge\mu\,\|\theta-\eta_*\|^2>0$, so both quantities are well defined and positive. By construction the two inequalities quoted above become *identities* when μ and R^2 are replaced by $\mu_{\mathrm{eff}}(\theta)$ and $L_{\mathrm{eff}}(\theta)$:

$$(\theta-\eta_*)^\top F'(\theta)=\mu_{\mathrm{eff}}(\theta)\,\|\theta-\eta_*\|^2,\qquad \frac{1}{n}\sum_{i=1}^n\left\|f_i'(\theta)-f_i'(\eta_*)\right\|^2=L_{\mathrm{eff}}(\theta)\,(\theta-\eta_*)^\top F'(\theta).$$

We call $\mu_{\mathrm{eff}}(\theta)$ the *effective curvature* and $L_{\mathrm{eff}}(\theta)$ the *effective co-coercivity (or effective smoothness) constant* at θ.

2.2 General properties

We first record the two-point inequalities that we use, with proofs, since the whole note rests on them.

Lemma 2.3 (two-point inequalities)**.** *Let $f : \mathbb{R}^d \to \mathbb{R}$ be convex and differentiable.*

(a) (Monotonicity) For all θ, η: $(f'(\theta) - f'(\eta))^\top (\theta - \eta) \geq 0$.

(b) (Strong monotonicity) If f is μ-strongly convex, then for all θ, η: $(f'(\theta) - f'(\eta))^\top (\theta - \eta) \geq \mu \|\theta - \eta\|^2$.

(c) (Co-coercivity, Proposition 5.4 of [6]) If f is L_f-smooth, then for all θ, η: $\|f'(\theta) - f'(\eta)\|^2 \leq L_f (f'(\theta) - f'(\eta))^\top (\theta - \eta)$, and consequently $(f'(\theta) - f'(\eta))^\top (\theta - \eta) \leq L_f \|\theta - \eta\|^2$.

Proof. (b) By definition of μ-strong convexity (Definition 5.2 of the book, with $\mu = 0$ for plain convexity, which gives (a)), $f(\eta) \geq f(\theta) + f'(\theta)^\top (\eta - \theta) + \frac{\mu}{2} \|\eta - \theta\|^2$ and $f(\theta) \geq f(\eta) + f'(\eta)^\top (\theta - \eta) + \frac{\mu}{2} \|\eta - \theta\|^2$. Summing the two inequalities and cancelling $f(\theta) + f(\eta)$ gives $0 \geq -(f'(\theta) - f'(\eta))^\top (\theta - \eta) + \mu \|\theta - \eta\|^2$, which is the claim.

(c) Fix θ and let $\varphi(\eta) = f(\eta) - f'(\theta)^\top \eta$. The function φ is convex, L_f-smooth, and $\varphi'(\theta) = 0$, so θ minimizes φ. The descent lemma for L_f-smooth functions (Definition 5.3 of the book: $\varphi(\eta') \leq \varphi(\eta) + \varphi'(\eta)^\top (\eta' - \eta) + \frac{L_f}{2} \|\eta' - \eta\|^2$) applied with $\eta' = \eta - \frac{1}{L_f} \varphi'(\eta)$ gives $\varphi(\theta) \leq \varphi(\eta') \leq \varphi(\eta) - \frac{1}{2L_f} \|\varphi'(\eta)\|^2$, that is,

$$f(\theta) \;\leq\; f(\eta) + f'(\theta)^\top (\theta - \eta) - \frac{1}{2L_f} \left\| f'(\eta) - f'(\theta) \right\|^2 .$$

This is equation (5.12) of the book. Writing it also with θ and η exchanged and adding the two inequalities yields $0 \leq (f'(\theta) - f'(\eta))^\top (\theta - \eta) - \frac{1}{L_f} \|f'(\theta) - f'(\eta)\|^2$, the co-coercivity inequality. The consequence follows from the Cauchy–Schwarz inequality $(f'(\theta) - f'(\eta))^\top (\theta - \eta) \leq \|f'(\theta) - f'(\eta)\| \, \|\theta - \eta\|$ together with the Lipschitz property $\|f'(\theta) - f'(\eta)\| \leq L_f \|\theta - \eta\|$. □

Lemma 2.4 (the effective curvature lies between μ and L)**.** *Under Assumption 2.1, for all $\theta \neq \eta_*$,*

$$\mu \|\theta - \eta_*\|^2 \;\leq\; (\theta - \eta_*)^\top F'(\theta) \qquad \textit{(strong convexity at the iterate; i.e. } \mu \leq \mu_{\text{eff}}(\theta)\textit{)}, \tag{25}$$

$$(\theta - \eta_*)^\top F'(\theta) \;\leq\; L \|\theta - \eta_*\|^2 \qquad \textit{(smoothness at the iterate; i.e. } \mu_{\text{eff}}(\theta) \leq L\textit{)}. \tag{26}$$

Proof. Apply Lemma 2.3(b) and (c) to $f = F$ with $\eta = \eta_*$ and use $F'(\eta_*) = 0$. □

The tightness ratio of (25) at θ is exactly $\mu / \mu_{\text{eff}}(\theta) \in [\mu/L, 1]$; this is the quantity the experiments of the B3 note record under the name r_{sc} (Definition 1.8 of that note, which likewise calls $r_{\text{coco}} = L_{\text{eff}}/R^2$ the tightness ratio of the co-coercivity step, see Lemma 2.6); we use the same two names below.

Lemma 2.5 (integral form of the effective curvature)**.** *Assume F is twice continuously differentiable and let, for $\theta \neq \eta_*$,*

$$\bar{H}(\theta) \;=\; \int_0^1 F''(\eta_* + s\, v)\, ds \qquad (v = \theta - \eta_*).$$

Then $\bar{H}(\theta)$ is symmetric, $\mu I \preccurlyeq \bar{H}(\theta) \preccurlyeq L I$, $F'(\theta) = \bar{H}(\theta)\, v$, and

$$\mu_{\text{eff}}(\theta) \;=\; \frac{v^\top \bar{H}(\theta)\, v}{\|v\|^2} \;=\; u^\top \bar{H}(\theta)\, u, \tag{27}$$

i.e. $\mu_{\text{eff}}(\theta)$ *is the Rayleigh quotient, in the direction of the error, of the Hessian averaged along the segment* $[\eta_*, \theta]$.

Proof. Let $g(s) = F'(\eta_* + sv)$ for $s \in [0,1]$. Since F is C^2, g is C^1 with $g'(s) = F''(\eta_* + sv)\, v$, and the fundamental theorem of calculus gives $F'(\theta) - F'(\eta_*) = g(1) - g(0) = \int_0^1 F''(\eta_* + sv) v \, ds = \bar{H}(\theta) v$; as $F'(\eta_*) = 0$, this is $F'(\theta) = \bar{H}(\theta) v$. The integral of a continuous function with values in the set $\{A \text{ symmetric} : \mu I \preccurlyeq A \preccurlyeq LI\}$ (a closed convex set, since it is defined by linear inequalities $\mu \|w\|^2 \le w^\top A w \le L \|w\|^2$ for all w) stays in that set, because Riemann sums are convex combinations; and for a twice differentiable function, μ-strong convexity and L-smoothness are equivalent to $\mu I \preccurlyeq F''(\theta) \preccurlyeq LI$ for all θ (Section 5.2.3 of [6]). Finally $\mu_{\text{eff}}(\theta) = v^\top F'(\theta) / \|v\|^2 = v^\top \bar{H}(\theta) v / \|v\|^2$. □

Lemma 2.6 (the effective co-coercivity constant lies between μ_{eff} and R^2)**.** *Under Assumption 2.1, for all* $\theta \neq \eta_*$*, with* $v = \theta - \eta_*$*,*

$$\left\|F'(\theta)\right\|^2 \;\le\; \frac{1}{n}\sum_{i=1}^{n} \left\|f_i'(\theta) - f_i'(\eta_*)\right\|^2 \qquad \textit{(Jensen / gradient diversity)}, \tag{28}$$

$$\frac{1}{n}\sum_{i=1}^{n} \left\|f_i'(\theta) - f_i'(\eta_*)\right\|^2 \;\le\; R^2\, v^\top F'(\theta) \qquad \textit{(averaged co-coercivity, eq. (5.31) of the book)}, \tag{29}$$

$$\mu \;\le\; \mu_{\text{eff}}(\theta) \;\le\; \frac{\|F'(\theta)\|}{\|v\|} \;\le\; \frac{\|F'(\theta)\|^2}{v^\top F'(\theta)}. \tag{30}$$

Consequently

$$\mu \;\le\; \mu_{\text{eff}}(\theta) \;\le\; \frac{\|F'(\theta)\|^2}{v^\top F'(\theta)} \;\le\; L_{\text{eff}}(\theta) \;\le\; R^2, \tag{31}$$

and, more precisely, with $a_i = f_i'(\theta) - f_i'(\eta_*)$ *and* $\bar{a} = \frac{1}{n}\sum_i a_i = F'(\theta)$*,*

$$L_{\text{eff}}(\theta) \;=\; \frac{\|F'(\theta)\|^2 + \frac{1}{n}\sum_{i=1}^{n} \|a_i - \bar{a}\|^2}{v^\top F'(\theta)}. \tag{32}$$

If moreover each f_i *is* L_i*-smooth, then* $L_{\text{eff}}(\theta) \le \max_i L_i$ *(so one may take* $R^2 = \max_i L_i$*).*

Proof. (32) is the variance decomposition $\frac{1}{n}\sum_i \|a_i\|^2 = \|\bar{a}\|^2 + \frac{1}{n}\sum_i \|a_i - \bar{a}\|^2$ (expand the squares and use $\sum_i (a_i - \bar{a}) = 0$), divided by $v^\top F'(\theta) > 0$; dropping the nonnegative variance term gives (28). For (29), apply Lemma 2.3(c) to each f_i (convex, R^2-smooth) with $\eta = \eta_*$: $\|a_i\|^2 \le R^2 a_i^\top v$; averaging over i gives $\frac{1}{n}\sum_i \|a_i\|^2 \le R^2 \bar{a}^\top v = R^2 v^\top F'(\theta)$. If f_i is L_i-smooth, the same argument gives $\frac{1}{n}\sum_i \|a_i\|^2 \le \frac{1}{n}\sum_i L_i a_i^\top v \le (\max_i L_i)\, \bar{a}^\top v$, using $a_i^\top v \ge 0$ (Lemma 2.3(a)). For (30): the first inequality is Lemma 2.4; the second is Cauchy–Schwarz, $v^\top F'(\theta) \le \|v\|\, \|F'(\theta)\|$, divided by $\|v\|^2$; the third is the same Cauchy–Schwarz inequality in the form $\|F'(\theta)\| / \|v\| \le \|F'(\theta)\|^2 / (v^\top F'(\theta))$. Combining (28)–(30) with the definition of L_{eff} gives (31). □

Remark 2.7. The chain (31) shows that the "effective condition number" $L_{\text{eff}}(\theta)/\mu_{\text{eff}}(\theta)$ is always at least 1, and that the two sources of slack in the co-coercivity step are separated: (28) measures the *diversity* of the individual gradient differences (its tightness ratio $\|F'(\theta)\|^2 / \frac{1}{n}\sum_i \|a_i\|^2$ equals 1 iff all a_i coincide), while the remaining factor $\|F'(\theta)\|^2 / (v^\top F'(\theta)) \le L_{\text{eff}}(\theta) \le R^2$ measures how much of the bound R^2 is actually used by the full gradient in the direction of the error. Both ratios are cheap to compute once η_* is known (one pass over the data).

Proposition 2.8 (limits at the optimum)**.** *Assume each f_i is twice continuously differentiable, and write $H_* = F''(\eta_*)$ and $M_* = \frac{1}{n}\sum_{i=1}^n f_i''(\eta_*)^2$. For a unit vector u define*

$$\Lambda(u) \;=\; \frac{u^\top M_* u}{u^\top H_* u}.$$

Then, as $\theta \to \eta_*$ (with $u = (\theta-\eta_*)/\|\theta-\eta_*\|$ allowed to vary),

$$\mu_{\mathrm{eff}}(\theta) - u^\top H_* u \;\to\; 0, \qquad L_{\mathrm{eff}}(\theta) - \Lambda(u) \;\to\; 0,$$

uniformly in the direction. In particular, if $\theta \to \eta_*$ along a fixed direction u, then $\mu_{\mathrm{eff}}(\theta) \to u^\top H_* u$ (the Rayleigh quotient of the Hessian at the optimum in the direction of approach) and $L_{\mathrm{eff}}(\theta) \to \Lambda(u)$. Moreover, for every unit u,

$$u^\top H_* u \;\le\; \frac{u^\top H_*^2 u}{u^\top H_* u} \;\le\; \Lambda(u) \;\le\; R^2, \tag{33}$$

consistently with (31)*.*

Proof. Let $r = \|v\|$ and $\bar H_i(\theta) = \int_0^1 f_i''(\eta_* + sv)\,ds$, so that, exactly as in Lemma 2.5, $f_i'(\theta) - f_i'(\eta_*) = \bar H_i(\theta)v$, $0 \preccurlyeq \bar H_i(\theta) \preccurlyeq R^2 I$, and $\bar H(\theta) = \frac{1}{n}\sum_i \bar H_i(\theta)$. Set $\varepsilon_i(r) = \sup_{\|w\|\le r}\|f_i''(\eta_*+w) - f_i''(\eta_*)\|_{\mathrm{op}}$ and $\varepsilon(r) = \frac{1}{n}\sum_i \varepsilon_i(r)$; by continuity of f_i'' at η_*, $\varepsilon(r) \to 0$ as $r \to 0$. Since every point $\eta_* + sv$ of the segment is within distance r of η_*, $\big\|\bar H_i(\theta) - f_i''(\eta_*)\big\|_{\mathrm{op}} \le \varepsilon_i(r)$, hence $\big\|\bar H(\theta) - H_*\big\|_{\mathrm{op}} \le \varepsilon(r)$ and, by (27), $\big|\mu_{\mathrm{eff}}(\theta) - u^\top H_* u\big| = \big|u^\top(\bar H(\theta) - H_*)u\big| \le \varepsilon(r)$, which proves the first limit.

For the second, write $L_{\mathrm{eff}}(\theta) = A(\theta)/B(\theta)$ with $A(\theta) = \frac{1}{r^2}\cdot\frac{1}{n}\sum_i \big\|\bar H_i(\theta)v\big\|^2 = \frac{1}{n}\sum_i u^\top \bar H_i(\theta)^2 u$ and $B(\theta) = \frac{1}{r^2}v^\top F'(\theta) = u^\top \bar H(\theta)u = \mu_{\mathrm{eff}}(\theta)$. For symmetric matrices $0 \preccurlyeq P, Q \preccurlyeq R^2 I$ one has $P^2 - Q^2 = P(P-Q) + (P-Q)Q$, so $\big\|P^2 - Q^2\big\|_{\mathrm{op}} \le 2R^2\|P-Q\|_{\mathrm{op}}$; therefore $\big|A(\theta) - u^\top M_* u\big| \le 2R^2\varepsilon(r)$, while $\big|B(\theta) - u^\top H_* u\big| \le \varepsilon(r)$. Both denominators are at least μ (Lemma 2.4 and $H_* \succcurlyeq \mu I$) and both numerators are at most R^4, so

$$|L_{\mathrm{eff}}(\theta) - \Lambda(u)| = \left|\frac{A}{B} - \frac{A_*}{B_*}\right| \le \frac{|A - A_*|}{B} + A_*\frac{|B - B_*|}{BB_*} \le \frac{2R^2\varepsilon(r)}{\mu} + \frac{R^4\varepsilon(r)}{\mu^2} \;\xrightarrow[r\to 0]{}\; 0,$$

uniformly in u (here $A_* = u^\top M_* u$, $B_* = u^\top H_* u$).

For (33): with $A_i = f_i''(\eta_*)$ and $\bar A = H_*$, the matrix $\frac{1}{n}\sum_i (A_i - \bar A)^2 = \frac{1}{n}\sum_i A_i^2 - \bar A^2 = M_* - H_*^2$ is positive semidefinite (each $(A_i-\bar A)^2 = (A_i-\bar A)^\top(A_i-\bar A) \succcurlyeq 0$), which gives the middle inequality; the first one is Cauchy–Schwarz, $(u^\top H_* u)^2 \le \|H_* u\|^2\,\|u\|^2 = u^\top H_*^2 u$; the last one uses $A_i^2 \preccurlyeq R^2 A_i$ for $0 \preccurlyeq A_i \preccurlyeq R^2 I$ (indeed $R^2 A_i - A_i^2 = A_i^{1/2}(R^2 I - A_i)A_i^{1/2} \succcurlyeq 0$), so that $M_* \preccurlyeq R^2 H_*$ and $\Lambda(u) \le R^2$. □

Remark 2.9. Proposition 2.8 says that, close to the optimum, μ_{eff} and L_{eff} are not numbers but functions of the direction of the error. This is the reason why the direction in which an algorithm approaches η_* (Section 2.4) matters for the observed contraction rate. The "local strong convexity constant" $\lambda_{\min}(H_*)$ of [5] is the worst case of $u^\top H_* u$ over directions, and the quantity $u^\top H_*^2 u/(u^\top H_* u)$ in (33) is the Rayleigh-type quotient that governs gradient descent on a quadratic.

2.3 Logistic regression

Setting and constants. Let $\ell(a) = \log(1 + e^{-a})$ and $\sigma(a) = 1/(1 + e^{-a})$. Then

$$\ell'(a) = \sigma(a) - 1 = -\sigma(-a), \qquad \ell''(a) = \sigma(a)\sigma(-a) = \sigma(a)\big(1 - \sigma(a)\big) = \sigma'(a) \in (0, \tfrac{1}{4}],$$

$$\ell'''(a) = \ell''(a)\big(1 - 2\sigma(a)\big),$$

so that ℓ'' is even, maximal at 0 with $\ell''(0) = 1/4$, and non-increasing on $[0, \infty)$ (because $1-2\sigma(a) \le 0$ for $a \ge 0$). The data are $(x_i, y_i) \in \mathbb{R}^d \times \{-1, 1\}$, $i = 1, \dots, n$; we write $R_x = \max_i \|x_i\|$, $\hat{\Sigma} = \frac{1}{n}\sum_i x_i x_i^\top$ (empirical second-moment matrix of the features), $a_i(\theta) = y_i x_i^\top \theta$, and

$$f_i(\theta) = \ell(a_i(\theta)) + \frac{\lambda}{2}\|\theta\|^2, \qquad f_i'(\theta) = \ell'(a_i(\theta))\, y_i x_i + \lambda\theta, \qquad f_i''(\theta) = \ell''(a_i(\theta))\, x_i x_i^\top + \lambda I,$$

using $y_i^2 = 1$. Hence

$$F''(\theta) = \frac{1}{n}\sum_{i=1}^n \ell''\big(y_i x_i^\top \theta\big)\, x_i x_i^\top + \lambda I. \tag{34}$$

The function f_i is L_i-smooth with $L_i = \|x_i\|^2/4 + \lambda$ (the largest eigenvalue of $f_i''(\theta)$ is $\ell''(a_i(\theta))\,\|x_i\|^2 + \lambda \le L_i$, with equality when $a_i(\theta) = 0$), so the per-sample smoothness constant of the project is $R^2 = \max_i L_i = R_x^2/4 + \lambda$. The smoothness constant of F is $L = \lambda + \lambda_{\max}(\hat{\Sigma})/4$: indeed $F''(\theta) \preccurlyeq \frac{1}{4}\hat{\Sigma} + \lambda I$ for all θ, with equality at $\theta = 0$. Finally $F''(\theta) \succcurlyeq \lambda I$ for all θ, so F is λ-strongly convex; Proposition 2.12 shows that $\mu = \lambda$ is the exact global constant.

Lemma 2.10 (mean-value weights and explicit formulas)**.** *Let $\theta \ne \eta_*$, $v = \theta - \eta_*$, $u = v/\|v\|$, $c_i = x_i^\top u$, $a_i = a_i(\theta)$, $a_i^* = a_i(\eta_*)$, and define the mean-value weights*

$$w_i(\theta) \;=\; \int_0^1 \ell''\big(a_i^* + s(a_i - a_i^*)\big)\, ds \;\in\; (0, \tfrac{1}{4}]. \tag{35}$$

By the fundamental theorem of calculus (see the proof), the weights have the closed form

$$w_i(\theta) = \frac{\ell'(a_i) - \ell'(a_i^*)}{a_i - a_i^*} = \frac{\sigma(a_i) - \sigma(a_i^*)}{a_i - a_i^*} \quad \text{if } a_i \ne a_i^*, \qquad w_i(\theta) = \ell''(a_i^*) \quad \text{if } a_i = a_i^*, \tag{36}$$

so that the formulas below are exact and computable in $O(nd)$ operations without any quadrature (in floating point, when $|a_i - a_i^*|$ is below the square root of the machine precision, use $\ell''(\frac{1}{2}(a_i + a_i^*))$, which agrees with $w_i(\theta)$ up to $O(|a_i - a_i^*|^2)$, to avoid cancellation; alternatively μ_{eff} and L_{eff} can be computed directly from Definition 2.2). Then

$$f_i'(\theta) - f_i'(\eta_*) = w_i(\theta)\,(x_i^\top v)\, x_i + \lambda v, \qquad \bar{H}(\theta) = \frac{1}{n}\sum_{i=1}^n w_i(\theta)\, x_i x_i^\top + \lambda I, \tag{37}$$

and, with $A(\theta) = \frac{1}{n}\sum_i w_i(\theta) c_i^2$ and $B(\theta) = \frac{1}{n}\sum_i w_i(\theta)^2 c_i^2 \|x_i\|^2$,

$$\mu_{\text{eff}}(\theta) = \lambda + A(\theta) = \lambda + \frac{1}{n}\sum_{i=1}^n w_i(\theta)\,(x_i^\top u)^2, \tag{38}$$

$$L_{\text{eff}}(\theta) = \frac{\lambda^2 + 2\lambda A(\theta) + B(\theta)}{\lambda + A(\theta)} = \mu_{\text{eff}}(\theta) + \frac{B(\theta) - A(\theta)^2}{\mu_{\text{eff}}(\theta)}, \qquad B(\theta) \ge A(\theta)^2. \tag{39}$$

Proof. The map $s \mapsto \ell'(a_i^* + s(a_i - a_i^*))$ is C^1 with derivative $\ell''(a_i^* + s(a_i - a_i^*))(a_i - a_i^*)$, so the fundamental theorem of calculus gives $\ell'(a_i) - \ell'(a_i^*) = w_i(\theta)(a_i - a_i^*) = w_i(\theta)\, y_i x_i^\top v$, which is (36) (when $a_i = a_i^*$ the integrand of (35) is the constant $\ell''(a_i^*)$; $\ell' = \sigma - 1$ gives the second expression). Therefore $f_i'(\theta) - f_i'(\eta_*) = [\ell'(a_i) - \ell'(a_i^*)] y_i x_i + \lambda v = w_i(\theta)(x_i^\top v) x_i + \lambda v$ (using $y_i^2 = 1$). Averaging over i gives $F'(\theta) = [\frac{1}{n}\sum_i w_i(\theta) x_i x_i^\top + \lambda I]v$; by linearity of the integral this matrix is

exactly $\int_0^1 F''(\eta_* + sv)ds$ computed from (34), i.e. the matrix $\bar{H}(\theta)$ of Lemma 2.5. Formula (38) is the Rayleigh quotient (27). For (39), $f_i'(\theta) - f_i'(\eta_*) = \|v\|\,(w_i c_i x_i + \lambda u)$ and $\|w_i c_i x_i + \lambda u\|^2 = w_i^2 c_i^2 \|x_i\|^2 + 2\lambda w_i c_i^2 + \lambda^2$ (because $x_i^\top u = c_i$ and $\|u\| = 1$), so $\frac{1}{n}\sum_i \|f_i'(\theta) - f_i'(\eta_*)\|^2 = \|v\|^2 (B + 2\lambda A + \lambda^2)$, while $v^\top F'(\theta) = \|v\|^2(\lambda + A)$; dividing gives the first expression, and $\lambda^2 + 2\lambda A + B = (\lambda + A)^2 + (B - A^2)$ gives the second. Finally $\|x_i\|^2 \geq c_i^2$ (Cauchy–Schwarz) gives $B \geq \frac{1}{n}\sum_i (w_i c_i^2)^2 \geq (\frac{1}{n}\sum_i w_i c_i^2)^2 = A^2$ (Jensen). □

Lemma 2.11 (two-sided bounds on the effective curvature). *Let $\theta \neq \eta_*$ and $m_i(\theta) = \min\{\ell''(a_i^*), \ell''(a_i)\}$, $m(\theta) = \min_i m_i(\theta)$. Then $m_i(\theta) \leq w_i(\theta) \leq \frac{1}{4}$, and*

$$\lambda + m(\theta)\, u^\top \hat{\Sigma} u \;\leq\; \lambda + \frac{1}{n}\sum_{i=1}^n m_i(\theta)(x_i^\top u)^2 \;\leq\; \mu_{\mathrm{eff}}(\theta), \tag{40}$$

$$\mu_{\mathrm{eff}}(\theta) \;\leq\; \lambda + \frac{1}{4}\, u^\top \hat{\Sigma} u \;\leq\; \lambda + \frac{1}{4}\lambda_{\max}(\hat{\Sigma}) = L. \tag{41}$$

In particular $\mu_{\mathrm{eff}}(\theta) \geq \lambda + m(\theta)\lambda_{\min}(\hat{\Sigma})$, where $m(\theta) = \min_i \min\{\sigma'(y_i x_i^\top \eta_*), \sigma'(y_i x_i^\top \theta)\}$ is the minimum of $\ell'' = \sigma'$ over the n segments $[a_i^*, a_i]$ and is computable in $O(nd)$ operations.

Proof. Every point $a_i^* + s(a_i - a_i^*)$, $s \in [0,1]$, of the segment satisfies $|a_i^* + s(a_i - a_i^*)| \leq \max\{|a_i^*|, |a_i|\}$; since ℓ'' is even and non-increasing in $|a|$, ℓ'' on the segment is at least $\min\{\ell''(a_i^*), \ell''(a_i)\} = m_i(\theta)$ and at most $1/4$. Integrating over s gives $m_i(\theta) \leq w_i(\theta) \leq 1/4$, and inserting these bounds into (38), with $\frac{1}{n}\sum_i (x_i^\top u)^2 = u^\top \hat{\Sigma} u \in [\lambda_{\min}(\hat{\Sigma}), \lambda_{\max}(\hat{\Sigma})]$, gives (40)–(41). □

Global versus local strong convexity. The next proposition states precisely in which sense "the global strong convexity constant is exactly λ" for logistic regression. Recall that, for a twice continuously differentiable F, F is μ-strongly convex if and only if $\theta \mapsto F(\theta) - \frac{\mu}{2}\|\theta\|^2$ is convex (Exercise 5.5 of [6]), if and only if $F''(\theta) \succcurlyeq \mu I$ for all θ, so that the best (largest) strong convexity constant is

$$\mu_{\mathrm{glob}} \;=\; \sup\{\mu \geq 0 :\; F \text{ is } \mu\text{-strongly convex}\} \;=\; \inf_{\theta \in \mathbb{R}^d} \lambda_{\min}\big(F''(\theta)\big).$$

Proposition 2.12 (the global strong convexity constant of ℓ_2-regularized logistic regression is λ). *Let $I_0 = \{i : x_i \neq 0\}$ (if $I_0 = \emptyset$ then $F'' \equiv \lambda I$ and there is nothing to prove).*

(a) There exists a unit vector $d \in \mathbb{R}^d$ with $x_i^\top d \neq 0$ for every $i \in I_0$; in fact Lebesgue-almost every direction has this property.

(b) For every such d, with $\delta(d) = \min_{i \in I_0} |x_i^\top d| > 0$, and every $t \geq 0$,

$$\lambda \;\leq\; \lambda_{\min}\big(F''(t\,d)\big) \;\leq\; \lambda + R_x^2\, e^{-t\,\delta(d)}. \tag{42}$$

(c) Consequently $\mu_{\mathrm{glob}} = \inf_\theta \lambda_{\min}(F''(\theta)) = \lambda$: the global strong convexity constant of F is exactly λ, whatever the data.

(d) The infimum is attained at some θ if and only if $\mathrm{span}\{x_1, \ldots, x_n\} \neq \mathbb{R}^d$; in that case $\lambda_{\min}(F''(\theta)) = \lambda$ for every θ. If the features span $\mathbb{R}^d$, then $\lambda_{\min}(F''(\theta)) > \lambda$ for every θ and the value λ is only approached as $\|\theta\| \to \infty$; it is in fact approached along every *ray: for every unit vector d and every $t \geq 0$,*

$$\lambda_{\min}\big(F''(t\,d)\big) \;\leq\; \lambda + \frac{1}{n}\sum_{i=1}^n \ell''\big(t\, y_i x_i^\top d\big)(x_i^\top d)^2 \;\leq\; \lambda + R_x^2\, e^{-t\,\delta'(d)}, \tag{43}$$

where $\delta'(d) = \min\{|x_i^\top d| : x_i^\top d \neq 0\}$, *with the conventions* $\delta'(d) = +\infty$ *and* $e^{-\infty} = 0$ *when* $x_i^\top d = 0$ *for all* i; *hence* $\lim_{t\to\infty} \lambda_{\min}(F''(td)) = \lambda$ *for every direction* d. *Directions as in (a) are those along which the whole matrix* $F''(td) - \lambda I$ *tends to* 0; *along a general ray,*

$$F''(td) - \lambda I \xrightarrow[t\to\infty]{} \frac{1}{4n} \sum_{i:\, x_i^\top d = 0} x_i x_i^\top,$$

a matrix which is singular because d *lies in its kernel.*

Proof. (a) For $i \in I_0$ the set $\{d : x_i^\top d = 0\}$ is a hyperplane, hence has Lebesgue measure zero; a finite union of null sets is null, so its complement is nonempty (indeed of full measure); normalize any element.

(b) The lower bound is $F'' \succcurlyeq \lambda I$. For the upper bound, $F''(td) - \lambda I = \frac{1}{n} \sum_{i\in I_0} \ell''(t\, y_i x_i^\top d)\, x_i x_i^\top$ (the terms with $x_i = 0$ vanish), and this matrix is positive semidefinite with

$$\big\|F''(td) - \lambda I\big\|_{\mathrm{op}} \le \frac{1}{n} \sum_{i\in I_0} \ell''(t\, y_i x_i^\top d)\, \|x_i\|^2 \le R_x^2 \max_{i\in I_0} \ell''(t\, y_i x_i^\top d) \le R_x^2 e^{-t\delta(d)},$$

where we used $\ell''(a) = \sigma(a)\sigma(-a) \le \sigma(-|a|) = \frac{1}{1+e^{|a|}} \le e^{-|a|}$ (because $\sigma \le 1$ and ℓ'' is even) and $|t\, y_i x_i^\top d| = t\, |x_i^\top d| \ge t\delta(d)$. Hence $\lambda_{\min}(F''(td)) \le \lambda + \|F''(td) - \lambda I\|_{\mathrm{op}} \le \lambda + R_x^2 e^{-t\delta(d)}$.

(c) Letting $t \to \infty$ in (42) gives $\inf_\theta \lambda_{\min}(F''(\theta)) \le \lambda$; the reverse inequality is $F'' \succcurlyeq \lambda I$.

(d) $\lambda_{\min}(F''(\theta)) = \lambda$ iff the positive semidefinite matrix $\frac{1}{n}\sum_i \ell''(a_i(\theta)) x_i x_i^\top$ is singular, iff there is $w \neq 0$ with $\frac{1}{n}\sum_i \ell''(a_i(\theta))(x_i^\top w)^2 = 0$, iff $x_i^\top w = 0$ for all i (since $\ell'' > 0$), iff the x_i do not span $\mathbb{R}^d$. This condition does not depend on θ. For (43): $\lambda_{\min}(F''(td)) \le d^\top F''(td)\, d = \lambda + \frac{1}{n}\sum_i \ell''(t\, y_i x_i^\top d)(x_i^\top d)^2$ (Rayleigh quotient with $\|d\| = 1$, using $y_i^2 = 1$); the terms with $x_i^\top d = 0$ vanish, and for the others $\ell''(t\, y_i x_i^\top d)(x_i^\top d)^2 \le e^{-t|x_i^\top d|}\, \|x_i\|^2 \le R_x^2 e^{-t\delta'(d)}$ by the bound $\ell''(a) \le e^{-|a|}$ used in (b). The limit of $F''(td) - \lambda I$ follows termwise from $\ell''(t\, y_i x_i^\top d) \to 0$ when $x_i^\top d \neq 0$ and $\ell''(0) = \frac{1}{4}$ when $x_i^\top d = 0$. □

Corollary 2.13 (strong convexity on a bounded region)**.** *For every* $D > 0$ *and every* θ *with* $\|\theta\| \le D$,

$$\lambda_{\min}\big(F''(\theta)\big) \;\ge\; \lambda + \ell''(R_x D)\, \lambda_{\min}(\hat\Sigma) \;\ge\; \lambda + \tfrac{1}{4} e^{-R_x D} \lambda_{\min}(\hat\Sigma). \tag{44}$$

Moreover $\|\eta_*\| \le \sqrt{2\log 2/\lambda}$, *so that the ball* $\{\|\theta\| \le 2\sqrt{2\log 2/\lambda}\}$ *contains the ball of radius* $\sqrt{2\log 2/\lambda}$ *around* η_*.

Proof. If $\|\theta\| \le D$ then $|a_i(\theta)| \le \|x_i\|\, \|\theta\| \le R_x D$, hence $\ell''(a_i(\theta)) \ge \ell''(R_x D)$ (ℓ'' even, non-increasing in $|a|$) and $F''(\theta) \succcurlyeq \lambda I + \ell''(R_x D)\hat\Sigma$. The second inequality uses $\ell''(a) = \sigma(|a|)\sigma(-|a|) \ge \frac{1}{2} \cdot \frac{1}{1+e^{|a|}} \ge \frac{1}{4} e^{-|a|}$. For the norm of η_*: $\frac{\lambda}{2}\|\eta_*\|^2 \le F(\eta_*) \le F(0) = \log 2$. □

Thus the global constant λ is only "seen" at infinity: on any bounded region the Hessian is uniformly larger than λI by a data-dependent amount, and an algorithm whose iterates stay in a bounded region never experiences the curvature λ unless $\lambda_{\min}(\hat\Sigma) = 0$ (for instance when $d > n$; Lemma 2.16 below identifies which trajectories then actually see it) or in directions where $\hat\Sigma$ is nearly singular. Gradient descent with $\gamma \le 1/L$ stays in the ball of radius $\|\theta_0 - \eta_*\|$ around η_*: with $v = \theta_{t-1} - \eta_*$, $\|\theta_t - \eta_*\|^2 = \|v\|^2 - 2\gamma v^\top F'(\theta_{t-1}) + \gamma^2 \|F'(\theta_{t-1})\|^2 \le \|v\|^2 - \gamma(2 - \gamma L)\, v^\top F'(\theta_{t-1}) \le \|v\|^2$ by co-coercivity of F (Lemma 2.3(c) with $\eta = \eta_*$) and $v^\top F'(\theta_{t-1}) \ge 0$. For SAGA the statement is only probabilistic: under Proposition 5.9, $V_t \ge 0$ and $\mathbb{E}[V_t \mid \mathcal{F}_{t-1}] \le V_{t-1}$, so $(V_t)_{t\ge 0}$ is a nonnegative supermartingale; it converges almost surely, hence $\sup_t V_t < \infty$ almost surely, and Ville's maximal

inequality gives $\mathbb{P}(\sup_{t\geq 0} V_t \geq c) \leq V_0/c$ for every $c > 0$. Since $\|\theta_t - \eta_*\|^2 \leq V_t$, the SAGA iterates are almost surely bounded, but they are not confined to a deterministic ball. The next lemma, which is the scalar case of the generalized self-concordance property of the logistic loss ([?], Lemma 1 and Proposition 1, eq. (6); Assumption (A2) in [5]), turns this into a quantitative comparison between $\mu_{\text{eff}}(\theta)$ and the curvature at the optimum.

Lemma 2.14 (generalized self-concordance of the logistic loss)**.** *For all $a, b \in \mathbb{R}$,*

$$e^{-|b-a|}\,\ell''(a) \;\leq\; \ell''(b) \;\leq\; e^{|b-a|}\,\ell''(a).$$

Consequently, for $\theta \neq \eta_*$ and all i, with $p_i = \ell''(a_i^*) = \sigma'(y_i x_i^\top \eta_*)$,

$$e^{-|x_i^\top v|}\,p_i \;\leq\; w_i(\theta) \;\leq\; e^{|x_i^\top v|}\,p_i, \qquad \textit{so that} \qquad e^{-R_x\|v\|}p_i \leq w_i(\theta) \leq e^{R_x\|v\|}p_i. \tag{45}$$

Proof. Since $\ell'' > 0$ and $\ell''' = \ell''(1-2\sigma)$, the function $\log \ell''$ is differentiable with $|(\log \ell'')'(a)| = |1 - 2\sigma(a)| \leq 1$; integrating between a and b gives $|\log \ell''(b) - \log \ell''(a)| \leq |b-a|$, which is the claim. For (45), every point of the segment $[a_i^*, a_i]$ is at distance at most $|a_i - a_i^*| = \left|y_i x_i^\top v\right| = \left|x_i^\top v\right| \leq R_x\,\|v\|$ from a_i^*; apply the first claim with $a = a_i^*$ and integrate over $s \in [0,1]$ in (35). □

Corollary 2.15 (effective curvature versus curvature at the optimum)**.** *Let $H_* = F''(\eta_*) = \frac{1}{n}\sum_i p_i x_i x_i^\top + \lambda I$ and $p_{\min} = \min_i p_i = \min_i \sigma'(y_i x_i^\top \eta_*)$. For every $\theta \neq \eta_*$, with $u = (\theta - \eta_*)/\,\|\theta - \eta_*\|$,*

$$\lambda + e^{-R_x\|\theta-\eta_*\|}\left(u^\top H_* u - \lambda\right) \;\leq\; \mu_{\text{eff}}(\theta), \tag{46}$$

$$\mu_{\text{eff}}(\theta) \;\leq\; \lambda + e^{R_x\|\theta-\eta_*\|}\left(u^\top H_* u - \lambda\right), \tag{47}$$

and

$$u^\top H_* u \;\geq\; \lambda + p_{\min}\,u^\top \hat{\Sigma} u \;\geq\; \lambda + p_{\min}\lambda_{\min}(\hat{\Sigma}). \tag{48}$$

In particular, $\liminf_{\theta\to\eta_*} \mu_{\text{eff}}(\theta) \geq \lambda + p_{\min}\lambda_{\min}(\hat{\Sigma})$*, and for $\|\theta - \eta_*\| \leq 1/R_x$ one has $\mu_{\text{eff}}(\theta) \geq \lambda + e^{-1}p_{\min}\lambda_{\min}(\hat{\Sigma})$. If moreover $\theta \in S = \operatorname{span}\{x_1, \ldots, x_n\}$ (which holds along the whole trajectory of gradient descent and of SAGA when $\theta_0 \in S$, Lemma 2.16 below), then $u \in S$, and $\lambda_{\min}(\hat{\Sigma})$ may be replaced, in* (48) *and in the two consequences, by $\lambda^+_{\min}(\hat{\Sigma})$, the smallest nonzero eigenvalue of $\hat{\Sigma}$ (which is positive even when $d > n$).*

Proof. By (38) and (45), $\mu_{\text{eff}}(\theta) - \lambda = \frac{1}{n}\sum_i w_i(\theta) c_i^2$ lies between $e^{\mp R_x\|v\|}\frac{1}{n}\sum_i p_i c_i^2$, and $\frac{1}{n}\sum_i p_i c_i^2 = u^\top H_* u - \lambda$. Inequality (48) is $p_i \geq p_{\min}$ and $\frac{1}{n}\sum_i c_i^2 = u^\top \hat{\Sigma} u \geq \lambda_{\min}(\hat{\Sigma})$. For the last claim, $\eta_* \in S$ (Lemma 2.16(a)), so $v = \theta - \eta_* \in S$ and $u \in S$; the kernel of the positive semidefinite matrix $\hat{\Sigma}$ is $\{w : x_i^\top w = 0\ \forall i\} = S^\perp$, so its range is S and its restriction to S has exactly the nonzero eigenvalues of $\hat{\Sigma}$, whence $u^\top \hat{\Sigma} u \geq \lambda^+_{\min}(\hat{\Sigma})$ for unit $u \in S$. □

How the data spectrum and λ control the gap μ_{eff}/μ. Since $\mu = \lambda$ exactly (Proposition 2.12), the ratio measuring the slack of the strong-convexity step along a trajectory is $\mu_{\text{eff}}(\theta_t)/\lambda$. Corollary 2.15 shows that, once $R_x\,\|\theta_t - \eta_*\| \lesssim 1$ (which happens after the first few passes of SAGA), this ratio is within a factor $e^{\pm 1}$ of

$$\frac{u_t^\top H_* u_t}{\lambda} \;=\; 1 + \frac{1}{\lambda}\cdot\frac{1}{n}\sum_{i=1}^n p_i\,(x_i^\top u_t)^2, \qquad u_t = \frac{\theta_t - \eta_*}{\|\theta_t - \eta_*\|}, \tag{49}$$

which, as u_t ranges over the unit sphere, spans the interval $[1 + \lambda_{\min}(H_* - \lambda I)/\lambda,\ 1 + \lambda_{\max}(H_* - \lambda I)/\lambda]$, where the "data Hessian" $H_* - \lambda I = \frac{1}{n}\sum_i p_i x_i x_i^\top$ satisfies $p_{\min}\hat{\Sigma} \preccurlyeq H_* - \lambda I \preccurlyeq \frac{1}{4}\hat{\Sigma}$. Three regimes follow.

- *Strong regularization,* $\lambda \gg \lambda_{\max}(\hat{\Sigma})/4$: the ratio is at most $1 + \lambda_{\max}(\hat{\Sigma})/(4\lambda) \approx 1$ in every direction, the strong-convexity step is essentially tight, but then $\kappa = R^2/\mu = 1 + R_x^2/(4\lambda)$ is small and the problem is uninteresting for (A4).

- *Intermediate regularization,* $\lambda_{\min}(H_* - \lambda I) \ll \lambda \ll \lambda_{\max}(H_* - \lambda I)$ (the typical situation with anisotropic features: a decaying spectrum of $\hat{\Sigma}$, or $d > n$, in which case $\lambda_{\min}(\hat{\Sigma}) = 0$ and $\lambda_{\min}(H_*) = \lambda$ exactly, the λ-eigenspace of H_* being the orthogonal complement $S^\perp$ of the feature span $S = \operatorname{span}\{x_1, \dots, x_n\}$; by Lemma 2.16 below this value λ is experienced only by trajectories having a component outside S, hence never when $\theta_0 \in S$, e.g. $\theta_0 = 0$, in which case the relevant bottom eigenvalue is $\lambda_{\min}(H_*|_S) = \lambda + \lambda^+_{\min}(H_* - \lambda I)$, with $\lambda^+_{\min}$ the smallest nonzero eigenvalue): the gap is *direction dependent*, close to 1 along the bottom eigenvectors of H_* and of order $\lambda_{\max}(H_*)/\lambda \gg 1$ along the top ones. Whether the slack is observed is then decided by the direction of the error u_t, which is the object of Section 2.4.

- *Weak regularization with well-conditioned features,* $\lambda \ll p_{\min}\lambda_{\min}(\hat{\Sigma})$: the gap is at least $1 + p_{\min}\lambda_{\min}(\hat{\Sigma})/\lambda \gg 1$ in *every* direction, and the μ-term of Proposition 5.9 is pessimistic, whatever the algorithm, by at least the factor $1 + e^{-1} p_{\min}\lambda_{\min}(\hat{\Sigma})/\lambda$ as soon as $R_x \|\theta_t - \eta_*\| \le 1$ and by at least $1 + p_{\min}\lambda_{\min}(\hat{\Sigma})/\lambda$ in the limit $t \to \infty$ (Corollary 2.15); far from η_* the data term carries the weight $e^{-R_x\|\theta_t - \eta_*\|}$ (or $m(\theta_t)$ in Lemma 2.11), which can be tiny, so the factor is not uniform along the whole trajectory. Here $p_{\min}$ is controlled by the margins at the optimum: $p_i \ge \frac{1}{4} e^{-|a_i^*|}$ with $|a_i^*| \le R_x \|\eta_*\| \le R_x\sqrt{2\log 2/\lambda}$ (Corollary 2.13), so $p_{\min}$ is of order $1/4$ when $\lambda \gtrsim R_x^2$ but may be exponentially small when λ is small and the data are nearly separable; in the latter case $H_* - \lambda I$ is dominated by the points with $|a_i^*| = O(1)$ (the points near the decision boundary), and the relevant quantity is the spectrum of $\frac{1}{n}\sum_i p_i x_i x_i^\top$ rather than that of $\hat{\Sigma}$.

In all cases, the statement "the gap is small when $\lambda \gg \lambda_{\min}(H_* - \lambda I)$" is correct only for the worst direction (the bottom eigenvector of H_*); in a typical (e.g. isotropically random) direction the ratio (49) is $1 + \operatorname{tr}(H_* - \lambda I)/(d\lambda)$, which can be large even when $\lambda \gg \lambda_{\min}(H_* - \lambda I)$. The tightness ratios to record are therefore $(\lambda + m(\theta_t) u_t^\top \hat{\Sigma} u_t)/\mu_{\text{eff}}(\theta_t)$ for (40), $\mu_{\text{eff}}(\theta_t)/(\lambda + \frac{1}{4} u_t^\top \hat{\Sigma} u_t)$ for (41), and the two self-concordance ratios of (46)–(47), all of which only require η_* (from a high-precision solver), one pass over the data and the eigen-decomposition of H_*.

2.4 Direction dependence: gradient descent versus SAGA

Proposition 2.8 shows that near the optimum $\mu_{\text{eff}}(\theta_t)$ is the Rayleigh quotient of $H_* = F''(\eta_*)$ in the direction u_t of the error. Which direction an algorithm selects is therefore part of the answer to (B2). We treat the quadratic case exactly and use it as a model for the logistic case.

An exact invariant subspace: the feature span. Before the quadratic model, we record an exact constraint on the directions that gradient descent and SAGA can explore for logistic regression (the argument only uses $f_i'(\theta) - \lambda\theta \in \operatorname{span}\{x_i\}$, so it applies verbatim to ridge regression and to every ℓ_2-regularized generalized linear model $f_i(\theta) = \varphi_i(x_i^\top \theta) + \frac{\lambda}{2}\|\theta\|^2$). Below, $z_j^{(t)}$ is the SAGA memory, ω_t the SAGA direction, and, for $i = 1, \dots, n$,

$$g_i^{(t+1)} \;=\; F'(\theta_t) - f_i'(\theta_t) + z_i^{(t)} - \frac{1}{n}\sum_{j=1}^{n} z_j^{(t)}$$

are the n possible values of the estimator error $\xi_{t+1} = F'(\theta_t) - \omega_{t+1}$ at the state $(\theta_t, z^{(t)})$ (for the quadratic model of Lemma 2.20 below these are the vectors $g_i^{(t+1)}$ defined there), and $C_{t+1} = \frac{1}{n}\sum_i g_i^{(t+1)} g_i^{(t+1)\top}$ is the conditional covariance of ω_{t+1} given the state.

Lemma 2.16 (invariance of the feature span)**.** *Let $S = \operatorname{span}\{x_1, \dots, x_n\} \subseteq \mathbb{R}^d$, let $S^\perp$ be its orthogonal complement, and write $\theta = \theta_S + \theta_\perp$ for the orthogonal decomposition of a vector θ (similarly $v_S, v_\perp$). For ℓ_2-regularized logistic regression:*

(a) $\eta_* \in S$, and $f_i'(\theta) - \lambda\theta \in S$ for every θ and every i; in particular $f_i'(\theta)_\perp = \lambda\theta_\perp$ for all i, and $F'(\theta)_\perp = \lambda\theta_\perp$.

(b) (Invariance) If $\theta_0 \in S$, then, for gradient descent and for SAGA with the initialization $z_i^{(0)} = f_i'(\theta_0)$ of Proposition 5.9 (or with $z_i^{(0)} = 0$), all iterates θ_t, all memories $z_i^{(t)}$, all directions ω_t and all error vectors $g_i^{(t)}$ lie in S, for every t and every realization of the indices. The same holds for the quadratic model of Lemma 2.20 whenever $H_i S \subseteq S$ for every i and $v_0, e_1^{(0)}, \dots, e_n^{(0)} \in S$.

(c) (The complementary component is autonomous) For an arbitrary θ_0, the $S^\perp$-component of the error $v_t = \theta_t - \eta_*$ evolves as if F were the quadratic $\frac{\lambda}{2}\|\theta_\perp\|^2$. For gradient descent, $v_{t,\perp} = (1-\gamma\lambda)^t v_{0,\perp}$. For SAGA, let $s_j(t)$ be the time at which the current memory was computed, $z_j^{(t)} = f_j'(\theta_{s_j(t)})$ (so $s_j(0) = 0$, $s_{i(t)}(t) = t-1$ and $s_j(t) = s_j(t-1)$ for $j \neq i(t)$); then

$$v_{t,\perp} = (1-\gamma\lambda)\, v_{t-1,\perp} - \gamma\lambda\Big(\frac{1}{n}\sum_{j=1}^n v_{s_j(t-1),\perp} - v_{s_{i(t)}(t-1),\perp}\Big),$$

and the error vectors satisfy

$$g_{i,\perp}^{(t+1)} = \lambda\Big(v_{s_i(t),\perp} - \frac{1}{n}\sum_{j=1}^n v_{s_j(t),\perp}\Big),$$

so that $\mathbb{E}[v_{t,\perp}] = (1-\gamma\lambda)^t v_{0,\perp}$, and the noise injected by SAGA in a unit direction $q \in S^\perp$ is

$$q^\top C_{t+1}\, q \;=\; \lambda^2\, \operatorname{Var}_j\left(q^\top v_{s_j(t)}\right) \;:=\; \lambda^2 \cdot \frac{1}{n}\sum_{i=1}^n \Big(q^\top v_{s_i(t)} - \frac{1}{n}\sum_{j=1}^n q^\top v_{s_j(t)}\Big)^2,$$

the empirical variance, over the memory, of the q-component of the stored errors.

Proof. (a) $f_i'(\theta) = \ell'(a_i(\theta))\, y_i x_i + \lambda\theta$ and $\ell'(a_i(\theta))\, y_i x_i \in S$; averaging over i, $F'(\theta) - \lambda\theta \in S$. At $\theta = \eta_*$, $F'(\eta_*) = 0$ gives $\lambda\eta_* = -\frac{1}{n}\sum_i \ell'(a_i^*)\, y_i x_i \in S$, hence $\eta_* \in S$ because $\lambda > 0$. Projecting $f_i'(\theta) = [f_i'(\theta) - \lambda\theta] + \lambda\theta$ onto $S^\perp$ gives $f_i'(\theta)_\perp = \lambda\theta_\perp$.

(b) Induction on t. At $t = 0$, $\theta_0 \in S$ and $z_i^{(0)} = f_i'(\theta_0) \in S$ by (a), since $\lambda\theta_0 \in S$ (and trivially if $z_i^{(0)} = 0$). If $\theta_{t-1} \in S$ and $z_j^{(t-1)} \in S$ for all j, then $f_i'(\theta_{t-1}) \in S$ for all i by (a), hence $\omega_t = f_{i(t)}'(\theta_{t-1}) + \frac{1}{n}\sum_j z_j^{(t-1)} - z_{i(t)}^{(t-1)} \in S$ (for gradient descent, $F'(\theta_{t-1}) \in S$), $\theta_t = \theta_{t-1} - \gamma\omega_t \in S$, the updated memory $z_{i(t)}^{(t)} = f_{i(t)}'(\theta_{t-1})$ lies in S and the other memories are unchanged. Finally each $g_i^{(t+1)}$ is a linear combination of vectors of S. For the quadratic model, $f_i'(\theta) - f_i'(\eta_*) = H_i v$ with $H_i S \subseteq S$, and the same induction applies to v_t, $e_i^{(t)}$ and $g_i^{(t+1)}$ (Lemma 2.20(a)).

(c) By (a) and $\eta_* \in S$, $f_i'(\theta)_\perp = \lambda\theta_\perp = \lambda(\theta - \eta_*)_\perp$ for every θ and i. For gradient descent, $v_{t,\perp} = v_{t-1,\perp} - \gamma F'(\theta_{t-1})_\perp = (1-\gamma\lambda)v_{t-1,\perp}$. For SAGA, $z_j^{(t-1)} = f_j'(\theta_{s_j(t-1)})$ has $S^\perp$-component $\lambda v_{s_j(t-1),\perp}$,

so the $S^\perp$-component of ω_t is $\lambda v_{t-1,\perp} + \lambda \frac{1}{n}\sum_j v_{s_j(t-1),\perp} - \lambda v_{s_{i(t)}(t-1),\perp}$, and $v_{t,\perp} = v_{t-1,\perp} - \gamma\,\omega_{t,\perp}$ is the displayed recursion. Since $i(t)$ is uniform and independent of the past, the bracket has conditional mean zero, so $\mathbb{E}[v_{t,\perp} \mid \mathcal{F}_{t-1}] = (1-\gamma\lambda)v_{t-1,\perp}$ and $\mathbb{E}[v_{t,\perp}] = (1-\gamma\lambda)^t v_{0,\perp}$. In $g_i^{(t+1)}$ the components $F'(\theta_t)_\perp$ and $f_i'(\theta_t)_\perp$ are both equal to $\lambda v_{t,\perp}$ and cancel, leaving $z_{i,\perp}^{(t)} - \frac{1}{n}\sum_j z_{j,\perp}^{(t)} = \lambda(v_{s_i(t),\perp} - \frac{1}{n}\sum_j v_{s_j(t),\perp})$. For $q \in S^\perp$, $q^\top g_i^{(t+1)} = q^\top g_{i,\perp}^{(t+1)}$, and $q^\top C_{t+1} q = \frac{1}{n}\sum_i (q^\top g_i^{(t+1)})^2$ gives the last formula. □

The lemma has two consequences for the measurements of this section. First, with the standard initialization $\theta_0 = 0$ (or any $\theta_0 \in S$) the error v_t never leaves S, so only the restriction $H_*|_S$ of the Hessian at the optimum to S is explored; S is H_*-invariant (as $H_* = \frac{1}{n}\sum_i p_i x_i x_i^\top + \lambda I$), and when $d > n$ the λ-eigenspace $S^\perp$ of H_* is exactly the part that is never explored. Second, for a generic $\theta_0 \notin S$ the $S^\perp$-component of the error is governed by the smallest possible curvature λ, exactly as for the quadratic $\frac{\lambda}{2}\|\theta\|^2$, for both algorithms.

Gradient descent on a quadratic. Let $F(\theta) = F(\eta_*) + \frac{1}{2}(\theta-\eta_*)^\top H(\theta-\eta_*)$ with H symmetric positive definite, eigenvalues $\lambda_1 \geq \cdots \geq \lambda_d = \lambda_{\min} > 0$ and an orthonormal basis of eigenvectors $q_1, \ldots, q_d$. Let $E_{\min}$ be the eigenspace of $\lambda_{\min}$, P the orthogonal projector onto $E_{\min}$, $K_{\min} = \{k : \lambda_k = \lambda_{\min}\}$, and $\lambda' = \min\{\lambda_k : \lambda_k > \lambda_{\min}\}$ (second smallest distinct eigenvalue; if $H = \lambda_{\min} I$ everything below is trivial). Gradient descent is $\theta_t = \theta_{t-1} - \gamma F'(\theta_{t-1})$. For a quadratic, $\bar{H}(\theta) = H$ for all θ, so by Lemma 2.5 $\mu_{\text{eff}}(\theta) = v^\top H v / \|v\|^2$ exactly, and the spectral profile of the error

$$\pi_k(t) \;=\; \frac{(q_k^\top v_t)^2}{\|v_t\|^2}, \qquad k = 1, \ldots, d, \qquad \sum_k \pi_k(t) = 1, \tag{50}$$

determines it: $\mu_{\text{eff}}(\theta_t) = \sum_k \lambda_k \pi_k(t)$.

Proposition 2.17 (alignment of gradient descent with the flattest direction)**.** *Let $0 < \gamma \leq 1/\lambda_1$. Then:*

(a) $v_t = (I - \gamma H)^t v_0$, i.e. $q_k^\top v_t = (1-\gamma\lambda_k)^t\, q_k^\top v_0$ for all k, and $\mu_{\text{eff}}(\theta_t) = \sum_k \lambda_k \pi_k(t)$ with $\pi_k(t) \propto (1-\gamma\lambda_k)^{2t}(q_k^\top v_0)^2$.

(b) (Monotonicity) $t \mapsto \mu_{\text{eff}}(\theta_t)$ is non-increasing.

(c) (Alignment) If $Pv_0 \neq 0$ (true for Lebesgue-almost every v_0), then with $\rho_{\text{gap}} = (1-\gamma\lambda')/(1-\gamma\lambda_{\min}) \in [0,1)$,

$$\frac{\|v_t - Pv_t\|}{\|Pv_t\|} \;\leq\; \rho_{\text{gap}}^t \,\frac{\|v_0 - Pv_0\|}{\|Pv_0\|},$$

so that $v_t/\|v_t\|$ converges to $E_{\min}$ at the geometric rate ρ_{gap}, and

$$0 \;\leq\; \mu_{\text{eff}}(\theta_t) - \lambda_{\min} \;\leq\; (\lambda_1 - \lambda_{\min})\,\rho_{\text{gap}}^{2t}\,\frac{\|v_0 - Pv_0\|^2}{\|Pv_0\|^2} \;\xrightarrow[t\to\infty]{}\; 0. \tag{51}$$

(d) (Invariant subspaces) If S is a linear subspace with $HS \subseteq S$ and $v_0 \in S$, then $v_t \in S$ for all t, and (a)–(c) hold with H replaced by its restriction $H|_S$: the sums run over the eigenvectors $q_k \in S$ only, $\lambda_{\min}$ is replaced by $\lambda_{\min}(H|_S) = \min\{\lambda_k : q_k \in S\}$, $E_{\min}$ and P by the corresponding eigenspace within S and its orthogonal projector, λ' by the second smallest distinct eigenvalue of $H|_S$, and the hypothesis $Pv_0 \neq 0$ by its analogue for the restricted projector.

Proof. (a) $F'(\theta) = H(\theta - \eta_*)$, so $v_t = v_{t-1} - \gamma H v_{t-1} = (I - \gamma H)v_{t-1}$, and $q_k^\top (I - \gamma H)^t v_0 = (1 - \gamma\lambda_k)^t q_k^\top v_0$ since $Hq_k = \lambda_k q_k$. The formula for μ_{eff} follows from $v_t^\top H v_t = \sum_k \lambda_k (q_k^\top v_t)^2$.

(b) Put $w_k(t) = (q_k^\top v_t)^2 = (1 - \gamma\lambda_k)^{2t}(q_k^\top v_0)^2 \geq 0$, $W(t) = \sum_k w_k(t) = \|v_t\|^2 > 0$, and $\varphi(\lambda) = (1 - \gamma\lambda)^2$, so that $w_k(t+1) = \varphi(\lambda_k) w_k(t)$. On $[0, 1/\gamma] \supset [\lambda_{\min}, \lambda_1]$ the function φ is non-increasing, hence $(\lambda_k - \lambda_j)(\varphi(\lambda_k) - \varphi(\lambda_j)) \leq 0$ for all k, j. Writing $\varphi_k = \varphi(\lambda_k)$,

$$\begin{aligned} W(t)W(t+1)\big[\mu_{\text{eff}}(\theta_{t+1}) - \mu_{\text{eff}}(\theta_t)\big] &= \sum_{k,j} w_k w_j \lambda_k \varphi_k - \sum_{k,j} w_k w_j \lambda_k \varphi_j \\ &= \frac{1}{2} \sum_{k,j} w_k w_j (\lambda_k - \lambda_j)(\varphi_k - \varphi_j) \;\leq\; 0, \end{aligned}$$

where the middle equality is obtained by symmetrizing the sum in (k, j) (Chebyshev's sum inequality).

(c) $\|Pv_t\|^2 = \sum_{k \in K_{\min}} (1 - \gamma\lambda_{\min})^{2t} (q_k^\top v_0)^2 = (1 - \gamma\lambda_{\min})^{2t} \|Pv_0\|^2$, where $1 - \gamma\lambda_{\min} > 0$ because $\gamma \leq 1/\lambda_1 < 1/\lambda_{\min}$. For $k \notin K_{\min}$, $\lambda' \leq \lambda_k \leq \lambda_1 \leq 1/\gamma$ gives $0 \leq 1 - \gamma\lambda_k \leq 1 - \gamma\lambda'$, hence $\|v_t - Pv_t\|^2 = \sum_{k \notin K_{\min}} (1 - \gamma\lambda_k)^{2t} (q_k^\top v_0)^2 \leq (1 - \gamma\lambda')^{2t} \|v_0 - Pv_0\|^2$. Dividing gives the first claim. Then $\mu_{\text{eff}}(\theta_t) - \lambda_{\min} = \sum_k (\lambda_k - \lambda_{\min}) \pi_k(t) \leq (\lambda_1 - \lambda_{\min}) \sum_{k \notin K_{\min}} \pi_k(t) = (\lambda_1 - \lambda_{\min}) \|v_t - Pv_t\|^2 / \|v_t\|^2 \leq (\lambda_1 - \lambda_{\min}) \|v_t - Pv_t\|^2 / \|Pv_t\|^2$, using $\|v_t\| \geq \|Pv_t\|$; the lower bound 0 is $\lambda_k \geq \lambda_{\min}$. The set $\{v_0 : Pv_0 = 0\}$ is a proper linear subspace, hence Lebesgue-null.

(d) Since H is symmetric and $HS \subseteq S$, also $HS^\perp \subseteq S^\perp$ (for $w \in S^\perp$ and $s \in S$, $s^\top H w = (Hs)^\top w = 0$), so an orthonormal eigenbasis (q_k) of H can be chosen with every q_k in S or in $S^\perp$. Then $q_k^\top v_0 = 0$ for $q_k \in S^\perp$, hence by (a) $q_k^\top v_t = 0$ for all t and $v_t \in S$; all the sums in (a)–(c) run over the k with $q_k \in S$ only, and the proofs apply verbatim to this restricted spectrum. □

Remark 2.18 (scope of Proposition 2.17). (i) The alignment time is of order $1/\log(1/\rho_{\text{gap}}) \approx 1/(\gamma(\lambda' - \lambda_{\min}))$ iterations; when the spectrum of H is dense near $\lambda_{\min}$ this can exceed the horizon of an experiment, in which case $\mu_{\text{eff}}(\theta_t)$ decreases monotonically towards $\lambda_{\min}$ without reaching it: monotonicity (b) is the robust prediction, the limit (c) the asymptotic one. (ii) The case $Pv_0 = 0$ is not exceptional: by Lemma 2.16, for logistic regression with $d > n$ and $\theta_0 \in S$ the bottom eigenspace $E_{\min} = S^\perp$ of H_* is orthogonal to the whole trajectory. Rounding errors do re-inject a component along $E_{\min}$, but of relative size about $\varepsilon_{\text{mach}} \approx 10^{-16}$; by (c) it can become dominant only after about $\log(1/\varepsilon_{\text{mach}})/\log(1/\rho_{\text{gap}}) \approx 37/(\gamma(\lambda' - \lambda_{\min}))$ iterations, whereas the error reaches the precision floor after about $37/(\gamma\lambda')$ iterations, which is fewer since $\lambda' - \lambda_{\min} < \lambda'$. In floating point one therefore observes the alignment with the flattest direction of $H|_S$, as in (d), and not with $E_{\min}$. (iii) For $\gamma \in (1/\lambda_1, 2/\lambda_1)$ the contraction factors are $|1 - \gamma\lambda_k|$ and the largest one may be attained at λ_1 (for $\gamma = 2/(\lambda_1 + \lambda_{\min})$ both extreme components survive with alternating signs), so (b)–(c) fail; the project uses $\gamma_{\text{GD}} \leq 1/L = 1/\lambda_1$, as in the book.

Remark 2.19 (non-quadratic F: a heuristic, with a quantitative linearization error for logistic regression). For a general C^2 function, Lemma 2.5 gives $v_t = (I - \gamma H_*)v_{t-1} - \gamma(\bar{H}(\theta_{t-1}) - H_*)v_{t-1}$, and $\big\|\bar{H}(\theta) - H_*\big\|_{\text{op}} \to 0$ as $\theta \to \eta_*$. Gradient descent is thus asymptotically the linear recursion of Proposition 2.17 with $H = H_*$, and we expect, for a generic θ_0, $v_t/\|v_t\| \to E_{\min}(H_*)$ and $\mu_{\text{eff}}(\theta_t) \to \lambda_{\min}(H_*)$ (for logistic regression, $\lambda_{\min}(H_*) = \lambda + \lambda_{\min}(\frac{1}{n}\sum_i p_i x_i x_i^\top)$, which equals λ when $d > n$). When $\theta_0 \in S = \operatorname{span}\{x_1, \ldots, x_n\}$, e.g. $\theta_0 = 0$, Lemma 2.16 confines the whole trajectory to S, which is H_*-invariant, and the expectation becomes, as in Proposition 2.17(d),

$$\mu_{\text{eff}}(\theta_t) \;\to\; \lambda_{\min}(H_*|_S) \;=\; \lambda + \lambda_{\min}^+\Big(\frac{1}{n}\sum_{i=1}^n p_i x_i x_i^\top\Big),$$

where $\lambda_{\min}^{+}$ denotes the smallest nonzero eigenvalue: the matrix $\frac{1}{n}\sum_i p_i x_i x_i^\top$ has range S (all $p_i > 0$) and its restriction to S is positive definite. For $d > n$ this limit is strictly larger than $\lambda_{\min}(H_*) = \lambda$, and the value λ is approached only from a $\theta_0 \notin S$, through the component $v_{t,\perp} = (1-\gamma\lambda)^t v_{0,\perp}$ of Lemma 2.16(c), which then dominates the error for large t. (Sanity check on a random instance with $n = 20$, $d = 40$, unit-norm Gaussian features, $\lambda = 10^{-4}$ and $\gamma = 1/L$: from $\theta_0 = 0$, $\mu_{\text{eff}}(\theta_t)$ decreases monotonically to $1.36 \cdot 10^{-4}$ after 3000 iterations, against $\lambda + \lambda_{\min}^{+} = 1.35 \cdot 10^{-4}$, while the $S^\perp$-component of $\theta_t - \eta_*$ stays at 10^{-14}; from a Gaussian θ_0 it reaches λ to within $3 \cdot 10^{-5}$ in relative terms.) We state this as a heuristic: a rigorous proof requires the stable-manifold theory of the C^1 map $T(\theta) = \theta - \gamma F'(\theta)$ at its fixed point η_*, whose Jacobian $I - \gamma H_*$ has eigenvalues in $[0, 1)$, and is not needed here. For logistic regression the linearization error is explicit: by (45), $|w_i(\theta) - p_i| \le (e^{R_x\|v\|} - 1)p_i$, hence for every unit vector w, $\left|w^\top(\bar{H}(\theta) - H_*)w\right| \le \frac{1}{n}\sum_i |w_i(\theta) - p_i|\,(x_i^\top w)^2 \le (e^{R_x\|v\|} - 1)\, w^\top(H_* - \lambda I)w$, i.e. $\left\|\bar{H}(\theta) - H_*\right\|_{\text{op}} \le (e^{R_x\|\theta-\eta_*\|} - 1)\,\|H_* - \lambda I\|_{\text{op}}$, and likewise $\left|\mu_{\text{eff}}(\theta) - u^\top H_* u\right| \le (e^{R_x\|\theta-\eta_*\|} - 1)(u^\top H_* u - \lambda)$ (Corollary 2.15). Once $R_x\,\|\theta_t - \eta_*\| \le 0.1$, the quadratic model is accurate to about 10% in relative terms, and the measured $\mu_{\text{eff}}(\theta_t)$ can be compared with the "quadratic proxy" $\sum_k \lambda_k \pi_k(t)$ built from the eigenpairs (λ_k, q_k) of H_*.

SAGA on a quadratic: a linear random recursion. Let now $f_i(\theta) = \frac{1}{2}\theta^\top H_i \theta - b_i^\top \theta + c_i$ with H_i symmetric, $0 \preccurlyeq H_i \preccurlyeq R^2 I$, and $H = \frac{1}{n}\sum_i H_i \succcurlyeq \mu I$ (for instance ridge regression, $H_i = x_i x_i^\top + \lambda I$; logistic regression is of this form to first order near η_* with $H_i = f_i''(\eta_*)$). Then $f_i'(\theta) - f_i'(\eta_*) = H_i v$ exactly. Consider SAGA with step γ and memory $z_i^{(t)}$ as in the book, and define the memory errors $e_i^{(t)} = z_i^{(t)} - f_i'(\eta_*)$ and $\bar{e}^{(t)} = \frac{1}{n}\sum_i e_i^{(t)}$.

Lemma 2.20 (SAGA on a quadratic: exact first- and second-moment recursions)**.** *Let $i(t)$ be uniform on $\{1, \dots, n\}$ and independent of the past $\mathcal{F}_{t-1} = \sigma(i(1), \dots, i(t-1))$.*

(a) (State recursion) $v_t = (I - \gamma H)v_{t-1} + \gamma\xi_t$ with

$$\xi_t = \big(e_{i(t)}^{(t-1)} - \bar{e}^{(t-1)}\big) - \big(H_{i(t)} - H\big)v_{t-1}, \qquad \mathbb{E}_{i(t)}[\xi_t] = 0,$$

and $e_{i(t)}^{(t)} = H_{i(t)}v_{t-1}$, $e_j^{(t)} = e_j^{(t-1)}$ for $j \neq i(t)$. Moreover $\xi_t = F'(\theta_{t-1}) - \omega_t$ is minus the error of the SAGA gradient estimator ω_t.

(b) (Mean) $\mathbb{E}[v_t] = (I - \gamma H)^t v_0$: the mean error follows exactly the gradient descent recursion.

(c) (Second moment) With $C_t = \mathbb{E}_{i(t)}[\xi_t \xi_t^\top] = \frac{1}{n}\sum_{i=1}^n g_i^{(t)} g_i^{(t)\top}$, where $g_i^{(t)} = (e_i^{(t-1)} - \bar{e}^{(t-1)}) - (H_i - H)v_{t-1}$ (so that C_t is the conditional covariance of ω_t given $\mathcal{F}_{t-1}$),

$$\mathbb{E}[v_t v_t^\top] = (I - \gamma H)\,\mathbb{E}[v_{t-1}v_{t-1}^\top]\,(I - \gamma H) + \gamma^2 \mathbb{E}[C_t],$$

and therefore, for every eigenpair (λ_k, q_k) of H,

$$\mathbb{E}\big[(q_k^\top v_t)^2\big] = (1 - \gamma\lambda_k)^{2t}(q_k^\top v_0)^2 + \gamma^2 \sum_{s=1}^{t} (1 - \gamma\lambda_k)^{2(t-s)}\, q_k^\top \mathbb{E}[C_s]\, q_k. \tag{52}$$

Proof. (a) By definition $\omega_t = f_{i(t)}'(\theta_{t-1}) + \frac{1}{n}\sum_j z_j^{(t-1)} - z_{i(t)}^{(t-1)}$. Write $f_{i(t)}'(\theta_{t-1}) = f_{i(t)}'(\eta_*) + H_{i(t)}v_{t-1}$, $z_j^{(t-1)} = f_j'(\eta_*) + e_j^{(t-1)}$ and $\frac{1}{n}\sum_j f_j'(\eta_*) = F'(\eta_*) = 0$. Then $\omega_t = H_{i(t)}v_{t-1} + \bar{e}^{(t-1)} - e_{i(t)}^{(t-1)} = Hv_{t-1} - \xi_t$, and $v_t = v_{t-1} - \gamma\omega_t$ gives the recursion. Since $i(t)$ is uniform and independent

of $\mathcal{F}_{t-1}$, $\mathbb{E}_{i(t)}[e_{i(t)}^{(t-1)}] = \bar{e}^{(t-1)}$ and $\mathbb{E}_{i(t)}[H_{i(t)}] = H$, so $\mathbb{E}_{i(t)}[\xi_t] = 0$. The memory update is $z_{i(t)}^{(t)} = f'_{i(t)}(\theta_{t-1})$, i.e. $e_{i(t)}^{(t)} = f'_{i(t)}(\theta_{t-1}) - f'_{i(t)}(\eta_*) = H_{i(t)}v_{t-1}$.

(b) Take expectations in (a): $\mathbb{E}[v_t] = (I-\gamma H)\mathbb{E}[v_{t-1}] + \gamma\mathbb{E}[\mathbb{E}_{i(t)}[\xi_t]] = (I-\gamma H)\mathbb{E}[v_{t-1}]$.

(c) Expand $v_t v_t^\top = (I-\gamma H)v_{t-1}v_{t-1}^\top(I-\gamma H) + \gamma\big[(I-\gamma H)v_{t-1}\xi_t^\top + \xi_t v_{t-1}^\top(I-\gamma H)\big] + \gamma^2\xi_t\xi_t^\top$. Since v_{t-1} is $\mathcal{F}_{t-1}$-measurable and $\mathbb{E}_{i(t)}[\xi_t] = 0$, the conditional expectation of the bracket vanishes; $\mathbb{E}_{i(t)}[\xi_t\xi_t^\top] = \frac{1}{n}\sum_i g_i^{(t)}g_i^{(t)\top} = C_t$ because $\xi_t = g_{i(t)}^{(t)}$. Taking total expectations gives the matrix recursion; multiplying it on both sides by q_k and unrolling from t down to 0 gives (52). □

Reading of (52) and a quasi-stationary heuristic. The first term of (52) is the gradient-descent term of Proposition 2.17, which concentrates the error on $E_{\min}$. The second term is the gradient-estimator noise accumulated along the trajectory: the noise injected at time s in direction q_k is damped by $(1-\gamma\lambda_k)^{2(t-s)}$, i.e. it is remembered for about $1/(2\gamma\lambda_k)$ iterations, longer in flat directions. The noise covariance C_s is itself a quadratic function of the state $(v_{s-1}, e^{(s-1)})$, with $\operatorname{tr} C_s = \frac{1}{n}\sum_i \left\|g_i^{(s)}\right\|^2 \le \frac{2}{n}\sum_i \left\|e_i^{(s-1)}\right\|^2 + \frac{2}{n}\sum_i \|H_i v_{s-1}\|^2$ (the two quantities bounded in Step 1 of the proof of Proposition 5.9), so it decays at the rate of the Lyapunov function. To extract a prediction, assume (quasi-stationarity) that in the linear-convergence regime $\mathbb{E}[C_s] \approx \rho^{s-t}\mathbb{E}[C_t]$ for $s \le t$, where ρ is the contraction factor of the second moments. For the proved factor $\rho = 1 - \min\{1/(3n), 3\mu/(16R^2)\}$, in the regime $\kappa \gg n$ with $\gamma = 1/(4R^2)$ one has $1-\rho = \frac{3}{4}\gamma\mu$ while $1-(1-\gamma\lambda_k)^2 = \gamma\lambda_k(2-\gamma\lambda_k) \ge \gamma\lambda_k \ge \gamma\mu$, so $\rho > (1-\gamma\lambda_k)^2$ for every k: the noise term decays more slowly than the gradient-descent term in *every* direction and dominates for large t. For the observed factor ρ_{obs} this strict comparison is not established; what is guaranteed is the weaker inequality $\rho_{\text{obs}} \ge (1-\gamma\lambda_k)^2$ for every k with $q_k^\top v_0 \neq 0$, because by (b) and Jensen's inequality $\mathbb{E}[(q_k^\top v_t)^2] \ge (q_k^\top \mathbb{E}v_t)^2 = (1-\gamma\lambda_k)^{2t}(q_k^\top v_0)^2$, so that the second moment along q_k cannot contract faster than $(1-\gamma\lambda_k)^2$ per step; in particular, if $Pv_0 \neq 0$, $1-\rho_{\text{obs}} \le 2\gamma\lambda_{\min} - \gamma^2\lambda_{\min}^2$, which keeps the denominators $2\lambda_k - (1-\rho_{\text{obs}})/\gamma \ge 2(\lambda_k - \lambda_{\min}) + \gamma\lambda_{\min}^2 > 0$ appearing below positive (with $\lambda_{\min}$ read as $\lambda_{\min}(H|_S)$ in the situation of Proposition 2.17(d)). Summing the geometric series,

$$\mathbb{E}\big[(q_k^\top v_t)^2\big] \;\approx\; \gamma^2\,\frac{c_k(t)}{1-(1-\gamma\lambda_k)^2/\rho} \;\approx\; \frac{\gamma\, c_k(t)}{2\lambda_k - (1-\rho)/\gamma}, \qquad c_k(t) = q_k^\top \mathbb{E}[C_t] q_k, \tag{53}$$

using $(1-\gamma\lambda_k)^2 \approx 1 - 2\gamma\lambda_k$ and $\rho \approx 1$. The predicted spectral profile of SAGA is therefore

$$\pi_k^{\text{SAGA}}(t) \;\propto\; \frac{c_k(t)}{2\lambda_k - (1-\rho)/\gamma}, \qquad \mu_{\text{eff}}(\theta_t) \;\approx\; \frac{\sum_k \lambda_k\, c_k(t)/(2\lambda_k - (1-\rho)/\gamma)}{\sum_k c_k(t)/(2\lambda_k - (1-\rho)/\gamma)}, \tag{54}$$

a weighted mean of the eigenvalues in which direction k receives the weight $c_k/(2\lambda_k - (1-\rho)/\gamma)$: the noise re-injects *all* directions it feeds, and the weight of a flat direction is larger only by the factor $1/\lambda_k$, not by the exponentially growing factor $((1-\gamma\lambda_{\min})/(1-\gamma\lambda_k))^{2t}$ of gradient descent. Two illustrative noise shapes: (i) isotropic noise, $c_k \equiv c$, gives $\mu_{\text{eff}} \approx \big[\sum_k \lambda_k/(2\lambda_k - \frac{3}{4}\mu)\big]/\big[\sum_k 1/(2\lambda_k - \frac{3}{4}\mu)\big]$; with one eigenvalue equal to μ and $d-1$ eigenvalues $L \gg d\mu$ this is $\approx \mu\,(0.8 + (d-1)/2)/0.8$, i.e. larger than $\lambda_{\min} = \mu$ by a factor of order d; (ii) "data-shaped" noise, $c_k \propto \lambda_k - \lambda$ (for logistic or ridge regression started at $\theta_0 \in S = \operatorname{span}\{x_1, \dots, x_n\}$, e.g. $\theta_0 = 0$, all vectors $g_i^{(t)}$ lie in S by Lemma 2.16(b), so the noise has no energy in the pure-regularization directions $S^\perp$; for a generic θ_0 these directions carry, by Lemma 2.16(c), the noise $c_q = \lambda^2 \operatorname{Var}_j(q^\top v_{s_j})$, $q \in S^\perp$, i.e. λ^2 times the spread of the $S^\perp$-errors stored in the memory, which is negligible compared with the data-direction noise unless λ is comparable to the data curvature), gives weights $\propto (\lambda_k - \lambda)/(2\lambda_k - \frac{3}{4}\mu)$, which vanish on the directions with $\lambda_k \approx \lambda$ and are $\approx \frac{1}{2}$ on all directions with $\lambda_k \gg \lambda$: $\mu_{\text{eff}}(\theta_t)$ is then of the

order of the *average* curvature of the data directions, far above $\lambda_{\min}(H_*) = \lambda + \lambda_{\min}(\frac{1}{n}\sum_i p_i x_i x_i^\top)$ (and above $\lambda_{\min}(H_*|_S)$ when $\theta_0 \in S$).

Testable predictions and what to measure. Let (λ_k, q_k) be the eigenpairs of $H_* = F''(\eta_*)$ (with η_* from a high-precision solver), and record along each run: the profile $\pi_k(t)$ of (50), its effective number of directions $d_{\text{eff}}(t) = 1/\sum_k \pi_k(t)^2 \in [1, d]$, the exact $\mu_{\text{eff}}(\theta_t)$ (formula (38) with the closed-form weights (36), or directly Definition 2.2) and its quadratic proxy $\sum_k \lambda_k \pi_k(t)$, and, for SAGA, the noise spectrum $c_k(t) = q_k^\top C_t q_k$ where, for a general (non-quadratic) F,

$$C_{t+1} = \frac{1}{n}\sum_{i=1}^n g_i g_i^\top, \qquad g_i = F'(\theta_t) - f_i'(\theta_t) + z_i^{(t)} - \frac{1}{n}\sum_j z_j^{(t)},$$

is the conditional covariance of the SAGA estimator ω_{t+1} over the n possible draws of $i(t+1)$, given the state $(\theta_t, z^{(t)})$ (Lemma 2.20(c), Lemma 2.16). Forming all the $c_k(t)$ costs $O(nd^2)$ per evaluation (the product $Q^\top G$ with $G = [g_1, \dots, g_n] \in \mathbb{R}^{d\times n}$ and $Q = [q_1, \dots, q_d]$, or n rank-one updates of a $d \times d$ matrix); it is $O(nd + d^2)$ only if a single direction q_k is monitored. Done once per pass, this is affordable for d up to a few thousands. The initial point must be part of the protocol, since it selects the subspace explored (Lemma 2.16): run every experiment both from $\theta_0 = 0 \in S$ and from a generic θ_0 (e.g. standard Gaussian), and compare $\mu_{\text{eff}}(\theta_t)$ with $\lambda_{\min}(H_*|_S) = \lambda + \lambda_{\min}^+(H_* - \lambda I)$ in the first case and with $\lambda_{\min}(H_*)$ in the second (the two coincide when the features span $\mathbb{R}^d$).

(P1) *Gradient descent*: $\mu_{\text{eff}}(\theta_t)$ is non-increasing (exactly for a quadratic, Proposition 2.17(b)) and converges to $\lambda_{\min}(H_*)$ for a generic θ_0, and to $\lambda_{\min}(H_*|_S) = \lambda + \lambda_{\min}^+(H_* - \lambda I)$ when $\theta_0 \in S$, e.g. $\theta_0 = 0$ (Proposition 2.17(d), Lemma 2.16, Remark 2.19); under alignment, $\pi(t)$ concentrates on the corresponding bottom eigenspace, so that $d_{\text{eff}}(t)$ converges to a value in $[1, m]$, m being the multiplicity of that bottom eigenvalue: the limit is $1/\sum_{k\in K_{\min}}(\pi_k^\infty)^2$ with $\pi_k^\infty = (q_k^\top P v_0)^2/\|Pv_0\|^2$, which equals m only when Pv_0 is equidistributed over the m eigenvectors (generically $m = 1$ and $d_{\text{eff}}(t) \to 1$); the convergence speed is governed by the spectral gap $\lambda' - \lambda_{\min}$ of the explored spectrum.

(P2) *SAGA*: after the initial transient, $\pi_k(t)$ remains spread over the directions fed by the noise, with $\pi_k(t) \propto c_k(t)/(2\lambda_k - (1-\rho)/\gamma)$ up to fluctuations (54); $d_{\text{eff}}(t)$ stays of the order of the number of data directions; $\mu_{\text{eff}}(\theta_t)$ fluctuates around a level strictly above $\lambda_{\min}(H_*)$ (resp. $\lambda_{\min}(H_*|_S)$ when $\theta_0 \in S$) and does not decrease monotonically; for the same problem and large t, $\mu_{\text{eff}}^{\text{SAGA}}(t) > \mu_{\text{eff}}^{\text{GD}}(t)$.

(P3) *Consequence for (B2)*: the slack of the strong-convexity step (25), measured by $\mu_{\text{eff}}(\theta_t)/\mu$, is larger for SAGA than the asymptotic slack $\lambda_{\min}(H_*)/\mu$ (resp. $\lambda_{\min}(H_*|_S)/\mu$ for $\theta_0 \in S$) of gradient descent; in the "intermediate regularization" regime of Section 2.3 this is precisely the situation where the data anisotropy makes the bottom eigenvalue of the explored spectrum close to λ while the trajectory-averaged curvature is much larger. A violation of (P2) (e.g. $\pi(t)$ concentrating on $E_{\min}$ for SAGA as well) would mean that the noise term in (52) decays faster than assumed, which can be checked directly on $c_k(t)$.

2.5 How μ_{eff} and L_{eff} enter the SAGA bound

We use the setting of Proposition 5.9 of [6] (SAGA [7] with uniform sampling): step $\gamma = 1/(4R^2)$, $\Delta = 3\gamma^2$, and the Lyapunov function

$$V_t = \|\theta_t - \eta_*\|^2 + \Delta \sum_{i=1}^n \left\| z_i^{(t)} - f_i'(\eta_*) \right\|^2.$$

The B3 note derives the following trajectory-dependent proposition by instrumenting each inequality of the book's proof ("see the B3 note"); we include the short derivation because it shows exactly where μ_{eff} and L_{eff} enter.

Proposition 2.21 (one-step contraction with trajectory-dependent constants)**.** *Under Assumption 2.1, for every $t \geq 1$, conditionally on the past and on the event $\theta_{t-1} \neq \eta_*$,*

$$\mathbb{E}_{i(t)}\big[V_t\big] \;\leq\; \Big(1 - \min\Big\{\frac{1}{3n},\, a(\theta_{t-1})\Big\}\Big) V_{t-1}, \qquad a(\theta) = \frac{\mu_{\text{eff}}(\theta)\,\big(8R^2 - 5L_{\text{eff}}(\theta)\big)}{16R^4}, \tag{55}$$

and, since $L_{\text{eff}}(\theta) \leq R^2$ implies $a(\theta) \geq 3\mu_{\text{eff}}(\theta)/(16R^2)$,

$$\mathbb{E}_{i(t)}\big[V_t\big] \;\leq\; \Big(1 - \min\Big\{\frac{1}{3n},\, \frac{3\mu_{\text{eff}}(\theta_{t-1})}{16R^2}\Big\}\Big) V_{t-1}. \tag{56}$$

On the event $\theta_{t-1} = \eta_*$ one has $\mathbb{E}_{i(t)}[V_t] \leq (1 - \frac{1}{3n})V_{t-1}$. The bound of Proposition 5.9, $\mathbb{E}_{i(t)}[V_t] \leq (1 - \min\{\frac{1}{3n}, \frac{3\mu}{16R^2}\})V_{t-1}$, follows from (56) and $\mu_{\text{eff}} \geq \mu$.

Proof. Write $v = \theta_{t-1} - \eta_*$, $e_i = z_i^{(t-1)} - f_i'(\eta_*)$, and ω_t for the SAGA direction, so that $\theta_t - \eta_* = v - \gamma\omega_t$ and $\mathbb{E}_{i(t)}[\omega_t] = F'(\theta_{t-1})$. Step 1 of the book's proof gives, using only the expansion of the square, unbiasedness, $\|a+b\|^2 \leq 2\|a\|^2 + 2\|b\|^2$ and $\mathbb{E}_{i(t)}\big\|Z - \mathbb{E}_{i(t)}Z\big\|^2 \leq \mathbb{E}_{i(t)}\|Z\|^2$,

$$\mathbb{E}_{i(t)}\|\theta_t - \eta_*\|^2 \leq \|v\|^2 - 2\gamma\, v^\top F'(\theta_{t-1}) + 2\gamma^2\,\frac{1}{n}\sum_i \big\|f_i'(\theta_{t-1}) - f_i'(\eta_*)\big\|^2 + \frac{2\gamma^2}{n}\sum_i \|e_i\|^2\,,$$

and Step 2 gives the *identity* $\mathbb{E}_{i(t)}\sum_i \Big\|z_i^{(t)} - f_i'(\eta_*)\Big\|^2 = (1-\frac{1}{n})\sum_i\|e_i\|^2 + \frac{1}{n}\sum_i\|f_i'(\theta_{t-1}) - f_i'(\eta_*)\|^2$. Instead of bounding $\frac{1}{n}\sum_i\|f_i'(\theta_{t-1}) - f_i'(\eta_*)\|^2$ by $R^2 v^\top F'(\theta_{t-1})$ (co-coercivity), we use its exact value $L_{\text{eff}}(\theta_{t-1})\, v^\top F'(\theta_{t-1})$ (Definition 2.2). Adding Δ times the second relation to the first,

$$\mathbb{E}_{i(t)}[V_t] \leq \|v\|^2 - 2\gamma\Big(1 - \gamma L_{\text{eff}} - \frac{L_{\text{eff}}\Delta}{2\gamma}\Big) v^\top F'(\theta_{t-1}) + \Big(\frac{2\gamma^2}{n\Delta} + 1 - \frac{1}{n}\Big)\Delta\sum_i\|e_i\|^2\,,$$

with $L_{\text{eff}} = L_{\text{eff}}(\theta_{t-1})$. With $\Delta = 3\gamma^2$ and $\gamma = 1/(4R^2)$: $1 - \gamma L_{\text{eff}} - \frac{3}{2}\gamma L_{\text{eff}} = 1 - \frac{5}{2}\gamma L_{\text{eff}} = 1 - \frac{5L_{\text{eff}}}{8R^2} \geq \frac{3}{8} > 0$ (as $L_{\text{eff}} \leq R^2$), and $\frac{2\gamma^2}{n\Delta} + 1 - \frac{1}{n} = 1 - \frac{1}{3n}$. Now, instead of $v^\top F'(\theta_{t-1}) \geq \mu\|v\|^2$, we use the exact value $v^\top F'(\theta_{t-1}) = \mu_{\text{eff}}(\theta_{t-1})\|v\|^2$, and the coefficient of $\|v\|^2$ becomes $1 - 2\gamma(1 - \frac{5}{2}\gamma L_{\text{eff}})\mu_{\text{eff}} = 1 - \frac{\mu_{\text{eff}}(8R^2 - 5L_{\text{eff}})}{16R^4} = 1 - a(\theta_{t-1})$, which is nonnegative because $a \leq \mu_{\text{eff}}/(2R^2) \leq L/(2R^2) \leq \frac{1}{2}$. Hence $\mathbb{E}_{i(t)}[V_t] \leq (1-a)\|v\|^2 + (1 - \frac{1}{3n})\Delta\sum_i\|e_i\|^2 \leq (1 - \min\{\frac{1}{3n}, a\})V_{t-1}$. If $v = 0$, the gradient-difference term vanishes and the same computation gives the factor $1 - \frac{1}{3n}$. □

Remark 2.22 (what each constant can explain). With the step size fixed at $\gamma = 1/(4R^2)$, the gain coming from $L_{\text{eff}} < R^2$ is the factor $(8R^2 - 5L_{\text{eff}})/(3R^2) \in [1, 8/3]$ on the parameter-term exponent: the co-coercivity slack can explain at most a factor $8/3$ of the gap, however small L_{eff} is (a smaller L_{eff} would allow a larger step, but the algorithm does not take it). In contrast, the gain from $\mu_{\text{eff}} > \mu$ is the factor μ_{eff}/μ, which can be as large as L/μ (Lemma 2.4) and is typically of the order described in Sections 2.3–2.4. The parameter term is the active one in the minimum as long as $a(\theta_{t-1}) < 1/(3n)$, i.e. (with $L_{\text{eff}} = R^2$) $\mu_{\text{eff}}(\theta_{t-1}) < 16R^2/(9n)$; in the regime $\kappa \gg n$ of hypothesis (A4) this holds for μ and for moderate values of μ_{eff}, and the bound saturates at $1 - \frac{1}{3n}$ once μ_{eff} exceeds $16R^2/(9n)$: beyond that point no improvement of the curvature term can be "explained", and the residual gap belongs to the memory term (inequality $\mathbb{E}_{i(t)}\big\|Z - \mathbb{E}_{i(t)}Z\big\|^2 \leq \mathbb{E}_{i(t)}\|Z\|^2$ and the choice of Δ), instrumented in the B3 note.

Explained fraction of the theory–practice gap. Fix a window $W = \{t_0, \dots, t_1 - 1\}$ of $T = t_1 - t_0$ iterations inside the linear-convergence regime (e.g. one effective pass, $T = n$), and let `V[t]`, `mu_eff[t]`, `L_eff[t]` be the recorded values of V_t, $\mu_{\text{eff}}(\theta_t)$, $L_{\text{eff}}(\theta_t)$ (computed with η_* from a high-precision solver). Define the per-iteration exponents

$$\nu_{\text{obs}} = \frac{1}{T} \log \frac{V_{t_0}}{V_{t_1}}, \tag{57}$$

$$\nu_{\text{th}} = -\log\Big(1 - \min\Big\{\frac{1}{3n}, \frac{3\mu}{16R^2}\Big\}\Big), \tag{58}$$

$$r_\mu = -\frac{1}{T} \sum_{t \in W} \log\Big(1 - \min\Big\{\frac{1}{3n}, \frac{3\mu_{\text{eff}}(\theta_t)}{16R^2}\Big\}\Big), \tag{59}$$

$$\nu_{\mu L} = -\frac{1}{T} \sum_{t \in W} \log\Big(1 - \min\Big\{\frac{1}{3n}, \frac{\mu_{\text{eff}}(\theta_t)\,(8R^2 - 5L_{\text{eff}}(\theta_t))}{16R^4}\Big\}\Big), \tag{60}$$

and the explained fractions

$$\phi_\mu = \frac{\nu_\mu - \nu_{\text{th}}}{\nu_{\text{obs}} - \nu_{\text{th}}}, \qquad \phi_L = \frac{\nu_{\mu L} - \nu_\mu}{\nu_{\text{obs}} - \nu_{\text{th}}}, \qquad \phi_{\text{rest}} = 1 - \phi_\mu - \phi_L = \frac{\nu_{\text{obs}} - \nu_{\mu L}}{\nu_{\text{obs}} - \nu_{\text{th}}}. \tag{61}$$

By Lemmas 2.4 and 2.6, $a(\theta) \geq 3\mu_{\text{eff}}(\theta)/(16R^2) \geq 3\mu/(16R^2)$ and the monotonicity of $x \mapsto -\log(1 - \min\{\frac{1}{3n}, x\})$, one has $\nu_{\text{th}} \leq \nu_\mu \leq \nu_{\mu L}$, so $\phi_\mu, \phi_L \geq 0$. The quantity $e^{-T\nu_{\mu L}} = \prod_{t \in W}(1 - \min\{\frac{1}{3n}, a(\theta_t)\})$ is exactly the contraction guaranteed over the window by Proposition 2.21 along the realized trajectory, so that $\nu_{\mu L} \leq \nu_{\text{obs}}$ and $\phi_{\text{rest}} \geq 0$ up to the stochastic fluctuations of V (the bound holds in conditional expectation while ν_{obs} is a realized quantity; average over seeds, or use windows of several passes). Thus ϕ_μ is the fraction of the gap (in exponent) explained by the trajectory-dependent curvature alone, ϕ_L the additional fraction explained by the effective co-coercivity constant (at most a factor 8/3 by Remark 2.22), and ϕ_{rest} the fraction left to the other inequalities of the proof (square-norm inequality, variance versus second moment, the minimum of the two rates, the choice of Δ), each of which the B3 note measures separately. The same decomposition can be reported instantaneously through the gain factors

$$g_\mu(t) = \frac{\min\{\frac{1}{3n}, \frac{3\mu_{\text{eff}}(\theta_t)}{16R^2}\}}{\min\{\frac{1}{3n}, \frac{3\mu}{16R^2}\}} \geq 1, \qquad g_L(t) = \frac{\min\{\frac{1}{3n}, a(\theta_t)\}}{\min\{\frac{1}{3n}, \frac{3\mu_{\text{eff}}(\theta_t)}{16R^2}\}} \in \Big[1, \frac{8}{3}\Big],$$

whose product is the ratio between the guaranteed per-step exponent along the trajectory and the one of Proposition 5.9. Finally, the per-step tightness ratios of the two inequalities themselves are, in the notation of Definition 1.8 of the B3 note,

$$r_{\text{sc}}(t) = \frac{\mu \|\theta_t - \eta_*\|^2}{(\theta_t - \eta_*)^\top F'(\theta_t)} = \frac{\mu}{\mu_{\text{eff}}(\theta_t)} \in \Big[\frac{\mu}{L}, 1\Big],$$

$$r_{\text{coco}}(t) = \frac{\frac{1}{n}\sum_i \|f_i'(\theta_t) - f_i'(\eta_*)\|^2}{R^2(\theta_t - \eta_*)^\top F'(\theta_t)} = \frac{L_{\text{eff}}(\theta_t)}{R^2} \in \Big[\frac{\mu}{R^2}, 1\Big],$$

for (25) and (29) respectively.

2.6 Related work

Local strong convexity and self-concordance. The observation that the curvature actually experienced by an algorithm is the local one is classical; the book itself notes that "locally around the

global optimum, the strong convexity constant converges to the minimal eigenvalue of the Hessian at η_*, which can be significantly larger than μ (the global constant)" ([6], Section 5.2.3, paragraph "Adaptivity"). For logistic regression, [5] proves that averaged SGD with constant step size proportional to $1/(R^2\sqrt{N})$ has an $O(1/\sqrt{N})$ rate which "improves to $O(R^2/(\mu N))$ where μ is the lowest eigenvalue of the Hessian at the global optimum", without knowledge of μ (adaptivity to unknown local strong convexity); the tool is the generalized self-concordance of the logistic loss, Assumption (A2) there: along every line, $|\varphi'''(t)| \leq R\,\|\theta_1 - \theta_2\|\,\varphi''(t)$, a notion introduced in [**?**], whose Proposition 1, eq. (6), gives the Hessian comparison $e^{-R\|v\|}F''(w) \preccurlyeq F''(w+v) \preccurlyeq e^{R\|v\|}F''(w)$. Lemma 2.14 and Corollary 2.15 are the per-sample, directional version of this comparison, and $\mu_{\text{eff}}(\theta_t)$ is the trajectory-wise counterpart of the local constant $\lambda_{\min}(H_*)$: by Proposition 2.8 it converges to the Rayleigh quotient $u_t^\top H_* u_t \geq \lambda_{\min}(H_*)$, with equality only when the error aligns with the flattest direction, which Section 2.4 predicts for gradient descent but not for SAGA.

Expected smoothness. The constant L_{eff} is a trajectory-wise version of the *expected smoothness* constant of [13] (Assumption 3.1 there: there is $\mathcal{L}_1 > 0$ with $\mathbb{E}_{\mathcal{D}}\,\|\nabla f_S(x) - \nabla f_S(x^*)\|^2 \leq 2\mathcal{L}_1(f(x) - f(x^*))$ for all x), which governs the rates of SAGA and of its JacSketch generalizations; for minibatch sketches with uniform sampling, Theorem 4.16 there shows that Assumption 3.1 is satisfied with $\mathcal{L}_1 = L^{\mathcal{G}}_{\max}$ (eq. (85) there), which for single-element uniform sampling is $\max_i L_i$, a valid constant rather than the smallest one; the same assumption is the basis of the general SGD analysis of [12] (Assumption 2.1: $\mathbb{E}_{\mathcal{D}}\,\|\nabla f_v(x) - \nabla f_v(x^*)\|^2 \leq 2\mathcal{L}(f(x) - f(x^*))$). Defining the trajectory-wise expected smoothness $\mathcal{L}_{\text{ES}}(\theta) = \frac{1}{n}\sum_i \|f_i'(\theta) - f_i'(\eta_*)\|^2 / \big(2(F(\theta) - F(\eta_*))\big)$, the convexity inequality $F(\theta) - F(\eta_*) \leq (\theta - \eta_*)^\top F'(\theta)$ (eq. (5.7) of the book) gives $\mathcal{L}_{\text{ES}}(\theta) \geq L_{\text{eff}}(\theta)/2$, with $\mathcal{L}_{\text{ES}} = L_{\text{eff}}$ for quadratics (where $2(F(\theta) - F(\eta_*)) = (\theta - \eta_*)^\top F'(\theta)$). We work with L_{eff} because the proof of Proposition 5.9 is written with the inner product $(\theta - \eta_*)^\top F'(\theta)$ rather than with function values; both constants measure how much of the worst-case $R^2 = \max_i L_i$ is used along the trajectory, and both can be recorded at the same cost.

3 Preconditioned (quadratic-metric) SAGA (hypothesis B1)

Throughout, $F(\theta) = \frac{1}{n}\sum_{i=1}^n f_i(\theta)$ on $\mathbb{R}^d$, each f_i is convex and R^2-smooth, F is μ-strongly convex with minimizer η_*, $\kappa = R^2/\mu$, and $P \in \mathbb{R}^{d\times d}$ is a *fixed* symmetric positive definite (SPD) matrix; $P^{1/2}$ denotes its SPD square root and $P^{-1/2} = (P^{1/2})^{-1}$. Whenever a Hessian is written we assume the corresponding function is twice continuously differentiable (this is the case for logistic regression); every statement that only involves gradients holds without this assumption (Remark 3.10). We write $\mathbb{E}_{i(t)}$ for the expectation with respect to the index $i(t)$ drawn at iteration t, conditionally on the past.

The content of this note is the following chain of facts. (1) SAGA run in the metric of P ("P-SAGA") is *exactly* Euclidean SAGA on a linearly transformed problem (Proposition 3.5), so Proposition 5.9 of [6] transfers verbatim, with (R^2, μ) replaced by the relative constants (R_P^2, μ_P) of (67) (Corollary 3.12). (2) Redoing the proof in the P-metric shows which of the six proof inequalities change (Section 3.3.1); this defines the six tightness ratios to be instrumented in the P-geometry (Definition 3.14). (3) On ℓ_2-regularized logistic regression, *no* quadratic metric improves the certified global condition number, $\kappa_P \geq \kappa$ for every P (Proposition 3.21): the benefit of preconditioning is entirely a trajectory-dependent effect, captured by the effective constants of the B2 note; for the statistical preconditioner $P = \hat{\Sigma} + \lambda I$ the trajectory-dependent condition number is controlled by ridge leverage scores and the effective dimension, and the anisotropy of $\hat{\Sigma}$ disappears (Corollary 3.23). (4) Cost and the case of a non-constant metric (Section 3.5), and a prediction table for the experiments (Section 3.6).

3.1 The P-geometry and the algorithm

Definition 3.1 (P-norm and its dual norm)**.** *For $v, g \in \mathbb{R}^d$ let $\|v\|_P^2 := v^\top P v$ (norm on parameters) and $\|g\|_{P^{-1}}^2 := g^\top P^{-1} g$ (norm on gradients and memory vectors).*

Lemma 3.2 (elementary facts on the P-geometry)**.** *Let P be SPD. Then:*

(i) (generalized Cauchy–Schwarz) $|\langle g, v\rangle| \le \|g\|_{P^{-1}} \|v\|_P$ for all $g, v \in \mathbb{R}^d$, with equality if and only if g and Pv are collinear; consequently $\|g\|_{P^{-1}} = \sup_{\|v\|_P \le 1} \langle g, v\rangle$, i.e. $\|\cdot\|_{P^{-1}}$ is the dual norm of $\|\cdot\|_P$;

(ii) $\|P^{1/2} v\| = \|v\|_P$, $\|P^{-1/2} g\| = \|g\|_{P^{-1}}$ and $\|P^{-1} g\|_P = \|g\|_{P^{-1}}$;

(iii) (square-norm inequality) $\|a+b\|_{P^{-1}}^2 \le 2\|a\|_{P^{-1}}^2 + 2\|b\|_{P^{-1}}^2$, with equality iff $a = b$;

(iv) (variance $\le$ second moment) for a random vector Z with $\mathbb{E}\|Z\|^2 < \infty$, $\mathbb{E}\|Z - \mathbb{E}Z\|_{P^{-1}}^2 = \mathbb{E}\|Z\|_{P^{-1}}^2 - \|\mathbb{E}Z\|_{P^{-1}}^2 \le \mathbb{E}\|Z\|_{P^{-1}}^2$;

(v) $\lambda_{\min}(P)\|v\|^2 \le \|v\|_P^2 \le \lambda_{\max}(P)\|v\|^2$ and $\lambda_{\max}(P)^{-1}\|g\|^2 \le \|g\|_{P^{-1}}^2 \le \lambda_{\min}(P)^{-1}\|g\|^2$.

Proof. (i) $\langle g, v\rangle = \langle P^{-1/2} g, P^{1/2} v\rangle$, and the Euclidean Cauchy–Schwarz inequality gives $|\langle g, v\rangle| \le \|P^{-1/2} g\|\,\|P^{1/2} v\| = \|g\|_{P^{-1}}\|v\|_P$, with equality iff $P^{-1/2} g$ and $P^{1/2} v$ are collinear, i.e. iff g and Pv are; the choice $v = P^{-1} g / \|g\|_{P^{-1}}$ (for $g \ne 0$) has $\|v\|_P = 1$ and attains the supremum. (ii) $\|P^{1/2} v\|^2 = v^\top P v$, $\|P^{-1/2} g\|^2 = g^\top P^{-1} g$ and $(P^{-1}g)^\top P (P^{-1} g) = g^\top P^{-1} g$. (iii) $\|a+b\|_{P^{-1}}^2 = \|a\|_{P^{-1}}^2 + 2a^\top P^{-1} b + \|b\|_{P^{-1}}^2$ and $2a^\top P^{-1} b \le \|a\|_{P^{-1}}^2 + \|b\|_{P^{-1}}^2$ because $\|a-b\|_{P^{-1}}^2 \ge 0$, with equality iff $a = b$. (iv) Expand $\mathbb{E}\|Z - \mathbb{E}Z\|_{P^{-1}}^2 = \mathbb{E}\|Z\|_{P^{-1}}^2 - 2(\mathbb{E}Z)^\top P^{-1} \mathbb{E}Z + \|\mathbb{E}Z\|_{P^{-1}}^2$. (v) These are the extremal values of the Rayleigh quotients of P and of P^{-1}. □

Definition 3.3 (P-SAGA)**.** *Fix $\gamma > 0$, $\theta_0 \in \mathbb{R}^d$ and memory vectors $z_i^{(0)} \in \mathbb{R}^d$, $i = 1, \dots, n$. At iteration $t \ge 1$, draw $i(t)$ uniformly in $\{1, \dots, n\}$, independently of the past, and set*

$$g_t := f'_{i(t)}(\theta_{t-1}) + \frac{1}{n}\sum_{i=1}^n z_i^{(t-1)} - z_{i(t)}^{(t-1)}, \tag{62}$$

$$\theta_t := \arg\min_{\theta \in \mathbb{R}^d} \Big\{ \langle g_t, \theta\rangle + \frac{1}{2\gamma}(\theta - \theta_{t-1})^\top P (\theta - \theta_{t-1}) \Big\}, \tag{63}$$

$$z_{i(t)}^{(t)} := f'_{i(t)}(\theta_{t-1}), \qquad z_j^{(t)} := z_j^{(t-1)} \quad (j \ne i(t)). \tag{64}$$

The memory update (64) *is the one of SAGA [7]; only the step* (63) *depends on P.*

Lemma 3.4 (closed form, unbiasedness, Bregman form)**.** *The minimization* (63) *has the unique solution*

$$\theta_t = \theta_{t-1} - \gamma P^{-1} g_t, \tag{65}$$

which for $P = I$ is the SAGA iteration of [6, Sec. 5.4.4]. Moreover $\mathbb{E}_{i(t)}[g_t] = F'(\theta_{t-1})$. Finally, (63) *is the Bregman step $\arg\min_\theta\{\gamma\langle g_t, \theta\rangle + D_h(\theta, \theta_{t-1})\}$ for the quadratic reference function $h = \frac{1}{2}\|\cdot\|_P^2$, whose Bregman divergence is $D_h(\theta, \theta') = \frac{1}{2}\|\theta - \theta'\|_P^2$; i.e. P-SAGA is Algorithm 1 (Bregman-SAGA) of [8] with a quadratic h.*

Proof. The objective in (63) is a strictly convex quadratic (its Hessian is $P/\gamma \succ 0$), so it has a unique minimizer, characterized by a vanishing gradient: $g_t + \gamma^{-1} P(\theta - \theta_{t-1}) = 0$, i.e. $\theta = \theta_{t-1} - \gamma P^{-1} g_t$. Since $i(t)$ is uniform and independent of the past, $\mathbb{E}_{i(t)}[f'_{i(t)}(\theta_{t-1})] = \frac{1}{n}\sum_i f'_i(\theta_{t-1}) = F'(\theta_{t-1})$ and

$\mathbb{E}_{i(t)}[z_{i(t)}^{(t-1)}] = \frac{1}{n}\sum_i z_i^{(t-1)}$, so the two memory terms of (62) cancel in expectation. For $h = \frac{1}{2}\|\cdot\|_P^2$ one has $\nabla h(\theta') = P\theta'$ and $D_h(\theta,\theta') = h(\theta) - h(\theta') - \langle \nabla h(\theta'), \theta - \theta'\rangle = \frac{1}{2}\|\theta - \theta'\|_P^2$; multiplying the objective of (63) by γ gives the Bregman form. □

3.2 Change of variables: P-SAGA is SAGA on a transformed problem

Proposition 3.5 (transfer). *Let $u := P^{1/2}\theta$, $\tilde{f}_i(u) := f_i(P^{-1/2}u)$ and $\tilde{F} := \frac{1}{n}\sum_i \tilde{f}_i = F \circ P^{-1/2}$. Then:*

(i) $\nabla \tilde{f}_i(u) = P^{-1/2} f_i'(P^{-1/2}u)$ and, when f_i is C^2, $\nabla^2 \tilde{f}_i(u) = P^{-1/2} f_i''(P^{-1/2}u) P^{-1/2}$; the same formulas hold for $(\tilde{F}, F)$.

(ii) $\tilde{F}$ is strongly convex and $\tilde{u}_* := P^{1/2}\eta_*$ is its unique minimizer.

(iii) Let $(\theta_t, z^{(t)})_{t\geq 0}$ be generated by P-SAGA (Definition 3.3) with step γ and index sequence $(i(t))_{t\geq 1}$, and let $u_t := P^{1/2}\theta_t$, $\tilde{z}_i^{(t)} := P^{-1/2} z_i^{(t)}$. Then $(u_t, \tilde{z}^{(t)})_{t\geq 0}$ is generated by Euclidean SAGA ($P = I$) on $(\tilde{f}_i)_{i\leq n}$ with the same *step γ and the* same *index sequence, started from $(u_0, \tilde{z}^{(0)})$.*

(iv) For all t and i, $\|u_t - \tilde{u}_*\|^2 = \|\theta_t - \eta_*\|_P^2$ and $\left\|\tilde{z}_i^{(t)} - \nabla \tilde{f}_i(\tilde{u}_*)\right\|^2 = \left\|z_i^{(t)} - f_i'(\eta_*)\right\|_{P^{-1}}^2$; in particular $z_i^{(0)} = f_i'(\theta_0)$ for all i iff $\tilde{z}_i^{(0)} = \nabla \tilde{f}_i(u_0)$ for all i.

Proof. (i) $\tilde{f}_i = f_i \circ A$ with the linear map $A = P^{-1/2}$ (symmetric), so by the chain rule $\nabla \tilde{f}_i(u) = A^\top f_i'(Au) = P^{-1/2} f_i'(P^{-1/2}u)$ and $\nabla^2 \tilde{f}_i(u) = A^\top f_i''(Au) A$. Averaging over i gives the statements for $\tilde{F}$.

(ii) By (i), $\nabla \tilde{F}(\tilde{u}_*) = P^{-1/2} F'(\eta_*) = 0$. For $u, u' \in \mathbb{R}^d$ write $\theta = P^{-1/2}u$, $\theta' = P^{-1/2}u'$; by (i) and μ-strong convexity of F,

$$\tilde{F}(u') - \tilde{F}(u) - \left\langle \nabla \tilde{F}(u), u' - u \right\rangle = F(\theta') - F(\theta) - \left\langle F'(\theta), \theta' - \theta \right\rangle \geq \frac{\mu}{2}\left\|\theta' - \theta\right\|^2 \geq \frac{\mu}{2\lambda_{\max}(P)}\left\|u' - u\right\|^2,$$

where we used $\left\langle P^{-1/2}F'(\theta), u' - u\right\rangle = \langle F'(\theta), \theta' - \theta\rangle$ and Lemma 3.2(v), which gives $\|\theta' - \theta\|^2 = \left\|P^{-1/2}(u' - u)\right\|^2 \geq \|u' - u\|^2 / \lambda_{\max}(P)$. Hence $\tilde{F}$ is $(\mu/\lambda_{\max}(P))$-strongly convex and its stationary point $\tilde{u}_*$ is its unique minimizer.

(iii) By induction on t; the initialization is the definition of $(u_0, \tilde{z}^{(0)})$. Let $\tilde{g}_t := \nabla \tilde{f}_{i(t)}(u_{t-1}) + \frac{1}{n}\sum_i \tilde{z}_i^{(t-1)} - \tilde{z}_{i(t)}^{(t-1)}$ be the Euclidean SAGA direction built from $(u_{t-1}, \tilde{z}^{(t-1)})$ and the index $i(t)$. By (i) and the induction hypothesis, $\tilde{g}_t = P^{-1/2} f_{i(t)}'(\theta_{t-1}) + P^{-1/2}\big(\frac{1}{n}\sum_i z_i^{(t-1)} - z_{i(t)}^{(t-1)}\big) = P^{-1/2} g_t$. Hence, by (65), $u_t = P^{1/2}\theta_t = P^{1/2}\theta_{t-1} - \gamma P^{1/2} P^{-1} g_t = u_{t-1} - \gamma P^{-1/2} g_t = u_{t-1} - \gamma \tilde{g}_t$, which is the Euclidean SAGA step. For the memory, $\tilde{z}_{i(t)}^{(t)} = P^{-1/2} f_{i(t)}'(\theta_{t-1}) = \nabla \tilde{f}_{i(t)}(u_{t-1})$ and $\tilde{z}_j^{(t)} = \tilde{z}_j^{(t-1)}$ for $j \neq i(t)$: this is the SAGA memory update.

(iv) By Lemma 3.2(ii), $\|u_t - \tilde{u}_*\|^2 = \left\|P^{1/2}(\theta_t - \eta_*)\right\|^2 = \|\theta_t - \eta_*\|_P^2$ and $\left\|\tilde{z}_i^{(t)} - \nabla \tilde{f}_i(\tilde{u}_*)\right\|^2 = \left\|P^{-1/2}(z_i^{(t)} - f_i'(\eta_*))\right\|^2 = \left\|z_i^{(t)} - f_i'(\eta_*)\right\|_{P^{-1}}^2$, using (i) at $\tilde{u}_*$. The last claim is (i) at u_0, $P^{-1/2}$ being invertible. □

Lemma 3.6 (relative smoothness and strong convexity constants). *(i) $\tilde{f}_i$ is L-smooth (i.e. $\nabla \tilde{f}_i$ is L-Lipschitz) if and only if*

$$\left\|f_i'(\theta) - f_i'(\theta')\right\|_{P^{-1}} \leq L\left\|\theta - \theta'\right\|_P \qquad \forall \theta, \theta' \in \mathbb{R}^d; \tag{66}$$

when f_i is C^2 and convex, this is equivalent to $\sup_\theta \lambda_{\max}(P^{-1/2} f_i''(\theta) P^{-1/2}) \le L$.

(ii) $\tilde F$ is m-strongly convex if and only if $F(\theta') \ge F(\theta) + \langle F'(\theta), \theta' - \theta\rangle + \frac{m}{2}\|\theta' - \theta\|_P^2$ for all θ, θ'; when F is C^2, this is equivalent to $\inf_\theta \lambda_{\min}(P^{-1/2} F''(\theta) P^{-1/2}) \ge m$.

(iii) If each f_i is R^2-smooth and F is μ-strongly convex (Euclidean constants), then (66) holds with $L = R^2/\lambda_{\min}(P)$ and (ii) holds with $m = \mu/\lambda_{\max}(P)$. Conversely, if (66) holds with $L = R_P^2$ and (ii) with $m = \mu_P$, then each f_i is $\lambda_{\max}(P) R_P^2$-smooth and F is $\lambda_{\min}(P)\mu_P$-strongly convex.

Proof. (i) With $\theta = P^{-1/2}u$, $\theta' = P^{-1/2}u'$, Proposition 3.5(i) and Lemma 3.2(ii) give

$$\left\|\nabla \tilde f_i(u) - \nabla \tilde f_i(u')\right\| = \left\|P^{-1/2}(f_i'(\theta) - f_i'(\theta'))\right\| = \left\|f_i'(\theta) - f_i'(\theta')\right\|_{P^{-1}}, \qquad \left\|u - u'\right\| = \left\|\theta - \theta'\right\|_P;$$

as u ranges over $\mathbb{R}^d$ so does θ. For a C^2 convex function, L-smoothness is equivalent to all Hessians being $\preceq LI$ [6, Def. 5.3 and Sec. 5.2]; apply this to $\tilde f_i$, whose Hessian at u is $P^{-1/2} f_i''(P^{-1/2}u) P^{-1/2}$. (ii) Same substitution in the definition of strong convexity, using $\left\langle \nabla \tilde F(u), u' - u\right\rangle = \langle F'(\theta), \theta' - \theta\rangle$ and $\|u' - u\|^2 = \|\theta' - \theta\|_P^2$; for C^2 functions m-strong convexity is equivalent to all Hessians being $\succeq mI$ [6, Sec. 5.2]. (iii) By Lemma 3.2(v),

$$\begin{aligned}\left\|f_i'(\theta) - f_i'(\theta')\right\|_{P^{-1}} &\le \lambda_{\min}(P)^{-1/2}\left\|f_i'(\theta) - f_i'(\theta')\right\| \le \lambda_{\min}(P)^{-1/2} R^2 \left\|\theta - \theta'\right\| \\ &\le \lambda_{\min}(P)^{-1} R^2 \left\|\theta - \theta'\right\|_P,\end{aligned}$$

and $\frac{\mu}{2}\|\theta' - \theta\|^2 \ge \frac{\mu}{2\lambda_{\max}(P)}\|\theta' - \theta\|_P^2$. The converse uses the reverse inequalities of Lemma 3.2(v) in the same way. □

Definition 3.7 (relative constants and relative condition number)**.** *We set*

$$R_P^2 := \max_{1\le i\le n} \sup_{\theta\in\mathbb{R}^d} \lambda_{\max}\big(P^{-1/2} f_i''(\theta) P^{-1/2}\big), \qquad \mu_P := \inf_{\theta\in\mathbb{R}^d} \lambda_{\min}\big(P^{-1/2} F''(\theta) P^{-1/2}\big), \qquad \kappa_P := \frac{R_P^2}{\mu_P}. \tag{67}$$

By Lemma 3.6, every $\tilde f_i$ is R_P^2-smooth and $\tilde F$ is μ_P-strongly convex; in the terminology of [16, Def. 1] and [24], f_i is R_P^2-smooth and F is μ_P-strongly convex relative to *the reference function $h = \frac12\|\cdot\|_P^2$ (whose Hessian is P). For $P = I$ these are the usual constants; in the rest of the note R^2, μ, κ always denote the Euclidean ($P = I$) constants.*

Lemma 3.8 (general comparison with the Euclidean constants)**.** *With* $\operatorname{cond}(P) := \lambda_{\max}(P)/\lambda_{\min}(P)$,

$$\mu_P \le \sup_\theta \lambda_{\max}\big(P^{-1/2} F''(\theta) P^{-1/2}\big) \le R_P^2, \qquad R_P^2 \le \frac{R^2}{\lambda_{\min}(P)}, \qquad \mu_P \ge \frac{\mu}{\lambda_{\max}(P)},$$
$$\frac{\kappa}{\operatorname{cond}(P)} \le \kappa_P \le \kappa \operatorname{cond}(P). \tag{68}$$

In particular $\kappa_P \ge 1$, $\kappa_{cP} = \kappa_P$ for every $c > 0$, and a badly chosen metric can degrade the condition number by at most $\operatorname{cond}(P)$.

Proof. $F'' = \frac1n\sum_i f_i''$ and $\lambda_{\max}$ is subadditive on symmetric matrices, so $\lambda_{\max}(P^{-1/2}F''(\theta)P^{-1/2}) \le \frac1n\sum_i \lambda_{\max}(P^{-1/2} f_i''(\theta) P^{-1/2}) \le R_P^2$ for every θ, while $\lambda_{\min} \le \lambda_{\max}$ at every θ gives the first inequality. The two middle inequalities are Lemma 3.6(iii) combined with the Hessian characterizations; dividing them gives the upper bound on κ_P, and the converse part of Lemma 3.6(iii) gives $\kappa \le \kappa_P \operatorname{cond}(P)$. Finally $R_{cP}^2 = R_P^2/c$ and $\mu_{cP} = \mu_P/c$ by (67). □

Remark 3.9 (scaling of P and of the step). (cP)-SAGA with step $c\gamma$ produces exactly the same iterates as P-SAGA with step γ, since $c\gamma(cP)^{-1} = \gamma P^{-1}$; consistently, the normalized step $\gamma = 1/(4R_P^2)$ of Corollary 3.12 is multiplied by c when P is. Only κ_P is scale free; the values of R_P^2 and μ_P quoted below for $P = \hat{\Sigma} + \lambda I$ depend on the normalization $\hat{\Sigma} = \frac{1}{n} X^\top X$ (with $P' = nP = X^\top X + n\lambda I$ one would get $R_{P'}^2 = R_P^2/n$).

Remark 3.10 (C^1 functions). If the f_i are only C^1, define R_P^2 as the smallest constant L in (66) (over all i) and μ_P as the largest m in Lemma 3.6(ii). Everything below that does not display a Hessian remains valid, because the transfer argument and the proof of Proposition 5.9 only use the gradient forms (co-coercivity, Proposition 5.4 of [6], is proved from L-smoothness and convexity alone).

Lemma 3.11 (ℓ_2-regularized logistic regression). *Let $f_i(\theta) = \ell(y_i x_i^\top \theta) + \frac{\lambda}{2}\|\theta\|^2$ with $\ell(u) = \log(1 + e^{-u})$, $y_i \in \{-1, 1\}$, $x_i \in \mathbb{R}^d$, $\lambda > 0$, and let $\sigma(u) = (1 + e^{-u})^{-1}$. Then $\ell'(u) = -\sigma(-u)$ and $\ell''(u) = \sigma'(u) = \sigma(u)\sigma(-u) = \left(2 + e^u + e^{-u}\right)^{-1} \in (0, \frac{1}{4}]$, which equals $\frac{1}{4}$ at $u = 0$ and is strictly decreasing in $|u|$. Writing $\sigma_i'(\theta) := \sigma'(y_i x_i^\top \theta)$,*

$$f_i'(\theta) = \ell'(y_i x_i^\top \theta)\, y_i x_i + \lambda\theta, \qquad f_i''(\theta) = \sigma_i'(\theta)\, x_i x_i^\top + \lambda I, \qquad F''(\theta) = \frac{1}{n}\sum_{i=1}^n \sigma_i'(\theta)\, x_i x_i^\top + \lambda I. \tag{69}$$

For every SPD P,

$$\max\left\{\frac{1}{4}\max_i \|x_i\|_{P^{-1}}^2,\ \frac{\max_i \|x_i\|^2/4 + \lambda}{\lambda_{\max}(P)}\right\} \ \le\ R_P^2 \ \le\ \frac{1}{4}\max_i \|x_i\|_{P^{-1}}^2 + \frac{\lambda}{\lambda_{\min}(P)}, \tag{70}$$

and more precisely $R_P^2 \ge \frac{1}{4}\|x_i\|_{P^{-1}}^2 + \lambda\,\frac{x_i^\top P^{-2} x_i}{x_i^\top P^{-1} x_i}$ for every i with $x_i \ne 0$;

$$\mu_P = \lambda\,\lambda_{\min}(P^{-1}) = \frac{\lambda}{\lambda_{\max}(P)}. \tag{71}$$

In particular, for $P = I$: $R^2 := \frac{1}{4}\max_i \|x_i\|^2 + \lambda$ is a valid smoothness constant of every f_i, $\mu = \lambda$, and $\kappa = R^2/\mu = \max_i \|x_i\|^2/(4\lambda) + 1$; these are the Euclidean constants used in the whole project.

Proof. $\ell'(u) = -e^{-u}/(1 + e^{-u}) = -\sigma(-u)$ and $\ell''(u) = \sigma'(u) = \sigma(u)(1 - \sigma(u)) = \sigma(u)\sigma(-u) = 1/(2 + e^u + e^{-u})$, which is maximal at $u = 0$ (value $\frac{1}{4}$) and decreasing in $|u|$ since $e^u + e^{-u} = 2\cosh u$ is increasing in $|u|$. The gradient and Hessians follow from the chain rule and $y_i^2 = 1$. Upper bound in (70): by Weyl's inequality $\lambda_{\max}(A + B) \le \lambda_{\max}(A) + \lambda_{\max}(B)$ with $A = \sigma_i'(\theta) P^{-1/2} x_i x_i^\top P^{-1/2}$ (rank one, with nonzero eigenvalue $\sigma_i'(\theta)\left\|P^{-1/2}x_i\right\|^2 \le \frac{1}{4}\|x_i\|_{P^{-1}}^2$) and $B = \lambda P^{-1}$. Lower bounds: at $\theta = 0$, $\sigma_i'(0) = \frac{1}{4}$; for $x_i \ne 0$ the Rayleigh quotient of $P^{-1/2} f_i''(0) P^{-1/2}$ along $v = P^{-1/2}x_i$ equals $\frac{1}{4}(x_i^\top P^{-1} x_i)^2 / \left\|P^{-1/2}x_i\right\|^2 + \lambda\left\|P^{-1}x_i\right\|^2 / \left\|P^{-1/2}x_i\right\|^2 = \frac{1}{4}\|x_i\|_{P^{-1}}^2 + \lambda\,(x_i^\top P^{-2} x_i)/(x_i^\top P^{-1} x_i)$, which is the "more precise" bound; it is $\ge \frac{1}{4}\|x_i\|_{P^{-1}}^2$, and also $\ge (\|x_i\|^2/4 + \lambda)/\lambda_{\max}(P)$ because, with $y := P^{-1/2}x_i$, the fraction $(x_i^\top P^{-2} x_i)/(x_i^\top P^{-1} x_i) = y^\top P^{-1} y / \|y\|^2$ is a Rayleigh quotient of P^{-1}, hence $\ge \lambda_{\min}(P^{-1}) = 1/\lambda_{\max}(P)$, while $\|x_i\|_{P^{-1}}^2 \ge \|x_i\|^2/\lambda_{\max}(P)$ by Lemma 3.2(v). If $x_i = 0$ then $f_i'' \equiv \lambda I$ and $\sup_\theta \lambda_{\max}(P^{-1/2} f_i''(\theta) P^{-1/2}) = \lambda/\lambda_{\min}(P) \ge (\|x_i\|^2/4 + \lambda)/\lambda_{\max}(P)$, the first lower bound being trivial. Taking the maximum over i gives the two lower bounds in (70). For (71): $F''(\theta) \succeq \lambda I$ for all θ gives $P^{-1/2} F''(\theta) P^{-1/2} \succeq \lambda P^{-1} \succeq \lambda\lambda_{\min}(P^{-1}) I$, hence $\mu_P \ge \lambda\lambda_{\min}(P^{-1})$. Conversely, let $I_0 = \{i : x_i \ne 0\}$ and pick v with $v^\top x_i \ne 0$ for all $i \in I_0$ (the excluded set is a finite union of hyperplanes). For $s \to \infty$, $|y_i x_i^\top (sv)| \to \infty$ hence $\sigma_i'(sv) \to 0$ for $i \in I_0$, while the terms $i \notin I_0$ vanish identically; thus $F''(sv) \to \lambda I$ and, by continuity of $\lambda_{\min}$, $\lambda_{\min}(P^{-1/2} F''(sv) P^{-1/2}) \to \lambda\lambda_{\min}(P^{-1})$, so the infimum in (67) is $\le \lambda\lambda_{\min}(P^{-1})$. The $P = I$ statements are the special case $\lambda_{\min}(P) = \lambda_{\max}(P) = 1$. □

3.3 Convergence of P-SAGA: Proposition 5.9 in the P-metric

Corollary 3.12 (P-SAGA; Proposition 5.9 of [6] transferred). *Let P be SPD and let R_P^2, μ_P be the constants of (67) (or any constants such that each $\tilde{f}_i$ is R_P^2-smooth and $\tilde{F}$ is μ_P-strongly convex). Run P-SAGA with $\gamma = 1/(4R_P^2)$ and define*

$$V_t^P := \|\theta_t - \eta_*\|_P^2 + 3\gamma^2 \sum_{i=1}^n \left\| z_i^{(t)} - f_i'(\eta_*) \right\|_{P^{-1}}^2,$$

$$\rho_P := 1 - \min\Big\{ \frac{1}{3n}, \frac{3\mu_P}{16 R_P^2} \Big\} = 1 - \min\Big\{ \frac{1}{3n}, \frac{3}{16\kappa_P} \Big\}. \tag{72}$$

Then for every $t \geq 1$,

$$\mathbb{E}_{i(t)}\big[V_t^P\big] \leq \rho_P \, V_{t-1}^P, \tag{73}$$

and if $z_i^{(0)} = f_i'(\theta_0)$ for all i,

$$\mathbb{E}\,\|\theta_t - \eta_*\|_P^2 \leq \rho_P^t \Big(1 + \frac{n}{4}\Big)\, \|\theta_0 - \eta_*\|_P^2 . \tag{74}$$

Consequently

$$\mathbb{E}\,\|\theta_t - \eta_*\|^2 \leq \mathrm{cond}(P)\, \rho_P^t \Big(1 + \frac{n}{4}\Big)\, \|\theta_0 - \eta_*\|^2$$

(the condition number of P enters the constant, not the rate), and $t \geq \max\{3n, \frac{16}{3}\kappa_P\} \log \frac{2n}{\varepsilon}$ iterations guarantee $\mathbb{E}\,\|\theta_t - \eta_*\|_P^2 \leq \varepsilon\, \|\theta_0 - \eta_*\|_P^2$, i.e. a running time of order $c_P\,(3n + \frac{16}{3}\kappa_P) \log \frac{2n}{\varepsilon}$, where c_P is the cost of one iteration (Section 3.5).

Proof by transfer. By Proposition 3.5(iii), $(u_t, \tilde{z}^{(t)}) = (P^{1/2}\theta_t, P^{-1/2} z^{(t)})$ is the Euclidean SAGA sequence on $(\tilde{f}_i)$ with step γ; the $\tilde{f}_i$ are convex and R_P^2-smooth and $\tilde{F}$ is μ_P-strongly convex with minimizer $\tilde{u}_*$ (Lemma 3.6, Proposition 3.5(ii)). The proof of Proposition 5.9 in [6] (transcribed in the B3 note) establishes, for Euclidean SAGA with $\gamma = 1/(4L)$ on any family of convex L-smooth functions whose average is m-strongly convex, the one-step contraction $\mathbb{E}_{i(t)}[\tilde{V}_t] \leq (1 - \min\{\frac{1}{3n}, \frac{3m}{16L}\})\tilde{V}_{t-1}$ of the Lyapunov function $\tilde{V}_t = \|u_t - \tilde{u}_*\|^2 + 3\gamma^2 \sum_i \left\| \tilde{z}_i^{(t)} - \nabla \tilde{f}_i(\tilde{u}_*) \right\|^2$, and then the bound $\mathbb{E}\,\|u_t - \tilde{u}_*\|^2 \leq (1 - \min\{\frac{1}{3n}, \frac{3m}{16L}\})^t (1 + \frac{n}{4})\, \|u_0 - \tilde{u}_*\|^2$ when $\tilde{z}_i^{(0)} = \nabla \tilde{f}_i(u_0)$. By Proposition 3.5(iv), $\tilde{V}_t = V_t^P$ and $\|u_t - \tilde{u}_*\|^2 = \|\theta_t - \eta_*\|_P^2$ identically, and the initialization conditions correspond. The Euclidean consequence follows from Lemma 3.2(v) applied on both sides; the iteration count follows from $\rho_P^t \leq \exp(-t \min\{\frac{1}{3n}, \frac{3}{16\kappa_P}\})$ and $1 + \frac{n}{4} \leq 2n$. □

3.3.1 The same proof written in the P-metric: which inequalities change

To see exactly where P enters, we rewrite the proof of Proposition 5.9 in the original coordinates; every inequality is labelled, and these labels are the ones instrumented by the experiments (Definition 3.14). Fix $t \geq 1$ and write $\delta := \theta_{t-1} - \eta_*$, $\omega_t := g_t$, $\bar{z} := \frac{1}{n}\sum_i z_i^{(t-1)}$ and, for each i,

$$a_{t,i} := f_i'(\theta_{t-1}) - f_i'(\eta_*), \qquad b_{t,i} := f_i'(\eta_*) - z_i^{(t-1)} + \bar{z}, \qquad \text{so that} \quad \omega_t = a_{t,i(t)} + b_{t,i(t)}.$$

Step 1. Expanding the square in the P-inner product and using (65),

$$\begin{aligned} \|\theta_t - \eta_*\|_P^2 = \big\|\delta - \gamma P^{-1}\omega_t\big\|_P^2 &= \|\delta\|_P^2 - 2\gamma\, \delta^\top P P^{-1}\omega_t + \gamma^2 \omega_t^\top P^{-1} P P^{-1}\omega_t \\ &= \|\delta\|_P^2 - 2\gamma\, \langle \delta, \omega_t \rangle + \gamma^2\, \|\omega_t\|_{P^{-1}}^2 . \end{aligned} \tag{75}$$

The pairing $\langle \delta, \omega_t \rangle$ is the same number for every P; taking $\mathbb{E}_{i(t)}$ and using Lemma 3.4 (unbiasedness, also metric independent), $\mathbb{E}_{i(t)} \|\theta_t - \eta_*\|_P^2 = \|\delta\|_P^2 - 2\gamma \langle \delta, F'(\theta_{t-1}) \rangle + \gamma^2 \mathbb{E}_{i(t)} \|\omega_t\|_{P^{-1}}^2$. The three inequalities of Step 1 are

$$\text{(I1}_P\text{, square norm)} \quad \mathbb{E}_{i(t)} \|\omega_t\|_{P^{-1}}^2 \le 2\,\mathbb{E}_{i(t)} \left\|a_{t,i(t)}\right\|_{P^{-1}}^2 + 2\,\mathbb{E}_{i(t)} \left\|b_{t,i(t)}\right\|_{P^{-1}}^2, \tag{76}$$

$$\text{(I2}_P\text{, co-coercivity)} \quad \mathbb{E}_{i(t)} \left\|a_{t,i(t)}\right\|_{P^{-1}}^2 = \frac{1}{n} \sum_{i=1}^n \left\|f_i'(\theta_{t-1}) - f_i'(\eta_*)\right\|_{P^{-1}}^2$$

$$\le R_P^2 \left\langle F'(\theta_{t-1}), \theta_{t-1} - \eta_* \right\rangle, \tag{77}$$

$$\text{(I3}_P\text{, variance} \le \text{2nd moment)} \quad \mathbb{E}_{i(t)} \left\|b_{t,i(t)}\right\|_{P^{-1}}^2 \le \frac{1}{n} \sum_{i=1}^n \left\|f_i'(\eta_*) - z_i^{(t-1)}\right\|_{P^{-1}}^2. \tag{78}$$

(76) is Lemma 3.2(iii) applied for each value of $i(t)$. For (77): each $\tilde{f}_i$ is convex and R_P^2-smooth, so co-coercivity (Proposition 5.4 of [6]) gives $\left\|\nabla \tilde{f}_i(u) - \nabla \tilde{f}_i(u')\right\|^2 \le R_P^2 \left\langle \nabla \tilde{f}_i(u) - \nabla \tilde{f}_i(u'), u - u' \right\rangle$; with $u = P^{1/2}\theta_{t-1}$, $u' = P^{1/2}\eta_*$ and Proposition 3.5(i), this reads

$$\left\|f_i'(\theta_{t-1}) - f_i'(\eta_*)\right\|_{P^{-1}}^2 \le R_P^2 \left\langle f_i'(\theta_{t-1}) - f_i'(\eta_*), \theta_{t-1} - \eta_* \right\rangle \qquad (i = 1, \dots, n), \tag{79}$$

because $\left\langle P^{-1/2}(f_i'(\theta) - f_i'(\eta_*)), P^{1/2}(\theta - \eta_*) \right\rangle = \langle f_i'(\theta) - f_i'(\eta_*), \theta - \eta_* \rangle$; averaging over i and using $\sum_i f_i'(\eta_*) = nF'(\eta_*) = 0$ gives (77), which is (5.31) of the book in the P-metric. For (78): with $Z := f_{i(t)}'(\eta_*) - z_{i(t)}^{(t-1)}$ one has $\mathbb{E}_{i(t)} Z = -\bar{z}$ (again $\sum_i f_i'(\eta_*) = 0$), so $Z - \mathbb{E}_{i(t)} Z = b_{t,i(t)}$ and Lemma 3.2(iv) applies. Combining, Step 1 yields

$$\mathbb{E}_{i(t)} \|\theta_t - \eta_*\|_P^2 \le \|\delta\|_P^2 - 2\gamma(1 - \gamma R_P^2) \left\langle F'(\theta_{t-1}), \delta \right\rangle + \frac{2\gamma^2}{n} \sum_{i=1}^n \left\|f_i'(\eta_*) - z_i^{(t-1)}\right\|_{P^{-1}}^2. \tag{80}$$

Step 2. Only the $i(t)$-th memory vector changes, so, exactly,

$$\sum_{i=1}^n \left\|f_i'(\eta_*) - z_i^{(t)}\right\|_{P^{-1}}^2 = \sum_{i=1}^n \left\|f_i'(\eta_*) - z_i^{(t-1)}\right\|_{P^{-1}}^2 - \left\|f_{i(t)}'(\eta_*) - z_{i(t)}^{(t-1)}\right\|_{P^{-1}}^2$$

$$+ \left\|f_{i(t)}'(\eta_*) - f_{i(t)}'(\theta_{t-1})\right\|_{P^{-1}}^2;$$

taking $\mathbb{E}_{i(t)}$ (each index has probability $1/n$) and using (77) once more,

$$\mathbb{E}_{i(t)} \sum_{i=1}^n \left\|f_i'(\eta_*) - z_i^{(t)}\right\|_{P^{-1}}^2 = \left(1 - \frac{1}{n}\right) \sum_{i=1}^n \left\|f_i'(\eta_*) - z_i^{(t-1)}\right\|_{P^{-1}}^2 + \frac{1}{n} \sum_{i=1}^n \left\|f_i'(\eta_*) - f_i'(\theta_{t-1})\right\|_{P^{-1}}^2$$

$$\le \left(1 - \frac{1}{n}\right) \sum_{i=1}^n \left\|f_i'(\eta_*) - z_i^{(t-1)}\right\|_{P^{-1}}^2 + R_P^2 \left\langle F'(\theta_{t-1}), \delta \right\rangle. \tag{81}$$

Combination. Multiply (81) by $\Delta = 3\gamma^2$ and add to (80). With $\gamma = 1/(4R_P^2)$: $1 - \gamma R_P^2 - R_P^2 \Delta/(2\gamma) = 1 - \frac{1}{4} - \frac{3}{8} = \frac{3}{8}$ and $\frac{2\gamma^2}{n\Delta} + 1 - \frac{1}{n} = 1 - \frac{1}{3n}$, so

$$\mathbb{E}_{i(t)}[V_t^P] \le \|\delta\|_P^2 - \frac{3\gamma}{4} \left\langle F'(\theta_{t-1}), \delta \right\rangle + \left(1 - \frac{1}{3n}\right) \Delta \sum_{i=1}^n \left\|f_i'(\eta_*) - z_i^{(t-1)}\right\|_{P^{-1}}^2. \tag{82}$$

The last two inequalities are

$$\text{(I4}_P\text{, strong convexity)} \quad \big\langle F'(\theta_{t-1}), \theta_{t-1} - \eta_* \big\rangle \;\geq\; \mu_P \, \|\theta_{t-1} - \eta_*\|_P^2 \,, \tag{83}$$

$$\text{(I5}_P\text{, min)} \quad \Big(1 - \frac{3\mu_P}{16 R_P^2}\Big) \, \|\delta\|_P^2 + \Big(1 - \frac{1}{3n}\Big) \Delta \sum_{i=1}^n \Big\| f_i'(\eta_*) - z_i^{(t-1)} \Big\|_{P^{-1}}^2 \;\leq\; \rho_P \, V_{t-1}^P. \tag{84}$$

For (83): $\tilde{F}$ is μ_P-strongly convex, so adding the inequalities of Lemma 3.6(ii) written at $(\theta, \theta') = (\theta_{t-1}, \eta_*)$ and at (η_*, θ_{t-1}), and using $F'(\eta_*) = 0$, gives $\langle F'(\theta_{t-1}), \theta_{t-1} - \eta_* \rangle \geq \mu_P \, \|\theta_{t-1} - \eta_*\|_P^2$; inserting it in (82) (the coefficient $-\frac{3\gamma}{4}$ is negative, so the inequality direction is preserved) produces the left-hand side of (84), since $\frac{3\gamma}{4}\mu_P = \frac{3\mu_P}{16 R_P^2}$. (84) is the scalar inequality $(1-\alpha)x + (1-\beta)y \leq (1 - \min\{\alpha, \beta\})(x+y)$ for $x, y \geq 0$, with $\alpha = \frac{3\mu_P}{16 R_P^2}$, $\beta = \frac{1}{3n}$. This proves (73). Iterating with the tower property, $\mathbb{E} V_t^P \leq \rho_P^t V_0^P$, and

$$\text{(I6}_P\text{, initialization)} \qquad \sum_{i=1}^n \big\| f_i'(\eta_*) - f_i'(\theta_0) \big\|_{P^{-1}}^2 \leq n \, R_P^4 \, \|\theta_0 - \eta_*\|_P^2 \tag{85}$$

(each term is bounded by the square of (66)) gives, when $z_i^{(0)} = f_i'(\theta_0)$,

$$V_0^P \leq (1 + 3\gamma^2 n R_P^4) \, \|\theta_0 - \eta_*\|_P^2 = \Big(1 + \frac{3n}{16}\Big) \, \|\theta_0 - \eta_*\|_P^2 \leq \Big(1 + \frac{n}{4}\Big) \, \|\theta_0 - \eta_*\|_P^2 \,,$$

which is (74) because $\|\theta_t - \eta_*\|_P^2 \leq V_t^P$. This completes the second proof of Corollary 3.12. □

Remark 3.13 (comparison with Bregman-SAGA). Corollary 1 of [8] states, for Bregman-SAGA with a quadratic reference function ($\nabla^2 h$ constant, which is exactly P-SAGA by Lemma 3.4) and step $1/(8L_{f/h})$, the contraction $\mathbb{E}[\psi_t] \leq (1 - \min\{\frac{1}{8\kappa_{f/h}}, \frac{1}{2n}\})^t \psi_0$ of their potential $\psi_t = \frac{1}{\eta} D_h(x^\star, x_t) + \frac{n}{2} \cdot \frac{1}{n} \sum_i D_{f_i}(\phi_i^t, x^\star)$, where $L_{f/h}$ is the relative smoothness of every f_i (their Assumption 3) and $\kappa_{f/h} = L_{f/h}/\mu_{f/h}$ the relative condition number: these are our R_P^2 and κ_P. The rate has the same structure $1 - \min\{c_1/n, \, c_2/\kappa_P\}$ as (72) (with $(c_1, c_2) = (\frac{1}{2}, \frac{1}{8})$ instead of $(\frac{1}{3}, \frac{3}{16})$), the difference coming from the step and from a potential expressed with Bregman distances of the f_i instead of squared norms of memory vectors).

3.3.2 Metric dependence of each step, and the six tightness ratios in the P-geometry

Table 1 summarizes where P enters. Three kinds of objects appear. (a) Quantities that are the same numbers for every P, given the current state $(\theta_{t-1}, z^{(t-1)})$: the pairing $\langle \theta_{t-1} - \eta_*, \omega_t \rangle$ and its conditional expectation $\langle \theta_{t-1} - \eta_*, F'(\theta_{t-1}) \rangle$, the unbiasedness, the memory bookkeeping identity of Step 2 (an identity, valid for any norm), the scalar inequality (I5), and the algebra of the combination (the constants $\frac{3}{8}$, $\frac{2}{3n}$, $\frac{3}{16}$ are unchanged because γ and Δ are normalized by R_P^2). (b) Inequalities whose *form* is metric independent but which are instantiated in the P^{-1}-inner product: (I1) and (I3), both valid in every Hilbert norm (Lemma 3.2). (c) Inequalities whose norm *and* constant change: (I2) (left side in $\|\cdot\|_{P^{-1}}$, right side metric free, constant R_P^2), (I4) (left side metric free, right side in $\|\cdot\|_P$, constant μ_P), (I6) (both norms and R_P^4). Of course the iterates themselves depend on P, so the numerical values of all ratios depend on P; the point of the classification is that the *same six ratios* can be computed, by Proposition 3.5(iv), either in θ-coordinates with P-norms (one linear solve with P per vector), or in u-coordinates with Euclidean norms on the transformed problem (whose regularizer is the non-isotropic quadratic $\frac{\lambda}{2} u^\top P^{-1} u$ for logistic regression, so the B3 code must accept a general quadratic regularizer if the second route is taken).

Step	Inequality	What changes with P	Metric-free part
(I1) (76)	square norm	norm $\|\cdot\|_{P^{-1}}$ only	form of the inequality
(I2) (77)	co-coercivity	norm $\|\cdot\|_{P^{-1}}$, constant R_P^2	pairing $\langle F'(\theta_{t-1}), \theta_{t-1} - \eta_* \rangle$
(I3) (78)	variance $\leq$ 2nd moment	norm $\|\cdot\|_{P^{-1}}$ only	form of the inequality
(I4) (83)	strong convexity	norm $\|\cdot\|_P$, constant μ_P	pairing $\langle F'(\theta_{t-1}), \theta_{t-1} - \eta_* \rangle$
(I5) (84)	min	only through the value of $3\mu_P/(16R_P^2)$	scalar inequality
(I6) (85)	initialization	both norms, constant R_P^4	—
exact steps	(75), Step 2 identity, combination	norms only	unbiasedness, pairing, algebra

Table 1: Metric dependence of the steps of the proof of Proposition 5.9 (Section 3.3.1).

Definition 3.14 (tightness ratios in the P-geometry)**.** *The effective (trajectory-dependent) constants of the B2 note, written in the P-metric, are, for $\theta \neq \eta_*$,*

$$\mu_{\text{eff}}^P(\theta) := \frac{\langle F'(\theta), \theta - \eta_* \rangle}{\|\theta - \eta_*\|_P^2}, \qquad L_{\text{eff}}^P(\theta) := \frac{\frac{1}{n}\sum_i \|f_i'(\theta) - f_i'(\eta_*)\|_{P^{-1}}^2}{\langle F'(\theta), \theta - \eta_* \rangle}. \tag{86}$$

Convention on indices. *As in the B3 note, the ratios of iteration t are evaluated at the state $(\theta_{t-1}, z^{(t-1)})$ seen before $i(t)$ is drawn, so they involve $\mu_{\text{eff}}^P(\theta_{t-1})$ and $L_{\text{eff}}^P(\theta_{t-1})$; the B2 note abbreviates $\mu_{\text{eff}}(t) := \mu_{\text{eff}}(\theta_t)$, which for $P = I$ is the value entering the ratios of iteration $t+1$. To avoid any off-by-one error in code shared by the three notes, we never write $\mu_{\text{eff}}^P(t)$ but always display the point, $\mu_{\text{eff}}^P(\theta_{t-1})$. With the notation of Section 3.3.1 ($\delta = \theta_{t-1} - \eta_*$, $a_{t,i}$, $b_{t,i}$, $\bar{z}$), $\alpha = \frac{3\mu_P}{16R_P^2}$, $\beta = \frac{1}{3n}$ and $\Delta = 3\gamma^2$, define, for $\theta_{t-1} \neq \eta_*$,*

$$\begin{aligned}
r_{\text{sq}}^P(t) &:= \frac{\frac{1}{n}\sum_i \|a_{t,i} + b_{t,i}\|_{P^{-1}}^2}{\frac{2}{n}\sum_i \left(\|a_{t,i}\|_{P^{-1}}^2 + \|b_{t,i}\|_{P^{-1}}^2 \right)}, \\
r_{\text{coco}}^P(t) &:= \frac{\frac{1}{n}\sum_i \|f_i'(\theta_{t-1}) - f_i'(\eta_*)\|_{P^{-1}}^2}{R_P^2 \, \langle F'(\theta_{t-1}), \delta \rangle} = \frac{L_{\text{eff}}^P(\theta_{t-1})}{R_P^2}, \\
r_{\text{var}}^P(t) &:= \frac{\frac{1}{n}\sum_i \|b_{t,i}\|_{P^{-1}}^2}{\frac{1}{n}\sum_i \left\| f_i'(\eta_*) - z_i^{(t-1)} \right\|_{P^{-1}}^2} = 1 - \frac{\|\bar{z}\|_{P^{-1}}^2}{\frac{1}{n}\sum_i \left\| f_i'(\eta_*) - z_i^{(t-1)} \right\|_{P^{-1}}^2}, \\
r_{\text{sc}}^P(t) &:= \frac{\mu_P \|\delta\|_P^2}{\langle F'(\theta_{t-1}), \delta \rangle} = \frac{\mu_P}{\mu_{\text{eff}}^P(\theta_{t-1})}, \\
r_{\text{min}}^P(t) &:= \frac{(1-\alpha)\|\delta\|_P^2 + (1-\beta)\Delta \sum_i \left\| f_i'(\eta_*) - z_i^{(t-1)} \right\|_{P^{-1}}^2}{\rho_P \, V_{t-1}^P},
\end{aligned}$$

with the convention $r_{\text{var}}^P(t) := 1$ when the memory is exact ($z_i^{(t-1)} = f_i'(\eta_*)$ for all i, the only case in which its denominator vanishes; (78) is then the equality $0 \leq 0$), and, for $\theta_0 \neq \eta_*$,

$$r_{\text{init}}^P := \frac{\sum_i \|f_i'(\eta_*) - f_i'(\theta_0)\|_{P^{-1}}^2}{nR_P^4 \|\theta_0 - \eta_*\|_P^2} = \frac{L_{\text{eff}}^P(\theta_0)\, \mu_{\text{eff}}^P(\theta_0)}{R_P^4} = r_{\text{coco}}^P(1)\, \frac{\mu_{\text{eff}}^P(\theta_0)}{R_P^2}. \tag{87}$$

Each ratio is the smaller side of one of (76)–(85) divided by the larger side, the expectations over $i(t)$ being exact averages over i. The identities in (87) are immediate from (86), the pairing

$\langle F'(\theta_0), \theta_0 - \eta_* \rangle$ *cancelling in the product* $L^P_{\text{eff}}(\theta_0)\mu^P_{\text{eff}}(\theta_0)$*:* r^P_{init} *is a derived ratio, not an independent one. The one-step ratio* $r^P_{\text{tot}}(t) := \mathbb{E}_{i(t)}[V_t^P]/(\rho_P V_{t-1}^P)$*, defined when* $V_{t-1}^P > 0$*, is also computable exactly (average of* V_t^P *over the* n *possible draws). For* $P = I$ *all of them reduce to the ratios of the B3 note.*

Lemma 3.15 (ranges and degenerate cases)**.** *Let* $\theta_{t-1} \neq \eta_*$*. Then*

$$0 < \mu_P \leq \mu^P_{\text{eff}}(\theta_{t-1}) \leq L^P_{\text{eff}}(\theta_{t-1}) \leq R_P^2, \tag{88}$$

all the denominators in Definition 3.14 are positive (that of r^P_{var} *when the memory is not exact), and* $r^P_{\text{sc}}(t), r^P_{\text{coco}}(t), r^P_{\text{sq}}(t), r^P_{\min}(t) \in (0,1]$*,* $r^P_{\text{var}}(t) \in [0,1]$*; if* $\theta_0 \neq \eta_*$ *then* $r^P_{\text{init}} \in (0,1]$*. At* $\theta_{t-1} = \eta_*$ *the ratios* r^P_{sc} *and* r^P_{coco} *are of the form* $0/0$ *(both numerator and denominator vanish since* $F'(\eta_*) = 0$*), as are* r^P_{sq} *when in addition all memory errors* $z_i^{(t-1)} - f_i'(\eta_*)$ *coincide, and* $r^P_{\min}$*,* r^P_{tot} *when in addition the memory is exact; they are left undefined there, as in the B3 note.*

Proof. By Proposition 3.5(i),(iv), every quantity above is the corresponding Euclidean quantity of the B3 note computed on the transformed problem $(\tilde{f}_i)$ at the state $(u_{t-1}, \tilde{z}^{(t-1)})$, for which the hypotheses of the B3 note hold (convex R_P^2-smooth $\tilde{f}_i$, μ_P-strongly convex $\tilde{F}$, $u_{t-1} \neq \tilde{u}_*$); the statement is the B3 lemma on ranges and degenerate cases, transferred. For completeness, directly: (83) gives $\langle F'(\theta_{t-1}), \delta \rangle \geq \mu_P \|\delta\|_P^2 > 0$, i.e. $\mu^P_{\text{eff}}(\theta_{t-1}) \geq \mu_P > 0$, so $F'(\theta_{t-1}) \neq 0$; by convexity of the squared norm (Jensen) and Lemma 3.2(i), $\frac{1}{n}\sum_i \|f_i'(\theta_{t-1}) - f_i'(\eta_*)\|_{P^{-1}}^2 \geq \|F'(\theta_{t-1})\|_{P^{-1}}^2 \geq \langle F'(\theta_{t-1}), \delta \rangle^2 / \|\delta\|_P^2 > 0$, which is $L^P_{\text{eff}} \geq \mu^P_{\text{eff}}$ together with the positivity of the numerator of r^P_{coco}; $L^P_{\text{eff}} \leq R_P^2$ is (77). The numerator of r^P_{sq} is $\geq \left\|\frac{1}{n}\sum_i (a_{t,i} + b_{t,i})\right\|_{P^{-1}}^2 = \|F'(\theta_{t-1})\|_{P^{-1}}^2 > 0$ (Jensen again, and $\sum_i b_{t,i} = 0$ because $\sum_i f_i'(\eta_*) = 0$); the numerator of $r^P_{\min}$ is $\geq (1-\alpha)\|\delta\|_P^2 > 0$ as $\alpha \leq \frac{3}{16}$, and its denominator is $\rho_P V_{t-1}^P \geq \rho_P \|\delta\|_P^2 > 0$. The upper bounds are the inequalities (76)–(84) themselves, and $r^P_{\text{var}} \geq 0$ is clear. For r^P_{init}, apply (88) at θ_0 in (87). □

Remark 3.16 (effective one-step contraction; link with the B2 note). Keep $\gamma = 1/(4R_P^2)$ and $\Delta = 3\gamma^2$ but, in Steps 1–2, replace the two uses of (77) and the use of (83) by the identities

$$\frac{1}{n}\sum_i \left\|f_i'(\theta_{t-1}) - f_i'(\eta_*)\right\|_{P^{-1}}^2 = L^P_{\text{eff}}(\theta_{t-1})\left\langle F'(\theta_{t-1}), \delta\right\rangle, \qquad \left\langle F'(\theta_{t-1}), \delta\right\rangle = \mu^P_{\text{eff}}(\theta_{t-1})\,\|\delta\|_P^2\,.$$

The coefficient of $\langle F'(\theta_{t-1}), \delta \rangle$ in the combination becomes (writing L^P_{eff} for $L^P_{\text{eff}}(\theta_{t-1})$)

$$-2\gamma\big(1 - \gamma L^P_{\text{eff}} - L^P_{\text{eff}}\Delta/(2\gamma)\big) = -2\gamma\big(1 - \tfrac{5}{2}\gamma L^P_{\text{eff}}\big) = -2\gamma\big(1 - \tfrac{5}{8} r^P_{\text{coco}}(t)\big),$$

whence, with only (76) and (78) still used,

$$\mathbb{E}_{i(t)}[V_t^P] \leq \Big(1 - \Big(1 - \frac{5}{8} r^P_{\text{coco}}(t)\Big)\frac{\mu^P_{\text{eff}}(\theta_{t-1})}{2R_P^2}\Big)\,\|\theta_{t-1} - \eta_*\|_P^2 + \Big(1 - \frac{1}{3n}\Big)\Delta \sum_{i=1}^n \left\|f_i'(\eta_*) - z_i^{(t-1)}\right\|_{P^{-1}}^2. \tag{89}$$

Since $r^P_{\text{coco}} \in (0,1]$ (Lemma 3.15), the contraction factor of the parameter part lies between $1 - \frac{\mu^P_{\text{eff}}(\theta_{t-1})}{2R_P^2}$ and $1 - \frac{3\mu^P_{\text{eff}}(\theta_{t-1})}{16R_P^2}$: the rate "seen by the proof" is governed by the effective relative condition number $R_P^2/\mu^P_{\text{eff}}(\theta_{t-1})$ (the B2 quantity in the P-metric), the looseness of co-coercivity affecting only the constant, between $\frac{3}{16}$ and $\frac{1}{2}$. This is the quantity whose behaviour under the four metrics is predicted in Section 3.6.

3.4 Choice of P on ℓ_2-regularized logistic regression

Setting and notation. $X \in \mathbb{R}^{n\times d}$ has rows $x_i^\top$, $\hat\Sigma := \frac{1}{n}X^\top X$ has eigenvalues $\sigma_1 \ge \cdots \ge \sigma_d \ge 0$ and eigenvectors $v_1, \dots, v_d$, and f_i is as in Lemma 3.11. We use the ridge leverage scores and the effective dimension ([?, Def. 1] with the linear kernel $K = XX^\top$; [?, Sec. 4.1]):

$$\ell_i(\lambda) := x_i^\top (X^\top X + n\lambda I)^{-1} x_i = \big[X(X^\top X + n\lambda I)^{-1} X^\top\big]_{ii}, \tag{90}$$

$$d_{\mathrm{eff}}(\lambda) := \operatorname{tr}\big(\hat\Sigma(\hat\Sigma + \lambda I)^{-1}\big) = \sum_{k=1}^{d} \frac{\sigma_k}{\sigma_k + \lambda}, \qquad d_\infty(\lambda) := n \max_i \ell_i(\lambda),$$

d_∞ being the "maximal marginal degrees of freedom" $n\,\|\operatorname{diag}(K(K+n\lambda I)^{-1})\|_\infty$ of [?, Sec. 4.1, eq. (5)].

Lemma 3.17 (leverage scores). *(i)* $\|x_i\|^2_{(\hat\Sigma+\lambda I)^{-1}} = n\,\ell_i(\lambda)$ *for every* i.

(ii) $\sum_{i=1}^n \ell_i(\lambda) = d_{\mathrm{eff}}(\lambda)$; *hence* $d_{\mathrm{eff}}(\lambda) \le d_\infty(\lambda)$.

(iii) $0 \le \ell_i(\lambda) < 1$ *for every* i, *hence* $d_\infty(\lambda) < n$; *for every* i *the map* $\lambda \mapsto \ell_i(\lambda)$ *is nonincreasing on* $(0,\infty)$, *hence so are* d_∞ *and* d_{eff}; *moreover* $d_{\mathrm{eff}}(\lambda) \le \min\{\operatorname{rank}(X), \operatorname{tr}(\hat\Sigma)/\lambda\}$.

(iv) $X(X^\top X + n\lambda I)^{-1}X^\top = K(K+n\lambda I)^{-1}$ *with* $K = XX^\top$, *so* $\ell_i(\lambda)$ *is the* λ*-ridge leverage score of [?, Def. 1].*

Proof. (i) $(\hat\Sigma + \lambda I)^{-1} = \big(\frac{1}{n}(X^\top X + n\lambda I)\big)^{-1} = n(X^\top X + n\lambda I)^{-1}$. (ii) By cyclicity of the trace, $\sum_i \ell_i = \operatorname{tr}\big(X(X^\top X + n\lambda I)^{-1}X^\top\big) = \operatorname{tr}\big((X^\top X + n\lambda I)^{-1}X^\top X\big) = \operatorname{tr}\big((\hat\Sigma + \lambda I)^{-1}\hat\Sigma\big)$, which equals $\sum_k \sigma_k/(\sigma_k+\lambda)$ in the eigenbasis of $\hat\Sigma$; the inequality is $\max_i \ell_i \ge \frac{1}{n}\sum_i \ell_i$. (iii) Let $X = \sum_{k=1}^r s_k u_k w_k^\top$ be a thin SVD ($r = \operatorname{rank} X$, $s_k > 0$, $u_k \in \mathbb{R}^n$ orthonormal). Then $X(X^\top X + n\lambda I)^{-1}X^\top = \sum_k \frac{s_k^2}{s_k^2+n\lambda} u_k u_k^\top$, whose i-th diagonal entry is $\ell_i = \sum_k \frac{s_k^2}{s_k^2+n\lambda} u_{ki}^2$. If $\sum_k u_{ki}^2 = 0$ then $\ell_i = 0$; otherwise $\frac{s_k^2}{s_k^2+n\lambda} < 1$ gives $\ell_i < \sum_k u_{ki}^2 \le 1$, the last inequality because the u_k are orthonormal columns of an $n \times r$ matrix. Each coefficient $s_k^2/(s_k^2+n\lambda)$ is decreasing in λ and $u_{ki}^2 \ge 0$, so $\ell_i(\lambda)$ is nonincreasing in λ, and so are $d_\infty = n\max_i \ell_i$ and $d_{\mathrm{eff}} = \sum_i \ell_i$. The bound on d_{eff} follows from $\sigma_k/(\sigma_k+\lambda) \le \min\{\mathbf{1}_{\sigma_k>0}, \sigma_k/\lambda\}$. (iv) Push-through identity: $X(X^\top X + n\lambda I) = (XX^\top + n\lambda I)X$, hence $(XX^\top + n\lambda I)^{-1}X = X(X^\top X + n\lambda I)^{-1}$; multiply by $X^\top$ on the right. □

Assumption 3.18 (coherence). *There is* $C \ge 1$ *such that* $\max_i \ell_i(\lambda) \le C\,d_{\mathrm{eff}}(\lambda)/n$, *i.e.* $d_\infty(\lambda) \le C\,d_{\mathrm{eff}}(\lambda)$: *no data point has a leverage score larger than* C *times the average leverage score.*

(a) $P = I$ **(SAGA).** By Lemma 3.11, $R^2 = \frac{1}{4}\max_i \|x_i\|^2 + \lambda$, $\mu = \lambda$ and

$$\kappa = \frac{\max_i \|x_i\|^2}{4\lambda} + 1 \;\ge\; \frac{\operatorname{tr}(\hat\Sigma)}{4\lambda} + 1 = \frac{\sum_{k=1}^d \sigma_k}{4\lambda} + 1, \tag{91}$$

since $\max_i \|x_i\|^2 \ge \frac{1}{n}\sum_i \|x_i\|^2 = \operatorname{tr}\hat\Sigma$. Thus κ contains the whole spectrum of $\hat\Sigma$ through its trace (not only σ_1, which is what the condition number L/μ of F alone would contain, since $L \le \sigma_1/4 + \lambda$), divided by λ.

(b) $P = D := \operatorname{diag}(\hat{\Sigma}) + \lambda I$ **(feature scaling, Jacobi preconditioning).** Write $D_0 := \operatorname{diag}(\hat{\Sigma})$, $D_{0,jj} = \frac{1}{n}\sum_i x_{ij}^2$, assume no feature is identically zero (otherwise drop it), and let $C_{\mathrm{corr}} := D_0^{-1/2}\hat{\Sigma}D_0^{-1/2}$ be the correlation matrix (unit diagonal, so $\lambda_{\min}(C_{\mathrm{corr}}) \le 1 \le \lambda_{\max}(C_{\mathrm{corr}}) \le d$).

Proposition 3.19 (Jacobi preconditioner). *(i)* $\|x_i\|_{D^{-1}}^2 = \sum_{j=1}^d \frac{x_{ij}^2}{D_{0,jj}+\lambda}$ *and* $\frac{1}{n}\sum_i \|x_i\|_{D^{-1}}^2 = \sum_{j=1}^d \frac{D_{0,jj}}{D_{0,jj}+\lambda} =: d_{\mathrm{eff}}^{\mathrm{diag}}(\lambda) \ge d_{\mathrm{eff}}(\lambda)$.

(ii) $R_D^2 \le \frac{1}{4}\max_i \|x_i\|_{D^{-1}}^2 + \frac{\lambda}{\min_j D_{0,jj}+\lambda}$ *and* $\mu_D = \frac{\lambda}{\max_j D_{0,jj}+\lambda}$.

(iii) $\lambda_{\min}(C_{\mathrm{corr}})\, D \preceq \hat{\Sigma} + \lambda I \preceq \lambda_{\max}(C_{\mathrm{corr}})\, D$; *consequently* $\lambda_{\min}(C_{\mathrm{corr}})\, n\ell_i(\lambda) \le \|x_i\|_{D^{-1}}^2 \le \lambda_{\max}(C_{\mathrm{corr}})\, n\ell_i(\lambda)$ *for all* i.

Proof. (i) D is diagonal, so $x_i^\top D^{-1} x_i = \sum_j x_{ij}^2/(D_{0,jj}+\lambda)$; averaging over i and using $\frac{1}{n}\sum_i x_{ij}^2 = D_{0,jj}$ gives $d_{\mathrm{eff}}^{\mathrm{diag}}$. For the inequality, $g(s) = s/(s+\lambda)$ is concave on $[0,\infty)$, the vector $(D_{0,jj})_j$ of diagonal entries of $\hat{\Sigma}$ is majorized by the vector of its eigenvalues (Schur–Horn theorem), and $x \mapsto \sum_j g(x_j)$ is Schur-concave when g is concave [**?**]; hence $\sum_j g(D_{0,jj}) \ge \sum_k g(\sigma_k)$. (ii) Lemma 3.11 with $\lambda_{\min}(D) = \min_j D_{0,jj} + \lambda$ and $\lambda_{\max}(D) = \max_j D_{0,jj} + \lambda$. (iii) $\hat{\Sigma} = D_0^{1/2} C_{\mathrm{corr}} D_0^{1/2}$, so $\lambda_{\min}(C_{\mathrm{corr}})D_0 \preceq \hat{\Sigma} \preceq \lambda_{\max}(C_{\mathrm{corr}})D_0$; adding λI and using $\lambda_{\min}(C_{\mathrm{corr}}) \le 1 \le \lambda_{\max}(C_{\mathrm{corr}})$ gives the sandwich. Inverting it (the map $A \mapsto A^{-1}$ is order reversing on SPD matrices) gives

$$\lambda_{\max}(C_{\mathrm{corr}})^{-1} D^{-1} \preceq (\hat{\Sigma} + \lambda I)^{-1} \preceq \lambda_{\min}(C_{\mathrm{corr}})^{-1} D^{-1},$$

which with Lemma 3.17(i) yields the bounds on $\|x_i\|_{D^{-1}}^2$. □

Thus the Jacobi metric removes the *scale* anisotropy of the features (up to the λ terms, it is invariant under a rescaling of each feature) but keeps their *correlation* anisotropy: every constant of the statistical preconditioner $\hat{\Sigma} + \lambda I$ derived below transfers to D at the price of the factor $\operatorname{cond}(C_{\mathrm{corr}}) = \lambda_{\max}(C_{\mathrm{corr}})/\lambda_{\min}(C_{\mathrm{corr}})$ (the argument of Lemma 3.8, which applies verbatim to any pair of metrics, applied to $(D, \hat{\Sigma} + \lambda I)$ through (iii)), and $D = \hat{\Sigma} + \lambda I$ exactly when the features are uncorrelated ($C_{\mathrm{corr}} = I$).

(c) $P = \hat{\Sigma} + \lambda I$ **(statistical preconditioning).** Here $\lambda_{\max}(P) = \sigma_1 + \lambda$, $\lambda_{\min}(P) = \sigma_d + \lambda$ and, by Lemma 3.17(i), $\|x_i\|_{P^{-1}}^2 = n\,\ell_i(\lambda)$.

Proposition 3.20 (global constants of the statistical preconditioner). *For* $P = \hat{\Sigma} + \lambda I$,

$$\frac{d_\infty(\lambda)}{4} \le R_P^2 \le \frac{d_\infty(\lambda)}{4} + \frac{\lambda}{\sigma_d + \lambda} \le \frac{d_\infty(\lambda)}{4} + 1 \le \frac{C\, d_{\mathrm{eff}}(\lambda)}{4} + 1 \quad \textit{(the last under Assumption 3.18)}, \tag{92}$$

$$\mu_P = \frac{\lambda}{\sigma_1 + \lambda}, \qquad \kappa_P = \frac{R_P^2}{\mu_P} \in \Big[\frac{d_\infty(\lambda)}{4} \cdot \frac{\sigma_1 + \lambda}{\lambda},\ \Big(\frac{d_\infty(\lambda)}{4} + \frac{\lambda}{\sigma_d + \lambda}\Big)\frac{\sigma_1 + \lambda}{\lambda}\Big]. \tag{93}$$

Proof. Insert $\max_i \|x_i\|_{P^{-1}}^2 = n \max_i \ell_i(\lambda) = d_\infty(\lambda)$ and $\lambda_{\min}(P) = \sigma_d + \lambda$ in (70), and $\lambda_{\max}(P) = \sigma_1 + \lambda$ in (71); $\lambda/(\sigma_d + \lambda) \le 1$ and Assumption 3.18 give the last two bounds. □

The smoothness side is now controlled by leverage scores, i.e. by an effective dimension, with no trace of the spectrum of $\hat{\Sigma}$; but the *global* strong convexity side has become $\lambda/(\sigma_1 + \lambda)$, and the following statement shows that this is not an artefact of the proof: with global constants, no quadratic metric can beat $P = I$ on this problem.

Proposition 3.21 (no free lunch for global constants)**.** *For ℓ_2-regularized logistic regression and* every *SPD matrix P,*

$$\kappa_P \;\geq\; \frac{\max_i \|x_i\|^2}{4\lambda} + 1 = \kappa, \tag{94}$$

with κ the Euclidean condition number (91)*; consequently $\rho_P \geq \rho_I$ for every P. In particular, for $P = \hat{\Sigma} + \lambda I$, $\kappa \leq \kappa_P \leq \kappa\,\frac{\sigma_1+\lambda}{\sigma_d+\lambda}$.*

Proof. By the second lower bound in (70), $R_P^2 \geq (\max_i \|x_i\|^2/4+\lambda)/\lambda_{\max}(P) = R^2/\lambda_{\max}(P)$, while $\mu_P = \lambda/\lambda_{\max}(P)$ by (71); divide. Since $\kappa_P \mapsto \rho_P = 1 - \min\{\frac{1}{3n}, \frac{3}{16\kappa_P}\}$ is nondecreasing, $\rho_P \geq \rho_I$. The upper bound for $P = \hat{\Sigma} + \lambda I$ is (68). □

The mechanism is transparent: the supremum over θ in R_P^2 is attained on the decision boundary of sample i ($x_i^\top\theta = 0$, curvature $\frac{1}{4}x_ix_i^\top$), whereas the infimum over θ in μ_P is approached far from all decision boundaries, where $F'' \to \lambda I$ is isotropic; no single metric can be adapted both to a rank-one curvature $\frac{1}{4}x_ix_i^\top$ and to an isotropic one. (For the square loss the curvature is constant, $F'' \equiv \hat{\Sigma} + \lambda I = P$, every $f_i'' = x_ix_i^\top + \lambda I$ has relative smoothness in $[n\ell_i, n\ell_i + 1]$, and the global κ_P lies in $[d_\infty(\lambda), d_\infty(\lambda) + 1]$: a genuine gain. This is the ridge-regression setting of [10], whose Theorem 4 bounds the *average* condition number of the preconditioned problem by $\mathrm{tr}(P^{-1/2}(\hat{\Sigma} + \lambda I)P^{-1/2})/\lambda_{\min}(P^{-1/2}(\hat{\Sigma} + \lambda I)P^{-1/2})$, equal to d for $P = \hat{\Sigma} + \lambda I$; with uniform sampling and the maximum in (67), d is replaced by $d_\infty(\lambda)$.) Hence, for logistic regression, any benefit of the metric must come from the *trajectory-dependent* constants, which we now quantify; this is exactly the point of view of the B2 note.

Proposition 3.22 (trajectory-dependent strong convexity: exact formula)**.** *For logistic regression, every $\theta \neq \eta_*$, $\delta := \theta - \eta_*$, and $\bar{\sigma}_i'(\theta) := \int_0^1 \sigma'\big(y_ix_i^\top(\eta_* + s\delta)\big)\,ds \in (0, \frac{1}{4}]$,*

$$f_i'(\theta) - f_i'(\eta_*) = \bar{\sigma}_i'(\theta)\,(x_i^\top\delta)\,x_i + \lambda\delta, \qquad \big\langle F'(\theta), \theta - \eta_* \big\rangle = \frac{1}{n}\sum_{i=1}^n \bar{\sigma}_i'(\theta)\,(x_i^\top\delta)^2 + \lambda\,\|\delta\|^2\,. \tag{95}$$

Consequently, with $m(\theta) := \min_i \bar{\sigma}_i'(\theta) \;\geq\; \min_i \sigma'\big(\max\{|x_i^\top\eta_*|, |x_i^\top\theta|\}\big) > 0$:

(i) for $P = \hat{\Sigma} + \lambda I$, the effective strong convexity $\mu_{\mathrm{eff}}^P(\theta) := \langle F'(\theta), \delta\rangle \,/\, \|\delta\|_P^2$ is the convex combination

$$\mu_{\mathrm{eff}}^P(\theta) = \frac{\frac{1}{n}\sum_i \bar{\sigma}_i'(\theta)(x_i^\top\delta)^2 + \lambda\,\|\delta\|^2}{\frac{1}{n}\sum_i (x_i^\top\delta)^2 + \lambda\,\|\delta\|^2} \;\in\; [\,m(\theta),\,1\,] \tag{96}$$

of the numbers $\bar{\sigma}_1'(\theta), \ldots, \bar{\sigma}_n'(\theta), 1$, and more precisely

$$\mu_{\mathrm{eff}}^P(\theta) \;\geq\; \frac{m(\theta)\,\sigma_1 + \lambda}{\sigma_1 + \lambda} \;\geq\; \max\Big\{m(\theta),\; \frac{\lambda}{\sigma_1 + \lambda}\Big\} = \max\{m(\theta), \mu_P\}; \tag{97}$$

(ii) for $P = I$, $\mu_{\mathrm{eff}}(\theta) = \frac{1}{n}\sum_i \bar{\sigma}_i'(\theta)\,(x_i^\top\delta/\,\|\delta\|)^2 + \lambda \in [\,m(\theta)\sigma_d + \lambda,\; \sigma_1/4 + \lambda\,]$.

Proof. By the fundamental theorem of calculus applied to $s \mapsto \ell'(y_ix_i^\top(\eta_* + s\delta))$, $\ell'(y_ix_i^\top\theta) - \ell'(y_ix_i^\top\eta_*) = \int_0^1 \ell''(y_ix_i^\top(\eta_* + s\delta))\,ds \cdot y_ix_i^\top\delta = \bar{\sigma}_i'(\theta)\,y_ix_i^\top\delta$; multiplying by y_ix_i ($y_i^2 = 1$) and adding $\lambda\delta$ gives the first identity in (95) by (69). Pairing with δ, averaging over i and using $F'(\eta_*) = 0$ gives the second. Lower bound on $m(\theta)$: $s \mapsto x_i^\top(\eta_* + s\delta)$ is affine, so $|x_i^\top(\eta_* + s\delta)| \leq \max\{|x_i^\top\eta_*|, |x_i^\top\theta|\}$ on $[0,1]$, and σ' is decreasing in $|u|$ (Lemma 3.11). (i) $\|\delta\|_P^2 = \delta^\top\hat{\Sigma}\delta + \lambda\,\|\delta\|^2 = \frac{1}{n}\sum_i (x_i^\top\delta)^2 + \lambda\,\|\delta\|^2$, which gives (96); it is a convex combination with (normalized) weights $(x_i^\top\delta)^2/n$ and $\lambda\,\|\delta\|^2$, whence

$m(\theta) \le \mu^P_{\text{eff}} \le 1$ because $\bar\sigma'_i \le \frac14 < 1$. For (97): the numerator is $\ge \delta^\top(m\hat\Sigma + \lambda I)\delta$ with $m = m(\theta)$, and since $\hat\Sigma$ and P are simultaneously diagonalizable, $\delta^\top(m\hat\Sigma + \lambda I)\delta \ge \min_k \frac{m\sigma_k+\lambda}{\sigma_k+\lambda}\,\delta^\top P\delta = \frac{m\sigma_1+\lambda}{\sigma_1+\lambda}\,\|\delta\|^2_P$, because $s \mapsto \frac{ms+\lambda}{s+\lambda}$ is nonincreasing on $[0,\infty)$ when $m \le 1$ (its derivative is $\lambda(m-1)/(s+\lambda)^2$). Finally $\frac{m\sigma_1+\lambda}{\sigma_1+\lambda}$ is a convex combination of m and 1 with weights $\frac{\sigma_1}{\sigma_1+\lambda}$ and $\frac{\lambda}{\sigma_1+\lambda}$, hence it is $\ge m$ and $\ge \frac{\lambda}{\sigma_1+\lambda}$. (ii) Divide (95) by $\|\delta\|^2$; the bounds use $m \le \bar\sigma'_i \le \frac14$ and $\sigma_d\|\delta\|^2 \le \delta^\top\hat\Sigma\delta \le \sigma_1\|\delta\|^2$. □

Note that (97) is the segment-averaged, *per-sample* version of the pointwise inequality $F''(\theta') \succeq \min_i \sigma'_i(\theta')\,\hat\Sigma + \lambda I$: integrating the latter along the segment $[\eta_*, \theta]$ would only produce the constant $\int_0^1 \min_i \sigma'_i(\eta_* + s\delta)\,ds$, which is $\le \min_j \int_0^1 \sigma'_j(\eta_* + s\delta)\,ds = m(\theta)$ (for each j, $\min_i \sigma'_i \le \sigma'_j$ pointwise), so (97), which uses the minimum of the per-sample segment averages, is at least as strong as the integrated pointwise bound. Its global version only gives $\mu_P = \lambda/(\sigma_1+\lambda)$ because $\inf_\theta \min_i \sigma'_i(\theta) = 0$. The experiments can test (97) directly (its tightness ratio is $\frac{m(\theta_{t-1})\sigma_1+\lambda}{\sigma_1+\lambda} / \mu^P_{\text{eff}}(\theta_{t-1})$), and (96) shows that $\mu^P_{\text{eff}}(\theta_{t-1})$ is a direction-weighted *average* of the per-sample curvatures along the segment, typically much larger than the minimum m: a single sample with a huge margin makes $m(\theta)$ tiny but has a negligible weight in (96) unless the error δ is aligned with it.

Corollary 3.23 (trajectory-dependent condition number: the anisotropy of $\hat\Sigma$ disappears)**.** *Let $\kappa^P_{\text{eff}}(\theta) := R^2_P/\mu^P_{\text{eff}}(\theta)$ be the effective relative condition number at θ (the quantity governing the effective contraction* (89)*). Then, for $P = \hat\Sigma + \lambda I$,*

$$\kappa^P_{\text{eff}}(\theta) \;\le\; \Big(\frac{d_\infty(\lambda)}{4} + \frac{\lambda}{\sigma_d+\lambda}\Big)\frac{\sigma_1+\lambda}{m(\theta)\sigma_1+\lambda} \;\le\; \frac{d_\infty(\lambda)/4+1}{m(\theta)} \;\le\; \frac{C\,d_{\text{eff}}(\lambda)/4+1}{m(\theta)}, \tag{98}$$

the last inequality under Assumption 3.18, whereas for $P = I$ the same argument only gives

$$\frac{\max_i\|x_i\|^2/4+\lambda}{\sigma_1/4+\lambda} \;\le\; \kappa_{\text{eff}}(\theta) \;\le\; \frac{\max_i\|x_i\|^2/4+\lambda}{m(\theta)\sigma_d+\lambda}, \qquad \textit{with} \qquad \frac{\max_i\|x_i\|^2/4+\lambda}{\sigma_1/4+\lambda} \;\ge\; \frac{\operatorname{tr}\hat\Sigma+4\lambda}{\sigma_1+4\lambda}. \tag{99}$$

Proof. Divide (92) by (97), then use $\lambda/(\sigma_d+\lambda) \le 1$ and $\frac{\sigma_1+\lambda}{m\sigma_1+\lambda} \le \frac1m$ (as $m \le 1$). For $P = I$, divide R^2 by the bounds of Proposition 3.22(ii), and use $\max_i\|x_i\|^2 \ge \operatorname{tr}\hat\Sigma$. □

In words: along the trajectory, the certified condition number in the statistical metric is (effective dimension)/(typical curvature of the logistic loss on the segment $[\eta_*, \theta]$); the eigenvalues $\sigma_1, \dots, \sigma_d$ of $\hat\Sigma$ have disappeared, except through $d_{\text{eff}}(\lambda) = \sum_k \sigma_k/(\sigma_k+\lambda) \le d$, which is a *count* of the directions not killed by the regularization and is insensitive to the dynamic range of the spectrum. In the Euclidean metric, by contrast, even the trajectory-dependent bound keeps the factor $\max_i\|x_i\|^2/(m\sigma_d+\lambda) \ge \operatorname{tr}\hat\Sigma/(\sigma_d/4+\lambda)$, the trace of $\hat\Sigma$ divided by its smallest eigenvalue (or by λ when $\lambda \gg m\sigma_d$); and the lower bound in (99) shows that no trajectory can make the Euclidean κ_{eff} smaller than $(\operatorname{tr}\hat\Sigma)/\sigma_1$ (up to the λ terms), the participation ratio of the spectrum. The price paid in the P-metric is the coherence constant C and the curvature $m(\theta)$, which is small when some margins $|x_i^\top\theta|$ are large ($\sigma'(u) \approx e^{-|u|}$); the Newton metric (d) addresses this second issue.

Remark 3.24 (when the memory term is the bottleneck). The certified contraction is $1 - \min\{\frac{1}{3n}, \frac{3}{16\kappa'}\}$ for a condition number κ' (global or effective); the memory term $\frac{1}{3n}$ is the binding one iff $\kappa' \le \frac{9n}{16}$. For $P = \hat\Sigma + \lambda I$ and the trajectory constants of Corollary 3.23, a sufficient condition is therefore

$$n \;\ge\; \frac{16}{9}\cdot\frac{d_\infty(\lambda)/4+1}{m} = \frac{4d_\infty(\lambda)+16}{9m} \qquad \Big(\le \frac{4C\,d_{\text{eff}}(\lambda)+16}{9m} \;\text{ under Assumption 3.18}\Big), \tag{100}$$

with m a lower bound on $m(\theta_t)$ along the trajectory. In this regime ($n \gg d_{\text{eff}}/m$) the effective rate of P-SAGA is $1 - \frac{1}{3n}$ whatever λ and the spectrum of $\hat{\Sigma}$: one needs $3\log(2n/\varepsilon)$ effective passes (the $\log(2n)$ coming from the prefactor $1 + \frac{n}{4}$ of (74)), which is the $n\log(1/\varepsilon)$ term of the lower bound of hypothesis (A1); no metric and no acceleration (A3) can improve the leading term further. For $P = I$ the corresponding condition $n \geq \frac{16}{9}\frac{\max_i \|x_i\|^2/4+\lambda}{m\sigma_d+\lambda}$ fails by a factor of order $\max_i \|x_i\|^2 / (d_\infty(\lambda)\,\sigma_d)$ when $\lambda \ll m\sigma_d$; and with global constants both conditions read $n \geq \frac{16}{9}\kappa_P \geq \frac{16}{9}\kappa$ by Proposition 3.21, which fails for the usual choice $\lambda \asymp R^2/n$ of [6, Sec. 5.4.4] (then $\kappa \asymp n$).

Remark 3.25 (the variant $\frac{1}{4}\hat{\Sigma} + \lambda I$). $P' = \frac{1}{4}\hat{\Sigma} + \lambda I = \frac{1}{4}(\hat{\Sigma} + 4\lambda I)$ is the exact global upper envelope of F'' ($F''(\theta) \preceq P'$ for all θ by (69)), so that F is 1-smooth relative to P'. It is the statistical preconditioner $P_{4\lambda} := \hat{\Sigma} + 4\lambda I$ at regularization 4λ, rescaled by $\frac{1}{4}$ (Remark 3.9; the loss keeps its regularization λ), so that $\kappa_{P'} = \kappa_{P_{4\lambda}}$ and $\kappa_{\text{eff}}^{P'}(\theta) = \kappa_{\text{eff}}^{P_{4\lambda}}(\theta)$ (both R_P^2 and $\mu_{\text{eff}}^P(\theta)$ are multiplied by $1/c$ under $P \mapsto cP$). By Lemma 3.17(i) at regularization 4λ, $\|x_i\|^2_{P_{4\lambda}^{-1}} = n\ell_i(4\lambda)$, so Lemma 3.11 gives $R^2_{P_{4\lambda}} \leq \frac{d_\infty(4\lambda)}{4} + \frac{\lambda}{\sigma_d+4\lambda}$ and $\mu_{P_{4\lambda}} = \frac{\lambda}{\sigma_1+4\lambda}$; moreover $\mu_{\text{eff}}^{P'}(\theta)$ is a convex combination of $4\bar{\sigma}'_1, \dots, 4\bar{\sigma}'_n, 1$ (weights $(x_i^\top \delta)^2/(4n)$ and $\lambda\|\delta\|^2$), hence lies in $[4m(\theta), 1]$, and the argument of (97), now with the function $s \mapsto \frac{ms+\lambda}{s+4\lambda}$ (nonincreasing since $m \leq \frac{1}{4}$), gives $\mu_{\text{eff}}^{P_{4\lambda}}(\theta) \geq \frac{m(\theta)\sigma_1+\lambda}{\sigma_1+4\lambda}$. Hence

$$\begin{aligned} \kappa_{P'} &\leq \Big(\frac{d_\infty(4\lambda)}{4} + \frac{\lambda}{\sigma_d+4\lambda}\Big)\frac{\sigma_1+4\lambda}{\lambda}, \\ \kappa_{\text{eff}}^{P'}(\theta) &\leq \Big(\frac{d_\infty(4\lambda)}{4} + \frac{\lambda}{\sigma_d+4\lambda}\Big)\frac{\sigma_1+4\lambda}{m(\theta)\sigma_1+\lambda} \leq \frac{d_\infty(4\lambda)/4+1}{m(\theta)}, \end{aligned} \tag{101}$$

the last step because $m(\sigma_1 + 4\lambda) \leq m\sigma_1 + \lambda$ when $m \leq \frac{1}{4}$. Since $\lambda \mapsto \ell_i(\lambda)$ is nonincreasing (Lemma 3.17(iii)), $d_\infty(4\lambda) \leq d_\infty(\lambda)$, and $\frac{\lambda}{\sigma_d+4\lambda} \leq \frac{\lambda}{\sigma_d+\lambda}$: the brackets in (101) are never larger than those of (93) and (98), but the factor $\sigma_1 + 4\lambda$ replaces $\sigma_1 + \lambda$. Thus the variant is never worse than $P = \hat{\Sigma} + \lambda I$ in the global and in the finer trajectory bounds *up to the factor* $(\sigma_1+4\lambda)/(\sigma_1+\lambda) \leq 4$, which is negligible when $\lambda \ll \sigma_1$, and never worse at all in the crude bound $(d_\infty/4 + 1)/m$; the experiments may use either.

(d) $P = H := F''(\eta_*)$ (oracle Newton metric). Let $w_i := \sigma_i'(\eta_*) = \sigma'(y_i x_i^\top \eta_*) \in (0, \frac{1}{4}]$, $W := \operatorname{diag}(w_1, \dots, w_n)$ and $\hat{\Sigma}_w := \frac{1}{n}X^\top W X$ (eigenvalues $\sigma_1^w \geq \cdots \geq \sigma_d^w \geq 0$), so that $H = \hat{\Sigma}_w + \lambda I$ by (69). Define the w-weighted ridge leverage scores and effective dimension

$$\begin{aligned} \ell_i^w(\lambda) &:= w_i\, x_i^\top (X^\top W X + n\lambda I)^{-1} x_i = \big[W^{1/2}X(X^\top W X + n\lambda I)^{-1}X^\top W^{1/2}\big]_{ii}, \\ d_{\text{eff}}^w(\lambda) &:= \operatorname{tr}\big(\hat{\Sigma}_w(\hat{\Sigma}_w + \lambda I)^{-1}\big), \qquad d_\infty^w(\lambda) := n \max_i \ell_i^w(\lambda), \end{aligned} \tag{102}$$

i.e. the ridge leverage scores of the weighted design $W^{1/2}X$ (rows $\sqrt{w_i}\,x_i^\top$). Lemma 3.17 applied to $W^{1/2}X$ gives $\big\|\sqrt{w_i}\,x_i\big\|^2_{H^{-1}} = n\ell_i^w(\lambda)$, $\sum_i \ell_i^w = d_{\text{eff}}^w$ and $0 \leq \ell_i^w < 1$.

Proposition 3.26 (constants in the Newton metric)**.** *For $P = H = F''(\eta_*)$:*

(i) $\|x_i\|^2_{H^{-1}} = n\,\ell_i^w(\lambda)/w_i$, hence

$$\max_i \frac{n\ell_i^w(\lambda)}{4w_i} \;\leq\; R_H^2 \;\leq\; \max_i \frac{n\ell_i^w(\lambda)}{4w_i} + \frac{\lambda}{\sigma_d^w+\lambda} \;\leq\; \frac{d_\infty^w(\lambda)}{4\min_i w_i} + 1, \qquad \mu_H = \frac{\lambda}{\sigma_1^w+\lambda}. \tag{103}$$

(ii) For $\theta \neq \eta_*$, with $\bar\sigma_i'(\theta)$ as in Proposition 3.22,

$$\mu_{\text{eff}}^H(\theta) = \frac{\frac{1}{n}\sum_i \bar\sigma_i'(\theta)(x_i^\top\delta)^2 + \lambda\|\delta\|^2}{\frac{1}{n}\sum_i w_i(x_i^\top\delta)^2 + \lambda\|\delta\|^2} \;\in\; \Big[\min\Big\{1, \min_i \frac{\bar\sigma_i'(\theta)}{w_i}\Big\},\ \max\Big\{1, \max_i \frac{\bar\sigma_i'(\theta)}{w_i}\Big\}\Big], \quad (104)$$

a convex combination of $\bar\sigma_1'/w_1, \dots, \bar\sigma_n'/w_n, 1$; in particular $\mu_{\text{eff}}^H(\theta) \to 1$ as $\theta \to \eta_*$, and $\mu_{\text{eff}}^H(\theta) \geq \mu_H$ always.

Proof. (i) $x_i^\top H^{-1} x_i = w_i^{-1}(\sqrt{w_i}x_i)^\top H^{-1}(\sqrt{w_i}x_i) = n\ell_i^w/w_i$ by Lemma 3.17(i) for the design $W^{1/2}X$ (whose empirical covariance is $\hat\Sigma_w$); then (70)–(71) with $\lambda_{\min}(H) = \sigma_d^w + \lambda$, $\lambda_{\max}(H) = \sigma_1^w + \lambda$, and $\lambda/(\sigma_d^w + \lambda) \leq 1$. (ii) Divide (95) by $\|\delta\|_H^2 = \frac{1}{n}\sum_i w_i(x_i^\top\delta)^2 + \lambda\|\delta\|^2$; the (normalized) weights are $w_i(x_i^\top\delta)^2/n$ for the values $\bar\sigma_i'/w_i$ and $\lambda\|\delta\|^2$ for the value 1. As $\theta \to \eta_*$, $\bar\sigma_i'(\theta) \to \sigma_i'(\eta_*) = w_i$ by continuity of σ', so every value tends to 1. The last claim is (83). □

Thus the Newton metric makes the strong convexity side perfect near the optimum ($\mu_{\text{eff}}^H \to 1$), but the smoothness side still pays the supremum over θ of the per-sample curvature: sample i enters R_H^2 with the weight $\sup_\theta \sigma_i'(\theta) = \frac{1}{4}$ in the numerator and with its curvature at the optimum w_i in the metric, hence the factor $1/w_i$ in (103); a sample with a large margin at η_* (small w_i) but a non-negligible weighted leverage is the worst case. Under a *homogeneous curvature* assumption $\min_i w_i \geq w_{\min}$ and coherence of the weighted leverage scores ($d_\infty^w \leq C\, d_{\text{eff}}^w$), one gets $R_H^2 \leq C\, d_{\text{eff}}^w(\lambda)/(4w_{\min}) + 1$ and $\kappa_{\text{eff}}^H(\theta) = R_H^2/\mu_{\text{eff}}^H(\theta) \to R_H^2 \leq C\, d_{\text{eff}}^w/(4w_{\min}) + 1$ as $\theta \to \eta_*$. The next lemma shows that this $1/w_{\min}$ is again an artefact of the global supremum: the *realized* smoothness constant L_{eff}^H is leverage-controlled, with no curvature factor.

Lemma 3.27 (asymptotic effective constants)**.** *Let the f_i be C^2, $H_i := f_i''(\eta_*)$, $H := F''(\eta_*) = \frac{1}{n}\sum_i H_i \succ 0$, and let $\theta \to \eta_*$ with $\delta/\|\delta\| \to v$, $\|v\| = 1$. Then, for every SPD P,*

$$\begin{aligned} L_{\text{eff}}^P(\theta) &\longrightarrow \frac{\frac{1}{n}\sum_i v^\top H_i P^{-1} H_i v}{v^\top H v}, \\ \mu_{\text{eff}}^P(\theta) &\longrightarrow \frac{v^\top H v}{v^\top P v} \in \big[\lambda_{\min}(P^{-1/2}HP^{-1/2}), \lambda_{\max}(P^{-1/2}HP^{-1/2})\big]. \end{aligned} \quad (105)$$

Moreover $\frac{1}{n}\sum_i H_i P^{-1} H_i \succeq HP^{-1}H$, so the limit of L_{eff}^P is at least $\lambda_{\min}(P^{-1/2}HP^{-1/2})$, and for $P = H$

$$\lim L_{\text{eff}}^H(\theta) = 1 + \frac{\frac{1}{n}\sum_i \big\|H^{-1/2}(H_i - H)v\big\|^2}{\big\|H^{1/2}v\big\|^2} \;\geq\; 1, \quad (106)$$

with equality iff $H_i v = Hv$ for all i (homogeneous per-sample curvature in the direction v). For logistic regression, in addition, $\lim L_{\text{eff}}^H(\theta) \leq \max\{d_\infty^w(\lambda), 2\}$ for $P = H$, $\lim L_{\text{eff}}^P(\theta) \leq d_\infty(\lambda)/4 + \frac{3}{2}$ for $P = \hat\Sigma + \lambda I$, and, for $P = I$, with $c_i := x_i^\top v$ and $B_v := \frac{1}{n}\sum_i w_i^2 c_i^2 \|x_i\|^2$,

$$\lim L_{\text{eff}}(\theta) = \frac{\lambda^2 + 2\lambda\, v^\top\hat\Sigma_w v + B_v}{\lambda + v^\top\hat\Sigma_w v}, \qquad (v^\top\hat\Sigma_w v)^2 \leq B_v \leq \frac{\max_i\|x_i\|^2}{4}\, v^\top\hat\Sigma_w v = (\kappa - 1)\lambda\, v^\top\hat\Sigma_w v \quad (107)$$

(this is the exact formula $L_{\text{eff}} = (\lambda^2 + 2\lambda A + B)/(\lambda + A)$ of the B2 note, in the limit); in particular $\lambda + v^\top\hat\Sigma_w v \leq \lim L_{\text{eff}}(\theta) \leq \lambda + (\kappa + 1)\, v^\top\hat\Sigma_w v$.

Proof. Since f_i is C^2, $f_i'(\theta) - f_i'(\eta_*) = H_i\delta + o(\|\delta\|)$ and $F'(\theta) = H\delta + o(\|\delta\|)$ (as $F'(\eta_*) = 0$). Hence $\frac{1}{n}\sum_i \|f_i'(\theta) - f_i'(\eta_*)\|_{P^{-1}}^2 = \frac{1}{n}\sum_i \delta^\top H_i P^{-1} H_i \delta + o(\|\delta\|^2)$, $\langle F'(\theta), \delta\rangle = \delta^\top H\delta + o(\|\delta\|^2)$ and $\|\delta\|_P^2 = \delta^\top P\delta$; dividing numerators and denominators by $\|\delta\|^2$ and letting $\delta/\|\delta\| \to v$ gives (105),

because $v^\top Hv > 0$ and $v^\top Pv > 0$; the bracket is the range of the Rayleigh quotient of $P^{-1/2}HP^{-1/2}$. The Loewner inequality: $0 \preceq \frac{1}{n}\sum_i (H_i - H)P^{-1}(H_i - H) = \frac{1}{n}\sum_i H_i P^{-1} H_i - HP^{-1}H$, since the cross terms give $-\frac{1}{n}\sum_i (H_i P^{-1}H + HP^{-1}H_i) = -2HP^{-1}H$. Then $v^\top HP^{-1}Hv = \left\| M\, P^{1/2} v \right\|^2$ with $M := P^{-1/2}HP^{-1/2}$, and for an SPD matrix M and any w, $\|Mw\|^2 = w^\top M^2 w \geq \lambda_{\min}(M)\, w^\top M w$ (because $M^2 - \lambda_{\min}(M) M = M^{1/2}(M - \lambda_{\min}(M) I) M^{1/2} \succeq 0$), with $w^\top M w = v^\top H v$ here; with $P = H$ the same decomposition gives (106). For logistic regression, $H_i = w_i x_i x_i^\top + \lambda I$ and $v^\top H_i P^{-1} H_i v = w_i^2 (x_i^\top v)^2 \|x_i\|^2_{P^{-1}} + 2\lambda w_i (x_i^\top v)(x_i^\top P^{-1} v) + \lambda^2 v^\top P^{-1} v$. For $P = H$: $w_i \|x_i\|^2_{H^{-1}} = n\ell_i^w$, so the first terms average to $\frac{1}{n}\sum_i n\ell_i^w\, w_i (x_i^\top v)^2 \leq d_\infty^w\, v^\top \hat\Sigma_w v$; the remaining terms average to $2\lambda v^\top \hat\Sigma_w H^{-1} v + \lambda^2 v^\top H^{-1} v = 2\lambda v^\top (H - \lambda I) H^{-1} v + \lambda^2 v^\top H^{-1} v = 2\lambda \|v\|^2 - \lambda^2 v^\top H^{-1} v \leq 2\lambda \|v\|^2$; hence the numerator is $\leq d_\infty^w\, v^\top \hat\Sigma_w v + 2\lambda \|v\|^2 \leq \max\{d_\infty^w, 2\}\, v^\top H v$. For $P = \hat\Sigma + \lambda I$: $\|x_i\|^2_{P^{-1}} = n\ell_i$ and $w_i \leq \frac{1}{4}$ give $\frac{1}{n}\sum_i w_i^2 (x_i^\top v)^2\, n\ell_i \leq \frac{d_\infty}{4}\, v^\top \hat\Sigma_w v$; next $P^{-1} \preceq \lambda^{-1} I$ gives $\lambda^2 v^\top P^{-1} v \leq \lambda \|v\|^2$; finally, by Cauchy–Schwarz, $\hat\Sigma_w \preceq \frac{1}{4}\hat\Sigma$ and $\hat\Sigma \preceq P$,

$$
\begin{aligned}
2\lambda\, v^\top \hat\Sigma_w P^{-1} v &\leq 2\lambda \left\| \hat\Sigma_w^{1/2} v \right\| \left\| \hat\Sigma_w^{1/2} P^{-1} v \right\| \leq \lambda \left\| \hat\Sigma_w^{1/2} v \right\| \left\| \hat\Sigma^{1/2} P^{-1} v \right\| \leq \lambda \left\| \hat\Sigma_w^{1/2} v \right\| \left\| P^{-1/2} v \right\| \\
&\leq \sqrt{\lambda} \left\| \hat\Sigma_w^{1/2} v \right\| \|v\| \leq \tfrac{1}{2}\big(v^\top \hat\Sigma_w v + \lambda \|v\|^2 \big),
\end{aligned}
$$

where we used $\left\| \hat\Sigma^{1/2} P^{-1} v \right\|^2 = v^\top P^{-1} \hat\Sigma P^{-1} v \leq v^\top P^{-1} v = \left\| P^{-1/2} v \right\|^2 \leq \|v\|^2 / \lambda$ and $ab \leq \frac{1}{2}(a^2 + b^2)$. Summing the three bounds, the numerator is $\leq (\frac{d_\infty}{4} + \frac{1}{2}) v^\top \hat\Sigma_w v + \frac{3}{2}\lambda \|v\|^2 \leq (\frac{d_\infty}{4} + \frac{3}{2})\, v^\top H v$. For $P = I$: $v^\top H_i^2 v = \|H_i v\|^2 = \|w_i c_i x_i + \lambda v\|^2 = w_i^2 c_i^2 \|x_i\|^2 + 2\lambda w_i c_i^2 + \lambda^2$ (using $x_i^\top v = c_i$ and $\|v\| = 1$); averaging over i gives $B_v + 2\lambda v^\top \hat\Sigma_w v + \lambda^2$, and $v^\top H v = v^\top \hat\Sigma_w v + \lambda$, which is (107). For the bounds on B_v: $\|x_i\|^2 \geq c_i^2$ (Cauchy–Schwarz, $\|v\| = 1$) and the convexity of the square give $B_v \geq \frac{1}{n}\sum_i w_i^2 c_i^4 \geq \big(\frac{1}{n}\sum_i w_i c_i^2\big)^2 = (v^\top \hat\Sigma_w v)^2$, while $w_i \leq \frac{1}{4}$ and $\|x_i\|^2 \leq \max_j \|x_j\|^2 = 4(\kappa - 1)\lambda$ (by (91)) give $B_v \leq \frac{\max_j \|x_j\|^2}{4} \cdot \frac{1}{n}\sum_i w_i c_i^2$. The two-sided bound on the limit follows, with $A := v^\top \hat\Sigma_w v$, from $(\lambda^2 + 2\lambda A + B_v)/(\lambda + A) \geq (\lambda + A)^2/(\lambda + A)$ and $(\lambda^2 + 2\lambda A + B_v)/(\lambda + A) \leq (\lambda^2 + 2\lambda A + (\kappa - 1)\lambda A)/\lambda$. □

Global versus trajectory-dependent constants: where the supremum over θ is needed. Proposition 5.9 uses (I2) for each i at the pair (θ_{t-1}, η_*) for every t, and (I6) at (θ_0, η_*); co-coercivity of $\tilde f_i$ between two points is proved (Proposition 5.4 of [6]) from smoothness on a neighbourhood containing an auxiliary point, so a constant valid on the whole space, R_P^2 with its $\sup_\theta$, is what the proof certifies; for logistic regression this supremum equals $\frac{1}{4}$ for every sample, whatever P. Strong convexity (I4) is needed only at the pairs (θ_{t-1}, η_*), and the exact formula (95) shows that it is a direction-weighted average of curvatures on the segment $[\eta_*, \theta_{t-1}]$, which depends on the trajectory. The global constants (R_P^2, μ_P) are therefore pessimistic in two different ways: μ_P by the gap between $\inf_\theta$ and the segment averages (Propositions 3.22 and 3.26), and R_P^2 by the gap between $\max_i \sup_\theta$ and the realized L_{eff}^P (Lemma 3.27). The B2 note measures both gaps through $\mu_{\text{eff}}^P(\theta_{t-1})$ and $L_{\text{eff}}^P(\theta_{t-1})$; the statements above predict that preconditioning moves the gap from one inequality to the other (Section 3.6). A rigorous way to use trajectory-dependent constants is to restrict a priori to a ball $\mathcal{B}$ containing η_* and all iterates and to replace $\sup_\theta$ and $\inf_\theta$ by $\sup_{\mathcal{B}}$ and $\inf_{\mathcal{B}}$; this is what [16, Sec. 4.2] do for non-quadratic losses (uniform concentration of Hessians over a ball of radius D), the ball being needed precisely because the iterates depend on the metric.

3.5 Cost, and the case of a non-constant metric

Per iteration and per pass. One iteration of P-SAGA computes one gradient $f'_{i(t)}(\theta_{t-1})$ ($O(d)$, or $O(\text{nnz}(x_i))$ for sparse data), updates the memory vector $z_{i(t)}$ and the running sum $\sum_i z_i$ ($O(d)$),

and applies P^{-1} to g_t: $O(d)$ if P is diagonal, $O(d^2)$ if P is dense and a Cholesky factor $P = LL^\top$ (or P^{-1} itself) has been precomputed (two triangular solves). Hence the cost of one iteration is $c_I = O(d)$, $c_D = O(d)$, $c_{\hat{\Sigma}+\lambda I} = c_H = O(d^2)$, and one effective pass (n iterations) costs $O(nd)$ for $P \in \{I, D\}$ and $O(nd^2)$ for a dense P. Precomputation: $\mathrm{diag}(\hat{\Sigma})$ costs $O(nd)$; $\hat{\Sigma}$ costs $O(nd^2)$ (one matrix product) and its Cholesky factorization $O(d^3)$; $F''(\eta_*)$ needs η_* (an oracle; in practice a Hessian at a reference point, same cost $O(nd^2)$). Memory: $O(nd)$ for the z_i as in Proposition 5.9 (the usual implementation for linear models stores one scalar $\ell'(y_i x_i^\top \theta_{\tau_i})$ per sample and keeps the regularizer outside the memory, $O(n)$ memory; this is a composite variant of SAGA [7] whose analysis differs slightly from Proposition 5.9), plus $O(d^2)$ for the factor of a dense P. A dense P also destroys sparsity ($P^{-1}x_i$ is dense), whereas a diagonal P preserves it.

Total complexity. By Corollary 3.12, the time to reach accuracy ε is

$$T_P \asymp c_P\Big(3n + \frac{16}{3}\kappa_P\Big)\log\frac{2n}{\varepsilon} \;+\; (\text{precomputation}), \tag{108}$$

to be compared with $T_I \asymp d\,(3n + \frac{16}{3}\kappa)\log\frac{2n}{\varepsilon}$ for SAGA [6, Sec. 5.4.4]. The diagonal metric is free. A dense metric multiplies the per-iteration cost by d and adds $O(nd^2 + d^3)$, so it pays off only if the number of iterations decreases by more than a factor d: dividing (108) by $d\log\frac{2n}{\varepsilon}$, the condition is

$$d\Big(3n + \frac{16}{3}\kappa_P\Big) + O\Big(\frac{nd + d^2}{\log(2n/\varepsilon)}\Big) \;\lesssim\; 3n + \frac{16}{3}\kappa, \qquad \text{i.e. roughly} \qquad \kappa \;\gtrsim\; d\,(n + \kappa_P),$$

the very ill-conditioned regime $\kappa \gg nd$ (the regime $\kappa \gg n$ of hypothesis (A4), with an extra factor d). With global constants this never happens for logistic regression ($\kappa_P \geq \kappa$, Proposition 3.21). With trajectory constants, *both* condition numbers must be replaced by their effective values: inserting the upper bounds (98) for $P = \hat{\Sigma} + \lambda I$ and (99) for $P = I$ (with the same lower bound m on the segment curvatures along both trajectories), the condition reads

$$\frac{\max_i \|x_i\|^2/4 + \lambda}{m\sigma_d + \lambda} \;\gtrsim\; d\left(n + \frac{d_\infty(\lambda)/4 + 1}{m}\right) \tag{109}$$

if both bounds are tight. Since the left-hand side of (109) is at most the global $\kappa = \max_i \|x_i\|^2/(4\lambda) + 1$, replacing it by κ (global-constant SAGA against effective-constant P-SAGA, an asymmetric comparison) gives a condition, $\kappa \gtrsim d(n + (d_\infty(\lambda)/4 + 1)/m)$, which is only *necessary*: it is a lower bound on the ill-conditioning required for a dense metric to pay off, and it overstates the benefit of the metric when $\lambda \ll m\sigma_d$ (then the left-hand side of (109) is of order $\max_i \|x_i\|^2/(4m\sigma_d) \ll \kappa$). In the memory-bottleneck regime of Remark 3.24, $T_P \asymp 3c_P n\log(2n/\varepsilon)$: $3\log(2n/\varepsilon)$ passes, each d times more expensive than a SAGA pass when P is dense.

A non-constant metric breaks the transfer argument. Suppose the step uses P_t at iteration t: $\theta_t = \theta_{t-1} - \gamma_t P_t^{-1} g_t$ (variable-metric or quasi-Newton variance reduction). Proposition 3.5 no longer applies: the transformed functions $\tilde{f}_i^{(t)} = f_i \circ P_t^{-1/2}$ and the transformed memory $P_t^{-1/2} z_i$ change with t, so there is no single Euclidean SAGA run to which Proposition 5.9 could be applied. In the P-metric proof of Section 3.3.1, the one-step inequality (73) still holds at iteration t for the Lyapunov function $V_t^{P_t}$ built with P_t and $\gamma_t = 1/(4R_{P_t}^2)$, *provided* P_t (hence γ_t, $R_{P_t}^2$, ρ_{P_t}) is measurable with respect to the past $\mathcal{F}_{t-1} := \sigma(i(1), \ldots, i(t-1))$, i.e. is computed before $i(t)$ is drawn: conditionally on $\mathcal{F}_{t-1}$ the metric is then fixed, the unbiasedness $\mathbb{E}[g_t \mid \mathcal{F}_{t-1}] = F'(\theta_{t-1})$ can be used inside the P_t-norms, and the proof applies verbatim, giving $\mathbb{E}[V_t^{P_t} \mid \mathcal{F}_{t-1}] \leq \rho_{P_t} V_{t-1}^{P_t}$ (if P_t

depended on $i(t)$, then $\mathbb{E}[P_t^{-1} g_t \mid \mathcal{F}_{t-1}] \neq P_t^{-1} F'(\theta_{t-1})$ in general and the argument already breaks at (75)). But this compares $\mathbb{E}[V_t^{P_t} \mid \mathcal{F}_{t-1}]$ with $V_{t-1}^{P_t}$, not with $V_{t-1}^{P_{t-1}}$; one must pay the change of metric

$$V_{t-1}^{P_t} \le c_t \, V_{t-1}^{P_{t-1}}, \qquad c_t := \max\Big\{ \lambda_{\max}\big(P_{t-1}^{-1/2} P_t P_{t-1}^{-1/2}\big), \; \frac{\gamma_t^2}{\gamma_{t-1}^2} \lambda_{\max}\big(P_t^{-1/2} P_{t-1} P_t^{-1/2}\big) \Big\} \tag{110}$$

(Lemma 3.2(v) applied to the pairs (P_{t-1}, P_t) and (P_{t-1}^{-1}, P_t^{-1}); with the convention $P_0 := P_1$, $\gamma_0 := \gamma_1$, so that $c_1 = 1$). Both ρ_{P_t} and c_t are $\mathcal{F}_{t-1}$-measurable but random for an adaptive metric, and correlated with $V_{t-1}^{P_{t-1}}$, so they cannot be pulled out of the expectation; if however $\rho_{P_s} c_s \le \bar\rho_s$ almost surely for deterministic constants $\bar\rho_s$, then taking expectations and iterating gives $\mathbb{E} V_t^{P_t} \le \prod_{s=1}^t \bar\rho_s \, V_0^{P_0}$. Linear convergence is thus certified only if the metric drift is controlled by deterministic bounds, e.g. $c_t \le \bar c_t$ with $\sum_t \log \bar c_t < \infty$ (a stationary limit metric, reached fast enough), or $c_t \le 1 + \frac{1}{2}\min\{\frac{1}{3n}, \frac{3}{16\bar\kappa}\}$ for a deterministic $\bar\kappa \ge \sup_t \kappa_{P_t}$ (then $\rho_{P_t} c_t \le 1 - \frac{1}{2}\min\{\frac{1}{3n}, \frac{3}{16\bar\kappa}\}$), which is the analogue of the condition "η_t does not vary too much (not more than a factor $1 - O(1/n)$)" stated after Theorem 4 of [8] (there the gains G_t must be nonincreasing for the Bregman potential to be a supermartingale). A uniform sandwich $\gamma I \preceq P_t^{-1} \preceq \Gamma I$ alone gives only $c_t \le \Gamma/\gamma$, useless per step; this is why the stochastic quasi-Newton variance-reduced methods of [25] (stochastic L-BFGS with SVRG gradients; Lemma 4: $\gamma I \preceq H_r \preceq \Gamma I$) and [11] (stochastic block BFGS with SVRG gradients; Assumption 1 and Lemma 1: $\gamma I \preceq H_t \preceq \Gamma I$) are analysed in the Euclidean norm, the metric entering only through (γ, Γ): Theorem 7 of [25] gives, for $\lambda I \preceq \nabla^2 f_T \preceq \Lambda I$, step η and inner-loop length m, the rate $\alpha = \big(\frac{1}{2m\eta} + \eta\Gamma^2\Lambda^2\big)/\big(\gamma\lambda - \eta\Gamma^2\Lambda^2\big)$, and Theorem 1 of [11] the rate $\rho = \big(\frac{1}{2m\eta} + \eta\Gamma^2\Lambda(\Lambda - \lambda)\big)/\big(\gamma\lambda - \eta\Gamma^2\Lambda^2\big)$. In the first formula, $\alpha < 1$ iff $\frac{1}{2m\eta} + 2\eta\Gamma^2\Lambda^2 < \gamma\lambda$; minimizing the left-hand side over η (AM–GM: its minimum is $2\Gamma\Lambda/\sqrt{m}$, attained at $\eta = 1/(2\Gamma\Lambda\sqrt{m})$) shows that $\alpha < 1$ for some η if and only if $m > 4\big(\Gamma\Lambda/(\gamma\lambda)\big)^2 \ge 4(\Lambda/\lambda)^2$, since $\Gamma \ge \gamma$: the certified epoch length is at least $4(\Lambda/\lambda)^2$, the square of the Hessian condition number Λ/λ of f (not to be confused with our $\kappa = R^2/\mu$), whereas the epoch length of Euclidean SVRG is linear in the condition number; the benefit of the metric is not captured at all. Capturing it requires a relative-smoothness analysis with a fixed (or asymptotically fixed) reference function, as here ([8]: Corollary 1 for quadratic h, Theorem 4 with gains G_t otherwise), or a preconditioner fixed in advance from statistics of the data: a sub-sampled Hessian plus ridge as in [16], whose Theorem 2 shows that for quadratic losses with $\|a_i\| \le R$, a sub-sample of size $n > \frac{28}{3}\log\frac{2d}{\delta}$ and the ridge $\mu = \frac{1}{2}\big(\frac{28R^2}{3n}\log\frac{2d}{\delta} - \lambda\big)^+$ give, with probability $1 - \delta$, $(\frac{3}{2} + \frac{2\mu}{\lambda})^{-1}(H_f + \mu I) \preceq H_F \preceq 2(H_f + \mu I)$, i.e. a relative condition number $O\big(1 + \frac{\kappa_\ell}{n}\log\frac{d}{\delta}\big)$ with $\kappa_\ell = R^2/\lambda$; or a sketched preconditioner as in [10] (rank-k sketch of the covariance for ridge regression, computed in time $O(ndk\log n)$; their Theorem 1 replaces the average condition number $\sum_i(\sigma_i + \lambda)/(\sigma_d + \lambda)$ of SVRG by one of order $(k\sigma_k + \sum_{i>k}\sigma_i)/\lambda$). In all these cases the metric is constant during the variance-reduced run, which is exactly what Proposition 3.5 needs. (Non-uniform sampling of $i(t)$ is outside the uniform-sampling setting of Proposition 5.9 and of this note. For the record: Schmidt et al. [30, Sec. 4.8] conjectured, and Gower et al. [13, Thm. 5.2 and Table 1, row 8] proved (see also [29]), that SAGA with the reweighted unbiased estimator $\frac{1}{n}\sum_i z_i^{(t-1)} + \frac{1}{np_{i(t)}}\big(f'_{i(t)}(\theta_{t-1}) - z_{i(t)}^{(t-1)}\big)$ and probabilities $p_i \propto \mu n + 4L_i$ reaches accuracy ε in $O\big((n + 4\bar L/\mu)\log\frac{1}{\varepsilon}\big)$ iterations, $\bar L = \frac{1}{n}\sum_i L_i$ being the *average* of the per-sample smoothness constants. The change of variables of Proposition 3.5 applies verbatim to this variant, and for $P = \hat\Sigma + \lambda I$ the per-sample relative smoothness constants satisfy $L_i \le \frac{n\ell_i(\lambda)}{4} + \frac{\lambda}{\sigma_d + \lambda}$ by the Weyl bound in the proof of Lemma 3.11, so that $d_\infty(\lambda)$ would be replaced by $d_{\mathrm{eff}}(\lambda) = \sum_i \ell_i(\lambda)$ in the *global* bound (93); whether the same substitution holds in the trajectory bounds of Corollary 3.23 is not covered by those analyses and is left open.)

3.6 Predictions for the experiments

Tables 2 and 3 collect, for the four metrics, the certified constants (Lemma 3.11, Propositions 3.19, 3.20 and 3.26), the certified rate $\rho_P = 1 - \min\{\frac{1}{3n}, \frac{3}{16\kappa_P}\}$, and the predicted late-phase behaviour of the effective constants (Lemma 3.27). The late-phase predictions rest on the heuristic that, as for gradient descent on a quadratic, the late-phase error δ aligns with the bottom eigenspace of the relative Hessian $M_P := P^{-1/2}F''(\eta_*)P^{-1/2}$ (the most slowly contracted direction); $\bar{w}$ denotes a leverage-weighted average of the curvatures $w_i = \sigma_i'(\eta_*)$ in that direction and $\overline{n\ell}$ a weighted average of the $n\ell_i(\lambda)$ (equal to $d_{\text{eff}}(\lambda)$ when leverage scores are homogeneous).

	$P = I$	$P = D = \operatorname{diag}(\hat{\Sigma}) + \lambda I$	$P = \hat{\Sigma} + \lambda I$	$P = H = F''(\eta_*)$
R_P^2 (certified)	$\frac{1}{4}\max_i \|x_i\|^2 + \lambda$	$\frac{1}{4}\max_i \|x_i\|^2_{D^{-1}} + \frac{\lambda}{\min_j D_{0,jj}+\lambda}$	$\frac{d_\infty(\lambda)}{4} + \frac{\lambda}{\sigma_d+\lambda}$ $\le \frac{Cd_{\text{eff}}(\lambda)}{4} + 1$	$\max_i \frac{n\ell_i^w(\lambda)}{4w_i} + \frac{\lambda}{\sigma_d^w+\lambda}$
μ_P (global, exact)	λ	$\frac{\lambda}{\max_j D_{0,jj}+\lambda}$	$\frac{\lambda}{\sigma_1+\lambda}$	$\frac{\lambda}{\sigma_1^w+\lambda}$
κ_P (global)	κ	$\ge \kappa$	$\in [\kappa,\ \kappa\,\text{cond}(P)]$	$\ge \kappa$
certified ρ_P	$1 - \min\{\frac{1}{3n}, \frac{3}{16\kappa}\}$	$\ge \rho_I$: not better than $P = I$	$\ge \rho_I$: not better than $P = I$	$\ge \rho_I$: not better than $P = I$
cost per pass	$O(nd)$	$O(nd)$	$O(nd^2)$ $+O(nd^2 + d^3)$ once	$O(nd^2)$ $+O(nd^2 + d^3)$ once, needs η_*

Table 2: Certified (global) constants for ℓ_2-regularized logistic regression.

late phase	$P = I$	$P = D$	$P = \hat{\Sigma} + \lambda I$	$P = H = F''(\eta_*)$
$\mu_{\text{eff}}^P \to$	$\lambda_{\min}(H) = \sigma_d^w + \lambda$	$\lambda_{\min}(M_D)$	$\lambda_{\min}(M_P) \approx \bar{w}$, $\in \left[\frac{w_{\min}\sigma_1+\lambda}{\sigma_1+\lambda}, 1\right]$	1
$L_{\text{eff}}^P \to$	$\frac{\lambda^2+2\lambda\sigma_d^w+B_v}{\lambda+\sigma_d^w}$ $\in [\sigma_d^w + \lambda,\ \lambda + (\kappa+1)\sigma_d^w]$	same with $\|\cdot\|_{D^{-1}}$	$\le \frac{d_\infty(\lambda)}{4} + \frac{3}{2}$, $\approx \bar{w}\,\overline{n\ell}$	$\in [1, \max\{d_\infty^w(\lambda), 2\}]$
$r_{\text{sc}} \to$	$\frac{\lambda}{\sigma_d^w+\lambda}$ (≈ 1 if $\sigma_d^w \ll \lambda$)	$\mu_D/\lambda_{\min}(M_D)$	$\approx \frac{\lambda}{(\sigma_1+\lambda)\bar{w}} \ll 1$	$\frac{\lambda}{\sigma_1^w+\lambda} \ll 1$
$r_{\text{coco}} \to$	$\frac{\lambda^2+2\lambda\sigma_d^w+B_v}{(\lambda+\sigma_d^w)R^2}$ $\in \left[\frac{1}{\kappa},\ \frac{1}{\kappa} + \frac{2\sigma_d^w}{\lambda}\right]$ ($\approx \frac{1}{\kappa}$ if $\sigma_d^w \ll \lambda/\kappa$)	intermediate	$\approx 4\bar{w}\,\overline{n\ell}/d_\infty(\lambda) \le 1$	$\frac{1+\text{heterogeneity}}{R_H^2}$
loosest step	(I2) co-coercivity	(I2) and (I4)	(I4) strong convexity	(I4) strong convexity
$R_P^2/\mu_{\text{eff}}^P \to$	$\frac{\max_i\|x_i\|^2/4+\lambda}{\sigma_d^w+\lambda}$	between the two neighbours, factor $\text{cond}(C_{\text{corr}})$	$\lesssim \frac{Cd_{\text{eff}}(\lambda)/4+1}{\bar{w}}$	$R_H^2 \le \frac{Cd_{\text{eff}}^w(\lambda)}{4w_{\min}} + 1$

Table 3: Predicted late-phase behaviour ($M_P = P^{-1/2}F''(\eta_*)P^{-1/2}$, v the late-phase error direction, taken to be the bottom eigenvector of M_P; for $P = I$, $v^\top\hat{\Sigma}_w v = \sigma_d^w$ and $B_v = \frac{1}{n}\sum_i w_i^2(x_i^\top v)^2\|x_i\|^2 \le (\kappa-1)\lambda\sigma_d^w$ as in (107); "heterogeneity" is the second term of (106)).

In words, with the statements they follow from:

(P1) *Certified rates.* With global constants, none of the three preconditioners improves ρ_P for logistic regression ($\rho_P \ge \rho_I$ since $\kappa_P \ge \kappa$, Proposition 3.21); any observed speed-up must show up in the effective constants, not in ρ_P. For the square loss the statistical preconditioner does improve ρ_P, to $1 - \min\{\frac{1}{3n}, \frac{3}{16(d_\infty(\lambda)+1)}\}$.

(P2) *Looseness moves from (I2) to (I4).* Under $P = I$ the late-phase error lives in the low-curvature directions of H, where (I4) is nearly tight ($r_{\mathrm{sc}} \to \lambda/\lambda_{\min}(H) = \lambda/(\sigma_d^w + \lambda)$, close to 1 when $\sigma_d^w \ll \lambda$) and (I2) is loose: by (107) with v the bottom eigenvector of H (so $v^\top \hat{\Sigma}_w v = \sigma_d^w$),

$$r_{\mathrm{coco}} \;\to\; \frac{\lambda^2 + 2\lambda\sigma_d^w + B_v}{(\lambda + \sigma_d^w)R^2} \;\in\; \Big[\frac{\sigma_d^w + \lambda}{R^2},\; \frac{1}{\kappa} + \Big(1 + \frac{1}{\kappa}\Big)\frac{\sigma_d^w}{\lambda}\Big],$$

which is $\approx \lambda/R^2 = 1/\kappa$ when $\sigma_d^w \ll \lambda$ *and* $B_v \ll \lambda^2$ (sufficient: $\sigma_d^w \ll \lambda/\kappa$, since $B_v \le (\kappa - 1)\lambda\sigma_d^w$), and in any case $\ll 1$ as soon as $\sigma_d^w \ll \lambda$. The condition $\sigma_d^w \ll \lambda$ alone does not give $L_{\mathrm{eff}} \approx \lambda$: it kills the term $2\lambda\sigma_d^w$ but not B_v, which may be as large as $(\kappa - 1)\lambda\sigma_d^w \gg \lambda^2$ on the hard instances $\kappa \gg n$ of (A4); the experiments should therefore record B_v (or directly the limit value) rather than test $r_{\mathrm{coco}} \approx 1/\kappa$. Under $P = \hat{\Sigma} + \lambda I$ and $P = H$ the bottom eigenspace of M_P is a high-variance data direction with relative curvature $\approx \bar{w}$ (resp. 1), so (I4) with the global μ_P becomes the loosest step ($r_{\mathrm{sc}} \approx \lambda/((\sigma_1 + \lambda)\bar{w})$, resp. $\lambda/(\sigma_1^w + \lambda)$), while r_{coco} rises to $\approx 4\bar{w}\,\overline{n\ell}/d_\infty(\lambda)$ (Lemma 3.27): it moves closer to 1 when the per-sample curvature is homogeneous and close to its maximum $\frac{1}{4}$ (data near the decision boundary) *and* the leverage scores are homogeneous ($\overline{n\ell} \approx d_{\mathrm{eff}} \approx d_\infty$); it remains bounded by $4\bar{w} \le 1$.

(P3) *Effective rate.* By (89) the parameter part contracts at $1-(1-\frac{5}{8}r_{\mathrm{coco}})\mu_{\mathrm{eff}}^P/(2R_P^2)$; the predicted late-phase effective condition numbers $R_P^2/\mu_{\mathrm{eff}}^P$ are the last row of Table 3: of order $(\operatorname{tr}\hat{\Sigma}/4 + \lambda)/(\sigma_d^w + \lambda)$ for $P = I$ (the anisotropy of $\hat{\Sigma}$), $\lesssim (Cd_{\mathrm{eff}}/4 + 1)/\bar{w}$ for the statistical metric (no anisotropy), $\to R_H^2$ for the Newton metric, the Jacobi metric sitting in between by the factor $\mathrm{cond}(C_{\mathrm{corr}})$ (Proposition 3.19). On the hard instances of (A4) ($\kappa \gg n$), preconditioning should thus bring the *effective* regime back to $n \gg \kappa_{\mathrm{eff}}$, where the memory term $\frac{1}{3n}$ is the bottleneck (Remark 3.24).

(P4) *The other ratios.* $r_{\min} \to 1$ iff V^P is dominated by the term with the smaller certified contraction (the parameter term under global constants, since $\frac{3}{16\kappa_P} < \frac{1}{3n}$ for every P on the hard instances by Proposition 3.21). Heuristically, in the late phase the memory errors $z_i^{(t-1)} - f_i'(\eta_*) \approx H_i\delta_{\tau_i}$ are built from comparable past errors, so $\|\bar{z}\|_{P^{-1}}^2 / \frac{1}{n}\sum_i \Big\|z_i^{(t-1)} - f_i'(\eta_*)\Big\|_{P^{-1}}^2 \approx \mu_{\mathrm{eff}}^P/L_{\mathrm{eff}}^P$ and $r_{\mathrm{var}} \approx 1 - \mu_{\mathrm{eff}}^P/L_{\mathrm{eff}}^P$: close to 1 when the effective condition number is large, hence *smaller* under preconditioning. Similarly $r_{\mathrm{init}} = r_{\mathrm{coco}}(1)\,\mu_{\mathrm{eff}}^P(\theta_0)/R_P^2$ exactly, by (87), which increases under preconditioning, and $r_{\mathrm{sq}} \le \frac{1}{2}$ up to the correlation of $a_{t,i}$ and $b_{t,i}$, which is negative (both are built from $H_i\delta$ at different times) and less so when $M_P \approx I$ contracts all directions at the same speed.

(P5) *Check of the transfer.* Running Euclidean SAGA on $(\tilde{f}_i)$ with the same seed must reproduce P-SAGA exactly, up to rounding (Proposition 3.5(iii)); this is a sanity check of the implementation, and the six ratios computed either way must coincide.

4 Lower bounds for finite-sum optimization and the hard instance (hypotheses A1, A4)

Throughout, as in [6], Section 5.4.4, we consider

$$F(\theta) = \frac{1}{n}\sum_{i=1}^n f_i(\theta), \qquad \theta \in \mathbb{R}^d,$$

where every f_i is convex and R^2-smooth (its gradient $f_i' = \nabla f_i$ is R^2-Lipschitz), F is μ-strongly convex with minimizer η_*, and $\kappa = R^2/\mu$. We write $F' = \nabla F$ and L_F for the smoothness constant of F; since $L_F \le R^2$ one always has $L_F/\mu \le \kappa$, and the gap between these two numbers is the main theme of this note. The cost of a method is the number of individual gradients $f_i'(\theta)$ it evaluates (one per SAGA iteration, n per full-gradient step).

The note has four parts. Section 4.1 fixes the oracle models and the span assumption and gives Nesterov's single-function construction with a complete proof. Section 4.2 quotes, with their exact hypotheses, the finite-sum lower bounds of Agarwal–Bottou [1], Lan–Zhou [22], Woodworth–Srebro [32], Arjevani–Shamir [3], Hannah–Liu–O'Connor–Yin [15] and Han–Xie–Zhang [**?**], and settles the meaning of hypothesis (A1). Section 4.3 builds the block-separable hard instance used for (A4), with every constant derived in the notation above, and proves the lower bound on it. Section 4.4 derives the predictions for the numerics (exponents κ^1 versus $\kappa^{1/2}$, crossover, finite-dimension effects) and explains why logistic regression is not a hard instance.

4.1 Oracle models, the span assumption, and Nesterov's chain

4.1.1 Oracle models and classes of algorithms

Definition 4.1 (incremental first-order oracle)**.** *Following [1] (Definition 2), the* incremental first-order oracle *(IFO) takes as input a point $\theta \in \mathbb{R}^d$ and an index $i \in \{1, \dots, n\}$ and returns the pair $(f_i(\theta), f_i'(\theta))$. An IFO algorithm is an algorithm whose specification depends on F only through such calls; its cost is the number of calls. The* proximal *IFO (PIFO) of [32, **?**] additionally returns $\operatorname{prox}_{f_i}(\theta, \beta) = \arg\min_u \{f_i(u) + \frac{\beta}{2}\|u - \theta\|^2\}$ for any requested $\beta > 0$ (the parametrization of display (3) of [32]; since β ranges over all positive numbers, writing $\frac{1}{2\beta}$ in place of $\frac{\beta}{2}$ describes the same oracle).*

A full gradient $F'(\theta) = \frac{1}{n}\sum_i f_i'(\theta)$ costs n IFO calls at the same point. The lower bounds below differ in *how the algorithm may choose the sequence of indices* and in *how it may combine the answers.*

Definition 4.2 (linear-span protocol)**.** *An algorithm follows the* linear-span protocol *(Nesterov's "black-box" assumption, [26] Section 2.1, relation (3.41) of [22], relation (1.2) of [15]) if, when it has queried the oracle at the pairs (θ_{s-1}, i_s), $s = 1, \dots, t$, its iterate satisfies*

$$\theta_t \in \theta_0 + \operatorname{span}\big\{f_{i_1}'(\theta_0),\ f_{i_2}'(\theta_1),\ \dots,\ f_{i_t}'(\theta_{t-1})\big\}. \tag{111}$$

We distinguish three classes according to the index sequence (i_s):

(S1) span-based with i.i.d. sampling*: (i_s) is i.i.d. from a fixed distribution p on $\{1, \dots, n\}$ (uniform in the project), independent of the oracle answers [22, 15]. This contains SGD, SAG, Finito/MISO, SDCA-without-duality, SAGA started from the zero memory $z^{(0)} = 0$, and, on block-separable problems, randomized block-coordinate methods. With the book's initialization $z_i^{(0)} = f_i'(\theta_0)$, SAGA is an (S1) method* preceded by one full gradient at θ_0 *(n deterministic queries); Lemma 4.16(ii) and Corollary 4.21 treat this variant.*

(S2) span-based with an arbitrary index sequence*: (i_s) is any sequence, deterministic (cyclic, or n consecutive queries at the same point, which is a full gradient), random, or adaptive. This contains GD, Nesterov's accelerated gradient (AGD), cyclic incremental methods, and the hybrid methods SVRG, SARAH, L-SVRG, Katyusha and L-Katyusha, which mix i.i.d. single queries with full gradients.*

(S3) no span restriction*: arbitrary deterministic algorithms [1, 32] or arbitrary randomized algorithms [32], and the* oblivious canonical linear iterative *(CLI) class of [3], in which new iterates are linear combinations* $\sum_j O_F(\theta^{(j)};\alpha^{(j)})$ *of oracle outputs (gradients possibly multiplied by fixed matrices, coordinate steps, full gradients) whose coefficients* $\alpha^{(j)}$ *are chosen* without looking at the oracle answers *(they may be random). Fixed step sizes, fixed momentum and fixed sampling schemes are allowed, line searches and quasi-Newton updates are not.*

SAGA's update direction $f'_{i(t)}(\theta_{t-1}) - z_{i(t)}^{(t-1)} + \frac{1}{n}\sum_j z_j^{(t-1)}$ is a linear combination of the current gradient and of the stored gradients $z_j^{(t-1)}$, each of which is either its initial value or a gradient $f'_j(\theta_s)$ computed at an earlier iterate. Hence (111) holds when $z^{(0)} = 0$, whereas with the book's initialization $z_i^{(0)} = f'_i(\theta_0)$ the span must be enlarged by the n initial gradients $f'_1(\theta_0), \dots, f'_n(\theta_0)$, which are not sampled i.i.d.: this is the only point where SAGA, as analysed in Proposition 5.9 of [6], leaves class (S1). [**?**] make the same observation (SAGA and Point-SAGA "only calculate the full gradient at the first iteration") and let the full gradient at θ_0 be free in their class. L-Katyusha [21] is *not* in (S1) because of its full gradients at the reference points w_k (each costs n queries at one point); it is in (S2), in the oblivious CLI class (its coin flips and parameters do not depend on the answers) and in the class of [**?**].

Dimension-free versus span-based bounds. A lower bound is *dimension-free* if it holds in every dimension d [3]. Span-based constructions are not: they reveal at most one new coordinate per query, so they can only certify as many iterations as there are unrevealed coordinates. This is Nesterov's caveat $t < d$ (in [6], Section 15.2.1, the nonsmooth lower bounds "are valid for $k < d$ only" and, for the smooth convex construction F_{2k+1}, "the number of iterations has to be less than half the dimension for the lower bound to hold"; the strongly convex construction of that section lives in ℓ_2 and carries no such caveat), the condition $n \geq \underline{n}(m,k)$ in [22], and the condition $m \gtrsim \sqrt{\kappa/n}\log(1/\varepsilon)$ of [**?**]; in the words of [32], "if we do assume that the queries lie in that span, the necessary dimension is only on the order of the number of oracle queries needed". Removing the span assumption costs a much larger dimension (polynomial in $1/\varepsilon$ in [32]) or an obliviousness assumption [3]. Quantitatively, Section 4.3 shows that the block-separable instance certifies the bound at accuracy ε only when each block has at least $\frac{1}{4}\sqrt{1+(\kappa-1)/n}\,\log(1/\varepsilon)$ coordinates.

4.1.2 The chain matrix

For $m \geq 1$ and $\zeta \in [1,3]$ let $A_m(\zeta) \in \mathbb{R}^{m\times m}$ be the symmetric tridiagonal matrix with entries 2 on the diagonal except $(A_m(\zeta))_{mm} = \zeta$, and -1 on the two off-diagonals:

$$A_m(\zeta) = \begin{pmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & \zeta \end{pmatrix}. \tag{112}$$

The value $\zeta = 2$ is Nesterov's original finite-dimensional chain ([26], Section 2.1.2; [6], Section 15.2.1); the value $\zeta = \zeta_Q$ chosen in (121) below makes the minimizer an exact geometric sequence in finite dimension [22].

Lemma 4.3 (quadratic form of the chain). *For every $s \in \mathbb{R}^m$ and $\zeta \in [1,3]$,*

$$\langle A_m(\zeta)s, s\rangle = s_1^2 + \sum_{j=1}^{m-1}(s_j - s_{j+1})^2 + (\zeta-1)s_m^2, \tag{113}$$

so that $0 \preceq A_m(\zeta) \preceq 4I$; more precisely

$$\langle A_m(\zeta)s, s\rangle \le 4\,\|s\|^2. \tag{114}$$

Moreover, for $\zeta \in [1,2]$, $\lambda_{\min}(A_m(\zeta)) \in \big[\,4\sin^2\frac{\pi}{2(2m+1)},\ 4\sin^2\frac{\pi}{2(m+1)}\,\big]$, which is of order $1/m^2$.

Proof. Expanding the right-hand side of (113): $\sum_{j=1}^{m-1}(s_j - s_{j+1})^2 = s_1^2 + 2\sum_{j=2}^{m-1} s_j^2 + s_m^2 - 2\sum_{j=1}^{m-1} s_j s_{j+1}$, so the right-hand side equals $2s_1^2 + 2\sum_{j=2}^{m-1} s_j^2 + \zeta s_m^2 - 2\sum_{j=1}^{m-1} s_j s_{j+1} = \langle A_m(\zeta)s, s\rangle$ (for $m = 1$ both sides equal ζs_1^2). Nonnegativity follows from $\zeta \ge 1$. For the upper bound, $(s_j - s_{j+1})^2 \le 2s_j^2 + 2s_{j+1}^2$ gives $\langle A_m(\zeta)s, s\rangle \le 3s_1^2 + 4\sum_{j=2}^{m-1} s_j^2 + (\zeta+1)s_m^2 \le 4\,\|s\|^2$ because $\zeta \le 3$. For the eigenvalue bounds: $A_m(\zeta) - A_m(1) = (\zeta-1)e_m e_m^\top \succeq 0$ and $A_m(2) - A_m(\zeta) = (2-\zeta)e_m e_m^\top \succeq 0$, so $\lambda_{\min}(A_m(1)) \le \lambda_{\min}(A_m(\zeta)) \le \lambda_{\min}(A_m(2))$ by the Courant–Fischer theorem; the eigenvalues of $A_m(2)$ are $2 - 2\cos\frac{j\pi}{m+1} = 4\sin^2\frac{j\pi}{2(m+1)}$ and those of $A_m(1)$ are $2 - 2\cos\frac{(2j-1)\pi}{2m+1} = 4\sin^2\frac{(2j-1)\pi}{2(2m+1)}$, $j = 1, \dots, m$ (the vectors $(\sin\frac{jk\pi}{m+1})_k$, resp. $(\sin\frac{(2j-1)k\pi}{2m+1})_k$, are eigenvectors, as one checks row by row; the last row is where the two boundary values $\zeta = 2$, resp. $\zeta = 1$, enter). □

4.1.3 Nesterov's construction for a single function

Fix $L > \mu > 0$, $Q = L/\mu$, and define on ℓ_2 (square-summable sequences) the function of [26], Section 2.1.4 (also [6], Section 15.2.1):

$$f(\theta) = \frac{L-\mu}{4}\Big\{\frac{1}{2}\Big[\theta_1^2 + \sum_{j\ge 1}(\theta_j - \theta_{j+1})^2\Big] - \theta_1\Big\} + \frac{\mu}{2}\,\|\theta\|^2. \tag{115}$$

In the notation of (113) the bracket is $\langle A\theta, \theta\rangle$ for the infinite tridiagonal matrix A with 2 on the diagonal and -1 off it, so $f(\theta) = \frac{L-\mu}{4}\{\frac{1}{2}\langle A\theta,\theta\rangle - \theta_1\} + \frac{\mu}{2}\|\theta\|^2$, $f'(\theta) = \frac{L-\mu}{4}(A\theta - e_1) + \mu\theta$, and $f'' = \frac{L-\mu}{4}A + \mu I$ has spectrum in $[\mu, \mu + (L-\mu)] = [\mu, L]$ by (114): f is L-smooth and μ-strongly convex, with condition number exactly Q.

Proposition 4.4 (minimizer of the chain). *Let $q = \dfrac{\sqrt{Q}-1}{\sqrt{Q}+1} \in (0,1)$. The minimizer of (115) is $\theta_j^* = q^j$, $j \ge 1$, with $\|\theta^*\|^2 = \dfrac{q^2}{1-q^2}$ and $f(\theta^*) = -\dfrac{(L-\mu)q}{8}$.*

Proof. The optimality condition $f'(\theta) = 0$ reads $(A + \frac{4}{Q-1}I)\theta = e_1$, i.e.

$$\Big(2 + \tfrac{4}{Q-1}\Big)\theta_1 - \theta_2 = 1, \qquad -\theta_{j-1} + \Big(2 + \tfrac{4}{Q-1}\Big)\theta_j - \theta_{j+1} = 0 \quad (j \ge 2).$$

Since $q + q^{-1} = \frac{(\sqrt{Q}-1)^2 + (\sqrt{Q}+1)^2}{Q-1} = \frac{2(Q+1)}{Q-1} = 2 + \frac{4}{Q-1}$, the sequence $\theta_j = q^j$ satisfies the j-th equation, $q^{j-1}[-1 + (q+q^{-1})q - q^2] = 0$, and the first one, $(q+q^{-1})q - q^2 = 1$. It belongs to ℓ_2 and the strongly convex f has a unique minimizer, hence $\theta_j^* = q^j$. Then $\|\theta^*\|^2 = \sum_{j\ge1} q^{2j} = q^2/(1-q^2)$ and, using $A\theta^* = e_1 - \frac{4}{Q-1}\theta^*$, $f(\theta^*) = \frac{L-\mu}{4}\{\frac{1}{2}\theta_1^* - \frac{2}{Q-1}\|\theta^*\|^2 - \theta_1^*\} + \frac{\mu}{2}\|\theta^*\|^2 = -\frac{(L-\mu)q}{8}$ because $\frac{L-\mu}{4}\cdot\frac{2}{Q-1} = \frac{\mu}{2}$. □

Lemma 4.5 (one coordinate per gradient call)**.** *For $\theta \in \ell_2$ (or $\mathbb{R}^m$) let $N(\theta) = \max\{j : \theta_j \neq 0\}$ ($N(0) = 0$). Then $N(f'(\theta)) \leq N(\theta) + 1$ for the function* (115)*, and the same holds for $\theta \mapsto c(A_m(\zeta)\theta - e_1) + \mu\theta$ on $\mathbb{R}^m$ for any $c > 0$ (with $N \leq m$).*

Proof. $(A\theta)_j = -\theta_{j-1} + 2\theta_j - \theta_{j+1}$ (with the obvious boundary modifications) vanishes as soon as $j - 1 > N(\theta)$, so $N(A\theta) \leq N(\theta) + 1$; $-e_1$ only touches coordinate 1, and $\mu\theta$ does not enlarge the support. □

Theorem 4.6 (Nesterov's lower bound, [26] Theorem 2.1.13)**.** *Let $\theta_0 = 0$ and let (θ_t) be produced by any method following the span protocol* (111) *for the single function* (115) *(here all $i_s = 1$). Then for every $t \geq 0$*

$$\|\theta_t - \theta^*\|^2 \;\geq\; q^{2t}\,\|\theta_0 - \theta^*\|^2\,, \qquad f(\theta_t) - f(\theta^*) \;\geq\; \frac{\mu}{2}\,q^{2t}\,\|\theta_0 - \theta^*\|^2\,. \tag{116}$$

Consequently, reaching $\|\theta_t - \theta^*\|^2 \leq \varepsilon\,\|\theta_0 - \theta^*\|^2$ requires $t \geq \dfrac{\log(1/\varepsilon)}{2\log(1/q)} \geq \dfrac{\sqrt{Q}-1}{4}\log\dfrac{1}{\varepsilon}$ gradient evaluations, i.e. $\Omega(\sqrt{Q}\log(1/\varepsilon))$, which is the rate $(1-\sqrt{\mu/L})^t$ of accelerated gradient descent ([6], Section 5.2.5) up to constants.

Proof. By induction, $N(\theta_t) \leq t$: this is true for $t = 0$, and if $N(\theta_{s-1}) \leq s-1$ for all $s \leq t$ then each $f'(\theta_{s-1})$ in (111) has $N \leq s \leq t$ by Lemma 4.5, so $N(\theta_t) \leq t$. Hence $\|\theta_t - \theta^*\|^2 \geq \sum_{j>t}(\theta_j^*)^2 = \sum_{j>t} q^{2j} = q^{2t}\frac{q^2}{1-q^2} = q^{2t}\,\|\theta^*\|^2 = q^{2t}\,\|\theta_0 - \theta^*\|^2$. The function-value bound follows from μ-strong convexity, $f(\theta) - f(\theta^*) \geq \frac{\mu}{2}\,\|\theta - \theta^*\|^2$ (since $f'(\theta^*) = 0$). For the iteration count, $q^{2t} \leq \varepsilon$ means $t \geq \log(1/\varepsilon)/(2\log(1/q))$ and $\log(1/q) = \log\left(1 + \frac{2}{\sqrt{Q}-1}\right) \leq \frac{2}{\sqrt{Q}-1}$. □

The mechanism is the whole story of this note: a gradient of the chain reveals one new coordinate, the minimizer has geometrically decaying coordinates with ratio $q \approx 1 - 2/\sqrt{Q}$, and the energy in the unrevealed coordinates is q^{2t} of the total. The finite-sum constructions below simply run n chains in parallel, one per component, so that a query to component i reveals a coordinate of chain i only.

4.2 Precise statements of the finite-sum lower bounds

We quote the theorems in the notation of their authors, then translate them into the book's notation (n components, per-component smoothness R^2, strong convexity μ of the average, $\kappa = R^2/\mu$). All sources were downloaded and read (arXiv 1410.0723, 1507.02000, 1605.08003, 1606.09333, 1805.07786, 2103.08280, and 1901.08689 for L-Katyusha); the arXiv identifier 1809.09608 given in the project description for Hannah et al. is wrong (it is an unrelated logic paper): the correct one is 1805.07786, and the paper has four authors (Hannah, Liu, O'Connor, Yin).

Deterministic IFO algorithms (Agarwal–Bottou). In [1] the problem is $f(x) = \frac{\mu}{2}\,\|x\|^2 + \frac{1}{n}\sum_{i=1}^n g_i(x)$ with each g_i convex and $(L-\mu)$-smooth on ℓ_2 (class $\mathcal{F}_n^{\mu,L}$), $\kappa = L/\mu$; so each component $\frac{\mu}{2}\,\|x\|^2 + g_i$ is L-smooth and f is μ-strongly convex: in our notation $R^2 = L$ and their κ is ours. The algorithms are deterministic IFO algorithms (Definition 4.1); the span assumption is *not* made: their Appendix A removes it in two steps, first by showing that for the worst case one may assume that the algorithm's final point lies in the span of its previous queries and of the oracle's answers (their Section A.1), then by a resisting oracle which builds, call after call, orthonormal families S_k of dimension at most $2k+2$ (Gram–Schmidt, their Definition 11) and applies Nesterov's function to the coordinates in these bases, so that the final point is confined to a $(2K+2)$-dimensional subspace while the minimizer of the constructed function stays far from it (their Sections A.2–A.3).

Theorem 4.7 ([1], Theorem 1 and Corollary 1). *Consider a deterministic IFO algorithm that makes $K \ge 0$ calls and outputs x_K. For any $\varepsilon > 0$ there is $f \in \mathcal{F}_n^{\mu,L}(\ell_2)$ with $\left\|x_f^*\right\| = \varepsilon$ and*

$$\left\|x_f^* - x_K\right\| \;\ge\; \varepsilon\, q^{2t}, \qquad q = \frac{\sqrt{1+\frac{\kappa-1}{n}}-1}{\sqrt{1+\frac{\kappa-1}{n}}+1}, \qquad t = \begin{cases} 0 & \text{if } K < n, \\ K/n & \text{otherwise.} \end{cases}$$

Hence an algorithm guaranteeing $\left\|x_f^* - x_K\right\| \le \varepsilon \left\|x_f^*\right\|$ for all f in the class, with $\varepsilon < 1$, must perform $K = \Omega\big(n + \sqrt{n(\kappa-1)}\log(1/\varepsilon)\big)$ IFO calls.

The number $1+\frac{\kappa-1}{n}$ is the condition number of f restricted to one of n disjoint coordinate blocks (their construction is block separable with the regularizer shared by all components, exactly as ours in Section 4.3); the bound is attained, on their own instance, by accelerated gradient descent, and [32] show that it is loose for deterministic algorithms in general (Theorem 4.9 below).

Span-based randomized incremental methods (Lan–Zhou). In [22], Section 3.3, the hard problem is

$$\min_{x_i \in \mathbb{R}^{\tilde n},\, i=1,\dots,m} \Psi(x) := \sum_{i=1}^{m} \Big[f_i(x_i) + \frac{\mu}{2}\left\|x_i\right\|^2 \Big], \qquad f_i(x_i) = \frac{\mu(\mathcal{Q}-1)}{4}\Big[\frac{1}{2}\left\langle Ax_i, x_i\right\rangle - \left\langle e_1, x_i\right\rangle\Big], \tag{117}$$

with $A = A_{\tilde n}(\zeta)$ as in (112) and $\zeta = \frac{\sqrt{\mathcal{Q}}+3}{\sqrt{\mathcal{Q}}+1}$ (their display (3.39), where the boundary entry is called κ); each f_i is convex with Lipschitz constant $L_i = \mu(\mathcal{Q}-1)$. The class is: i_t i.i.d. with $\mathbb{P}\{i_t = i\} = p_i$ for a fixed distribution p (their (3.40)), and $x^k \in x^0 + \mathrm{Lin}\{\nabla f_{i_1}(x^0), \dots, \nabla f_{i_k}(x^{k-1})\}$ (their (3.41)), i.e. class (S1).

Theorem 4.8 ([22], Theorem 3 and Corollary 3). *Let $q = \frac{\sqrt{\mathcal{Q}}-1}{\sqrt{\mathcal{Q}}+1}$. The iterates of any randomized incremental gradient method of class (S1) applied to* (117) *satisfy*

$$\frac{\mathbb{E}\left\|x^k - x^*\right\|^2}{\left\|x^0 - x^*\right\|^2} \;\ge\; \frac{1}{2}\exp\Big(-\frac{4k\sqrt{\mathcal{Q}}}{m(\sqrt{\mathcal{Q}}+1)^2 - 4\sqrt{\mathcal{Q}}}\Big) \qquad \text{for all } \; n \ge \underline{n}(m,k),$$

$$\underline{n}(m,k) := \frac{m\log\big[(1-(1-q^2)/m)^k/2\big]}{2\log q},$$

where $n = m\tilde n$ is the total dimension. Consequently the number of gradient evaluations needed by such a method to find $\bar x$ with $\mathbb{E}\left\|\bar x - x^*\right\|^2 \le \varepsilon$ cannot be smaller than

$$\Omega\Big\{\big(m + \sqrt{m\mathcal{C}}\big)\log\frac{\left\|x^0 - x^*\right\|^2}{\varepsilon}\Big\}, \qquad \mathcal{C} = \frac{L}{\mu},\; L = \sum_{i=1}^{m} L_i,$$

if n is sufficiently large; precisely (their (3.45)) $k \ge \big[\frac{m}{2}\big(\frac{\sqrt{\mathcal{Q}}}{2}+1\big) - 1\big]\log\frac{\left\|x^0-x^*\right\|^2}{2\varepsilon}$.

Translation. Write $\Psi = \frac{1}{m}\sum_i g_i$ with $g_i(x) = m[f_i(x_i) + \frac{\mu}{2}\left\|x_i\right\|^2]$. Then g_i is $m\mu\mathcal{Q}$-smooth (Lemma 4.3), Ψ is μ-strongly convex, so in the book's notation $R^2 = m\mu\mathcal{Q}$, $n_{\text{book}} = m$, $\kappa = R^2/\mu = m\mathcal{Q}$, and $\sqrt{m\mathcal{C}} = m\sqrt{\mathcal{Q}-1} \asymp \sqrt{n_{\text{book}}\,\kappa}$. Theorem 4.8 is therefore the bound

$$\Omega\big((n + \sqrt{n\kappa})\log(1/\varepsilon)\big) \qquad \text{for class (S1)},$$

which is the first half of hypothesis (A1). Note that the proof minimizes the bound over the sampling distribution p and that the minimum is the uniform distribution: the bound holds for every fixed p. Note also that on these block-separable instances $\kappa = m\mathcal{Q} \geq m$ always (the per-block condition number $\mathcal{Q} = \kappa/n_{\text{book}}$ is what the chain mechanism sees), so the regime $\kappa \ll n$ cannot be probed with (117); our instance in Section 4.3 shares the regularizer, as in [1], which gives $\mathcal{Q} = 1+(\kappa-1)/n$ and allows every $\kappa \geq 1$.

Arbitrary deterministic and randomized algorithms (Woodworth–Srebro). In [32] the components are γ-smooth and F is λ-strongly convex ($R^2 = \gamma$, $\mu = \lambda$, $m = n_{\text{book}}$ components), the oracle is the gradient-and-prox oracle h_F, and no span assumption is made.

Theorem 4.9 ([32], Theorems 4 and 8)**.** (Deterministic.) *For any $\gamma, \lambda > 0$ with $\gamma/\lambda > 73$, any $\varepsilon > 0$, any $\varepsilon_0 > 3\gamma\varepsilon/\lambda$, any $m \geq 2$ and any deterministic algorithm A with access to h_F, there exist a dimension $d = O\big(m\sqrt{\gamma/\lambda}\log\frac{\lambda\varepsilon_0}{\gamma\varepsilon}\big)$ and m functions f_i on $\mathcal{X} \subseteq \mathbb{R}^d$, γ-smooth and λ-strongly convex with $F(0) - F(x^*) = \varepsilon_0$, such that finding $\hat{x}$ with $F(\hat{x}) - F(x^*) < \varepsilon$ requires $\Omega\big(m\sqrt{\gamma/\lambda}\log\frac{\lambda\varepsilon_0}{\gamma\varepsilon}\big)$ queries.*

(Randomized.) *For any $m \geq 2$, any $\gamma, \lambda > 0$ with $\gamma/\lambda > 161m$, any $\varepsilon > 0$, any $\varepsilon_0 > 60\varepsilon\sqrt{\gamma/(\lambda m)}$ and any randomized algorithm A, there exist a dimension $d = O\big(\frac{\gamma m}{\lambda\varepsilon}\log^4\frac{\varepsilon_0^2\lambda m}{\gamma\varepsilon^2}\log\frac{\gamma m^2}{\lambda\varepsilon}\big)$, a domain $\mathcal{X} \subseteq \mathbb{R}^d$, $x_0 \in \mathcal{X}$ and m functions f_i, γ-smooth and λ-strongly convex with $F(x_0)-F(x^*) = \varepsilon_0$, such that finding $\hat{x}$ with $\mathbb{E}[F(\hat{x}) - F(x^*)] < \varepsilon$ requires $\Omega\big(m + \sqrt{m\gamma/\lambda}\log\big(\frac{\varepsilon_0}{\varepsilon}\sqrt{\frac{m\lambda}{\gamma}}\big)\big)$ queries.*

In our notation: deterministic algorithms need $\Omega\big(n\sqrt{\kappa}\log\frac{\varepsilon_0}{\kappa\varepsilon}\big)$ queries (the argument of the logarithm is $\frac{\lambda\varepsilon_0}{\gamma\varepsilon} = \frac{\varepsilon_0}{\kappa\varepsilon}$, so the bound is informative only for relative accuracies $\varepsilon/\varepsilon_0 \ll 1/\kappa$; up to the constant inside the logarithm it is attained by AGD on F, since $L_F \leq R^2$), and randomized algorithms with arbitrary adaptive queries need $\Omega\big(n + \sqrt{n\kappa}\log\big(\frac{\varepsilon_0}{\varepsilon}\sqrt{n/\kappa}\big)\big)$ in the regime $\kappa > 161n$. Both theorems use components that are individually λ-strongly convex; this is a subclass of the book's class (where only F is μ-strongly convex), so the lower bounds apply a fortiori to the book's class. Two facts matter for (A1): the $\sqrt{n\kappa}\log(1/\varepsilon)$ term survives without any span or i.i.d. assumption, and the $n\log(1/\varepsilon)$ term of Theorem 4.8 is replaced by a bare n (no accuracy dependence).

Oblivious CLI algorithms, dimension-free (Arjevani–Shamir).

Theorem 4.10 ([3], Theorem 2)**.** *The iteration complexity of oblivious (possibly stochastic) CLI algorithms for finite sums $\frac{1}{n}\sum_{i=1}^n f_i$ with L-smooth and μ-strongly convex components (their problem (1); the sum is then μ-strongly convex, and the bound transfers a fortiori to the book's class, which contains this one), $\kappa = L/\mu$, equipped with first-order and coordinate-descent oracles, is bounded from below by $\tilde{\Omega}\big(n + \sqrt{n(\kappa-1)}\ln(1/\varepsilon)\big)$, in every dimension d.*

The proof is not a chain argument: on a parametrized family of quadratics, the coordinates of the k-th iterate of an oblivious CLI are polynomials of degree $\leq k$ in the parameters, and approximating the minimizer well requires a high degree (Chebyshev-type approximation bounds). The "n" term holds for any algorithm using an incremental oracle (their Theorem 5), and the class covers deterministic schedules (cyclic coordinate/component orders), mini-batches and full-gradient steps.

Breaking the span assumption (Hannah–Liu–O'Connor–Yin). In [15], $F = \frac{1}{n}\sum f_i + \psi$ with f_i L-smooth (or L_i-smooth, $L_Q = \frac{1}{n}\sum L_i$) and F μ-strongly convex, $\kappa = L/\mu$ (so $R^2 = L$ and their κ is ours). Their span condition is relation (111) "for some fixed IID random variable i_k over the indices", i.e. class (S1); a full gradient step violates it because it uses all n indices at the same point. The hard functions of their Theorems 2 and 4 are individually L-smooth and μ-strongly convex, a subclass of the book's class to which the lower bounds therefore apply a fortiori.

Theorem 4.11 ([15], Corollary 3, Corollary 2, Theorem 2/Theorem 4 and Corollary 1)**.** *Let the components be L-smooth and F be μ-strongly convex, $\kappa = L/\mu$.*

(a) (Rewording of Theorem 4.8.) For every ε and every randomized algorithm following the span assumption there are a dimension d and L-smooth, μ-strongly convex f_i on $\mathbb{R}^d$ such that reaching $\mathbb{E}f(x^k) - f(x^*) < \varepsilon$ takes the algorithm at least $\Omega((n+\sqrt{\kappa n})\ln(1/\varepsilon))$ steps.

(b) (Upper bound.) With epoch length $m = n + 121\kappa_Q$ and step size $\eta = \kappa_Q^{1/2} m^{-1/2}/(2L_Q)$, Prox-SVRG needs

$$K(\varepsilon) = O\Big(\Big(\frac{n}{1+(\ln(n/\kappa_Q))_+} + \kappa_Q\Big)\ln\frac{1}{\varepsilon} + n + \kappa_Q\Big)$$

gradient evaluations to reach $\mathbb{E}f(x^K) - f(x^*) < \varepsilon$ (here $(a)_+ = \max\{a, 0\}$ and $\kappa_Q = L_Q/\mu$).

(c) (Lower bound.) For any oblivious p-CLI algorithm (in particular SVRG, SARAH, SAGA, SAG, Katyusha, L-Katyusha with fixed parameters) and all μ, L there exist L-smooth, μ-strongly convex f_i such that at least

$$K(\varepsilon) = \tilde{\Omega}\Big(\Big(\frac{n}{1+(\ln(n/\kappa))_+} + \sqrt{n\kappa}\Big)\ln\frac{1}{\varepsilon} + n\Big)$$

iterations are needed; equivalently (their (1.3)) $K(\varepsilon) = \Omega(n + \sqrt{n\kappa}\ln(1/\varepsilon))$ for $n = O(\kappa)$ and $K(\varepsilon) = \Omega\big(n + \frac{n}{1+(\ln(n/\kappa))_+}\ln(1/\varepsilon)\big)$ for $\kappa = O(n)$.

(d) (Example.) For $\varepsilon = 1/n^\alpha$ and $\kappa = n^\beta$ with $\alpha, \beta \in (0,1)$, modified SVRG needs $O(n)$ iterations whereas algorithms following the span assumption need $\Omega(n\ln n)$ (the source writes "require $O(n\ln(n))$ iterations", quoting Theorem 4.8; the content is a lower bound).

Their Section 3 explains the mechanism on the block instance (117): after k i.i.d. uniform queries, block i has been touched a Binomial$(k, 1/n)$ number of times, and the *average* over blocks of the unrevealed energy is $(1-(1-q^2)/n)^k \geq (1-1/n)^k$ whatever the conditioning, which forces $n\ln(1/\varepsilon)$ queries; one full-gradient pass (n queries) touches *every* block once, so with full gradients the energy decreases by a factor q^2 per block per n queries and the $n\ln(1/\varepsilon)$ term disappears when $q^2 \leq \varepsilon$ is reached quickly, i.e. when $\kappa \ll n$. Part (c) shows that the gain is at most the factor $1 + (\ln(n/\kappa))_+$, and part (b) that SVRG with a well-chosen epoch length achieves it.

Loopless methods and small dimension (Han–Xie–Zhang). The class $\mathcal{A}$ of [?] (their Definition 3.1) consists of randomized algorithms which at each step t sample i_t i.i.d. from a fixed distribution $\mathcal{D}$ and may query the PIFO of f_{i_t} *at all previous iterates* $x_0, \ldots, x_{t-1}$ ("simultaneous queries"; earlier answers may be reused), and with an independent probability $\bar{q} \leq c_0/n$ also query the full gradient at the current point; the set $\mathcal{H}$ of points where the full gradient is available is initialized with x_0, so the full gradient at x_0 is free. The iterate is then any point of the linear span of the previous iterates, of the oracle answers and of the full gradients at the points of $\mathcal{H}$, projected on the domain. By the authors' own account, this covers SAGA and Point-SAGA (whose only full gradient is the free one at x_0) and the loopless methods L-SVRG, L-Katyusha, KatyushaX, L2S, PAGE and Anita, but *not* the two-loop methods SVRG and Katyusha, whose full gradients are computed at deterministic times.

Theorem 4.12 ([?], Theorem E.6). *Let $n \geq 2$ and consider the class $\mathcal{F}^{\mathrm{C}}(R, L, \mu)$ of finite sums $f = \frac{1}{n}\sum_i f_i$ on a closed convex domain $\mathcal{X}$ of diameter at most $2R$, with L-smooth components and μ-strongly convex f, $\kappa = L/\mu \geq 2$, and let $\varepsilon \leq LR^2/4$ (the hard instance is constrained to the ball $\{\|x\| \leq R\}$, which contains $x_0 = 0$ and x^*; its components are individually μ-strongly convex, their Proposition E.13). The number of PIFO queries that algorithms of class $\mathcal{A}$ need to find $\hat{x} \in \mathcal{X}$ with $\mathbb{E}f(\hat{x}) - \min_{\mathcal{X}} f < \varepsilon$ is*

$$\mathfrak{m}_*^{\mathrm{C}}(\varepsilon, R, L, \mu) = \begin{cases} \Omega\big((n + \sqrt{\kappa n})\log(1/\varepsilon)\big) & \text{for } \kappa = \Omega(n), \\ \Omega\big(n + \frac{n}{1+(\log(n/\kappa))_+}\log(1/\varepsilon)\big) & \text{for } \kappa = O(n), \end{cases}$$

and the construction only requires the dimension to be $O\big(1 + \sqrt{\kappa/n}\,\log(1/\varepsilon)\big)$.

Their instance is a single chain of length $m = O(1 + \sqrt{\kappa/n}\log(1/\varepsilon))$ whose rows are distributed among the n components (component i owns the rows $l \equiv i - 1 \pmod{n}$, scaled by n): only the component owning the next row can extend the support, so the time to extend it is geometric with mean n, which gives the n factor without needing n separate chains.

Assessment of hypothesis (A1). Hypothesis (A1) reads: "lower bound $\Omega((n + \sqrt{n\kappa})\log(1/\varepsilon))$ for span-based methods with i.i.d. uniform sampling; for an arbitrary index sequence, $\sqrt{n\kappa}$ survives and $n\log(1/\varepsilon)$ weakens by $\log(n/\kappa)$ when $\kappa \ll n$."

(i) The first clause is exactly Theorem 4.8 (class (S1), any fixed sampling distribution, dimension $\gtrsim$ number of iterations) and is correct.

(ii) The second clause is correct *if* "arbitrary index sequence" means "index sequences that are not i.i.d. single queries, in particular full-gradient sweeps (SVRG, SARAH, L-SVRG, L-Katyusha)", and if it is understood within the oblivious CLI class and with the precise factor $1 + (\ln(n/\kappa))_+$: by Theorem 4.11(c) and Theorem 4.12 the bound becomes $\Omega\big(n + \big(\frac{n}{1+(\ln(n/\kappa))_+} + \sqrt{n\kappa}\big)\ln\frac{1}{\varepsilon}\big)$, the $\sqrt{n\kappa}\ln(1/\varepsilon)$ term is untouched, and the $n\ln(1/\varepsilon)$ term is divided by $1 + (\ln(n/\kappa))_+$ but never falls below the bare n (one query per component is unavoidable, Theorem 4.10). This is tight: Theorem 4.11(b) attains it when $\kappa = O(n)$, and Katyusha/L-Katyusha attain $(n + \sqrt{n\kappa})\log(1/\varepsilon)$ when $n = O(\kappa)$ [?, 21].

(iii) The clause is *not* correct for other readings. For deterministic (adaptive) index sequences the worst case is $\Omega(n\sqrt{\kappa}\log(1/\varepsilon))$ (Theorem 4.9, deterministic part): the $\sqrt{n\kappa}$ term does not merely survive, it deteriorates by $\sqrt{n}$, and the $\sqrt{n}$ saving of randomized methods is genuinely due to randomization. For randomized algorithms with fully adaptive queries and no span restriction, the known bound is $\Omega(n+\sqrt{n\kappa}\log(1/\varepsilon))$ (Theorem 4.9, randomized part, stated for $\kappa > 161n$), which says nothing about an $n\log(1/\varepsilon)$ term; whether cyclic (without-replacement) span-based methods can beat $n\log(1/\varepsilon)$ is not settled by these sources.

(iv) Finally, the phrase "$n\log(1/\varepsilon)$ weakens by $\log(n/\kappa)$" should be read multiplicatively, as a division by $1 + (\ln(n/\kappa))_+$, which is only a logarithmic gain and only in the big-data regime $\kappa \ll n$; in the regime $\kappa \gtrsim n$ of hypothesis (A4) the two bounds coincide, $\Theta((n+\sqrt{n\kappa})\log(1/\varepsilon))$.

On the hard instance of the next section, Proposition 4.18 gives a self-contained proof that the $\sqrt{n\kappa}\log(1/\varepsilon)$ term is unavoidable for *every* index sequence (class (S2)), and Theorem 4.19 proves the full $(n + \sqrt{n\kappa})\log(1/\varepsilon)$ bound for class (S1).

4.3 The block-separable hard instance (hypothesis A4)

4.3.1 Definition and constants

Fix $n \geq 1$ components, a block length $m \geq 2$, $\mu > 0$, a target condition number $\kappa > 1$ and $\zeta \in [1, 3]$. The variable is $\theta = (\theta^{(1)}, \dots, \theta^{(n)}) \in \mathbb{R}^{nm}$ with blocks $\theta^{(i)} \in \mathbb{R}^m$; $E_i : \mathbb{R}^m \to \mathbb{R}^{nm}$ denotes the embedding of block i (so $E_i^\top \theta = \theta^{(i)}$). Let

$$c = \frac{\mu(\kappa - 1)}{4}, \qquad \phi(x) = \frac{1}{2}\langle A_m(\zeta)x, x\rangle - x_1 \ (x \in \mathbb{R}^m), \qquad f_i(\theta) = c\,\phi(\theta^{(i)}) + \frac{\mu}{2}\|\theta\|^2, \quad i = 1, \dots, n. \tag{118}$$

Then

$$\begin{aligned} F(\theta) &= \frac{1}{n}\sum_{i=1}^n f_i(\theta) = \frac{c}{n}\sum_{i=1}^n \phi(\theta^{(i)}) + \frac{\mu}{2}\|\theta\|^2, \\ f_i'(\theta) &= c\,E_i\big(A_m(\zeta)\theta^{(i)} - e_1\big) + \mu\theta, \qquad F'(\theta) = \frac{c}{n}\sum_{i=1}^n E_i\big(A_m(\zeta)\theta^{(i)} - e_1\big) + \mu\theta. \end{aligned} \tag{119}$$

In numpy, with θ stored as an $n \times m$ array Θ and $A = A_m(\zeta)$: $f_i'(\theta)$ is $\mu\Theta$ plus, on row i only, $c\,(A\Theta_{i,:} - e_1)$; $F'(\theta) = \frac{c}{n}(\Theta A - \mathbf{1}e_1^\top) + \mu\Theta$ (row-wise matrix product, $\mathbf{1}$ the all-ones column); the products with the tridiagonal A cost $O(m)$ per row.

Proposition 4.13 (constants of the instance)**.** *Let $Q := 1 + \dfrac{\kappa - 1}{n}$. For the instance* (118)*:*

(a) each f_i is μ-strongly convex and R^2-smooth with $R^2 = \mu\kappa$ (its exact Lipschitz constant is $\mu + c\,\lambda_{\max}(A_m(\zeta)) \leq \mu\kappa$, with equality only in the limit $m \to \infty$);

(b) $\nabla^2 F = \operatorname{diag}\big(\frac{c}{n}A_m(\zeta) + \mu I, \dots, \frac{c}{n}A_m(\zeta) + \mu I\big)$ is block diagonal; F is μ-strongly convex, its smallest Hessian eigenvalue is $\mu_{\text{eff}} := \mu + \frac{c}{n}\lambda_{\min}(A_m(\zeta)) = \mu\big(1 + \frac{Q-1}{4}\lambda_{\min}(A_m(\zeta))\big) \geq \mu$ and its smoothness constant is $L_F = \mu + \frac{c}{n}\lambda_{\max}(A_m(\zeta)) \leq \mu Q$;

(c) hence $\kappa = R^2/\mu = 1 + n(Q-1)$ and

$$\frac{L_F}{\mu} \ \leq \ Q = 1 + \frac{\kappa - 1}{n} \ \leq \ \kappa, \tag{120}$$

so that $(L_F/\mu)/\kappa \leq \frac{1}{n} + \frac{1}{\kappa}$: the condition number of F is about n times smaller than κ as soon as $\kappa \gg n$.

Proof. (a) $\nabla^2 f_i = cE_iA_m(\zeta)E_i^\top + \mu I$ has eigenvalues μ (on the other blocks) and $\mu + c\lambda$ for λ an eigenvalue of $A_m(\zeta)$; by Lemma 4.3 these lie in $[\mu, \mu + 4c] = [\mu, \mu\kappa]$. (b) $\nabla^2 F = \frac{1}{n}\sum_i \nabla^2 f_i$ is block diagonal with blocks $\frac{c}{n}A_m(\zeta) + \mu I$, whose eigenvalues are $\mu + \frac{c}{n}\lambda \in [\mu, \mu + \frac{4c}{n}]$ and $\mu + \frac{4c}{n} = \mu(1 + \frac{\kappa-1}{n}) = \mu Q$. (c) is arithmetic. □

Why the chain of f_i carries the factor n. The restriction of F to block i is $\frac{c}{n}\phi + \frac{\mu}{2}\|\cdot\|^2$: the chain that the i-th block of F sees is the chain of f_i divided by n. Nesterov's per-block function with condition number Q is $\frac{\mu(Q-1)}{4}\phi + \frac{\mu}{2}\|\cdot\|^2$ (compare (115) with $L = \mu Q$), so one needs $\frac{c}{n} = \frac{\mu(Q-1)}{4}$, i.e. $c = \frac{n\mu(Q-1)}{4} = \frac{\mu(\kappa-1)}{4}$: the component must carry n times the chain of the block. Conversely, if one naively takes $f_i(\theta) = \psi(\theta^{(i)})$ with a single μ-strongly convex, L-smooth ψ (block-only regularizer, as in (117)), then $F = \frac{1}{n}\sum_i \psi(\theta^{(i)})$ is only μ/n-strongly convex, $R^2 = L$, and the book's κ equals nL/μ while the per-block condition number is $L/\mu = \kappa/n$. In both normalizations the quantity that drives the chain mechanism is the *per-block* condition number $Q \approx \kappa/n$, and $\sqrt{n\kappa} \approx n\sqrt{Q}$ is "n blocks times $\sqrt{Q}$ touches per block".

Proposition 4.14 (closed-form minimizer). *Let* $Q = 1 + \frac{\kappa-1}{n}$ *and* $q = \dfrac{\sqrt{Q}-1}{\sqrt{Q}+1}$.

(a) (Geometric chain.) If

$$\zeta = \zeta_Q := \frac{\sqrt{Q}+3}{\sqrt{Q}+1} \in (1,2), \tag{121}$$

then the minimizer is $\eta_* = (x^*, \dots, x^*)$ *with* $x_j^* = q^j$, $j = 1, \dots, m$, *so that*

$$\|\eta_*\|^2 = n\,\frac{q^2(1-q^{2m})}{1-q^2}, \qquad F(\eta_*) = -\frac{c\,q}{2} = -\frac{\mu(\kappa-1)\,q}{8}.$$

(b) (Nesterov's chain, $\zeta = 2$.*) The minimizer is* $\eta_* = (x^*, \dots, x^*)$ *with*

$$x_j^* = \frac{q^{\,j} - q^{\,2(m+1)-j}}{1-q^{\,2(m+1)}} = \frac{\sinh\big((m+1-j)\tau\big)}{\sinh\big((m+1)\tau\big)}, \qquad \tau = \log\frac{1}{q} = 2\,\operatorname{artanh}\frac{1}{\sqrt{Q}}, \quad j = 1, \dots, m.$$

(c) (Any $\zeta \in [1,3]$.*)* x^* *is the solution of the tridiagonal system* $\big(A_m(\zeta) + \frac{4}{Q-1}I\big)x = e_1$, *computable in* $O(m)$ *by the Thomas algorithm (forward elimination, backward substitution), which is numerically stable here because the matrix is symmetric positive definite and diagonally dominant.*

In all cases $F(\theta) - F(\eta_*) = \frac{1}{2}\left\langle \nabla^2 F(\theta - \eta_*), \theta - \eta_* \right\rangle$ *lies between* $\frac{\mu_{\text{eff}}}{2}\|\theta - \eta_*\|^2$ *and* $\frac{L_F}{2}\|\theta - \eta_*\|^2$.

Proof. Since F is block separable and the blocks play symmetric roles, $\eta_* = (x^*, \dots, x^*)$ where x^* minimizes $\frac{c}{n}\phi(x) + \frac{\mu}{2}\|x\|^2$, i.e. solves $\frac{c}{n}(A_m(\zeta)x - e_1) + \mu x = 0$, or $(A_m(\zeta) + \frac{n\mu}{c}I)x = e_1$; and $\frac{n\mu}{c} = \frac{4n}{\kappa-1} = \frac{4}{Q-1}$, which proves (c). Written row by row, with $a := 2 + \frac{4}{Q-1} = q + q^{-1}$ (see the proof of Proposition 4.4):

$$a\,x_1 - x_2 = 1, \qquad -x_{j-1} + a\,x_j - x_{j+1} = 0 \ (2 \le j \le m-1), \qquad -x_{m-1} + \Big(\zeta + \frac{4}{Q-1}\Big)x_m = 0.$$

(a) The first $m-1$ equations are satisfied by $x_j = q^j$ exactly as in Proposition 4.4. The last one holds iff $\zeta + \frac{4}{Q-1} = q^{-1}$; since $q^{-1} = \frac{\sqrt{Q}+1}{\sqrt{Q}-1} = \frac{(\sqrt{Q}+1)^2}{Q-1}$, this means $\zeta = \frac{(\sqrt{Q}+1)^2-4}{Q-1} = \frac{(\sqrt{Q}-1)(\sqrt{Q}+3)}{(\sqrt{Q}-1)(\sqrt{Q}+1)} = \zeta_Q$, and $1 < \zeta_Q < 2$ for $Q > 1$. Then $\|x^*\|^2 = \sum_{j=1}^m q^{2j} = q^2(1-q^{2m})/(1-q^2)$. For the value, $A_m(\zeta_Q)x^* = e_1 - \frac{4}{Q-1}x^*$ gives $\phi(x^*) = -\frac{x_1^*}{2} - \frac{2}{Q-1}\|x^*\|^2$ and $F(\eta_*) = c\,\phi(x^*) + \frac{n\mu}{2}\|x^*\|^2 = -\frac{cq}{2}$ because $\frac{2c}{Q-1} = \frac{\mu(\kappa-1)}{2(Q-1)} = \frac{n\mu}{2}$. (b) The general solution of the three-term recursion $x_{j+1} = a\,x_j - x_{j-1}$ is $x_j = \alpha q^j + \beta q^{-j}$ (q and q^{-1} are the roots of $z^2 - az + 1 = 0$). The first equation says that the recursion extends to $j = 1$ with $x_0 := 1$, and the last one, with $\zeta = 2$, that it extends to $j = m$ with $x_{m+1} := 0$. Hence $\alpha + \beta = 1$ and $\alpha q^{m+1} + \beta q^{-(m+1)} = 0$, i.e. $\beta = -\alpha q^{2(m+1)}$ and $\alpha = 1/(1-q^{2(m+1)})$, which is the stated formula; the sinh form follows by multiplying numerator and denominator by $q^{-(m+1)}$, and $\tau = \log\frac{\sqrt{Q}+1}{\sqrt{Q}-1} = 2\,\operatorname{artanh}(Q^{-1/2})$. The formula with q is the one to implement (all powers are in $(0,1]$). The last claim is Taylor's formula for the quadratic F with $F'(\eta_*) = 0$. □

4.3.2 The zero-chain property and the sequence-free bound

For $\theta \in \mathbb{R}^{nm}$ and $j \in \{1, \dots, n\}$ let $N_j(\theta) = \max\{l : \theta_l^{(j)} \neq 0\}$ ($N_j = 0$ if the block vanishes). Consider a method following the linear-span protocol (111) with $\theta_0 = 0$ and an *arbitrary* index sequence $(i_s)_{s \ge 1}$ (class (S2); full gradients are n consecutive queries at the same point, and the indices may be chosen adaptively). Write $\tau_j(t) = \#\{s \le t : i_s = j\}$ for the number of queries to component j among the first t queries, so that $\sum_j \tau_j(t) = t$. (Starting from $\theta_0 \neq 0$ amounts to translating the instance, $f_i(\cdot + \theta_0)$; the bounds below then hold with $\eta_* - \theta_0$ in place of η_*.)

Lemma 4.15 (zero chain, block version)**.** *For every θ and i, j: $N_j(f_i'(\theta)) \le N_j(\theta) + \mathbf{1}[i=j]$.*

Proof. By (119), the j-th block of $f_i'(\theta)$ is $\mu\theta^{(j)}$ if $j \neq i$ and $c(A_m(\zeta)\theta^{(i)} - e_1) + \mu\theta^{(i)}$ if $j = i$; apply Lemma 4.5. □

Lemma 4.16 (support under the span protocol)**.** *(i) For every $t \ge 0$ and j: $N_j(\theta_t) \le \tau_j(t)$. (ii) If the span in (111) is enlarged by the n initial gradients $f_1'(\theta_0), \dots, f_n'(\theta_0)$ (an initial full gradient, as for SAGA with $z_i^{(0)} = f_i'(\theta_0)$), then $N_j(\theta_t) \le 1 + \tau_j(t)$, where $\tau_j(t)$ counts the queries made after the initial full gradient; equivalently, $N_j(\theta_t) \le \tau_j(t)$ still holds if the initial full gradient is counted as one query to every component.*

Proof. (i) Induction on t; the case $t = 0$ is $\theta_0 = 0$. If the claim holds up to $t-1$, each generator $f_{i_s}'(\theta_{s-1})$, $s \le t$, of the span in (111) satisfies $N_j \le \tau_j(s-1) + \mathbf{1}[i_s = j] = \tau_j(s) \le \tau_j(t)$ by Lemma 4.15, and the support of a linear combination is contained in the union of the supports. (ii) Same induction: the n additional generators $f_l'(\theta_0) = f_l'(0)$ satisfy $N_j(f_l'(0)) \le \mathbf{1}[l=j] \le 1$ by Lemma 4.15, and if $N_j(\theta_{s-1}) \le 1 + \tau_j(s-1)$ for all $s \le t$ then each generator $f_{i_s}'(\theta_{s-1})$ has $N_j \le 1 + \tau_j(s-1) + \mathbf{1}[i_s = j] = 1 + \tau_j(s) \le 1 + \tau_j(t)$. □

Proposition 4.17 (tail bound)**.** *Let $\zeta = \zeta_Q$. If $N_j(\theta) \le \tau_j \le m$ for all j, then*

$$\frac{\|\theta - \eta_*\|^2}{\|\eta_*\|^2} \;\ge\; \frac{1}{n}\sum_{j=1}^{n} \frac{q^{2\tau_j} - q^{2m}}{1 - q^{2m}}. \tag{122}$$

Proof. $\|\theta - \eta_*\|^2 = \sum_j \left\|\theta^{(j)} - x^*\right\|^2 \ge \sum_j \sum_{l > \tau_j} (x_l^*)^2 = \sum_j \sum_{l=\tau_j+1}^{m} q^{2l} = \frac{q^2}{1-q^2} \sum_j \left(q^{2\tau_j} - q^{2m}\right)$, and $\|\eta_*\|^2 = n\frac{q^2(1-q^{2m})}{1-q^2}$ by Proposition 4.14. □

Proposition 4.18 (the $\sqrt{n\kappa}$ term for an arbitrary index sequence)**.** *Let $\zeta = \zeta_Q$ and consider any method of class (S2) started at $\theta_0 = 0$. After G gradient evaluations in total (full gradients counted as n), every iterate θ in the span of the answers satisfies*

$$\frac{1}{n}\sum_{j=1}^{n} q^{2\tau_j} \;\ge\; q^{2G/n}, \qquad \textit{hence} \qquad \frac{\|\theta - \eta_*\|^2}{\|\eta_*\|^2} \;\ge\; \frac{q^{2G/n} - q^{2m}}{1 - q^{2m}}. \tag{123}$$

Consequently, if $q^{2m} \le \varepsilon < \frac{1}{2}$, reaching $\|\theta - \eta_*\|^2 \le \varepsilon \|\eta_*\|^2$ requires

$$G \;\ge\; \frac{n \log \frac{1}{2\varepsilon}}{2\log(1/q)} \;\ge\; \frac{n(\sqrt{Q}-1)}{4} \log\frac{1}{2\varepsilon} \;\ge\; \frac{\sqrt{n(\kappa-1)}}{8} \log\frac{1}{2\varepsilon} \quad (\textit{the last step for } \kappa - 1 \ge 2n). \tag{124}$$

Proof. Let $\bar\tau_j = \min(\tau_j(G), m)$. By Lemma 4.16 and $N_j \le m$, the hypothesis of Proposition 4.17 holds with $\tau_j = \bar\tau_j$, and $q^{2\bar\tau_j} \ge q^{2\tau_j(G)}$. The map $\tau \mapsto q^{2\tau}$ is convex and $\sum_j \tau_j(G) = G$, so Jensen's inequality gives $\frac{1}{n}\sum_j q^{2\bar\tau_j} \ge \frac{1}{n}\sum_j q^{2\tau_j(G)} \ge q^{2G/n}$, and (122) yields the second inequality in (123). If $\|\theta - \eta_*\|^2 \le \varepsilon \|\eta_*\|^2$ then $q^{2G/n} \le \varepsilon(1-q^{2m}) + q^{2m} \le \varepsilon + q^{2m} \le 2\varepsilon$, i.e. $\frac{2G}{n}\log\frac{1}{q} \ge \log\frac{1}{2\varepsilon}$. Finally $\log\frac{1}{q} = \log\left(1 + \frac{2}{\sqrt{Q}-1}\right) \le \frac{2}{\sqrt{Q}-1}$, and for $y = \frac{\kappa-1}{n} \ge 2$ one has $\sqrt{1+y} - 1 \ge \frac{\sqrt{y}}{2}$ (the function $y \mapsto \sqrt{1+y} - 1 - \frac{\sqrt{y}}{2}$ is positive at $y = 2$ and increasing for $y > \frac{1}{3}$), so $n(\sqrt{Q}-1) \ge \frac{1}{2}\sqrt{n(\kappa-1)}$. □

Proposition 4.18 applies verbatim to GD, AGD, cyclic incremental methods, SAGA, SVRG, SARAH, Katyusha and L-Katyusha, whatever their randomization: on this instance none of them can beat $\Omega(\sqrt{n\kappa}\log(1/\varepsilon))$ gradient evaluations, and accelerated gradient descent on F attains this order (it needs $O(\sqrt{L_F/\mu}\log(1/\varepsilon)) \le O(\sqrt{Q}\log(1/\varepsilon))$ iterations of n gradients, i.e. $O(n\sqrt{Q}\log(1/\varepsilon))$ gradient evaluations, where $n\sqrt{Q} = \sqrt{n^2+n(\kappa-1)}$). What the sequence-free argument cannot produce is the $n\log(1/\varepsilon)$ term: a full-gradient sweep advances all n chains at once, and this is exactly the point of [15].

4.3.3 The i.i.d. bound

Theorem 4.19 (Lan–Zhou's bound on the instance, book notation). *Let $\zeta = \zeta_Q$, $\theta_0 = 0$, and let the method belong to class (S1) with a fixed sampling distribution p (uniform or not). Then for every $t \ge 0$,*

$$\frac{\mathbb{E}\,\|\theta_t - \eta_*\|^2}{\|\eta_*\|^2} \;\ge\; \frac{\big(1-\frac{1-q^2}{n}\big)^t - q^{2m}}{1-q^{2m}}, \tag{125}$$

and, whenever $q^{2m} \le \frac{1}{2}\big(1-\frac{1-q^2}{n}\big)^t$,

$$\frac{\mathbb{E}\,\|\theta_t - \eta_*\|^2}{\|\eta_*\|^2} \;\ge\; \frac{1}{2}\Big(1-\frac{1-q^2}{n}\Big)^t \;\ge\; \frac{1}{2}\exp\Big(-\frac{4\sqrt{Q}\,t}{n(\sqrt{Q}+1)^2-4\sqrt{Q}}\Big). \tag{126}$$

Proof. Under (S1), $\tau_j(t) \sim \mathrm{Binomial}(t,p_j)$, so $\mathbb{E}q^{2\tau_j(t)} = \sum_{k=0}^{t}\binom{t}{k}p_j^k(1-p_j)^{t-k}q^{2k} = \big(1-(1-q^2)p_j\big)^t$. Taking expectations in (122) (with $\tau_j = \min(\tau_j(t),m)$ as in the previous proof),

$$\frac{\mathbb{E}\,\|\theta_t - \eta_*\|^2}{\|\eta_*\|^2} \;\ge\; \frac{\frac{1}{n}\sum_j\big(1-(1-q^2)p_j\big)^t - q^{2m}}{1-q^{2m}} \;\ge\; \frac{\big(1-\frac{1-q^2}{n}\big)^t - q^{2m}}{1-q^{2m}},$$

where the last step is Jensen's inequality for the convex map $p \mapsto (1-(1-q^2)p)^t$ on $[0,1]$ and $\frac{1}{n}\sum_j p_j = \frac{1}{n}$: the uniform distribution is the most favourable one for the algorithm, and the bound is the same for all p. For (126), write $a = (1-\frac{1-q^2}{n})^t$ and $b = q^{2m}$; if $b \le a/2$ then $\frac{a-b}{1-b} \ge a-b \ge \frac{a}{2}$. Next, $1-q^2 = \frac{(\sqrt{Q}+1)^2-(\sqrt{Q}-1)^2}{(\sqrt{Q}+1)^2} = \frac{4\sqrt{Q}}{(\sqrt{Q}+1)^2}$, so $a = (1-x)^t$ with $x = \frac{4\sqrt{Q}}{n(\sqrt{Q}+1)^2} \in (0,1)$, and $\log(1-x) = -\sum_{k\ge1}x^k/k \ge -\sum_{k\ge1}x^k = -\frac{x}{1-x}$ gives $(1-x)^t \ge \exp(-\frac{tx}{1-x})$ with $\frac{x}{1-x} = \frac{4\sqrt{Q}}{n(\sqrt{Q}+1)^2-4\sqrt{Q}}$. □

Corollary 4.20 (iteration count and dimension requirement). *Let $\varepsilon \in (0,\frac{1}{2})$ and assume the block length satisfies*

$$m \;\ge\; \frac{\log(1/\varepsilon)}{2\log(1/q)}, \qquad \textit{for which it suffices that} \qquad m \;\ge\; \frac{\sqrt{Q}}{4}\log\frac{1}{\varepsilon} = \frac{1}{4}\sqrt{1+\frac{\kappa-1}{n}}\,\log\frac{1}{\varepsilon}. \tag{127}$$

Then any method of class (S1) needs

$$t \;\ge\; \Big(\frac{n(\sqrt{Q}+1)^2}{4\sqrt{Q}}-1\Big)\log\frac{1}{2\varepsilon} \;\ge\; \Big(\frac{n+\sqrt{n(\kappa-1)}}{8}+\frac{n}{2}-1\Big)\log\frac{1}{2\varepsilon} = \Omega\Big((n+\sqrt{n\kappa})\log\frac{1}{\varepsilon}\Big) \tag{128}$$

gradient evaluations to reach $\mathbb{E}\,\|\theta_t-\eta_*\|^2 \le \varepsilon\,\|\eta_*\|^2$. In particular $t \ge (n-1)\log\frac{1}{2\varepsilon}$ for every $\kappa > 1$, because $(\sqrt{Q}+1)^2 \ge 4\sqrt{Q}$.

Proof. Let $x = \frac{4\sqrt{Q}}{n(\sqrt{Q}+1)^2}$ and $t^* = (\frac{1}{x} - 1)\log\frac{1}{2\varepsilon}$. For $t \le t^*$ the condition $q^{2m} \le \frac{1}{2}(1-x)^t$ of (126) holds: indeed $t\log\frac{1}{1-x} \le t^*\frac{x}{1-x} = \log\frac{1}{2\varepsilon}$, so $\frac{1}{2}(1-x)^t \ge \varepsilon \ge q^{2m}$ by (127) (the sufficient form uses $\log\frac{1}{q} = 2\,\mathrm{artanh}(Q^{-1/2}) \ge 2Q^{-1/2}$). Hence for $t \le t^*$, (126) gives $\mathbb{E}\,\|\theta_t - \eta_*\|^2 / \|\eta_*\|^2 \ge \frac{1}{2}\exp(-\frac{tx}{1-x})$, which is $\le \varepsilon$ only if $t \ge t^*$: so at least t^* iterations are needed. Finally $\frac{n(\sqrt{Q}+1)^2}{4\sqrt{Q}} = \frac{n}{4}(\sqrt{Q} + 2 + Q^{-1/2}) \ge \frac{n\sqrt{Q}}{4} + \frac{n}{2}$ and $n\sqrt{Q} = \sqrt{n^2 + n(\kappa-1)} \ge \max\{n, \sqrt{n(\kappa-1)}\} \ge \frac{1}{2}(n + \sqrt{n(\kappa-1)})$. □

Corollary 4.21 (initial full gradient)**.** *Let $\zeta = \zeta_Q$, $\theta_0 = 0$, and let the method be of class (S1) preceded by one full gradient at θ_0 (Lemma 4.16(ii)), t denoting the number of i.i.d. queries made after this initial full gradient. Then for every $t \ge 0$*

$$\frac{\mathbb{E}\,\|\theta_t - \eta_*\|^2}{\|\eta_*\|^2} \;\ge\; \frac{q^2\big(1 - \frac{1-q^2}{n}\big)^t - q^{2m}}{1 - q^{2m}}, \tag{129}$$

and, under (127) and for $\varepsilon < q^2/2$, reaching $\mathbb{E}\,\|\theta_t - \eta_*\|^2 \le \varepsilon\,\|\eta_*\|^2$ requires

$$t \;\ge\; \Big(\frac{n(\sqrt{Q}+1)^2}{4\sqrt{Q}} - 1\Big)\log\frac{q^2}{2\varepsilon} \;\ge\; (n-1)\log\frac{q^2}{2\varepsilon}$$

i.i.d. queries, i.e. $G = n+t$ gradient evaluations in total. Compared with (128), ε is replaced by ε/q^2: the initial full gradient reveals the first coordinate of every block, which carries the fraction $1-q^2$ of the energy of each block, and this is worth about n i.i.d. queries since $(\frac{1}{x}-1)\cdot 2\log\frac{1}{q} \le \frac{n(\sqrt{Q}+1)^2}{\sqrt{Q}(\sqrt{Q}-1)}$, which is $n(1+O(Q^{-1/2}))$ for large Q (for $\kappa \le n$, however, $q^2 \le 0.03$ and the shift $(n-1)\log(1/q^2)$ is a noticeable part of the count).

Proof. By Lemma 4.16(ii) and $N_j \le m$, Proposition 4.17 applies with $\tau_j = \min(1+\tau_j(t), m)$, and $q^{2\min(1+\tau_j(t),m)} \ge q^{2(1+\tau_j(t))}$. Under (S1), $\tau_j(t) \sim \mathrm{Binomial}(t, p_j)$, so $\mathbb{E}q^{2(1+\tau_j(t))} = q^2(1-(1-q^2)p_j)^t$, and Jensen's inequality in p gives (129) exactly as in the proof of Theorem 4.19. For the count, let $x = \frac{1-q^2}{n}$ and $t_1^* = (\frac{1}{x}-1)\log\frac{q^2}{2\varepsilon} > 0$. For $t \le t_1^*$, $\frac{q^2}{2}(1-x)^t \ge \frac{q^2}{2}\exp(-\frac{tx}{1-x}) \ge \varepsilon \ge q^{2m}$ (the middle inequality is the definition of t_1^*, the last one is (127)), so the right-hand side of (129) is at least $q^2(1-x)^t - q^{2m} \ge \frac{q^2}{2}(1-x)^t \ge \varepsilon$; hence at least t_1^* i.i.d. queries are needed, and $\frac{1}{x} - 1 \ge n-1$ as in Corollary 4.20. For the last claim, $\log\frac{1}{q} \le \frac{2}{\sqrt{Q}-1}$ and $\frac{1}{x} = \frac{n(\sqrt{Q}+1)^2}{4\sqrt{Q}}$. □

The dimension requirement (127) is the precise form of "m must exceed the number of touches an accelerated method needs per block": $\frac{\log(1/\varepsilon)}{2\log(1/q)}$ is exactly the number of coordinates of a single chain that must be revealed before the unrevealed energy $q^{2\tau}$ drops below ε (Theorem 4.6), and it is of order $\frac{\sqrt{Q}}{4}\log\frac{1}{\varepsilon} = \frac{1}{4}\sqrt{\kappa/n}\log\frac{1}{\varepsilon}$ for $\kappa \gg n$ (the project's "$m \gtrsim \sqrt{\kappa}\log(1/\varepsilon)$" should read $\sqrt{\kappa/n}$, since each block only sees the per-block condition number Q). It is also necessary for the argument: since $\log(1/q) \le 2/(\sqrt{Q}-1)$, the condition (127) forces $m \ge \frac{\sqrt{Q}-1}{4}\log\frac{1}{\varepsilon}$. The total dimension is $nm \gtrsim \frac{1}{4}\sqrt{n\kappa}\log(1/\varepsilon)$, of the order of the number of iterations certified, in agreement with the discussion of Section 4.1.

4.4 Predictions for the numerics, finite-dimension effects, and logistic regression

4.4.1 Gradient counts on the instance

Write $T_{\mathrm{alg}}(\varepsilon)$ for the number of gradient evaluations an algorithm needs on the instance (118) to reach $\mathbb{E}\,\|\theta - \eta_*\|^2 \le \varepsilon\,\|\theta_0 - \eta_*\|^2$. For SAGA with the book's initialization $z_i^{(0)} = f_i'(\theta_0)$ the

n gradients of the initial sweep are counted; for L-Katyusha, whose iteration costs $1 + n\,\mathbf{1}[\text{coin}]$ gradients with $\mathbb{P}[\text{coin}] = p = \frac{1}{n}$, i.e. 2 in expectation, T_{LK} is twice the number of iterations. For the function-value criterion divide and multiply by $\mu_{\text{eff}}/2$ and $L_F/2$ (Proposition 4.14); for a guarantee stated for a Lyapunov function, such as Theorem 4.6 of [21], the conversion to the distance criterion puts the ratio $\Psi^0/\left\|\theta_0 - \eta_*\right\|^2$ inside the logarithm, see (131) below. The mechanism is the one of Lemma 4.16: a block only moves when its component is queried, and a block with per-block condition number $Q \approx \kappa/n$ needs about $\sqrt{Q}$ touches with an accelerated scheme and about Q touches with a plain gradient scheme (the per-touch contraction of the slow mode of a block is $1 - \gamma\mu_{\text{eff}}$ for a gradient step of size $\gamma \le 1/R^2$, and $1 - c/\sqrt{Q}$ for momentum schemes). With i.i.d. uniform sampling each block is touched once per n queries on average, and $n\log(1/\varepsilon)$ queries are needed before the *average* unrevealed energy $(1-(1-q^2)/n)^t$ is small (Theorem 4.19), whatever Q. This gives:

method	upper bound on T (gradient evaluations)	lower bound on T
SAGA, $\gamma = \frac{1}{4R^2}$, $z_i^{(0)} = f_i'(\theta_0)$	$n + \max\{3n, \frac{16\kappa_{\text{eff}}}{3}\}\log\frac{1+n/4}{\varepsilon}$ (Cor. 4.23)	$(\kappa - \frac{1}{2})\log\frac{\rho}{\varepsilon}$ with $\rho \ge \frac{1}{105}$ (Cor. 4.23, Lem. 4.24); $n + (n-1)\log\frac{q^2}{2\varepsilon}$ (Cor. 4.21)
L-Katyusha	$2\max\{8n, 2\sqrt{6}\sqrt{n\kappa}\}\log\frac{C_0}{\varepsilon}$, (132)	$\frac{\sqrt{n(\kappa-1)}}{8}\log\frac{1}{2\varepsilon}$ for $\kappa - 1 \ge 2n$, and $\frac{n\log(1/(2\varepsilon))}{2\log(1/q)}$ for all κ (Prop. 4.18)
GD on F, $\gamma = 1/L_F$	$\frac{nQ}{2}\log\frac{1}{\varepsilon}$ $(Q \ge L_F/\mu)$	$n(\frac{L_F}{4\mu} - \frac{1}{2})\log\frac{\rho}{\varepsilon}$ (slow modes, Lem. 4.24)
AGD on F	$O(n\sqrt{Q}\log\frac{1}{\varepsilon})$ ([6], Sec. 5.2.5)	$\frac{\sqrt{n(\kappa-1)}}{8}\log\frac{1}{2\varepsilon}$ (Prop. 4.18)
any method of class (S1)	–	$(\frac{n+\sqrt{n(\kappa-1)}}{8} + \frac{n}{2} - 1)\log\frac{1}{2\varepsilon}$ (Cor. 4.20)
any method of class (S2)	–	$\frac{\sqrt{n(\kappa-1)}}{8}\log\frac{1}{2\varepsilon}$ (Prop. 4.18)

The lower bounds of the first four rows are stated for $4n + 1 \le \kappa \le \kappa_{\max}(m, \varepsilon)$, the window (137) in which m is large enough both for the dimension condition (127) and for the slow-mode mass of Lemma 4.24, and for $\varepsilon < \frac{1}{105}$; the sequence-free bound $\frac{n\log(1/(2\varepsilon))}{2\log(1/q)}$ only needs $q^{2m} \le \varepsilon < \frac{1}{2}$; the slow-mode bound for GD uses the exact identity $e_t = (I - \gamma\nabla^2 F)^t e_0$ and $\nu' = 2\mu \le L_F$ (Proposition 4.26). In orders of magnitude, for $\kappa \gtrsim n$: SAGA and GD cost $\Theta((n+\kappa)\log\frac{1}{\varepsilon})$, L-Katyusha and AGD $\Theta((n+\sqrt{n\kappa})\log\frac{1}{\varepsilon})$, up to the logarithmic factors inside the bounds (Proposition 4.26).

Two corrections to the counts announced in the project description ("AGD $\sim n\sqrt{\kappa}\log(1/\varepsilon)$, GD $\sim n\kappa\log(1/\varepsilon)$") are necessary. Those figures are the worst-case costs over all instances with the same (n, R^2, μ), obtained from $L_F \le R^2$; they are attained only when $L_F \approx R^2$, i.e. when the stiff directions of the components are aligned (for instance identical components). On the block-separable instance $L_F \le \mu Q \approx R^2/n$ (Proposition 4.13), so GD costs about as much as SAGA and AGD about as much as L-Katyusha; AGD even attains the lower bound of Proposition 4.18. This is not a defect of the instance: the lower bound $\Omega((n+\sqrt{n\kappa})\log(1/\varepsilon))$ is a statement about the worst case over instances, and on the worst-case instance for variance-reduced methods full-gradient methods happen to be as good. What variance reduction buys is the *uniform* guarantee $(n+\sqrt{n\kappa})\log(1/\varepsilon)$ over all instances, including the aligned ones where AGD needs $n\sqrt{\kappa}\log(1/\varepsilon)$. To exhibit the four distinct orders κ, $\sqrt{n\kappa}$, $n\kappa$, $n\sqrt{\kappa}$ on one problem one can use the aligned variant $f_i(\theta) = c\,\phi(\theta) + \frac{\mu}{2}\left\|\theta\right\|^2$ for all i (one chain in $\mathbb{R}^m$, $L_F = R^2 = \mu\kappa$): there GD needs $\Theta(n\kappa\log(1/\varepsilon))$, AGD $\Theta(n\sqrt{\kappa}\log(1/\varepsilon))$, SAGA $\Theta(\kappa\log(1/\varepsilon))$ (Corollary 4.23 applies to every convex quadratic instance, and Lemma 4.24 with $n = 1$, $Q = \kappa$ provides the slow-mode mass ρ for this aligned instance, which has the block structure of (118) with a single block), and L-Katyusha $O((n+\sqrt{n\kappa})\log(\kappa/\varepsilon))$ by (132)–(131) with $L_F/\mu = \kappa$; but on that instance no chain argument

certifies more than $\Omega(\sqrt{\kappa}\log(1/\varepsilon))$ for randomized methods, so only the block instance certifies the exponent $1/2$ as optimal.

4.4.2 SAGA on quadratic instances: the exponent 1 is a theorem

Proposition 4.22 (mean iterate of SAGA on quadratics)**.** *Let the f_i be convex quadratics with Hessians $H_i \succeq 0$, let $\bar{H} = \frac{1}{n}\sum_i H_i = \nabla^2 F$ have eigenpairs $(\nu_l, w_l)_{l=1}^d$ (w_l orthonormal) and $\mu_{\text{eff}} = \lambda_{\min}(\bar{H})$, and for $\nu' \geq \mu_{\text{eff}}$ let $P_{\nu'} = \sum_{l:\nu_l \leq \nu'} w_l w_l^\top$ be the orthogonal projector onto the sum of the eigenspaces of $\bar{H}$ with eigenvalues at most ν'. For SAGA with step $\gamma > 0$, uniform sampling and any initial memory $z^{(0)}$ (deterministic, or random and independent of the sampling), $e_t = \theta_t - \eta_*$ satisfies $\mathbb{E}[e_t] = (I - \gamma\bar{H})^t e_0$; hence, for every $\nu' \in [\mu_{\text{eff}}, 1/\gamma]$,*

$$\mathbb{E}\,\|\theta_t - \eta_*\|^2 \;\geq\; \big\|(I-\gamma\bar{H})^t(\theta_0-\eta_*)\big\|^2 = \sum_{l=1}^{d}(1-\gamma\nu_l)^{2t}\,\langle \theta_0-\eta_*, w_l\rangle^2 \;\geq\; (1-\gamma\nu')^{2t}\,\|P_{\nu'}(\theta_0-\eta_*)\|^2 . \tag{130}$$

With $\nu' = \mu_{\text{eff}}$ and a simple smallest eigenvalue with unit eigenvector $v_{\min}$, the last term is $(1-\gamma\mu_{\text{eff}})^{2t}\,\langle\theta_0-\eta_*, v_{\min}\rangle^2$.

Proof. Since $f_i'(\theta) = f_i'(\eta_*) + H_i(\theta-\eta_*)$ and $\sum_i f_i'(\eta_*) = nF'(\eta_*) = 0$, the SAGA direction (Section 5.4.4 of [6]) at time t, with $i = i(t)$ and $u_j^{(t-1)} = z_j^{(t-1)} - f_j'(\eta_*)$, equals

$$\omega_t = f_i'(\theta_{t-1}) - z_i^{(t-1)} + \frac{1}{n}\sum_j z_j^{(t-1)} = H_i e_{t-1} - u_i^{(t-1)} + \frac{1}{n}\sum_j u_j^{(t-1)}.$$

Conditionally on the past (which determines θ_{t-1} and $z^{(t-1)}$), $i(t)$ is uniform and independent, so $\mathbb{E}_{i(t)}[H_i e_{t-1}] = \bar{H}e_{t-1}$ and $\mathbb{E}_{i(t)}[u_i^{(t-1)}] = \frac{1}{n}\sum_j u_j^{(t-1)}$: the memory terms cancel in expectation, whatever the memory, and $\mathbb{E}_{i(t)}[e_t] = (I-\gamma\bar{H})e_{t-1}$. Taking expectations and iterating gives $\mathbb{E}[e_t] = (I-\gamma\bar{H})^t e_0$. Then $\mathbb{E}\,\|e_t\|^2 \geq \|\mathbb{E}e_t\|^2$ (Jensen), and expanding e_0 in the eigenbasis of $\bar{H}$ gives the equality in (130). Finally, for $\nu_l \leq \nu' \leq 1/\gamma$ one has $0 \leq 1-\gamma\nu' \leq 1-\gamma\nu_l$, so $(1-\gamma\nu_l)^{2t} \geq (1-\gamma\nu')^{2t}$ for every l in the range of $P_{\nu'}$, and the remaining terms of the sum are nonnegative. □

Corollary 4.23 (SAGA sandwich)**.** *Consider an instance with convex quadratic components such that $\lambda_{\max}(H_i) \leq R^2$ for all i (each f_i is then convex and R^2-smooth), let $\kappa_{\text{eff}} = R^2/\mu_{\text{eff}}$ and $\gamma = 1/(4R^2)$.*

(a) (Lower bound, any initial memory.) If $\|P_{\nu'}(\theta_0-\eta_*)\|^2 \geq \rho\,\|\theta_0-\eta_*\|^2$ for some $\nu' \in [\mu_{\text{eff}}, 4R^2]$ and $\rho \in (0,1]$, then for $\varepsilon < \rho$

$$T_{\text{SAGA}}(\varepsilon) \;\geq\; \Big(\frac{2R^2}{\nu'} - \frac{1}{2}\Big)\log\frac{\rho}{\varepsilon}, \qquad \textit{i.e.} \quad \Big(\kappa - \frac{1}{2}\Big)\log\frac{\rho}{\varepsilon} \;\text{ for } \nu' = 2\mu, \qquad \Big(2\kappa_{\text{eff}} - \frac{1}{2}\Big)\log\frac{\rho}{\varepsilon} \;\text{ for } \nu' = \mu_{\text{eff}}.$$

(b) (Upper bound.) With the initialization $z_i^{(0)} = f_i'(\theta_0)$ of Proposition 5.9 of [6], the number of iterations needed is at most

$$\max\Big\{3n, \frac{16\kappa_{\text{eff}}}{3}\Big\}\log\frac{1+n/4}{\varepsilon} \qquad (\textit{add } n \textit{ gradient evaluations for the initial sweep}),$$

and with $z^{(0)} = 0$ the same bound holds with $1+n/4$ replaced by $1 + \frac{3\sum_i \|f_i'(\eta_*)\|^2}{16R^4\|\theta_0-\eta_*\|^2}$.

Proof. (a) By (130) with $x = \gamma\nu' = \nu'/(4R^2) \in (0,1]$, $\mathbb{E}\,\|e_t\|^2 \le \varepsilon\,\|e_0\|^2$ forces $(1-x)^{2t} \le \varepsilon/\rho$, i.e. $2t\log\frac{1}{1-x} \ge \log\frac{\rho}{\varepsilon}$, and $\log\frac{1}{1-x} \le \frac{x}{1-x}$ gives $t \ge \frac{1-x}{2x}\log\frac{\rho}{\varepsilon} = (\frac{2R^2}{\nu'} - \frac{1}{2})\log\frac{\rho}{\varepsilon}$; for $\nu' = 2\mu$, $\frac{2R^2}{\nu'} = \frac{R^2}{\mu} = \kappa$. (b) The proof of Proposition 5.9 of [6] (transcribed in the SAGA note) uses the convexity and R^2-smoothness of each f_i through cocoercivity (Proposition 5.4 of the book, which requires each f_i to be convex), and strong convexity only through $\langle \theta_{t-1} - \eta_*, F'(\theta_{t-1})\rangle \ge \mu\,\|\theta_{t-1} - \eta_*\|^2$; for a quadratic F the latter holds with $\mu_{\mathrm{eff}} = \lambda_{\min}(\nabla^2 F)$ in place of μ, and the proof gives, for any initial memory,

$$\mathbb{E}\,\|\theta_t - \eta_*\|^2 \;\le\; \Big(1 - \min\Big\{\frac{1}{3n}, \frac{3\mu_{\mathrm{eff}}}{16R^2}\Big\}\Big)^t \Big[\,\|\theta_0 - \eta_*\|^2 + \frac{3}{16R^4}\sum_i \Big\|f_i'(\eta_*) - z_i^{(0)}\Big\|^2\Big].$$

For $z_i^{(0)} = f_i'(\theta_0)$ the bracket is at most $(1+\frac{3n}{16})\,\|\theta_0 - \eta_*\|^2 \le (1+\frac{n}{4})\,\|\theta_0 - \eta_*\|^2$ by the R^2-Lipschitz continuity of each f_i', and for $z^{(0)} = 0$ it is the stated constant times $\|\theta_0 - \eta_*\|^2$; in both cases the right-hand side is $\le \varepsilon\,\|\theta_0 - \eta_*\|^2$ as soon as $t \ge \max\{3n, \frac{16\kappa_{\mathrm{eff}}}{3}\}\log\frac{C}{\varepsilon}$, C the constant in the bracket (using $(1-y)^t \le e^{-ty}$). □

On the instance (118), $\mu_{\mathrm{eff}} = \mu\big(1 + \frac{Q-1}{4}\lambda_{\min}(A_m(\zeta))\big)$ (Proposition 4.13), so for large m ($\lambda_{\min}(A_m) \approx \pi^2/m^2 \to 0$) $\kappa_{\mathrm{eff}} \approx \kappa$ and the upper bound of Corollary 4.23 is $O(\kappa\log(1/\varepsilon))$ for $\kappa \gtrsim n$. The matching lower bound requires the initial error to carry a fixed fraction of its energy on the eigenspaces of $\nabla^2 F$ with curvature at most 2μ; this is the object of Lemma 4.24 below, and the two together make the log-log slope 1 of hypothesis (A4) a theorem for SAGA with the book's step size. The exponent is *not* a consequence of the lower bound of Section 4.3 (which only says $\sqrt{n\kappa}$), but of the fact that the step must be $O(1/R^2)$ while a fixed fraction of the initial error sits on modes of curvature between μ and 2μ. Note also that on the instance, since $A_m(\zeta_Q)x^* = e_1 - \frac{4}{Q-1}x^*$ and $c\cdot\frac{4}{Q-1} = n\mu$, one has $f_i'(\eta_*) = \mu(\eta_* - nE_ix^*)$: its block i equals $-\mu(n-1)x^*$ and its other blocks equal μx^*, so $\|f_i'(\eta_*)\|^2 = \mu^2 n(n-1)\,\|x^*\|^2$ and $\sum_i \|f_i'(\eta_*)\|^2 = \mu^2 n(n-1)\,\|\eta_*\|^2$. The constant of Corollary 4.23(b) for the zero memory is therefore $1 + \frac{3n(n-1)}{16\kappa^2} \le 1 + \frac{3}{16}$ when $\kappa \ge n$: on this instance the zero memory, which keeps SAGA literally in class (S1) and dispenses with the initial sweep, is at least as good as the book's initialization.

For L-Katyusha, Theorem 4.6 of [21] (parameters $\theta_1 = \min\{\sqrt{2n\sigma/3}, \frac{1}{2}\}$, $\theta_2 = \frac{1}{2}$, $p = \frac{1}{n}$, $\sigma = \mu/L$, $\eta = \theta_2/((1+\theta_2)\theta_1) = 1/(3\theta_1)$, with $L = R^2$ the component smoothness of their Assumption 1.1 and μ the strong convexity of F of their Assumption 1.2) states that the Lyapunov function

$$\Psi^k = \frac{L(1+\eta\sigma)}{2\eta}\Big\|z^k - \eta_*\Big\|^2 + \frac{1}{\theta_1}\big(F(y^k) - F(\eta_*)\big) + \frac{\theta_2(1+\theta_1)}{p\,\theta_1}\big(F(w^k) - F(\eta_*)\big)$$

satisfies $\mathbb{E}\Psi^{k+1} \le (1-\theta)\mathbb{E}\Psi^k$ with $\theta = \min\{\frac{\sigma}{6\theta_1}, \frac{\theta_1}{2n}\}$ (their proof of Theorem 4.6; the second term is $\theta_1/(2n)$), whence $\mathbb{E}\Psi^k \le \varepsilon\Psi^0$ after $k = O((n + \sqrt{nL/\mu})\log(1/\varepsilon))$ iterations. Explicitly, $\frac{1}{\theta} = \max\{\frac{6\theta_1}{\sigma}, \frac{2n}{\theta_1}\}$ equals $\max\{3\kappa, 4n\} \le 8n$ when $\kappa \le \frac{8n}{3}$ (then $\theta_1 = \frac{1}{2}$), and $2\sqrt{6}\sqrt{n\kappa}$ when $\kappa > \frac{8n}{3}$ (then $\theta_1 = \sqrt{2n\sigma/3} < \frac{1}{2}$, $\frac{6\theta_1}{\sigma} = 2\sqrt{6}\sqrt{n\kappa}$ and $\frac{2n}{\theta_1} = \sqrt{6}\sqrt{n\kappa}$). For the distance criterion on the sequence z^k, started from $y^0 = z^0 = w^0 = \theta_0$, one has $\mathbb{E}\,\big\|z^k - \eta_*\big\|^2 \le \frac{2\eta}{L(1+\eta\sigma)}\mathbb{E}\Psi^k \le (1-\theta)^k C_0\,\|\theta_0 - \eta_*\|^2$ with, using $F(\theta_0) - F(\eta_*) \le \frac{L_F}{2}\,\|\theta_0 - \eta_*\|^2$ and $\frac{\eta}{\theta_1} = \frac{1}{3\theta_1^2}$,

$$C_0 = 1 + \frac{L_F}{3L\theta_1^2}\Big(1 + \frac{n(1+\theta_1)}{2}\Big) \;\le\; \begin{cases} 1 + \frac{5}{8}\,\frac{L_F}{\mu} \;\le\; 1 + \frac{5}{8}Q & \text{if } \kappa \ge \frac{8n}{3} \text{ and } n \ge 2, \\ n + \frac{7}{3} & \text{if } \kappa < \frac{8n}{3}, \end{cases} \tag{131}$$

because in the first case $\frac{1}{3\theta_1^2} = \frac{1}{2\sigma n} = \frac{\kappa}{2n}$, so that $\frac{L_F}{3L\theta_1^2}(1 + \frac{n(1+\theta_1)}{2}) = \frac{L_F}{\mu}(\frac{1}{2n} + \frac{1+\theta_1}{4}) \le \frac{L_F}{\mu}(\frac{1}{4} + \frac{3}{8})$, and in the second case $\frac{1}{3\theta_1^2} = \frac{4}{3}$ and $L_F \le L$. Hence, in expected gradient evaluations,

$$T_{\mathrm{LK}}(\varepsilon) \;\le\; \frac{2}{\theta}\log\frac{C_0}{\varepsilon} \;\le\; 2\max\{8n,\, 2\sqrt{6}\sqrt{n\kappa}\}\log\frac{C_0}{\varepsilon}, \tag{132}$$

where the logarithm carries the extra term $\log(1 + \frac{5}{8}Q) \approx \log(\kappa/n)$ in the regime $\kappa \gg n$ (for the sequence y^k, bounded through $F(y^k) - F(\eta_*) \le \theta_1\Psi^k$ and μ_{eff}-strong convexity, the constant is $O(n(1 + L_F/\mu)) = O(\kappa)$ instead); this must be kept in mind when reading slopes. The lower bounds of Proposition 4.18 and Theorem 4.19 then give $T_{\mathrm{LK}}(\varepsilon) = \Theta(\sqrt{n\kappa}\log(1/\varepsilon))$ for $\kappa \ge 3n$, up to the constants and the $\log(\kappa/n)$ just mentioned, as long as (127) holds: this is the slope $1/2$, and this exponent is certified optimal within classes (S1), (S2) and the class of [**?**].

4.4.3 The testable prediction (A4) and the finite-dimension effect

Corollary 4.23(a) needs the initial error to carry a fixed fraction of its energy on the slow modes of $\nabla^2 F$. On the instance, the eigenvectors of the chain are explicit enough for this fraction to be bounded below uniformly in κ, as long as the chain is long compared with the decay length $1/\log(1/q) \approx \sqrt{Q}/2$ of the minimizer (the slow modes are the sine waves whose wavelength exceeds this decay length; the slowest mode alone carries a fraction of order m^{-3} only when $m \gg \sqrt{Q}$, which is why the full slow subspace, and not a single eigenvector, must be used).

Lemma 4.24 (mass of the initial error on the slow modes)**.** *Let $m \ge 2$, $\zeta \in [1,2]$ and $s \in (0,1]$. Let $0 < \lambda_1 < \cdots < \lambda_m < 4$ be the eigenvalues of $A_m(\zeta)$ with orthonormal eigenvectors $v_1, \dots, v_m$, and let $\Pi_s = \sum_{k:\lambda_k \le s} v_k v_k^\top$ be the orthogonal projector onto the eigenspaces with eigenvalue at most s. For $x^* = (A_m(\zeta) + sI)^{-1}e_1$ and $K_0 := \lfloor (m+1)\sqrt{s}/\pi \rfloor$,*

$$\|\Pi_s x^*\|^2 \;\ge\; \frac{7\pi^2(1 - s/4)}{96\, m\, s^2(2m+1)^2}\, K_0\big(4K_0^2 - 1\big). \tag{133}$$

If moreover $\zeta = \zeta_Q$ and $s = \frac{4}{Q-1}$ (so that $x^* = (q^j)_{j \le m}$ is the block of η_*, Proposition 4.14(a)), the slow-mode fraction $\rho := \|\Pi_s x^*\|^2 / \|x^*\|^2$ satisfies, with $y := (m+1)\sqrt{s}/\pi$,

$$\rho \;\ge\; \frac{7(1 - s/4)}{192\pi} \cdot \frac{K_0(4K_0^2 - 1)}{y^3} \;\ge\; \begin{cases} 1/105 & \text{if } y \ge 2, \\ 1/310 & \text{if } y \ge 1. \end{cases} \tag{134}$$

On the instance (118) with $\theta_0 = 0$, $\|P_{2\mu}(\theta_0 - \eta_*)\|^2 = \rho\, \|\theta_0 - \eta_*\|^2$, where $P_{2\mu}$ is the projector of Proposition 4.22 for $\nu' = 2\mu$.

Proof. Step 1: the eigenvectors are sampled sines. Write $A = A_m(\zeta)$. By Lemma 4.3, $A_m(1) \preceq A \preceq A_m(2)$, so Weyl's monotonicity theorem (the Courant–Fischer characterization of the k-th eigenvalue) gives, with the eigenvalues of $A_m(1)$ and $A_m(2)$ computed in Lemma 4.3,

$$4\sin^2\frac{(2k-1)\pi}{2(2m+1)} \;\le\; \lambda_k \;\le\; 4\sin^2\frac{k\pi}{2(m+1)}, \qquad k = 1, \dots, m. \tag{135}$$

In particular $0 < \lambda_k < 4$, and we may write $\lambda_k = 2 - 2\cos\theta_k = 4\sin^2(\theta_k/2)$ with $\theta_k \in (0, \pi)$. Let $Av = \lambda v$, $v \ne 0$, $\lambda = 2 - 2\cos\theta$. Rows 1 to $m-1$ of this system read $v_2 = 2\cos\theta\, v_1$ and $v_{j+1} = 2\cos\theta\, v_j - v_{j-1}$ for $2 \le j \le m-1$; the sequence $u_j = \sin(j\theta)$ satisfies the same recursion with $u_0 = 0$, since $\sin((j+1)\theta) + \sin((j-1)\theta) = 2\cos\theta\sin(j\theta)$, and a solution of this second-order

recursion with prescribed $v_0 = 0$ is determined by v_1. Hence $v_j = (v_1/\sin\theta)\sin(j\theta)$ for $j = 1, \dots, m$; in particular $v_1 \neq 0$ and every eigenvalue is simple (the last row of the system is the equation that selects θ). Normalizing, $1 = \|v\|^2 = (v_1^2/\sin^2\theta)S(\theta)$ with $S(\theta) = \sum_{j=1}^{m} \sin^2(j\theta) \le m$, hence

$$(v_k)_1^2 = \frac{\sin^2\theta_k}{S(\theta_k)} \;\ge\; \frac{\sin^2\theta_k}{m} = \frac{\lambda_k(1-\lambda_k/4)}{m}, \tag{136}$$

because $\sin^2\theta = 4\sin^2\frac{\theta}{2}\cos^2\frac{\theta}{2} = \lambda(1-\frac{\lambda}{4})$.

Step 2: Fourier coefficients of x^*. With $B = A + sI$, $Bv_k = (\lambda_k + s)v_k$ and B symmetric, $\langle x^*, v_k\rangle = \langle e_1, B^{-1}v_k\rangle = (v_k)_1/(\lambda_k + s)$, so that

$$\|\Pi_s x^*\|^2 = \sum_{k:\lambda_k \le s} \frac{(v_k)_1^2}{(\lambda_k+s)^2}, \qquad \|x^*\|^2 = \sum_{k=1}^{m} \frac{(v_k)_1^2}{(\lambda_k+s)^2}.$$

By (135) and $\sin x \le x$, $\lambda_k \le (k\pi/(m+1))^2 \le s$ for $k \le K_0$: the slow set $\{k : \lambda_k \le s\}$ contains $\{1, \dots, K_0\}$. For such k, $(\lambda_k + s)^2 \le 4s^2$ and $1 - \lambda_k/4 \ge 1 - s/4$, so (136) gives

$$\|\Pi_s x^*\|^2 \;\ge\; \frac{1-s/4}{4ms^2}\sum_{k=1}^{K_0}\lambda_k.$$

Step 3: lower bound on the slow eigenvalues. For $k \le K_0$ put $x_k = \frac{(2k-1)\pi}{2(2m+1)}$. Then $x_k \le \frac{K_0\pi}{2m+1} \le \frac{(m+1)\sqrt{s}}{2m+1} \le \frac{3}{5}\sqrt{s} \le \frac{3}{5}$ (as $m \ge 2$ and $s \le 1$), and $\sin x \ge x - \frac{x^3}{6} \ge x\big(1 - \frac{9}{25\cdot 6}\big) = \frac{47}{50}x$ on $[0, \frac{3}{5}]$, so $\sin^2 x_k \ge \frac{2209}{2500}x_k^2 \ge \frac{7}{8}x_k^2$. By (135), $\lambda_k \ge 4\sin^2 x_k \ge \frac{7}{2}x_k^2 = \frac{7\pi^2(2k-1)^2}{8(2m+1)^2}$, and $\sum_{k=1}^{K_0}(2k-1)^2 = \frac{K_0(4K_0^2-1)}{3}$. Inserting this in Step 2 gives (133).

Step 4: the fraction for $\zeta = \zeta_Q$. By Proposition 4.14(a), $\|x^*\|^2 = \frac{q^2(1-q^{2m})}{1-q^2} \le \frac{q^2}{1-q^2} = \frac{(\sqrt{Q}-1)^2}{4\sqrt{Q}} \le \frac{Q-1}{4\sqrt{Q}} = \frac{1}{s\sqrt{Q}} \le \frac{1}{2\sqrt{s}}$, using $(\sqrt{Q}-1)^2 \le (\sqrt{Q}-1)(\sqrt{Q}+1) = Q - 1 = \frac{4}{s}$ and $\sqrt{Q} \ge \sqrt{4/s} = \frac{2}{\sqrt{s}}$. Hence $\rho \ge 2\sqrt{s}\cdot\frac{7\pi^2(1-s/4)K_0(4K_0^2-1)}{96ms^2(2m+1)^2}$; substituting $s = \pi^2y^2/(m+1)^2$, i.e. $s^{-3/2} = (m+1)^3/(\pi^3y^3)$, and using $\frac{(m+1)^3}{m(2m+1)^2} = \frac{m+1}{m}\big(\frac{m+1}{2m+1}\big)^2 \ge \frac{1}{4}$, we obtain the first inequality in (134). For $y \in [K, K+1)$ one has $K_0 = K$ and $\frac{K_0(4K_0^2-1)}{y^3} \ge g(K) := \frac{K(4K^2-1)}{(K+1)^3}$; writing $u = K+1$, $g = 4 - \frac{12}{u} + \frac{11}{u^2} - \frac{3}{u^3}$ has derivative $\frac{12u^2-22u+9}{u^4} > 0$ for $u \ge 2$ (the larger root of $12u^2-22u+9$ is $\frac{22+\sqrt{52}}{24} < 2$), so g is increasing on $K \ge 1$, with $g(1) = \frac{3}{8}$ and $g(2) = \frac{30}{27}$. With $1 - \frac{s}{4} \ge \frac{3}{4}$: for $y \ge 2$, $\rho \ge \frac{7}{192\pi}\cdot\frac{3}{4}\cdot\frac{30}{27} = \frac{35}{1152\pi} > \frac{1}{105}$, and for $y \ge 1$, $\rho \ge \frac{7}{192\pi}\cdot\frac{3}{4}\cdot\frac{3}{8} = \frac{63}{6144\pi} > \frac{1}{310}$.

Step 5: the instance. $\nabla^2 F = \operatorname{diag}(H_b, \dots, H_b)$ with $H_b = \frac{c}{n}A_m(\zeta_Q) + \mu I = \frac{\mu}{s}\big(A_m(\zeta_Q) + sI\big)$, because $\frac{c}{n} = \frac{\mu(Q-1)}{4} = \frac{\mu}{s}$; its eigenvalues are $\mu(1 + \lambda_k/s)$, each with the n eigenvectors $E_1v_k, \dots, E_nv_k$, and $\mu(1+\lambda_k/s) \le 2\mu$ iff $\lambda_k \le s$. Since $\theta_0 - \eta_* = -(x^*, \dots, x^*)$, $\|P_{2\mu}(\theta_0 - \eta_*)\|^2 = \sum_{i=1}^{n}\|\Pi_s x^*\|^2 = \rho\, n\,\|x^*\|^2 = \rho\,\|\theta_0 - \eta_*\|^2$. □

Remark 4.25 (sharpness of (134), and the exhausted chain). The constants in (134) are not sharp: the proof loses a factor 2 in $S(\theta) \le m$ (for all but the first mode $S(\theta_k) \approx m/2$), a factor up to 4 in $(\lambda_k+s)^2 \le 4s^2$, and the rounding in K_0. The exact ρ is computable from the $m\times m$ tridiagonal eigenproblem and is the quantity to report in the experiments; for $n = 100$, $m = 200$ and $\kappa \in [10^3, 3\cdot 10^5]$ we computed $\rho \in [0.18, 0.35]$, against 0.027–0.045 for the bound (134). For the infinite chain ($m = \infty$) the spectral measure of e_1 for A is $d\nu(\lambda) = \frac{1}{2\pi}\sqrt{\lambda(4-\lambda)}\,d\lambda$ on $[0,4]$ (the orthogonal polynomials of the recursion of Step 1 are Chebyshev polynomials of the second kind), $\|x^*\|^2 = \int_0^4 (\lambda+s)^{-2}d\nu(\lambda)$,

and the slow fraction $\int_0^s(\lambda+s)^{-2}d\nu/\int_0^4(\lambda+s)^{-2}d\nu$ tends, as $s\to 0$, to $\frac{\int_0^1\sqrt{u}\,(1+u)^{-2}du}{\int_0^\infty\sqrt{u}\,(1+u)^{-2}du}=\frac{\pi/4-1/2}{\pi/2}=\frac{1}{2}-\frac{1}{\pi}\approx 0.18$; this is the value approached by the exact ρ when $m\sqrt{s}\to\infty$ and then $s\to 0$, and it is not used in the proofs. At the other extreme, when $\kappa\to\infty$ with m fixed ($s\to 0$, $y\to 0$), eventually no eigenvalue of $A_m(\zeta_Q)$ lies below s and the slow set is empty; the relevant quantity is then the single-mode overlap $\delta:=\langle x^*,v_1\rangle^2/\|x^*\|^2$ of Corollary 4.23(a) with $\nu'=\mu_{\text{eff}}$ (the eigenspace of μ_{eff} is spanned by $E_1v_1,\dots,E_nv_1$, so $\|P_{\mu_{\text{eff}}}(\theta_0-\eta_*)\|^2=n\langle x^*,v_1\rangle^2=\delta\|\theta_0-\eta_*\|^2$). Since $\zeta_Q\to 1$, $x^*=(A_m(\zeta_Q)+sI)^{-1}e_1\to A_m(1)^{-1}e_1$ and $v_1(\zeta_Q)\to v_1(1)$ (simple eigenvalue, eigenvectors normalized with positive first coordinate), so $\delta\to\delta_\infty(m):=\langle A_m(1)^{-1}e_1,v_1(1)\rangle^2/\|A_m(1)^{-1}e_1\|^2$, which is positive because both vectors have positive entries: $v_1(1)$ by the Perron–Frobenius theorem applied to the irreducible nonnegative matrix $4I-A_m(1)$, whose top eigenvector is the bottom eigenvector of $A_m(1)$, and $A_m(1)^{-1}e_1$ because $A_m(1)=4I-N$ with $N\ge 0$ irreducible and $\lambda_{\max}(N)=4-\lambda_{\min}(A_m(1))<4$, so that $A_m(1)^{-1}=\frac{1}{4}\sum_{k\ge 0}(N/4)^k$ is entrywise positive. Numerically $\delta\approx 0.18$ for $m=200$, $n=100$ and $\kappa=10^6$.

Proposition 4.26 (testable prediction)**.** *Fix $n\ge 2$, $m\ge 2$, μ and a relative accuracy $\varepsilon\in(0,\frac{1}{105})$, start at $\theta_0=0$ (for SAGA with $z_i^{(0)}=f_i'(0)$, the n gradients of this sweep being counted in T_{SAGA}), and let κ vary. On the instance* (118) *with $\zeta=\zeta_Q$, with C_0 from* (131)*:*

(i) (Plateau, $\kappa\le n$.) For SAGA, under (127) *and $\varepsilon<q^2/2$,*

$$n+(n-1)\log\frac{q^2}{2\varepsilon}\;\le\;T_{\text{SAGA}}(\varepsilon)\;\le\;n+3n\log\frac{1+n/4}{\varepsilon},$$

both sides being $\Theta(n\log(1/\varepsilon))$ once $\varepsilon\ll q^2$ and $\varepsilon\ll 1/n$ (the upper bound is $\Theta(n\log(n/\varepsilon))$ in general): the plateau with slope ≈ 0 is a theorem for SAGA. For L-Katyusha only

$$\frac{n\log(1/(2\varepsilon))}{2\log(1/q)}\;\le\;T_{\text{LK}}(\varepsilon)\;\le\;16n\log\frac{C_0}{\varepsilon},\qquad C_0\le n+\tfrac{7}{3},$$

is proved (the lower bound under (127)*); the two sides are of the same order $n\log(1/\varepsilon)$ only when $\log(1/q)=O(1)$, i.e. when κ is of order n (for $\kappa=n$, $\log(1/q)\approx 1.76$). For $\kappa\ll n$, $\log(1/q)\approx\log\frac{4n}{\kappa-1}$, and a gain of this logarithmic order over $n\log(1/\varepsilon)$ is not excluded for L-Katyusha; it is realized by GD on F with step $1/L_F$, which contracts the error by $1-\frac{\mu_{\text{eff}}}{L_F}\le 1-\frac{1}{Q}\approx\frac{\kappa}{n}$ per full gradient, in agreement with Theorem 4.11(c) and with the assessment of (A1)(ii). In the upper bounds, the second term takes over at $\kappa=\frac{9}{16}n$ for SAGA ($\frac{16\kappa}{3}=3n$) and at $\kappa=\frac{8}{3}n$ for L-Katyusha (θ_1 reaches $\frac{1}{2}$).*

(ii) (Power laws.) Let

$$\kappa_{\max}(m,\varepsilon):=1+n\min\Big\{\frac{16m^2}{\log^2(1/\varepsilon)}-1,\;\frac{(m+1)^2}{\pi^2}\Big\},\tag{137}$$

where the first term is the sufficient form $m\ge\frac{\sqrt{Q}}{4}\log\frac{1}{\varepsilon}$ of the dimension condition (127) *solved for κ (so that $q^{2m}\le\varepsilon$), and the second is the condition $y\ge 2$ of Lemma 4.24 solved for κ. In the window $4n+1\le\kappa\le\kappa_{\max}(m,\varepsilon)$,*

$$\Big(\kappa-\frac{1}{2}\Big)\log\frac{\rho}{\varepsilon}\;\le\;T_{\text{SAGA}}(\varepsilon)\;\le\;n+\frac{16\kappa}{3}\log\frac{1+n/4}{\varepsilon},\qquad\rho=\rho(m,\kappa)\ge\frac{1}{105},\tag{138}$$

$$\frac{\sqrt{n(\kappa-1)}}{8}\log\frac{1}{2\varepsilon}\;\le\;T_{\text{LK}}(\varepsilon)\;\le\;4\sqrt{6}\,\sqrt{n\kappa}\,\log\frac{C_0}{\varepsilon},\qquad C_0\le 1+\frac{5}{8}Q,\tag{139}$$

so that $T_{\mathrm{SAGA}} = \Theta(\kappa \log(1/\varepsilon))$ *and* $T_{\mathrm{LK}} = \Theta(\sqrt{n\kappa}\log(1/\varepsilon))$ *up to the displayed constants and the* $\log(1+\frac{5}{8}Q)$ *of* (139)*: the log-log slopes of* T *against* κ *are* 1 *and* $\frac{1}{2}$*, and* $T_{\mathrm{SAGA}}/T_{\mathrm{LK}} \asymp \sqrt{\kappa/n}$*. GD on* F *(step* $1/L_F$*) and AGD on* F *follow the same two slopes:* $n(\frac{L_F}{4\mu} - \frac{1}{2})\log\frac{\rho}{\varepsilon} \le T_{\mathrm{GD}} \le \frac{nQ}{2}\log\frac{1}{\varepsilon}$ *with* $L_F \ge \mu(1+\frac{13}{20}(Q-1))$*, and* $\frac{\sqrt{n(\kappa-1)}}{8}\log\frac{1}{2\varepsilon} \le T_{\mathrm{AGD}} = O(n\sqrt{Q}\log\frac{1}{\varepsilon})$.

(iii) (Saturation.) For $\kappa \gg nm^2$ *the chain is exhausted:* $\mu_{\mathrm{eff}}/\mu = 1 + \frac{\kappa-1}{4n}\lambda_{\min}(A_m(\zeta_Q))$ *grows linearly in* κ*, so*

$$\kappa_{\mathrm{eff}} = \frac{R^2}{\mu_{\mathrm{eff}}} = \frac{\kappa}{1+\frac{\kappa-1}{4n}\lambda_{\min}(A_m(\zeta_Q))} \xrightarrow[\kappa\to\infty]{} \frac{4n}{\lambda_{\min}(A_m(1))} = \frac{n}{\sin^2\frac{\pi}{2(2m+1)}} \approx \frac{4n(2m+1)^2}{\pi^2}, \tag{140}$$

because $\zeta_Q \to 1$ *and* $\lambda_{\min}$ *is continuous in the matrix; for every finite* κ*, Lemma 4.3 gives the two-sided bound* $\frac{\kappa}{1+\frac{\kappa-1}{n}\sin^2\frac{\pi}{2(m+1)}} \le \kappa_{\mathrm{eff}} \le \frac{\kappa}{1+\frac{\kappa-1}{n}\sin^2\frac{\pi}{2(2m+1)}}$*. Then* $(2\kappa_{\mathrm{eff}} - \frac{1}{2})\log\frac{\delta}{\varepsilon} \le T_{\mathrm{SAGA}}(\varepsilon) \le n + \frac{16\kappa_{\mathrm{eff}}}{3}\log\frac{1+n/4}{\varepsilon}$ *for* $\varepsilon < \delta$*, with* $\delta \to \delta_\infty(m) > 0$ *(Remark 4.25):* T_{SAGA} *saturates at* $\Theta(\kappa_{\mathrm{eff}}\log(1/\varepsilon)) = \Theta(nm^2\log(1/\varepsilon))$*, while* T_{LK} *is still bounded above by* (132) *but no longer bounded below by* $\sqrt{n\kappa}$ *(the hypothesis* $q^{2m} \le \varepsilon$ *of Proposition 4.18 fails); the measured slopes decrease towards* 0.

Proof. (i) For SAGA, the lower bound is Corollary 4.21 (the sweep is one query to every component, Lemma 4.16(ii)) plus the n gradients of the sweep, and the upper bound is Corollary 4.23(b) with $\max\{3n, \frac{16\kappa_{\mathrm{eff}}}{3}\} = 3n$, since $\kappa_{\mathrm{eff}} \le \kappa \le n$. For L-Katyusha, the lower bound is (124), valid for every index sequence under (127), and the upper bound is (132) with $2\max\{8n, 2\sqrt{6}\sqrt{n\kappa}\} = 16n$ for $\kappa \le n$ and $C_0 \le n + \frac{7}{3}$ from (131). For $\kappa \ll n$: $Q - 1 = \frac{\kappa-1}{n}$, $\sqrt{Q} - 1 \approx \frac{Q-1}{2}$ and $q = \frac{\sqrt{Q}-1}{\sqrt{Q}+1} \approx \frac{Q-1}{4} = \frac{\kappa-1}{4n}$. The GD contraction follows from $\left\|(I - \frac{1}{L_F}\nabla^2 F)e\right\| \le (1 - \frac{\mu_{\mathrm{eff}}}{L_F})\|e\|$ and $\frac{L_F}{\mu_{\mathrm{eff}}} \le \frac{L_F}{\mu} \le Q$ (Proposition 4.13).

(ii) The window conditions: $\kappa \ge 4n+1$ means $Q \ge 5$, i.e. $s = \frac{4}{Q-1} \le 1$; $\kappa \le 1 + \frac{n(m+1)^2}{\pi^2}$ means $Q - 1 \le \frac{(m+1)^2}{\pi^2}$, i.e. $(m+1)\sqrt{s} \ge 2\pi$, i.e. $y \ge 2$; and $\kappa \le 1 + n(\frac{16m^2}{\log^2(1/\varepsilon)} - 1)$ is $m \ge \frac{\sqrt{Q}}{4}\log\frac{1}{\varepsilon}$, the sufficient form of (127). For SAGA, Lemma 4.24 gives $\|P_{2\mu}(\theta_0 - \eta_*)\|^2 = \rho\|\theta_0 - \eta_*\|^2$ with $\rho \ge \frac{1}{105} > \varepsilon$, so Corollary 4.23(a) with $\nu' = 2\mu \le 4R^2$ gives the lower bound, for any initial memory and in particular after the sweep (the mean-iterate identity of Proposition 4.22 holds whatever the memory, and the sweep only adds n to the count); the upper bound is Corollary 4.23(b) with $\kappa_{\mathrm{eff}} \le \kappa$ and $\frac{16\kappa}{3} \ge 3n$ for $\kappa \ge \frac{9}{16}n$. For L-Katyusha, the lower bound is (124), which needs $q^{2m} \le \varepsilon$ and $\kappa - 1 \ge 2n$, both true in the window, and the upper bound is (132)–(131) with $\kappa > \frac{8}{3}n$. For GD, $e_t = (I - \frac{1}{L_F}\nabla^2 F)^t e_0$ exactly, so the second inequality of (130) applies with $\gamma = 1/L_F$ and $\nu' = 2\mu$, which is $\le 1/\gamma = L_F$ because $\lambda_{\max}(A_m(\zeta_Q)) \ge \lambda_{\max}(A_m(1)) = 4\sin^2\frac{(2m-1)\pi}{2(2m+1)} \ge 4\sin^2\frac{3\pi}{10} > \frac{13}{5}$ for $m \ge 2$ (Lemma 4.3 and Weyl's inequality), whence $L_F = \mu(1 + \frac{Q-1}{4}\lambda_{\max}(A_m(\zeta_Q))) \ge \mu(1 + \frac{13}{20}(Q-1)) \ge \frac{18}{5}\mu$ for $Q \ge 5$; the argument of Corollary 4.23(a) then gives $t \ge (\frac{L_F}{4\mu} - \frac{1}{2})\log\frac{\rho}{\varepsilon}$ iterations of n gradients, while $(1 - \frac{\mu_{\mathrm{eff}}}{L_F})^{2t} \le \varepsilon$ for $t \ge \frac{L_F}{2\mu_{\mathrm{eff}}}\log\frac{1}{\varepsilon}$ with $\frac{L_F}{\mu_{\mathrm{eff}}} \le Q$. For AGD, Proposition 4.18 and Section 5.2.5 of [6] ($O(\sqrt{L_F/\mu_{\mathrm{eff}}}\log\frac{1}{\varepsilon})$ iterations of n gradients, and $L_F/\mu_{\mathrm{eff}} \le Q$) give the two sides. Since $\rho \ge \frac{1}{105}$ and $C_0 \le 1 + \frac{5}{8}Q$ depend on κ only through the displayed logarithm, all four pairs of bounds have the announced orders.

(iii) As $\kappa \to \infty$ with m fixed, $\zeta_Q = \frac{\sqrt{Q}+3}{\sqrt{Q}+1} \to 1$, the eigenvalues of $A_m(\zeta)$ depend continuously on ζ, and $\frac{\kappa}{1+\frac{\kappa-1}{4n}\lambda} \to \frac{4n}{\lambda}$; the two-sided bound is Lemma 4.3 inserted in Proposition 4.13(b). The bounds on T_{SAGA} are Corollary 4.23 with $\nu' = \mu_{\mathrm{eff}}$ and the value of $\|P_{\mu_{\mathrm{eff}}}(\theta_0 - \eta_*)\|^2$ computed in Remark 4.25, which also gives the limit of δ; for the upper bound, $\frac{16\kappa_{\mathrm{eff}}}{3} \ge 3n$ for κ large. For

T_{LK}, (132) holds for all κ (Theorem 4.6 of [21] has no dimension hypothesis), whereas (124) requires $q^{2m} \le \varepsilon$, which fails for $\kappa > \kappa_{\max}$. □

Remark 4.27 (what to expect in a plot). For $\varepsilon = 10^{-6}$ ($\log(1/\varepsilon) \approx 13.8$), $n = 100$ and $m = 200$: the dimension condition gives $1 + n(\frac{16m^2}{\log^2(1/\varepsilon)} - 1) \approx 3.4 \cdot 10^5$ and the slow-mode condition $1 + \frac{n(m+1)^2}{\pi^2} \approx 4.1 \cdot 10^5$, so $\kappa_{\max} \approx 3.4 \cdot 10^5$ and the window $401 \le \kappa \le 3.4 \cdot 10^5$ of (ii) spans almost three decades, enough to read the slopes 1 and $\frac{1}{2}$, with the plateau below $\kappa \approx n$ and the bend above $\kappa_{\max}$. Two caveats. First, a sandwich $c_1\kappa \le T \le c_2\kappa$ pins the slope of a line fitted over D decades only up to $\pm \log_{10}(c_2/c_1)/D$; for SAGA, $c_2/c_1 = \frac{16}{3} \cdot \frac{\log((1+n/4)/\varepsilon)}{\log(\rho/\varepsilon)} \approx 7$ with the exact $\rho \approx 0.2$, and for L-Katyusha the ratio of the two constants in (139) exceeds 10^2. The order of growth is a theorem; the numerical value of the slope must be read from exact quantities: the mean-iterate time $t^*(\kappa) = \min\{t : \left\|(I - \gamma\nabla^2 F)^t(\theta_0 - \eta_*)\right\|^2 \le \varepsilon \left\|\theta_0 - \eta_*\right\|^2\}$, computable from the m eigenpairs of $A_m(\zeta_Q)$, is a lower bound on T_{SAGA} (first inequality of (130)), and for these parameters we found $t^* \approx 1.4\,\kappa \log(1/\varepsilon)$ with log-log slope 0.99 over $\kappa \in [10^3, 3 \cdot 10^5]$. Second, with $m = 20$ the window collapses ($\kappa_{\max} \approx 3.3 \cdot 10^3$, less than a decade above $4n+1$): the slopes measured over $\kappa \in [10^3, 10^6]$ would be far below 1 and $\frac{1}{2}$. This finite-dimension effect is quantified by (140): plotting T_{SAGA} against κ_{eff} instead of κ restores a linear growth (Corollary 4.23 with $\nu' = \mu_{\mathrm{eff}}$), which is the cleanest test of the "effective constants" idea of hypothesis (B2) on an instance where $\mu_{\mathrm{eff}} = \lambda_{\min}(\nabla^2 F)$ is known exactly.

4.4.4 Logistic regression is not a hard instance

The lower bounds of Section 4.2 are statements about the worst case over all instances with given (n, R^2, μ); for a given instance only the upper bounds are informative, and the ℓ_2-regularized logistic problems of the project, $f_i(\theta) = \log(1 + e^{-y_i x_i^\top \theta}) + \frac{\lambda}{2}\left\|\theta\right\|^2$ with $R^2 = \max_i \left\|x_i\right\|^2/4 + \lambda$ and $\mu = \lambda$, are far from the worst case, for three reasons.

(a) *No chain.* Every component acts on the whole vector θ through x_i; after one pass the span of the gradients is typically all of $\mathbb{R}^d$ (generic x_i), so nothing prevents the iterates from moving in every direction. The "one coordinate per query" mechanism, which is the only source of the $\sqrt{n\kappa}$ term, is absent.

(b) *Slack in the constants.* The certified smoothness R^2 is reached only at $\sigma'(y_i x_i^\top \theta) = 1/4$, i.e. on the decision boundary, and only for the sample with the largest $\|x_i\|$; along the trajectory the per-sample curvature is $\sigma'(y_i x_i^\top \theta)\left\|x_i\right\|^2 + \lambda$, often much smaller (this is the cocoercivity slack measured in (B3)). Likewise the strong convexity $\mu = \lambda$ ignores the data term: the trajectory-dependent constant of (B2),

$$\mu_{\mathrm{eff}}(t) = \frac{\langle \theta_t - \eta_*, F'(\theta_t)\rangle}{\left\|\theta_t - \eta_*\right\|^2} \;\ge\; \lambda + \lambda_{\min}\Big(\frac{1}{n}\sum_i \bar{\sigma}'_{i,t}\, x_i x_i^\top\Big) \;\ge\; \lambda, \tag{141}$$

where $\bar{\sigma}'_{i,t} \in (0, \frac{1}{4}]$ is the average of σ' on the segment $[\eta_*, \theta_t]$ (mean-value form of $F'(\theta_t) - F'(\eta_*)$). When $n \ge d$ and the data are not separable, the data Hessian is positive definite and $\mu_{\mathrm{eff}}(t) \ge \lambda + \mu_{\mathrm{data}}$ with $\mu_{\mathrm{data}} > 0$ independent of λ; the proof of Proposition 5.9 of [6] then runs with $\mu_{\mathrm{eff}}(t)$ in place of μ, so SAGA converges at the rate dictated by $R^2/\min_t \mu_{\mathrm{eff}}(t)$ rather than by $\kappa = R^2/\lambda$.

(c) $L_F \ll R^2$ *by isotropy rather than by block structure.* For isotropic features (x_i with i.i.d. unit-variance coordinates), $\left\|x_i\right\|^2 \approx d$ so $R^2 \approx d/4$, while $L_F \le \lambda + \frac{1}{4}\lambda_{\max}(\frac{1}{n}\sum_i x_i x_i^\top) \approx \lambda + \frac{1}{4}(1 + \sqrt{d/n})^2$; thus $L_F/R^2 \approx 1/d$ for $n \gg d$ (compare with $1/n$ on the block instance).

Full-gradient methods are therefore much faster on these problems than the worst-case counts $n\kappa$, $n\sqrt{\kappa}$ suggest, by a factor about d.

Consequently, when $\kappa = R^2/\lambda$ is increased by decreasing λ at fixed data, the number of gradient evaluations of SAGA and L-Katyusha should grow like κ_{eff}^1 and $\kappa_{\text{eff}}^{1/2}$ at most, with $\kappa_{\text{eff}} = R^2/\mu_{\text{eff}}$ saturating as soon as $\lambda \ll \mu_{\text{data}}$: the empirical log-log slopes against the nominal κ are expected to be strictly smaller than 1 and 1/2 and to decrease towards 0 for small λ (for $d > n$ or separable data $\mu_{\text{data}} = 0$ and the slopes can stay close to their hard-instance values, with large constants). Measuring the slopes against R^2/μ_{eff} (with μ_{eff} the time average or the terminal value of (141)) is the natural way to connect (A4) with (B2); the hard instance, where $\mu_{\text{eff}} = \lambda_{\min}(\nabla^2 F)$ is known in closed form, serves as the calibration of that procedure.

4.4.5 Inequalities to instrument

For a lower-bound inequality LHS $\geq$ RHS the tightness ratio is RHS/LHS, which lies in $(0, 1]$ whenever RHS > 0; for an upper bound LHS $\leq$ RHS it is LHS/RHS. On the instance (118) every quantity below is computable exactly from the iterates (η_* is known in closed form, and the eigenpairs of the $m \times m$ tridiagonal matrix $A_m(\zeta_Q)$ are cheap). Every gradient query is counted, including the initial sweep of SAGA ($\tau_j = 1$ for all j and $G = n$ right after the sweep) and the full gradients of L-Katyusha (n queries each).

- (114) *chain smoothness*: ratio $\langle A_m(\zeta)s, s\rangle / (4\|s\|^2)$ at $s = \theta_t^{(i)} - x^*$ (how much of the error sits in the stiff directions; also certifies the slack between $R^2 = \mu\kappa$ and the true Lipschitz constant of f_i').
- (120) *condition number of F versus κ*: ratio $(L_F/\mu)/\kappa$ ($\approx 1/n$ on the instance; on logistic data compare with $1/d$).
- Lemma 4.16 *support growth*: ratio $N_j(\theta_t)/\min(\tau_j(t), m)$ per block, with τ_j counting the sweep (equal to 1 for SAGA and L-Katyusha generically, which validates the zero-chain mechanism).
- (122) *tail bound*: ratio $\left[\frac{1}{n}\sum_j \frac{q^{2\min(\tau_j(t),m)} - q^{2m}}{1-q^{2m}}\right] / \left[\|\theta_t - \eta_*\|^2 / \|\eta_*\|^2\right]$ (how much of the error is the unrevealed energy; a ratio near 1 means the method is optimal on the revealed coordinates; the cap at m keeps the right-hand side nonnegative once a block has been touched more than m times).
- (123) *Jensen over blocks*: ratio $q^{2G/n} / \frac{1}{n}\sum_j q^{2\tau_j}$ (equal to 1 for cyclic or full-gradient schedules, smaller for i.i.d. sampling: this is the price of i.i.d. sampling isolated by [15]).
- (125) and (129) *i.i.d. bound*: ratio of the right-hand side to the empirical mean of $\|\theta_t - \eta_*\|^2 / \|\eta_*\|^2$ over seeds, using (129) for SAGA with $z_i^{(0)} = f_i'(\theta_0)$ (t = number of i.i.d. iterations after the sweep) and (125) for $z^{(0)} = 0$; report the ratio only while the right-hand side is positive, i.e. for $t \leq t^*$ of Corollary 4.20 (this is the dimension condition), since beyond that point the bound is vacuous.
- (126) *exponential bound*: ratio $\exp(-\frac{tx}{1-x})/(1-x)^t$ with $x = \frac{1-q^2}{n}$.
- (127) *dimension condition*: the scalar q^{2m}/ε (≤ 1 required; report it next to each measured slope), together with $y = (m+1)\sqrt{s}/\pi$ (≥ 2 required for (138)).
- (134) *slow-mode mass*: the exact $\rho = \sum_{k:\lambda_k \leq s} \langle x^*, v_k\rangle^2 / \|x^*\|^2$ from the eigenpairs of $A_m(\zeta_Q)$, and the ratio of the bound (134) to it (expected between 0.1 and 0.2, Remark 4.25); also the single-mode overlap $\delta = \langle x^*, v_1\rangle^2 / \|x^*\|^2$ for the saturation regime.
- (130) *SAGA mean iterate*: ratio $\left\|(I - \gamma\nabla^2 F)^t(\theta_0 - \eta_*)\right\|^2 / \mathbb{E}\|\theta_t - \eta_*\|^2$ (exact mean iterate versus the empirical mean square error over seeds) and, for the second inequality, $(1 - 2\gamma\mu)^{2t}\rho\|\theta_0 - \eta_*\|^2 / \left\|(I - \gamma\nabla^2 F)^t(\theta_0 - \eta_*)\right\|^2$ with $\nu' = 2\mu$ (or $(1 - \gamma\mu_{\text{eff}})^{2t}\delta\|\theta_0 - \eta_*\|^2$ in the numerator with $\nu' = \mu_{\text{eff}}$); the mean-iterate time $t^*(\kappa)$ of Remark 4.27 is the natural reference

curve for $T_{\mathrm{SAGA}}(\kappa)$.

- (132) *L-Katyusha upper bound*: ratio of the measured T_{LK} to $\frac{2}{\theta}\log(C_0/\varepsilon)$, and the Lyapunov ratio $\Psi^{k+1}/((1-\theta)\Psi^k)$ along the run (the individual steps of that proof are instrumented in the L-Katyusha note).
- (141) *effective strong convexity*: the ratio $\mu/\mu_{\mathrm{eff}}(t)$, on the instance (it converges there to $\mu/\lambda_{\min}(\nabla^2 F)$) and on logistic data.
- (116) *single chain* ($n=1$ version of the instance, finite m): ratio of $\frac{q^{2t}-q^{2m}}{1-q^{2m}}\left\|\eta_*\right\|^2$, i.e. the right-hand side of (122) with $n=1$, to $\left\|\theta_t-\eta_*\right\|^2$, for AGD and for GD; the infinite-chain form $q^{2t}\left\|\eta_*\right\|^2$ of (116) is not a valid lower bound on the finite instance (at $t=m$ an optimal span method has zero error), so the ratio defined with it could exceed 1.

5 Loopless Katyusha: a Lyapunov proof in the style of the notes (hypotheses A2, A3)

This note reproduces, in the notation and in the style of Section 5.4.4 of the notes [6] (one Lyapunov function, a short list of elementary inequalities, explicit constants), the convergence proof of the *loopless Katyusha* method (L-Katyusha) of Kovalev, Horváth and Richtárik [21], i.e. the accelerated variance-reduced method of Allen-Zhu [**?**] in which the outer loop (a full gradient every m iterations) is replaced by a coin flip. The goal is twofold. (A3) We want to see *exactly* which term of the proof is paid by the "negative momentum", that is the pull of the extrapolation point towards the reference point w_t at which the full gradient is known. (A2) We want to see why the same proof does not close when Nesterov's momentum of Section 5.2.5 of the notes is combined with the variance-reduced estimator *without* this pull. Every inequality is labelled so that its tightness can be measured in the experiments (Section 5.5). The constants are those of [21] (Algorithm 2, Lemmas 4.1–4.5 and Theorem 4.6 there, proofs in their Appendix C); we quote them where we use them.

5.1 Setting, estimator and algorithm in the notation of the notes

Assumption 5.1. *$F(\theta)=\frac{1}{n}\sum_{i=1}^n f_i(\theta)$ on $\mathbb{R}^d$, where each f_i is convex and R^2-smooth; we write $L:=R^2$ so that L is the* per-sample *smoothness constant, as in Assumption 1.1 of [21]. F is μ-strongly convex, with minimizer η_* and optimal value $F_*:=F(\eta_*)$. We write $\kappa:=R^2/\mu=L/\mu\geq 1$ and $\sigma:=\mu/L=1/\kappa\in(0,1]$ (the quantity called σ in [21]).*

Since $F'=\frac{1}{n}\sum_i f_i'$, the function F is itself L-smooth (its own constant may be smaller, but we only use L). For ℓ_2-regularized logistic regression, $f_i(\theta)=\log(1+\exp(-y_i x_i^\top\theta))+\frac{\lambda}{2}\left\|\theta\right\|^2$, one can take $L=\max_i\left\|x_i\right\|^2/4+\lambda$ and $\mu=\lambda$ (hypothesis (B3)).

Toolbox. The proof uses only the following facts, all of which appear in Chapter 5 of the notes. For a convex L-smooth f and all $\theta,\eta\in\mathbb{R}^d$:

(T1) convexity (eq. (5.7) of the notes): $f(\theta)\geq f(\eta)+\langle f'(\eta),\theta-\eta\rangle$;

(T2) quadratic upper bound (eq. (5.10)): $f(\theta)\leq f(\eta)+\langle f'(\eta),\theta-\eta\rangle+\frac{L}{2}\left\|\theta-\eta\right\|^2$;

(T3) co-coercivity in its "Bregman" form (Prop. 5.4, eq. (5.12)): $f(\theta)\geq f(\eta)+\langle f'(\eta),\theta-\eta\rangle+\frac{1}{2L}\left\|f'(\theta)-f'(\eta)\right\|^2$;

(T4) strong convexity of F (eq. (5.9)): $F(\theta)\geq F(\eta)+\langle F'(\eta),\theta-\eta\rangle+\frac{\mu}{2}\left\|\theta-\eta\right\|^2$;

(T5) three-point identity: $2\langle a-b, b-c\rangle = \|a-c\|^2 - \|a-b\|^2 - \|b-c\|^2$;

(T6) Young's inequality: for $\beta > 0$, $\langle a, b\rangle \geq -\frac{1}{2\beta}\|a\|^2 - \frac{\beta}{2}\|b\|^2$;

(T7) variance is at most the second moment (used as inequality [I3] in the proof of Prop. 5.9): for a random vector X, $\mathbb{E}\|X - \mathbb{E}X\|^2 = \mathbb{E}\|X\|^2 - \|\mathbb{E}X\|^2 \leq \mathbb{E}\|X\|^2$.

We write $D_F(w,x) := F(w) - F(x) - \langle F'(x), w - x\rangle \geq 0$ for the Bregman divergence of F; by (T2), $D_F(w,x) \leq \frac{L}{2}\|w-x\|^2$.

Definition 5.2 (loopless SVRG estimator)**.** *Let $p \in (0,1]$. At iteration t the method holds a query point x_t and a* reference point *w_t at which the full gradient $F'(w_t)$ has been computed. An index $i(t)$ is drawn uniformly in $\{1,\dots,n\}$ and the estimator is*

$$g_t := f'_{i(t)}(x_t) - f'_{i(t)}(w_t) + F'(w_t). \tag{142}$$

Independently of $i(t)$ and of the past, a coin $b_t \sim \mathrm{Bernoulli}(p)$ decides whether the reference point is refreshed: $w_{t+1} = y_t$ if $b_t = 1$ and $w_{t+1} = w_t$ otherwise (with y_t the "gradient-step" sequence defined below). Refreshing costs one pass (n gradients f'_i) and happens with probability p; with $p = 1/n$ it costs one gradient per iteration on average.

Remark 5.3. We write $\mathbb{E}_{i(t)}$ for the conditional expectation given the state (y_t, z_t, w_t) of the algorithm, i.e. over $i(t)$ and b_t (this is the notation $\mathbb{E}[\,\cdot \mid y^k, z^k, w^k]$ of [21]). Since $i(t)$ is uniform, $\mathbb{E}_{i(t)}[g_t] = F'(x_t) - F'(w_t) + F'(w_t) = F'(x_t)$: the estimator is unbiased, exactly as the SAGA direction ω_t of Prop. 5.9. The difference with SAGA is that the memory is a single point w_t and a single vector $F'(w_t)$ instead of the table $(z_i^{(t)})_{i\leq n}$, and that each iteration costs two gradients $f'_{i(t)}(x_t), f'_{i(t)}(w_t)$ instead of one.

The algorithm. L-Katyusha (Algorithm 2 of [21]) keeps three sequences x_t, y_t, z_t and the reference point w_t. Parameters: $\theta_1, \theta_2 > 0$ with $\theta_1 + \theta_2 \leq 1$, $p \in (0,1]$, $\sigma = \mu/L$ and the step

$$\eta := \frac{\theta_2}{(1+\theta_2)\theta_1}. \tag{143}$$

Initialization $y_0 = w_0 = z_0 = \theta_0 \in \mathbb{R}^d$ (with $F'(w_0)$ computed). For $t = 0, 1, 2, \dots$:

$$x_t = \theta_1 z_t + \theta_2 w_t + (1 - \theta_1 - \theta_2) y_t, \tag{K1}$$

$$g_t = f'_{i(t)}(x_t) - f'_{i(t)}(w_t) + F'(w_t), \tag{K2}$$

$$z_{t+1} = \frac{1}{1+\eta\sigma}\Big(\eta\sigma\, x_t + z_t - \frac{\eta}{L}\, g_t\Big), \tag{K3}$$

$$y_{t+1} = x_t + \theta_1(z_{t+1} - z_t), \tag{K4}$$

$$w_{t+1} = \begin{cases} y_t & \text{with probability } p, \\ w_t & \text{with probability } 1-p. \end{cases} \tag{K5}$$

The reading of the three steps is the one of "linear coupling" [2]: (K3) is a *mirror* (or dual-averaging) step with step size η/L on a μ-strongly convex surrogate, namely $z_{t+1} = \arg\min_z\{\langle g_t, z\rangle + \frac{L}{2\eta}\|z - z_t\|^2 + \frac{\mu}{2}\|z - x_t\|^2\}$ (set the gradient to zero: $g_t + \frac{L}{\eta}(z - z_t) + \mu(z - x_t) = 0$, multiply by η/L and use $\eta\mu/L = \eta\sigma$); (K4) is a *gradient* step from x_t with step $\theta_1\eta/L$ up to a term of order σ: indeed by (K3), $y_{t+1} = x_t - \frac{\theta_1\eta}{L(1+\eta\sigma)} g_t + \frac{\theta_1\eta\sigma}{1+\eta\sigma}(x_t - z_t)$; and (K1) is the coupling. The term $\theta_2 w_t$ in (K1) is the *negative momentum* ("Katyusha momentum") of [**?**]: the query point is pulled towards the point where the estimator has zero variance.

Parameters of the paper. Theorem 4.6 of [21] uses

$$\theta_2 = \tfrac{1}{2}, \qquad \theta_1 = \min\Big\{\sqrt{\tfrac{2\sigma n}{3}},\, \tfrac{1}{2}\Big\} = \min\Big\{\sqrt{\tfrac{2n}{3\kappa}},\, \tfrac{1}{2}\Big\}, \qquad p = \tfrac{1}{n}, \qquad \eta = \frac{\theta_2}{(1+\theta_2)\theta_1} = \frac{1}{3\theta_1}. \tag{144}$$

With this choice the gradient step in (K4) has length $\theta_1\eta/L = 1/(3L)$ (up to the factor $1/(1+\eta\sigma)$), the same as in Katyusha and close to the $1/(4L)$ of SAGA in Prop. 5.9, while the mirror step $\eta/L = 1/(3\theta_1 L)$ is longer by the factor $1/\theta_1 = \max\{\sqrt{3\kappa/(2n)}, 2\}$: this is where the momentum lives.

The Lyapunov function. Equation (13) of [21] defines, verbatim in their notation, $\Psi^k = \mathcal{Z}^k + \mathcal{Y}^k + \mathcal{W}^k$ with $\mathcal{Z}^k = \frac{L(1+\eta\sigma)}{2\eta}\|z^k - x^*\|^2$, $\mathcal{Y}^k = \frac{1}{\theta_1}(f(y^k) - f(x^*))$, $\mathcal{W}^k = \frac{\theta_2(1+\theta_1)}{p\theta_1}(f(w^k) - f(x^*))$. In our notation:

$$\begin{aligned} &\Psi_t := Z_t + Y_t + W_t, \\ &Z_t := \frac{L(1+\eta\sigma)}{2\eta}\,\|z_t - \eta_*\|^2, \qquad Y_t := \frac{1}{\theta_1}\big(F(y_t) - F_*\big), \qquad W_t := \frac{\theta_2(1+\theta_1)}{p\theta_1}\big(F(w_t) - F_*\big). \end{aligned} \tag{145}$$

Compare with Prop. 5.9: $V_t = \|\theta_t - \eta_*\|^2 + \Delta\sum_i \Big\|z_i^{(t)} - f_i'(\eta_*)\Big\|^2$ is a distance plus a memory term; here Z_t is the distance (of the *mirror* sequence), Y_t a function value (of the *gradient* sequence, as in the potential of Exercise 5.14 of the notes for Nesterov's method) and W_t the memory term: it measures how good the reference point is, and it is the loopless substitute for the "aggregation across the inner loop" of the original Katyusha proof.

Theorem 5.4 (L-Katyusha; Lemma 4.5 and Theorem 4.6 of [21])**.** *Under Assumption 5.1, for any $\theta_1, \theta_2 > 0$ with $\theta_1 + \theta_2 \le 1$, any $p \in (0,1]$, $\sigma = \mu/L$ and η as in* (143)*, the iterates of* (K1)–(K5) *satisfy, for all $t \ge 0$,*

$$\mathbb{E}_{i(t)}\big[\Psi_{t+1}\big] \;\le\; \frac{1}{1+\eta\sigma}\,Z_t + \big(1-\theta_1(1-\theta_2)\big)\,Y_t + \Big(1 - \frac{p\theta_1}{1+\theta_1}\Big)\,W_t \qquad \textit{(one-step contraction)}. \tag{146}$$

With the parameters (144)*, for all $t \ge 0$,*

$$\mathbb{E}\big[\Psi_t\big] \le (1-\rho)^t\,\Psi_0, \qquad \rho := \min\Big\{\frac{\sigma}{6\theta_1}, \frac{\theta_1}{2n}\Big\} = \begin{cases} \dfrac{1}{\sqrt{24\,n\kappa}} & \text{if } \kappa \ge \frac{8n}{3}, \\ \min\Big\{\dfrac{1}{3\kappa}, \dfrac{1}{4n}\Big\} \;\Big(\ge \dfrac{1}{8n}\Big) & \text{if } \kappa < \frac{8n}{3}, \end{cases} \tag{147}$$

*so that in all cases $\rho \ge \min\{1/(4n), 1/\sqrt{24n\kappa}\}$ (*global rate*).*

The proof occupies Section 5.2. The complexity in gradient evaluations is derived in Corollary 5.12: $\mathcal{O}\big((n+\sqrt{n\kappa})\log(1/\varepsilon)\big)$, to be compared with $(3n + \frac{16}{3}\kappa)\log(2n/\varepsilon)$ for SAGA (Prop. 5.9) and with $n\sqrt{\kappa}\log(1/\varepsilon)$ for Nesterov's method of Section 5.2.5 applied to the full gradient: the condition number enters through $\sqrt{n\kappa}$ instead of κ (SAGA) or $n\sqrt{\kappa}$ (accelerated gradient descent).

Remark 5.5 (relation with the original Katyusha and with the notes). Theorem 2.1 of [**?**] states (their notation: σ is our μ, m the epoch length, and the objective of their Problem (1.1) is $F = \frac{1}{n}\sum_i f_i + \psi$ with a proximal term ψ) that "if each $f_i(x)$ is convex, L-smooth, and $\psi(x)$ is σ-strongly convex in Problem (1.1)", then Katyusha satisfies $\mathbb{E}[F(\tilde{x}^S)] - F(x^*) \le \mathcal{O}\big((1+\sqrt{\sigma/(3Lm)})^{-Sm}\big)(F(\tilde{x}^0) - F(x^*))$ when $m\sigma/L \le 3/4$ and $\mathcal{O}(1.5^{-S})(F(\tilde{x}^0) - F(x^*))$ otherwise, "choosing $m = \Theta(n)$, Katyusha achieves an ε-additive error using at most $\mathcal{O}\big((n+\sqrt{nL/\sigma})\log\frac{F(\tilde{x}^0)-F(x^*)}{\varepsilon}\big)$ iterations". The strong convexity

is thus assumed on the proximal term ψ (for ℓ_2-regularized logistic regression one can take $\psi = \frac{\lambda}{2}\|\cdot\|^2$), whereas L-Katyusha has no proximal term and assumes it on F (Assumption 1.2 of [21]), as in Assumption 5.1. The loopless version has the same complexity with a proof that needs a single iteration (no telescoping over an epoch). Allen-Zhu also explains the role of the three sequences: "In a classical accelerated gradient method, x^{k+1} is only a convex combination of y^k and z^k (or equivalently, $\tau_2 = 0$ in our formulation)"; we make this precise in Section 5.4 where we show that (K1)–(K4) with $\theta_2 = 0$ and exact gradients is an accelerated gradient method with the rate $(1 - \frac{1}{2}\sqrt{\mu/L})^t$, i.e. the method of eq. (5.16)–(5.17) of the notes up to the choice of constants.

5.2 The Lyapunov proof

Lemma 5.6 (L1: unbiasedness and variance of the estimator; Lemma 4.1 of [21])**.** *For all t, $\mathbb{E}_{i(t)}[g_t] = F'(x_t)$ and*

$$\mathbb{E}_{i(t)}\left\|g_t - F'(x_t)\right\|^2 = \frac{1}{n}\sum_{i=1}^n \left\|f_i'(x_t) - f_i'(w_t)\right\|^2 - \left\|F'(x_t) - F'(w_t)\right\|^2 \;\le\; \frac{1}{n}\sum_{i=1}^n \left\|f_i'(x_t) - f_i'(w_t)\right\|^2 \tag{148}$$

$$\le\; 2L\left(F(w_t) - F(x_t) - \left\langle F'(x_t), w_t - x_t\right\rangle\right) \;=\; 2L\, D_F(w_t, x_t). \tag{149}$$

We call (148) *the* centering inequality *and* (149) *the* co-coercivity bound*; their product is the* variance bound

$$\mathbb{E}_{i(t)}\left\|g_t - F'(x_t)\right\|^2 \le 2L\, D_F(w_t, x_t). \tag{150}$$

Proof. Given the state, x_t and w_t are fixed and $X := f_{i(t)}'(x_t) - f_{i(t)}'(w_t)$ is a random vector with $\mathbb{E}_{i(t)}X = F'(x_t) - F'(w_t)$, so that $g_t - F'(x_t) = X - \mathbb{E}_{i(t)}X$ (unbiasedness). By (T7), $\mathbb{E}_{i(t)}\left\|X - \mathbb{E}_{i(t)}X\right\|^2 = \mathbb{E}_{i(t)}\|X\|^2 - \left\|\mathbb{E}_{i(t)}X\right\|^2$, and $\mathbb{E}_{i(t)}\|X\|^2 = \frac{1}{n}\sum_i \|f_i'(x_t) - f_i'(w_t)\|^2$ because $i(t)$ is uniform: this is (148). For (149), apply (T3) to each convex L-smooth f_i with $\theta = w_t$, $\eta = x_t$: $\|f_i'(w_t) - f_i'(x_t)\|^2 \le 2L\big(f_i(w_t) - f_i(x_t) - \langle f_i'(x_t), w_t - x_t\rangle\big)$. Average over i and use $\frac{1}{n}\sum_i f_i = F$, $\frac{1}{n}\sum_i f_i' = F'$. This is the same use of co-coercivity as in eq. (5.31) of the notes (there with $\eta = \eta_*$, which gives the inner-product form $R^2\, F'(\theta)^\top(\theta - \eta_*)$). Here we keep the one-sided (Bregman) form (5.12): it is not stronger than the symmetric form (5.11) pointwise, since (5.11) follows from (5.12) by symmetrization, but it is the form that produces the divergence $D_F(w_t, x_t)$, which is exactly what the anchor term cancels in Step 6 of Lemma 5.11. □

Lemma 5.7 (L2: the mirror step; Lemma 4.2 of [21])**.** *For every realization of $i(t)$ (no expectation is needed),*

$$\langle g_t, \eta_* - z_{t+1}\rangle + \frac{\mu}{2}\|x_t - \eta_*\|^2 \;\ge\; \frac{L}{2\eta}\|z_t - z_{t+1}\|^2 + Z_{t+1} - \frac{1}{1+\eta\sigma}Z_t \qquad (\text{mirror-step inequality}), \tag{151}$$

and the slack is exactly $\frac{\mu}{2}\|x_t - z_{t+1}\|^2$ (i.e. (151) *becomes an identity if this term is added to the right-hand side).*

Proof. Multiplying (K3) by $1 + \eta\sigma$ gives $(1+\eta\sigma)z_{t+1} = \eta\sigma x_t + z_t - \frac{\eta}{L}g_t$, i.e.

$$\frac{\eta}{L}\, g_t = \eta\sigma\,(x_t - z_{t+1}) + (z_t - z_{t+1}).$$

Take the inner product with $z_{t+1} - \eta_*$ and multiply by L/η, using $L\sigma = \mu$:

$$\langle g_t, z_{t+1} - \eta_*\rangle = \mu\,\langle x_t - z_{t+1}, z_{t+1} - \eta_*\rangle + \frac{L}{\eta}\,\langle z_t - z_{t+1}, z_{t+1} - \eta_*\rangle\,.$$

Apply (T5) twice, with $(a,b,c)=(x_t,z_{t+1},\eta_*)$ and $(a,b,c)=(z_t,z_{t+1},\eta_*)$:

$$\begin{aligned}\langle g_t, z_{t+1}-\eta_*\rangle &= \frac{\mu}{2}\Big(\|x_t-\eta_*\|^2-\|x_t-z_{t+1}\|^2-\|z_{t+1}-\eta_*\|^2\Big)\\ &\quad+\frac{L}{2\eta}\Big(\|z_t-\eta_*\|^2-\|z_t-z_{t+1}\|^2-\|z_{t+1}-\eta_*\|^2\Big)\\ &=\frac{\mu}{2}\|x_t-\eta_*\|^2-\frac{\mu}{2}\|x_t-z_{t+1}\|^2+\frac{L}{2\eta}\|z_t-\eta_*\|^2\\ &\quad-\frac{L}{2\eta}(1+\eta\sigma)\|z_{t+1}-\eta_*\|^2-\frac{L}{2\eta}\|z_t-z_{t+1}\|^2,\end{aligned}$$

where we grouped $\frac{\mu}{2}+\frac{L}{2\eta}=\frac{L}{2\eta}(\eta\sigma+1)$. By definition (145), $\frac{L}{2\eta}(1+\eta\sigma)\|z_{t+1}-\eta_*\|^2=Z_{t+1}$ and $\frac{L}{2\eta}\|z_t-\eta_*\|^2=\frac{Z_t}{1+\eta\sigma}$. Moving $\langle g_t,z_{t+1}-\eta_*\rangle$ to the right and $\frac{\mu}{2}\|x_t-\eta_*\|^2$ to the left gives the claimed identity with the slack $\frac{\mu}{2}\|x_t-z_{t+1}\|^2\ge 0$; dropping it gives (151). □

Lemma 5.8 (L3: the gradient step; Lemma 4.3 of [21])**.** *For every realization of $i(t)$, the* gradient-step inequality *holds:*

$$\frac{1}{\theta_1}\Big(F(y_{t+1})-F(x_t)\Big)-\frac{\theta_2}{2L\theta_1}\big\|g_t-F'(x_t)\big\|^2\ \le\ \frac{L}{2\eta}\|z_{t+1}-z_t\|^2+\langle g_t,z_{t+1}-z_t\rangle. \tag{152}$$

It is the product of two inequalities, the smoothness bound

$$F(y_{t+1})\le F(x_t)+\big\langle F'(x_t),y_{t+1}-x_t\big\rangle+\frac{L}{2}\|y_{t+1}-x_t\|^2, \tag{153}$$

and the Young bound *on the cross term between the noise of the estimator and the step,*

$$\big\langle g_t-F'(x_t),y_{t+1}-x_t\big\rangle\ \ge\ -\frac{\theta_2}{2L}\big\|g_t-F'(x_t)\big\|^2-\frac{L}{2\theta_2}\|y_{t+1}-x_t\|^2. \tag{154}$$

Proof. By (K4), $y_{t+1}-x_t=\theta_1(z_{t+1}-z_t)$, hence

$$\frac{L}{2\eta}\|z_{t+1}-z_t\|^2+\langle g_t,z_{t+1}-z_t\rangle=\frac{1}{\theta_1}\Big(\frac{L}{2\eta\theta_1}\|y_{t+1}-x_t\|^2+\langle g_t,y_{t+1}-x_t\rangle\Big).$$

Write $g_t=F'(x_t)+(g_t-F'(x_t))$ and split the quadratic term as $\frac{L}{2\eta\theta_1}=\frac{L}{2}+\frac{L}{2}\big(\frac{1}{\eta\theta_1}-1\big)$. By the choice (143), $\eta\theta_1=\frac{\theta_2}{1+\theta_2}<1$ and $\frac{1}{\eta\theta_1}-1=\frac{1}{\theta_2}$. Therefore the bracket equals

$$\Big[\frac{L}{2}\|y_{t+1}-x_t\|^2+\big\langle F'(x_t),y_{t+1}-x_t\big\rangle\Big]+\frac{L}{2\theta_2}\|y_{t+1}-x_t\|^2+\big\langle g_t-F'(x_t),y_{t+1}-x_t\big\rangle.$$

The first bracket is at least $F(y_{t+1})-F(x_t)$ by (T2) applied to F ($\theta=y_{t+1}$, $\eta=x_t$), which is (153). The last two terms are at least $-\frac{\theta_2}{2L}\|g_t-F'(x_t)\|^2$ by (T6) with $\beta=L/\theta_2$, $a=g_t-F'(x_t)$, $b=y_{t+1}-x_t$, which is (154). Dividing by θ_1 gives (152). □

Remark 5.9 (the step size couples the two momenta). In the proof of L3 the coefficient of the variance term is $\frac{1}{2L}\cdot\frac{\eta\theta_1}{1-\eta\theta_1}$, a function of the step only. The choice (143) is precisely the one which makes this coefficient equal to $\frac{\theta_2}{2L}$, with the *same* θ_2 as the weight of w_t in (K1). This is what allows the exact cancellation in L5 below. For a general step we write $\nu:=\frac{\eta\theta_1}{1-\eta\theta_1}$ (so $\nu=\theta_2$ for L-Katyusha); L3 then holds with θ_2 replaced by ν in the variance term. We use this in Section 5.4.

Lemma 5.10 (L4: the reference point; Lemma 4.4 of [21]). *For all t,*

$$\mathbb{E}_{i(t)}\big[W_{t+1}\big] = (1-p)\,W_t + \theta_2(1+\theta_1)\,Y_t. \tag{155}$$

Proof. By (K5) and independence of the coin from $i(t)$, $\mathbb{E}_{i(t)}[F(w_{t+1})] = (1-p)F(w_t) + pF(y_t)$. Subtract F_* and multiply by $\frac{\theta_2(1+\theta_1)}{p\theta_1}$: the first term gives $(1-p)W_t$ and the second $\frac{\theta_2(1+\theta_1)}{\theta_1}(F(y_t) - F_*) = \theta_2(1+\theta_1)Y_t$. □

Lemma 5.11 (L5: combination; Lemma 4.5 of [21]). *Under the assumptions of Theorem 5.4,*

$$\mathbb{E}_{i(t)}\big[Z_{t+1} + Y_{t+1}\big] \le \frac{1}{1+\eta\sigma} Z_t + (1-\theta_1-\theta_2)\,Y_t + \frac{p}{1+\theta_1}\,W_t, \tag{156}$$

and consequently the one-step contraction (146) *holds.*

Proof. Step 1 (strong convexity at the query point). By (T4) with $\theta = \eta_*$ and $\eta = x_t$,

$$F_* \;\ge\; F(x_t) + \big\langle F'(x_t), \eta_* - x_t\big\rangle + \frac{\mu}{2}\,\|x_t - \eta_*\|^2 \qquad (\textit{strong convexity at } x_t). \tag{157}$$

Step 2 (decomposition through the coupling). By (K1), $\theta_2 w_t + (1-\theta_1-\theta_2)y_t = x_t - \theta_1 z_t$, hence

$$\theta_2(x_t - w_t) + (1-\theta_1-\theta_2)(x_t - y_t) = (1-\theta_1)x_t - (x_t - \theta_1 z_t) = \theta_1(z_t - x_t),$$

that is

$$z_t - x_t = \frac{\theta_2}{\theta_1}(x_t - w_t) + \frac{1-\theta_1-\theta_2}{\theta_1}(x_t - y_t). \tag{158}$$

Writing $\eta_* - x_t = (\eta_* - z_t) + (z_t - x_t)$ in (157) and using (158),

$$F_* \;\ge\; F(x_t) + \frac{\mu}{2}\,\|x_t - \eta_*\|^2 + \big\langle F'(x_t), \eta_* - z_t\big\rangle + \frac{\theta_2}{\theta_1}\,\big\langle F'(x_t), x_t - w_t\big\rangle + \frac{1-\theta_1-\theta_2}{\theta_1}\,\big\langle F'(x_t), x_t - y_t\big\rangle.$$

Step 3 (convexity towards y_t). Since $1-\theta_1-\theta_2 \ge 0$, we may use (T1) with $\theta = y_t$, $\eta = x_t$:

$$\big\langle F'(x_t), x_t - y_t\big\rangle \;\ge\; F(x_t) - F(y_t) \qquad (\textit{convexity towards } y_t). \tag{159}$$

Step 4 (unbiasedness). Since z_t is fixed given the state and $\mathbb{E}_{i(t)} g_t = F'(x_t)$,

$$\big\langle F'(x_t), \eta_* - z_t\big\rangle = \mathbb{E}_{i(t)}\,\langle g_t, \eta_* - z_t\rangle = \mathbb{E}_{i(t)}\Big[\,\langle g_t, \eta_* - z_{t+1}\rangle + \langle g_t, z_{t+1} - z_t\rangle\,\Big].$$

(The deterministic quantity $\frac{\mu}{2}\,\|x_t - \eta_*\|^2$ may be put inside $\mathbb{E}_{i(t)}$ as well.) Steps 1–4 give

$$\begin{aligned} F_* \;\ge\; & F(x_t) + \frac{\theta_2}{\theta_1}\,\big\langle F'(x_t), x_t - w_t\big\rangle + \frac{1-\theta_1-\theta_2}{\theta_1}\big(F(x_t) - F(y_t)\big) \\ & + \mathbb{E}_{i(t)}\Big[\frac{\mu}{2}\,\|x_t - \eta_*\|^2 + \langle g_t, \eta_* - z_{t+1}\rangle + \langle g_t, z_{t+1} - z_t\rangle\Big]. \end{aligned}$$

Step 5 (the three lemmas). Inside the expectation, L2 (151) bounds $\frac{\mu}{2}\,\|x_t - \eta_*\|^2 + \langle g_t, \eta_* - z_{t+1}\rangle$ from below by $\frac{L}{2\eta}\,\|z_t - z_{t+1}\|^2 + Z_{t+1} - \frac{Z_t}{1+\eta\sigma}$, then L3 (152) bounds $\frac{L}{2\eta}\,\|z_{t+1} - z_t\|^2 + \langle g_t, z_{t+1} - z_t\rangle$ from below by $\frac{1}{\theta_1}(F(y_{t+1}) - F(x_t)) - \frac{\theta_2}{2L\theta_1}\,\|g_t - F'(x_t)\|^2$; finally L1 (150) bounds $-\frac{\theta_2}{2L\theta_1}\mathbb{E}_{i(t)}\,\|g_t - F'(x_t)\|^2$ from below by $-\frac{\theta_2}{\theta_1} D_F(w_t, x_t)$. Altogether

$$\begin{aligned} F_* \;\ge\; & F(x_t) + \frac{\theta_2}{\theta_1}\,\big\langle F'(x_t), x_t - w_t\big\rangle + \frac{1-\theta_1-\theta_2}{\theta_1}\big(F(x_t) - F(y_t)\big) + \mathbb{E}_{i(t)}[Z_{t+1}] - \frac{Z_t}{1+\eta\sigma} \\ & + \frac{1}{\theta_1}\mathbb{E}_{i(t)}\big[F(y_{t+1}) - F(x_t)\big] - \frac{\theta_2}{\theta_1}\Big(F(w_t) - F(x_t) - \big\langle F'(x_t), w_t - x_t\big\rangle\Big). \end{aligned}$$

Step 6 (the cancellation: this is where the negative momentum acts). The inner products in w_t cancel *exactly*:

$$\underbrace{\frac{\theta_2}{\theta_1}\langle F'(x_t), x_t - w_t\rangle}_{\text{from } \theta_2 w_t \text{ in (K1), Step 2}} - \underbrace{\frac{\theta_2}{\theta_1}\Big(F(w_t) - F(x_t) - \langle F'(x_t), w_t - x_t\rangle\Big)}_{\text{variance term, L1 and L3}} = -\frac{\theta_2}{\theta_1}\big(F(w_t) - F(x_t)\big), \quad (160)$$

because $-\langle F'(x_t), w_t - x_t\rangle = \langle F'(x_t), x_t - w_t\rangle$ enters with the sign $-\frac{\theta_2}{\theta_1}$. What survives of the variance term is a *difference of function values* only.
Step 7 (bookkeeping). Collect the coefficients of $F(x_t)$ after (160): they sum to $1+\frac{\theta_2}{\theta_1}+\frac{1-\theta_1-\theta_2}{\theta_1}-\frac{1}{\theta_1} = 0$. Hence

$$F_* \;\geq\; -\frac{\theta_2}{\theta_1}F(w_t) - \frac{1-\theta_1-\theta_2}{\theta_1}F(y_t) + \mathbb{E}_{i(t)}[Z_{t+1}] - \frac{Z_t}{1+\eta\sigma} + \frac{1}{\theta_1}\mathbb{E}_{i(t)}[F(y_{t+1})].$$

Since $\frac{1}{\theta_1} - \frac{\theta_2}{\theta_1} - \frac{1-\theta_1-\theta_2}{\theta_1} = 1$, we may write $F_* = \frac{1}{\theta_1}F_* - \frac{\theta_2}{\theta_1}F_* - \frac{1-\theta_1-\theta_2}{\theta_1}F_*$ and rearrange:

$$\mathbb{E}_{i(t)}[Z_{t+1}] + \frac{1}{\theta_1}\mathbb{E}_{i(t)}\big[F(y_{t+1}) - F_*\big] \leq \frac{Z_t}{1+\eta\sigma} + \frac{1-\theta_1-\theta_2}{\theta_1}\big(F(y_t) - F_*\big) + \frac{\theta_2}{\theta_1}\big(F(w_t) - F_*\big).$$

By (145), $\frac{1-\theta_1-\theta_2}{\theta_1}(F(y_t) - F_*) = (1-\theta_1-\theta_2)Y_t$ and $\frac{\theta_2}{\theta_1}(F(w_t) - F_*) = \frac{p}{1+\theta_1}W_t$: this is (156).
Step 8 (add the reference-point term). Adding (155) to (156): the coefficient of Y_t becomes $1-\theta_1-\theta_2+\theta_2(1+\theta_1) = 1-\theta_1(1-\theta_2)$ and the coefficient of W_t becomes $\frac{p}{1+\theta_1} + 1 - p = 1 - \frac{p\theta_1}{1+\theta_1}$, which is (146). □

Proof of Theorem 5.4. (146) is Lemma 5.11. For the rate, (146) gives $\mathbb{E}_{i(t)}[\Psi_{t+1}] \leq \max\{c_Z, c_Y, c_W\}\Psi_t$ with $c_Z = \frac{1}{1+\eta\sigma}$, $c_Y = 1-\theta_1(1-\theta_2)$, $c_W = 1 - \frac{p\theta_1}{1+\theta_1}$, and by the tower property $\mathbb{E}[\Psi_{t+1}] \leq \max\{c_Z, c_Y, c_W\}\mathbb{E}[\Psi_t]$. With the parameters (144):

- $c_Z \leq 1 - \frac{\sigma}{6\theta_1}$. Indeed $\eta\sigma = \frac{\sigma}{3\theta_1} \leq 1$: if $\theta_1 = \frac{1}{2}$ then $\eta\sigma = \frac{2\sigma}{3} \leq \frac{2}{3}$; if $\theta_1 = \sqrt{2\sigma n/3}$ then $\eta\sigma = \frac{\sigma}{3\sqrt{2\sigma n/3}} = \sqrt{\frac{\sigma}{6n}} \leq 1$. And for $x \in [0,1]$, $\frac{1}{1+x} \leq 1 - \frac{x}{2}$ because $(1-\frac{x}{2})(1+x) = 1 + \frac{x}{2} - \frac{x^2}{2} \geq 1$.
- $c_Y = 1 - \frac{\theta_1}{2}$ since $\theta_2 = \frac{1}{2}$.
- $c_W \leq 1 - \frac{\theta_1}{2n}$: with $p = \frac{1}{n}$ and $\theta_1 \leq \frac{1}{2}$, $\frac{p\theta_1}{1+\theta_1} = \frac{\theta_1}{n(1+\theta_1)} \geq \frac{2\theta_1}{3n} \geq \frac{\theta_1}{2n}$.

Hence $\max\{c_Z, c_Y, c_W\} \leq 1 - \min\{\frac{\sigma}{6\theta_1}, \frac{\theta_1}{2}, \frac{\theta_1}{2n}\} = 1 - \rho$ with $\rho = \min\{\frac{\sigma}{6\theta_1}, \frac{\theta_1}{2n}\}$, which is the θ of the proof of Theorem 4.6 of [21] ("$\theta = \min\{\sigma/6\theta_1, \theta_1/2n\}$"). Explicitly, with $\sigma = 1/\kappa$: if $\kappa \geq \frac{8n}{3}$ then $\frac{2n}{3\kappa} \leq \frac{1}{4}$, $\theta_1 = \sqrt{2n/(3\kappa)}$, and $\frac{\sigma}{6\theta_1} = \frac{1}{6\kappa}\sqrt{\frac{3\kappa}{2n}} = \frac{1}{\sqrt{24n\kappa}}$, $\frac{\theta_1}{2n} = \frac{1}{2n}\sqrt{\frac{2n}{3\kappa}} = \frac{1}{\sqrt{6n\kappa}}$, so $\rho = \frac{1}{\sqrt{24n\kappa}}$; if $\kappa < \frac{8n}{3}$ then $\theta_1 = \frac{1}{2}$ and $\rho = \min\{\frac{1}{3\kappa}, \frac{1}{4n}\} \geq \frac{1}{8n}$. Finally $\frac{1}{3\kappa} \geq \frac{1}{\sqrt{24n\kappa}}$ iff $9\kappa^2 \leq 24n\kappa$ iff $\kappa \leq \frac{8n}{3}$, which gives the unified bound $\rho \geq \min\{\frac{1}{4n}, \frac{1}{\sqrt{24n\kappa}}\}$. Iterating, $\mathbb{E}[\Psi_t] \leq (1-\rho)^t\Psi_0$. □

Corollary 5.12 (gradient complexity)**.** *With the parameters* (144) *and* $y_0 = z_0 = w_0 = \theta_0$*, for all* $t \geq 0$*,*

$$\mathbb{E}\big[F(y_t) - F_*\big] \leq \theta_1(1-\rho)^t\Psi_0, \qquad \mathbb{E}\,\|z_t - \eta_*\|^2 \leq \frac{2\eta}{L(1+\eta\sigma)}(1-\rho)^t\Psi_0,$$
$$\theta_1\Psi_0 \leq \frac{3L(n+3)}{8}\,\|\theta_0 - \eta_*\|^2. \quad (161)$$

Consequently $\mathbb{E}[F(y_t) - F_*] \le \varepsilon$ *as soon as* $t \ge \big(4n + \sqrt{24n\kappa}\big) \log \frac{3L(n+3)\|\theta_0-\eta_*\|^2}{8\varepsilon}$. *An iteration evaluates* $f'_{i(t)}(x_t)$ *and* $f'_{i(t)}(w_t)$ *in* (K2) *and, with probability* $p = 1/n$, *the* n *gradients of a full pass:* $2 + pn = 3$ *evaluations of some* f'_i *in expectation. If the* n *gradients* $f'_i(w)$ *computed during the pass are stored* ($\mathcal{O}(nd)$ *memory, as for the table of SAGA),* $f'_{i(t)}(w_t)$ *is a look-up and the count is* $1 + pn = 2$ *(this is the "*$\mathcal{O}(1 + pn)$ *times per iteration" of [21]). Adding the* n *evaluations of* $F'(w_0)$ *at initialization, the expected number of gradient evaluations is at most*

$$\begin{aligned} n + \big(12\,n + 3\sqrt{24}\,\sqrt{n\kappa}\big) \log \frac{3L(n+3)\,\|\theta_0-\eta_*\|^2}{8\varepsilon} \;&\le\; n + \big(12\,n + 15\sqrt{n\kappa}\big) \log \frac{3L(n+3)\,\|\theta_0-\eta_*\|^2}{8\varepsilon} \quad \textit{(without stora} \\ n + \big(8\,n + 2\sqrt{24}\,\sqrt{n\kappa}\big) \log \frac{3L(n+3)\,\|\theta_0-\eta_*\|^2}{8\varepsilon} \;&\le\; n + \big(8\,n + 10\sqrt{n\kappa}\big) \log \frac{3L(n+3)\,\|\theta_0-\eta_*\|^2}{8\varepsilon} \quad \textit{(with storage),} \end{aligned} \tag{162}$$

and in both cases $\mathcal{O}\big((n + \sqrt{n\kappa}) \log \frac{nL\|\theta_0-\eta_*\|^2}{\varepsilon}\big)$.

Proof. The first two bounds follow from $Y_t \le \Psi_t$, $Z_t \le \Psi_t$ and (147). For Ψ_0: $Z_0 = \frac{L(1+\eta\sigma)}{2\eta}\|\theta_0 - \eta_*\|^2 = \frac{3\theta_1 L(1+\eta\sigma)}{2}\|\theta_0 - \eta_*\|^2$ since $1/\eta = 3\theta_1$; and $Y_0 + W_0 = \big(\frac{1}{\theta_1} + \frac{n(1+\theta_1)}{2\theta_1}\big)(F(\theta_0) - F_*) \le \frac{2+n(1+\theta_1)}{2\theta_1} \cdot \frac{L}{2}\|\theta_0 - \eta_*\|^2$ by (T2) at $\eta = \eta_*$ ($F'(\eta_*) = 0$). Using $\eta\sigma \le \frac{2}{3}$, $\theta_1 \le \frac{1}{2}$ and $1 \le \frac{1}{2\theta_1}$: $\Psi_0 \le \frac{L}{2}\big(\frac{5}{2} + \frac{2+3n/2}{2\theta_1}\big)\|\theta_0 - \eta_*\|^2 \le \frac{L}{2} \cdot \frac{9/2+3n/2}{2\theta_1}\|\theta_0 - \eta_*\|^2 = \frac{3L(n+3)}{8\theta_1}\|\theta_0 - \eta_*\|^2$. For the iteration count use $(1-\rho)^t \le e^{-\rho t}$ and $1/\rho \le \max\{4n, \sqrt{24n\kappa}\} \le 4n + \sqrt{24n\kappa}$. □

Remark 5.13 (reading of the rate). Per iteration, Ψ contracts by $1 - \rho$ with $\rho = \min\{\frac{\sigma}{6\theta_1}, \frac{\theta_1}{2n}\}$: the first term is the contraction of the mirror sequence (a strongly convex mirror step with the long step $\eta/L = 1/(3\theta_1 L)$ contracts the distance by $\frac{1}{1+\eta\sigma} \approx 1 - \frac{\sigma}{3\theta_1}$), the second the contraction of the reference term (w is refreshed with probability $p = 1/n$ and Y decreases by the fraction $\theta_1(1-\theta_2) = \theta_1/2$). Balancing the two gives $\theta_1 \asymp \sqrt{\sigma n}$ and $\rho \asymp \sqrt{\sigma/n} = 1/\sqrt{n\kappa}$: this is the $\sqrt{\kappa}$ scaling of hypothesis (A4). For SAGA (Prop. 5.9) the analogous balance is $\min\{\frac{1}{3n}, \frac{3\mu}{16R^2}\}$: the "parameter" contraction is $\asymp \sigma$ because the step is $\asymp 1/L$; here the mirror step is $\asymp 1/(\theta_1 L) \gg 1/L$ and the price is paid by the variance, which is cancelled by the negative momentum (Step 6) instead of being absorbed by the progress term $-2\gamma(1-\gamma R^2)F'(\theta)^\top(\theta - \eta_*)$ as in Step 1 of the proof of Prop. 5.9.

5.3 Where the negative momentum acts (A3)

The proof of Lemma 5.11 contains a single place where θ_2 is used for something else than bookkeeping: Step 6, equation (160). Let us isolate it.

Proposition 5.14 (the Bregman-gap cancellation)**.** *In the chain of Lemma 5.11, the variance of the estimator enters through L3 and L1 as the term* $-\frac{\theta_2}{\theta_1} D_F(w_t, x_t)$ *with* $D_F(w_t, x_t) = \big(F(w_t) - F(x_t)\big) + \langle F'(x_t), x_t - w_t\rangle$, *and the weight* θ_2 *of* w_t *in* (K1) *enters, through the decomposition* (158), *as the term* $+\frac{\theta_2}{\theta_1}\langle F'(x_t), x_t - w_t\rangle$. *The two inner products cancel identically, and the remainder* $-\frac{\theta_2}{\theta_1}(F(w_t) - F(x_t))$ *is absorbed as follows:* $F(x_t)$ *disappears because the coefficients of* $F(x_t)$ *sum to zero (Step 7), and* $\frac{\theta_2}{\theta_1}(F(w_t) - F_*) = \frac{p}{1+\theta_1} W_t$ *is paid by the reference-point potential, which contracts by* (155) *because* w *is refreshed with probability* p.

Proof. This is Steps 5–8 of the proof of Lemma 5.11; the identity $D_F(w_t, x_t) = (F(w_t) - F(x_t)) + \langle F'(x_t), x_t - w_t\rangle$ is the definition of D_F. □

Three comments make the mechanism explicit.

1. *Why an exact cancellation is needed.* The variance term is not small: by (T2), $D_F(w_t, x_t) \le \frac{L}{2}\|w_t - x_t\|^2$ is a squared distance at the scale L, whereas the decrease of the potential per step is $\rho\Psi_t$ with $\rho \asymp 1/\sqrt{n\kappa}$. To absorb a term $\asymp L\|w_t - x_t\|^2$ into $\rho\Psi_t$ one would have to convert a squared distance into function values, which by strong convexity costs the factor κ (we quantify this in Proposition 5.19). The cancellation avoids any such conversion: the inner-product part of D_F is removed identically, and the function-value part (a signed difference of function values; see the discussion of $c(t)$ in Table 5) is of the same nature as Y_t and W_t.

2. *What the negative momentum costs.* Because $\theta_2 w_t$ takes a fraction θ_2 of the weight in (K1), the function-value potential Y_t contracts by $\theta_1(1-\theta_2)$ instead of θ_1 (compare (146) with the deterministic Proposition 5.17 below), and the reference-point potential must carry the weight $\frac{\theta_2(1+\theta_1)}{p\theta_1} \asymp n/\theta_1$ in front of $F(w_t) - F_*$. With $\theta_2 = \frac{1}{2}$ this halves the momentum contraction, a constant factor.

3. *Why the step size is tied to θ_2.* By Remark 5.9, the coefficient of $\|g_t - F'(x_t)\|^2$ produced by Young's inequality in L3 is $\frac{\nu}{2L\theta_1}$ with $\nu = \frac{\eta\theta_1}{1-\eta\theta_1}$. For the cancellation (160) to be exact one needs $\nu = \theta_2$, i.e. (143). A larger step ($\nu > \theta_2$) leaves a residual $+\frac{\nu-\theta_2}{\theta_1} D_F(w_t, x_t)$ on the right-hand side of (156), which is exactly the uncontrolled term of the naive scheme below (with $\nu - \theta_2$ in place of ν). A smaller step ($\nu < \theta_2$) wastes part of the pull.

5.4 (A2) Naive acceleration: Nesterov's momentum with the SVRG estimator

Definition 5.15 (naive accelerated SVRG)**.** *The* naive scheme $\mathrm{N}(\theta_1, \eta, p)$ *is* (K1)–(K5) *with* $\theta_2 = 0$ *and the step* $\eta > 0$ *taken as a free parameter with* $\eta\theta_1 \le 1$*:*

$$\begin{aligned} x_t &= \theta_1 z_t + (1-\theta_1) y_t, \qquad g_t \text{ as in (K2)}, \qquad z_{t+1} = \frac{\eta\sigma x_t + z_t - \frac{\eta}{L} g_t}{1+\eta\sigma}, \\ y_{t+1} &= x_t + \theta_1(z_{t+1} - z_t), \qquad w_{t+1} = \begin{cases} y_t & w.p.\ p, \\ w_t & w.p.\ 1-p. \end{cases} \end{aligned} \tag{163}$$

We set $\nu := \frac{\eta\theta_1}{1-\eta\theta_1} \in (0, \infty]$*, with* $\nu = \infty$ *in the limiting case* $\eta\theta_1 = 1$*; this case is admissible only with exact gradients* $g_t = F'(x_t)$ *(Proposition 5.17), because the Young step of L3 requires* $\nu < \infty$ *(Remark 5.9). The potential is the same* Ψ_t *as in* (145) *(with* W_t *defined with an arbitrary constant* $c_W \ge 0$ *in place of* $\frac{\theta_2(1+\theta_1)}{p\theta_1}$*, which is* 0 *when* $\theta_2 = 0$*).*

Remark 5.16 (this is Nesterov's method of the notes with F' replaced by g_t). Eliminating z from (163): $y_{t+1} = \theta_1 z_{t+1} + (1-\theta_1) y_t$ and $x_{t+1} = \theta_1 z_{t+1} + (1-\theta_1) y_{t+1}$ give $x_{t+1} = y_{t+1} + (1-\theta_1)(y_{t+1} - y_t)$, which is the extrapolation (5.17) of the notes, $\eta_t = \theta_t + \beta(\theta_t - \theta_{t-1})$, with $\beta = 1 - \theta_1$. The notes take $\beta = \frac{1-\sqrt{\mu/L}}{1+\sqrt{\mu/L}} = 1 - 2\sqrt{\mu/L} + \mathcal{O}(\mu/L)$: with $\theta_1 = \sqrt{\mu/L}$ the two momentum coefficients agree at zeroth order (both are $1 - \Theta(\sqrt{\mu/L})$), but the momentum deficits differ by a factor 2 ($1 - \beta = \sqrt{\mu/L}$ here, against $2\sqrt{\mu/L} + \mathcal{O}(\mu/L)$ in the notes). The gradient step reads $y_{t+1} = x_t - \frac{\theta_1\eta}{L(1+\eta\sigma)} g_t + \frac{\theta_1\eta\sigma}{1+\eta\sigma}(x_t - z_t)$: it is (5.16) with $F'(\eta_{t-1})$ replaced by g_t, step $\frac{\theta_1\eta}{L(1+\eta\sigma)}$, and a drift which comes from putting the strong convexity in the mirror step. Since $x_t - z_t = -\frac{1-\theta_1}{\theta_1}(x_t - y_t)$, the drift equals $-\frac{\eta\sigma(1-\theta_1)}{1+\eta\sigma}(x_t - y_t)$, i.e. it is of relative size $\eta\sigma$ with respect to the momentum term $x_t - y_t = (1-\theta_1)(y_t - y_{t-1})$, where $\eta\sigma = \sqrt{\sigma}$ for the parameters of Proposition 5.17 and $\eta\sigma = \frac{\sigma}{3\theta_1}$ for (144) ($= \sqrt{\sigma/(6n)}$ when $\kappa \ge 8n/3$). The drift is therefore small compared with the momentum term, but of the same order $\mathcal{O}(\sqrt{\sigma})$ as the discrepancy between the two momentum coefficients,

which is the order that governs the accelerated rate: the two-sequence form of the notes and (163) coincide only up to $\mathcal{O}(\sqrt{\sigma})$ corrections of the momentum term. (The drift disappears when the strong convexity is carried by a proximal term instead of the pull towards x_t, as in Algorithm 1 of [9], whose z-step is $\operatorname{prox}_{\gamma g}(z_t - \gamma g_t)$.) The same scheme with g_t replaced by a mini-batch SVRG estimator is the method of Nitanda [27]; Allen-Zhu writes that it "corresponds to this paper but *without* Katyusha momentum (i.e., $\tau_2 = 0$)" [?, Appendix A], and Driggs et al. write their scheme in the same two-sequence form ("$x_{k+1} = y_k + (1-\tau_k)(y_k - y_{k-1})$") [9, Section 2].

The deterministic case closes. First we check that the proof machinery of Section 5.2 does prove acceleration for the scheme with $\theta_2 = 0$ when the gradient is exact; this is the analogue of Exercise 5.14 of the notes in the linear-coupling form of [2].

Proposition 5.17 (accelerated gradient descent via L2–L3 with $\theta_2 = 0$)**.** *Let $g_t = F'(x_t)$ in (163) (full gradient), $\theta_2 = 0$, $\eta = 1/\theta_1$ (so $\eta\theta_1 = 1$ and $\nu = \infty$: the limiting case of Definition 5.15) and $\theta_1 = \sqrt{\sigma} = \sqrt{\mu/L}$. Then for all t,*

$$Z_{t+1} + Y_{t+1} \le \frac{1}{1+\sqrt{\sigma}} Z_t + (1-\sqrt{\sigma})\, Y_t \le \Big(1 - \frac{1}{2}\sqrt{\frac{\mu}{L}}\Big)(Z_t + Y_t), \tag{164}$$

hence $F(y_t) - F_* \le \sqrt{\mu/L}\,(1 - \frac{1}{2}\sqrt{\mu/L})^t (Z_0 + Y_0)$: the accelerated rate, as for (5.16)–(5.17) in the notes.

Proof. L2 holds verbatim (it does not involve x_t's composition nor the nature of g_t). For L3 with $g_t = F'(x_t)$ and $\eta\theta_1 = 1$, the computation of Lemma 5.8 gives $\frac{L}{2\eta}\|z_{t+1} - z_t\|^2 + \langle F'(x_t), z_{t+1} - z_t\rangle = \frac{1}{\theta_1}\big[\frac{L}{2}\|y_{t+1} - x_t\|^2 + \langle F'(x_t), y_{t+1} - x_t\rangle\big] \ge \frac{1}{\theta_1}(F(y_{t+1}) - F(x_t))$ by (T2): no Young step and no variance term. Steps 1–5 and 7 of Lemma 5.11 with $\theta_2 = 0$ (so (158) reads $z_t - x_t = \frac{1-\theta_1}{\theta_1}(x_t - y_t)$ and no w-terms appear) give $Z_{t+1} + Y_{t+1} \le \frac{1}{1+\eta\sigma} Z_t + (1-\theta_1) Y_t$ exactly as in (156) with $\theta_2 = 0$, $p = 0$. With $\eta = 1/\theta_1$ and $\theta_1 = \sqrt{\sigma}$: $\eta\sigma = \sqrt{\sigma} \le 1$, hence $\frac{1}{1+\sqrt{\sigma}} \le 1 - \frac{\sqrt{\sigma}}{2}$ and $1 - \sqrt{\sigma} \le 1 - \frac{\sqrt{\sigma}}{2}$. $\square$

The stochastic case does not close. Now replace $F'(x_t)$ by the loopless SVRG estimator.

Proposition 5.18 (one-step inequality of the naive scheme)**.** *For the scheme $\mathrm{N}(\theta_1, \eta, p)$ of Definition 5.15 with $\eta\theta_1 < 1$ (so that $\nu < \infty$), for all t,*

$$\begin{aligned} \mathbb{E}_{i(t)}\big[Z_{t+1} + Y_{t+1}\big] &\le \frac{1}{1+\eta\sigma} Z_t + (1-\theta_1)\, Y_t + \frac{\nu}{2L\theta_1} \mathbb{E}_{i(t)} \big\| g_t - F'(x_t)\big\|^2 \\ &\le \frac{1}{1+\eta\sigma} Z_t + (1-\theta_1)\, Y_t + \frac{\nu}{\theta_1}\, D_F(w_t, x_t), \end{aligned} \tag{165}$$

where the last inequality is the variance bound (150). The extra term $\frac{\nu}{\theta_1} D_F(w_t, x_t)$ is nonnegative; it vanishes only when F is affine on the segment $[x_t, w_t]$.

Proof. L1 and L2 hold verbatim. L3 holds with θ_2 replaced by ν (Remark 5.9; the proof of Lemma 5.8 only used $\frac{1}{\eta\theta_1} - 1 = \frac{1}{\nu} > 0$ in Young's inequality with $\beta = L/\nu$). In the proof of Lemma 5.11, Step 2 now reads $z_t - x_t = \frac{1-\theta_1}{\theta_1}(x_t - y_t)$, so Steps 1–5 give

$$\begin{aligned} F_* \;\ge\; & F(x_t) + \frac{1-\theta_1}{\theta_1}\big(F(x_t) - F(y_t)\big) + \mathbb{E}_{i(t)}[Z_{t+1}] - \frac{Z_t}{1+\eta\sigma} \\ & + \frac{1}{\theta_1}\mathbb{E}_{i(t)}\big[F(y_{t+1}) - F(x_t)\big] - \frac{\nu}{2L\theta_1}\mathbb{E}_{i(t)}\big\|g_t - F'(x_t)\big\|^2. \end{aligned}$$

The coefficients of $F(x_t)$ sum to $1+\frac{1-\theta_1}{\theta_1}-\frac{1}{\theta_1}=0$, and $F_*=\frac{1}{\theta_1}F_*-\frac{1-\theta_1}{\theta_1}F_*$; rearranging as in Step 7 gives the first inequality of (165). There is no Step 6: nothing in the chain produces a term $+\frac{\nu}{\theta_1}\langle F'(x_t),x_t-w_t\rangle$, because w_t does not appear in x_t. Nonnegativity of D_F is (T1), and $D_F(w_t,x_t)=0$ iff $F(w_t)=F(x_t)+\langle F'(x_t),w_t-x_t\rangle$, which for a convex function means that F coincides with its tangent on the segment. □

Thus, compared with (156), the right-hand side of (165) contains the *uncontrolled term* $\frac{\nu}{\theta_1}D_F(w_t,x_t)$, where ν is tied to the step: for the Katyusha step ($\eta\theta_1=\frac{1}{3}$) $\nu=\frac{1}{2}$, and for the deterministic step of Proposition 5.17 ($\eta\theta_1\to 1$) $\nu\to\infty$. The potential Ψ decreases, per step, by at most $\frac{\eta\sigma}{1+\eta\sigma}Z_t$ (distance of the mirror sequence to η_*), $\theta_1 Y_t$ (function value at y_t) and, if one adds a reference term $c_W(F(w_t)-F_*)$, by $pc_W(F(w_t)-F_*)-pc_W(F(y_t)-F_*)$; none of these involves the query point x_t or $F'(x_t)$. The next statement quantifies what it costs to express $D_F(w_t,x_t)$ in terms of the potential.

Proposition 5.19 (closing (165) by smoothness and strong convexity costs a factor κ)**.** *Under Assumption 5.1, for the scheme* $\mathrm{N}(\theta_1,\eta,p)$*,*

$$\frac{\nu}{\theta_1}D_F(w_t,x_t)\ \le\ 2\kappa\nu\Big[\frac{1}{\theta_1}\big(F(w_t)-F_*\big)+\frac{\eta}{1+\eta\sigma}Z_t+(1-\theta_1)\,Y_t\Big],\tag{166}$$

so that, writing $W_t=c_W(F(w_t)-F_*)$ *with* $c_W>0$ *and using* (155) *with* $\theta_2(1+\theta_1)/(p\theta_1)$ *replaced by* c_W*,*

$$\mathbb{E}_{i(t)}[\Psi_{t+1}]\le\Big(\frac{1+2\kappa\nu\eta}{1+\eta\sigma}\Big)Z_t+\Big(1-\theta_1+2\kappa\nu(1-\theta_1)+pc_W\theta_1\Big)Y_t+\Big(1-p+\frac{2\kappa\nu}{\theta_1 c_W}\Big)W_t.\tag{167}$$

The coefficient of Z_t *is* <1 *iff* $2\kappa\nu\eta<\eta\sigma$*, i.e.* $\nu<\frac{1}{2\kappa^2}$*; the coefficient of* Y_t *is* <1 *only if* $2\kappa\nu(1-\theta_1)<\theta_1$*, i.e.* $\nu<\frac{\theta_1}{2\kappa(1-\theta_1)}$*; the coefficient of* W_t *is* <1 *iff* $c_W>\frac{2\kappa\nu}{p\theta_1}$ *(substituting this bound in the* Y*-coefficient sharpens the second condition to* $\nu<\frac{\theta_1}{2\kappa(2-\theta_1)}$*, which we do not need). Suppose all three coefficients are* <1*. Since* $\nu\ge\eta\theta_1$*, two cases: if* $\theta_1\le\frac{1}{2}$*, the* Y*-condition gives* $\eta\le\frac{\nu}{\theta_1}<\frac{1}{2\kappa(1-\theta_1)}\le\frac{1}{\kappa}$*, hence* $\eta\sigma<\kappa^{-2}$*; if* $\theta_1>\frac{1}{2}$*, the* Z*-condition gives* $\eta\le\frac{\nu}{\theta_1}<\frac{1}{2\kappa^2\theta_1}<\kappa^{-2}$*, hence* $\eta\sigma<\kappa^{-3}$*. In all cases* $\eta\sigma<\kappa^{-2}$*, and the certified contraction factor is at least the* Z*-coefficient,* $\frac{1+2\kappa\nu\eta}{1+\eta\sigma}\ge\frac{1}{1+\eta\sigma}>\frac{1}{1+\kappa^{-2}}\ge 1-\kappa^{-2}$*. Hence no choice of* (θ_1,η,p,c_W) *makes* (167) *certify a rate better than* $1-\kappa^{-2}$ *per iteration: worse than gradient descent with step* $1/L$*, which contracts by* $1-1/\kappa$ *per iteration (per gradient evaluation, with* $p=1/n$*, the comparison is* $1-\frac{1}{3\kappa^2}$ *against* $1-\frac{1}{n\kappa}$*, so the naive certificate is worse as soon as* $\kappa>n/3$*), let alone the accelerated* $1-\Theta(1/\sqrt{n\kappa})$*.*

Proof. By (T2), $D_F(w_t,x_t)\le\frac{L}{2}\|w_t-x_t\|^2\le L\|w_t-\eta_*\|^2+L\|x_t-\eta_*\|^2$ (triangle and $(a+b)^2\le 2a^2+2b^2$). By (T4) at $\eta=\eta_*$, $\|\theta-\eta_*\|^2\le\frac{2}{\mu}(F(\theta)-F_*)$ for all θ, so $L\|w_t-\eta_*\|^2\le 2\kappa(F(w_t)-F_*)$ and $L\|x_t-\eta_*\|^2\le 2\kappa(F(x_t)-F_*)$. By convexity of F and (163), $F(x_t)-F_*\le\theta_1(F(z_t)-F_*)+(1-\theta_1)(F(y_t)-F_*)$, and by (T2) at η_*, $F(z_t)-F_*\le\frac{L}{2}\|z_t-\eta_*\|^2=\frac{\eta}{1+\eta\sigma}Z_t$ by (145); also $F(y_t)-F_*=\theta_1 Y_t$. Multiplying by ν/θ_1 gives (166). Adding (166) to (165) and then $\mathbb{E}_{i(t)}[W_{t+1}]=(1-p)W_t+pc_W\theta_1 Y_t$ gives (167); the stated conditions are read off the coefficients, using $\nu\ge\eta\theta_1$ and $\sigma=1/\kappa$. □

Remark 5.20 (other routes, and what the statement does and does not say). (i) The route through Z_t instead of μ is no better: $L\|x_t-\eta_*\|^2\le L\theta_1\|z_t-\eta_*\|^2+L(1-\theta_1)\|y_t-\eta_*\|^2$ and $L\theta_1\|z_t-\eta_*\|^2=\frac{2\eta\theta_1}{1+\eta\sigma}Z_t$, so the Z-coefficient receives $\frac{2\nu\eta}{1+\eta\sigma}$ against the available $\frac{\eta\sigma}{1+\eta\sigma}$: again $\nu<\sigma/2=\frac{1}{2\kappa}$, a factor κ away from the $\nu=\Theta(1)$ needed for a step $\asymp 1/L$. (ii) Paying D_F by the reference term alone is impossible: the reference potential pays differences of function values, and $D_F(w_t,x_t)-(F(w_t)-$

$F(x_t)) = \langle F'(x_t), x_t - w_t\rangle$ is not sign-definite and not a difference of function values; this inner product is precisely what the anchor term produces in (160). (iii) The comparison with the *non-accelerated* proof of Prop. 5.9 is instructive: there the variance term $2\gamma^2 R^2 F'(\theta_{t-1})^\top(\theta_{t-1} - \eta_*)$ (eq. (5.31)) is absorbed by the progress term $-2\gamma F'(\theta_{t-1})^\top(\theta_{t-1} - \eta_*)$, which is possible because both scale with the *same* quantity and the step is $\gamma \le 1/(4R^2)$; the price is that the contraction is then $\asymp \gamma\mu \asymp \sigma$, not $\sqrt{\sigma}$. In the accelerated proof the progress is measured by Y_t and Z_t, which are not comparable with $D_F(w_t, x_t)$ without the factor κ. (iv) Propositions 5.18 and 5.19 show that *this* Lyapunov argument does not certify acceleration for the naive scheme; they do not prove that the scheme is not accelerated (no lower bound specific to the scheme is known to us).

What is known. We summarize the literature on the anchor-free scheme, quoting the statements we rely on.

- *Nitanda (2014, 2016): acceleration needs large mini-batches.* Corollary 1 and Table 1 of [27] give, for Acc-Prox-SVRG with mini-batch size b, the overall complexity $\mathcal{O}\big((n + \frac{n-b}{n-1}\kappa)\log\frac{1}{\varepsilon}\big)$ when $b < b_0$ and $\mathcal{O}\big((n + b\sqrt{\kappa})\log\frac{1}{\varepsilon}\big)$ otherwise (with a threshold b_0 depending on n and κ), and the paper comments: "if $b = 1$, then Acc-Prox-SVRG has the same complexity as that of Prox-SVRG, while if $b = n$ then the complexity of this method is equal to that of APG". For $b = 1$ this is $\mathcal{O}((n+\kappa)\log\frac{1}{\varepsilon})$: no acceleration. Allen-Zhu [**?**, Remark 5.7] summarizes the same bound as follows: "In the simple case $\bar{L} = L$, Nitanda [46] obtained a total work $\tilde{\mathcal{O}}\big(n + \frac{n-b}{n-1}\frac{L}{\sigma} + b\sqrt{L/\sigma}\big)$ (see their page 7), which also implies a phase transition for b. However, this result is no better than ours for all parameters $b \in \{1, 2, \dots, n\}$. In fact, in terms of total work, Nitanda's result is even slower than SVRG when $b \le n/2$, and slower than Nesterov's method when $b > n/2$." (Here σ is our μ.) Allen-Zhu adds (footnote 21): "Nitanda's method is usually not considered as an accelerated method, since it requires mini-batch size to be very large in order to be accelerated. If mini-batch is large then one can use full-gradient method directly and acceleration is trivial." Nitanda's later AMSVRG [28, Theorem 3] with *increasing* mini-batches has, under the assumptions stated there, an overall complexity "$\mathcal{O}\big((n + \frac{n\kappa}{n+\sqrt{\kappa}})\log\frac{1}{\varepsilon}\big)$"; since $\frac{n\kappa}{n+\sqrt{\kappa}} = \frac{1}{2}H(\kappa, n\sqrt{\kappa})$ with H the harmonic mean, this is $\tilde{\mathcal{O}}(n + \min\{\kappa, n\sqrt{\kappa}\})$, the entry of Table 1 there for both Acc-SVRG and AMSVRG in the strongly convex column, while the optimal methods (Acc-SDCA, APCG, SPDC) are listed with $\tilde{\mathcal{O}}(n + \min\{\kappa, \sqrt{n\kappa}\})$; the paper states that "for strongly convex problems, AMSVRG is slower than optimal methods". Neither bound is better than the best of SVRG and accelerated gradient descent.

- *Allen-Zhu's discussion and experiment.* Footnote 1 of [**?**]: "experimentalists have observed that momentums could sometimes help if stochastic gradient iterations are used. However, the so-obtained methods (1) sometimes fail to converge in an accelerated rate, (2) become unstable and hard to tune, and (3) have no support theory behind them." Section 7.1 ("Effectiveness of Katyusha momentum") compares Katyusha with $\tau_2 = \frac{1}{2}$ and $\tau_2 = 0$ on ridge regression and concludes: "the old heuristics (i.e., $\tau_2 = 0$) sometimes indeed make the method faster after careful parameter tuning. However, for certain tasks such as Figure 1(c), without Katyusha momentum the algorithm does not even enjoy an accelerated convergence rate."

- *Hu, Wright and Lessard (2018): the SDP certificate needs the anchor.* In [19] the one-iteration inequality of Katyusha is recovered as a dissipation inequality certified by a 6×6 linear matrix inequality (their Section 4.3 and Lemma 11). Generalizing the parameters they obtain: "the one-iteration convergence result (27) holds for any $\frac{1}{5} \le \tau_2 < 1$ and $\tau_1 \le \min\{\frac{5\tau_2 - 1}{9\alpha L\tau_2}, 1 -$

$\tau_2\}$". This is derived for one explicit storage function (their matrix P of eq. (32)) and fixed multipliers ($\rho^2 = \frac{1}{1+\alpha\sigma}$, $\lambda_1 = \alpha/\tau_1$): with this certificate the admissible momentum weight τ_1 shrinks to 0 as $\tau_2 \downarrow \frac{1}{5}$, and no certificate is given for $\tau_2 < \frac{1}{5}$, in particular not for the anchor-free scheme $\tau_2 = 0$. The paper does not claim that the LMI is infeasible for $\tau_2 < \frac{1}{5}$ (it stresses that "Lemma 11 works for general choices of ζ and τ_2") and reports no search over other certificates, so the absence of an anchor-free certificate in this family is consistent with, but does not prove, Propositions 5.18–5.19. (The companion jump-system/LMI analysis of SAGA and SVRG without momentum is [18].)

- *Driggs, Ehrhardt and Schönlieb (2022): anchor-free acceleration with mini-batches.* [9] analyze the anchor-free scheme (163) up to the placement of the strong convexity, for generic estimators: their Algorithm 1 is $x_{k+1} = \tau z_k + (1-\tau)y_k$, $z_{k+1} = \operatorname{prox}_{\gamma g}(z_k - \gamma\tilde{\nabla}_{k+1})$, $y_{k+1} = \tau z_{k+1} + (1-\tau)y_k$, with the strong convexity assumed on the proximal term g ("If g is μ-strongly convex"), so that the z-step is pulled towards the minimizer of g instead of towards x_t with weight $\eta\sigma$ as in (K3) (see Remark 5.16). They explain the difficulty in the same terms as above: "In the analysis of Katyusha, negative momentum cancels part of the variance term, leaving telescoping terms that decrease at an accelerated rate. Without negative momentum, we must handle the variance term differently." For the SVRG estimator with mini-batch size b and full gradient with probability $1/p$, their Theorem 14 gives, under the condition $b \le 32p^2$ and for g μ-strongly convex, $\mathbb{E}\,\|z_T - x^*\|^2 \le \big(1+\min\{\frac{\sqrt{b\mu}}{4p\sqrt{6L}}, \frac{b}{4p}\}\big)^{-T} K_2$, at an expected cost of "$n/p + b$" stochastic gradient evaluations per iteration; with $b = 1$ and $p = n$ (admissible since $1 \le 32n^2$; one full pass every n iterations on average, as in L-Katyusha) this is a contraction $1 - \Theta\big(\frac{1}{n\sqrt{\kappa}}\big)$ per iteration, i.e. $\mathcal{O}(n\sqrt{\kappa}\log\frac{1}{\varepsilon})$ gradient evaluations: worse than SVRG's $n+\kappa$ unless $\kappa \ge n^2$ and worse than $\sqrt{n\kappa}$ by the factor $\sqrt{n}$. Their Remark 4 is stated for the SAGA estimator and transferred to SVRG: "when using the SAGA gradient estimator on a strongly convex objective with $b = \mathcal{O}(n^{2/3})$ and $\gamma = \mathcal{O}(1/\sqrt{\mu L})$, Algorithm 1 finds a point satisfying $\mathbb{E}\|z_T - x^*\|^2 \le \epsilon$ in $\mathcal{O}(n^{2/3}\sqrt{\kappa}\log(1/\epsilon))$ iterations. This is compared to the complexity lower bound of $\mathcal{O}(\sqrt{n\kappa}\log(1/\epsilon))$ achieved by Katyusha [37]. SVRG has a similar complexity when $b = n^{2/3}$ and $p = n^{1/3}$" (again $b \le 32p^2$). So, as for Nitanda, acceleration without the anchor is proved only with mini-batches of size $b \asymp n^{2/3}$ and remains suboptimal by $n^{1/6}$.

- *Indirect acceleration.* Catalyst [23] accelerates SAGA/SVRG as inner solvers with complexity $\tilde{\mathcal{O}}(\sqrt{n\kappa}\log\frac{1}{\varepsilon})$ when $n \le \kappa$ (their Table 1), with an extra logarithmic factor; this is a different mechanism (proximal-point outer loop), not momentum on the estimator.

To our knowledge, no theorem shows that (163) with $b = 1$ attains $\mathcal{O}((n+\sqrt{n\kappa})\log\frac{1}{\varepsilon})$, and no lower bound excludes it; the honest statement of (A2) is therefore: *the Lyapunov proof of L-Katyusha closes only because of the cancellation* (160)*; without the anchor the best available certificates (Nitanda, Driggs et al.) give no improvement over* $\min\{n+\kappa, n\sqrt{\kappa}\}$ *for* $b = 1$.

Empirical test of (A2). We propose to compare, on the same instances as hypotheses (A4) and (B3) (the hard finite-sum instance with $\kappa \gg n$ of the lower-bound note, and ℓ_2-regularized logistic regression with $d = 40$ Gaussian inputs as in Fig. 5.4 of the notes, $\lambda \in \{R^2/n, R^2/(10n), R^2/(100n)\}$ to vary κ at fixed n):

1. SAGA with $\gamma = 1/(4R^2)$ (Prop. 5.9) and L-SVRG with $\eta = 1/(6L)$, $p = 1/n$ (Theorem 3.5 of [21]);

2. L-Katyusha with (144);

3. the naive scheme (163) with the *same* θ_1, η, p as L-Katyusha (so that the only difference is the anchor), with the smaller steps $\eta \in \{\frac{1}{3\theta_1}, \frac{1}{3}, \frac{1}{3\kappa\theta_1}\}$, and in the two-sequence form of the notes (Remark 5.16) with $\beta = \frac{1-\sqrt{\mu/L}}{1+\sqrt{\mu/L}}$ and step $1/L$;

4. the anchor-free scheme tuned as in Theorem 14 of [9] with $b = 1$ and $b = \lceil n^{2/3} \rceil$.

For each run we measure the linear rate per gradient evaluation, $\hat{\rho} := -$slope of $\log(F(y_t) - F_*)$ against the number of f_i' evaluations, and fit the exponent α in $\hat{\rho} \propto \kappa^{-\alpha}$ over the three values of λ. Predictions: (P1) $\alpha \approx 1$ for SAGA and L-SVRG, $\alpha \approx \frac{1}{2}$ for L-Katyusha once $\kappa \geq 8n/3$ (regime where $\rho = 1/\sqrt{24n\kappa}$), $\alpha \approx 0$ for all methods when $\kappa \ll n$ (the $n \log \frac{1}{\varepsilon}$ term dominates); (P2) the naive scheme with the Katyusha parameters ($\theta_1 = \sqrt{2n/(3\kappa)}$ small, mirror step $\frac{1}{3\theta_1 L}$ large) is non-monotone and may diverge for $\kappa \gg n$, because without the anchor $\|x_t - w_t\|$, hence the uncontrolled term $\frac{\nu}{\theta_1} D_F(w_t, x_t)$, is not kept small (this is the instability of Allen-Zhu's footnote 1); (P3) with a step small enough to be stable, $\alpha \approx 1$ for the naive scheme (no $\sqrt{\kappa}$ scaling), and its rate is at best that of L-SVRG; (P4) for the naive scheme the ratio $v(t)$ of the uncontrolled term of (165) to the certified decrease (Section 5.5) is frequently ≥ 1, i.e. the one-step inequality certifies no decrease at all, while for L-Katyusha the size $q(t)$ of the cancelled inner product relative to the certified decrease $\rho\Psi_t$ is of order one or larger, i.e. the term removed by the anchor could not have been absorbed into $\rho\Psi_t$; this is the mechanism behind (P2)–(P3). A positive result for the batch-$n^{2/3}$ variant of item 4 and a negative one for $b = 1$ would be consistent with [9, 27].

5.5 What to instrument

Each inequality of the proof is a pair (left-hand side, right-hand side) whose ratio can be computed along a run (with η_* obtained by a long run of L-Katyusha or SAGA; conditional expectations $\mathbb{E}_{i(t)}$ are computed exactly by one pass over the data, which is affordable on the instances of (B3)). Tables 4 and 5 list them with the ratio to record; a ratio close to 1 means the inequality is tight on the trajectory. The last two rows of Table 5 concern the naive scheme. Convention: a ratio is not recorded (left undefined) at the steps where its denominator vanishes. This happens systematically at $t = 0$ for $r_{\text{cen}}, r_{\text{coco}}, r_{\text{var}}, c, u$ and q, because $x_0 = w_0 = \theta_0$ gives $D_F(w_0, x_0) = 0$ and an exact estimator $g_0 = F'(x_0)$, and it happens for r_{sc} and μ_{eff} when $x_t = \eta_*$. For the Young row of Table 4, note that the step contains the noise: by (K3)–(K4), $y_{t+1} - x_t = -\alpha\big(g_t - \mu(x_t - z_t)\big)$ with $\alpha := \frac{\theta_1\eta}{L(1+\eta\sigma)}$ (the drift coefficient $\frac{\theta_1\eta\sigma}{1+\eta\sigma}$ equals $\alpha L\sigma = \alpha\mu$), so that, with $V_t := \mathbb{E}_{i(t)} \|g_t - F'(x_t)\|^2$, unbiasedness gives $\mathbb{E}_{i(t)}[-\langle g_t - F'(x_t), y_{t+1} - x_t\rangle] = \alpha V_t$ and $\mathbb{E}_{i(t)} \|y_{t+1} - x_t\|^2 = \alpha^2\big(V_t + \|F'(x_t) - \mu(x_t - z_t)\|^2\big)$; this gives the closed form of the averaged ratio $\bar{r}_{\text{Y}}(t)$ in the table.

5.6 Complexity comparison

Table 6 collects the number of evaluations of a single f_i' needed to reach accuracy ε (on $\mathbb{E}[F - F_*]$ or $\mathbb{E}\,\|\cdot - \eta_*\|^2$, which differ by factors L or μ inside the logarithm), with the constants of the cited statements; $\kappa = L/\mu$ with L the per-sample smoothness constant, as everywhere in this note. Counting convention: the evaluation of f_i' at a stored reference point (w_t for L-SVRG and L-Katyusha, $\tilde{x}$ for Katyusha) counts as one gradient unless the n reference gradients are stored when the full gradient is computed ($\mathcal{O}(nd)$ memory, as for the table of SAGA), in which case it is a look-up; where the two conventions differ we give both counts, the first one without storage.

Label	Inequality		Tightness ratio to measure (and expected behaviour)
(148) centering	$\mathbb{E}_{i(t)} \left\|g_t - F'(x_t)\right\|^2$	$\leq$ $\frac{1}{n}\sum_i \left\|f_i'(x_t) - f_i'(w_t)\right\|^2$	$r_{\mathrm{cen}}(t) = 1 - \frac{\left\|F'(x_t) - F'(w_t)\right\|^2}{\frac{1}{n}\sum_i \left\|f_i'(x_t) - f_i'(w_t)\right\|^2}$; close to 1 when the per-sample gradient differences are not aligned (expected for logistic regression).
(149) cocoercivity	$\frac{1}{n}\sum_i \left\|f_i'(x_t) - f_i'(w_t)\right\|^2$	$\leq$ $2L\, D_F(w_t, x_t)$	$r_{\mathrm{coco}}(t) = \frac{\frac{1}{n}\sum_i \left\|f_i'(x_t) - f_i'(w_t)\right\|^2}{2L\, D_F(w_t, x_t)}$; for logistic regression expected $\approx L_{\mathrm{eff}}/L \ll 1$ where L is the worst-case constant $\max_i \left\|x_i\right\|^2/4 + \lambda$ (link with (B2)).
(150) variance	$\mathbb{E}_{i(t)} \left\|g_t - F'(x_t)\right\|^2 \leq 2L\, D_F(w_t, x_t)$		$r_{\mathrm{var}}(t) = r_{\mathrm{cen}}(t)\, r_{\mathrm{coco}}(t)$.
(151) mirror step	slack $= \frac{\mu}{2}\left\|x_t - z_{t+1}\right\|^2$		$s_z(t) = \frac{\frac{\mu}{2}\left\|x_t - z_{t+1}\right\|^2}{\rho\, \Psi_t}$ (slack relative to the certified decrease; small means the dropped term is irrelevant).
(153) smoothness	$F(y_{t+1}) - F(x_t) - \langle F'(x_t), y_{t+1} - x_t\rangle \leq \frac{L}{2}\left\|y_{t+1} - x_t\right\|^2$		$r_{\mathrm{sm}}(t) = \frac{D_F(y_{t+1}, x_t)}{\frac{L}{2}\left\|y_{t+1} - x_t\right\|^2} \in [0, 1]$: local curvature along the step over L (this is the L_{eff} of (B2)).
(154) Young	$\langle g_t - F'(x_t), y_{t+1} - x_t\rangle$	$\geq$ $-\frac{\theta_2}{2L}\left\|g_t - F'(x_t)\right\|^2 - \frac{L}{2\theta_2}\left\|y_{t+1} - x_t\right\|^2$	$r_{\mathrm{Y}}(t) = \frac{-\langle g_t - F'(x_t), y_{t+1} - x_t\rangle}{\frac{\theta_2}{2L}\left\|g_t - F'(x_t)\right\|^2 + \frac{L}{2\theta_2}\left\|y_{t+1} - x_t\right\|^2} \in [-1, 1]$ per realization, and the ratio of conditional means $\bar{r}_{\mathrm{Y}}(t) = \frac{\alpha V_t}{\frac{\theta_2}{2L} V_t + \frac{L}{2\theta_2}\alpha^2\left(V_t + \left\|F'(x_t) - \mu(x_t - z_t)\right\|^2\right)}$, with V_t and α as in the text above. With (144), $\bar{r}_{\mathrm{Y}}(t) \leq \frac{12(1+\eta\sigma)}{9(1+\eta\sigma)^2+4} \leq \frac{12}{13}$, with near-equality in the noise-dominated regime $V_t \gg \left\|F'(x_t) - \mu(x_t - z_t)\right\|^2$: the Young step is then nearly tight, and it is loose only when the full gradient or the drift dominates the step.
(157) strong convexity	$F_* - F(x_t) - \langle F'(x_t), \eta_* - x_t\rangle \geq \frac{\mu}{2}\left\|x_t - \eta_*\right\|^2$		$r_{\mathrm{sc}}(t) = \frac{2D_F(\eta_*, x_t)}{\mu\left\|x_t - \eta_*\right\|^2} \geq 1$: the Bregman analogue of the μ_{eff}/μ of (B2). For comparability with (B2) also record the inner-product form $\mu_{\mathrm{eff}}(x_t)/\mu$ with $\mu_{\mathrm{eff}}(x_t) := \langle F'(x_t), x_t - \eta_*\rangle / \left\|x_t - \eta_*\right\|^2 \geq \mu$; the two differ, since $D_F(\eta_*, x_t) \leq \langle F'(x_t), x_t - \eta_*\rangle$ (because $F_* \leq F(x_t)$), so that $r_{\mathrm{sc}}(t) \leq 2\mu_{\mathrm{eff}}(x_t)/\mu$.
(159) convexity	$\langle F'(x_t), x_t - y_t\rangle \geq F(x_t) - F(y_t)$		$s_{\mathrm{cvx}}(t) = \frac{D_F(y_t, x_t)}{F(y_t) - F_*}$ (slack relative to the Y-potential).

Table 4: Inequalities of the one-step lemmas L1–L3 and of the convexity steps of L5, with the ratio to record along a trajectory.

Label	Inequality	Tightness ratio to measure (and expected behaviour)
(160) cancellation	identity	$c(t) = \dfrac{\langle F'(x_t), x_t - w_t\rangle}{D_F(w_t, x_t)} \in \mathbb{R}$ and $1 - c(t) = \dfrac{F(w_t) - F(x_t)}{D_F(w_t, x_t)}$: the *signed* shares of the Bregman gap removed by the anchor and left as a difference of function values. They sum to 1 by definition of D_F but are not fractions in $[0,1]$: $c(t) < 0$ when $\langle F'(x_t), x_t - w_t\rangle < 0$ (the anchor term then adds to the variance budget on that step), and $c(t) > 1$ when $F(w_t) < F(x_t)$ (the reference point is better than the query point and the function-value share is negative). Record $c(t)$ together with the sign of $F(w_t) - F(x_t)$, and the size of the cancelled term relative to the certified decrease, $q(t) = \dfrac{\left\lvert \frac{\theta_2}{\theta_1} \langle F'(x_t), x_t - w_t\rangle \right\rvert}{\rho\, \Psi_t}$ ($q(t) \gtrsim 1$ means that the cancelled term could not have been absorbed into the decrease); and $u(t) = \dfrac{\frac{\theta_2}{2L\theta_1}\mathbb{E}_{i(t)} \left\| g_t - F'(x_t)\right\|^2}{\frac{\theta_2}{\theta_1} D_F(w_t, x_t)} = r_{\mathrm{var}}(t)$, the fraction of the cancelled budget actually used by the noise.
(146) contraction	$\mathbb{E}_{i(t)}\Psi_{t+1} \le \max\{c_Z, c_Y, c_W\}\Psi_t$	$r_\Psi(t) = \dfrac{\mathbb{E}_{i(t)}\Psi_{t+1}}{(1-\rho)\Psi_t}$ and the three component ratios $\mathbb{E}_{i(t)}Z_{t+1}/Z_t$, $\mathbb{E}_{i(t)}Y_{t+1}/Y_t$, $\mathbb{E}_{i(t)}W_{t+1}/W_t$ against c_Z, c_Y, c_W (which of Z, Y, W is the bottleneck).
(147) global rate	$\mathbb{E}\Psi_t \le (1-\rho)^t \Psi_0$	$\hat\rho_T(t) = -\frac{1}{T}\log\frac{\Psi_{t+T}}{\Psi_t}$ against ρ (expected $\hat\rho \gg \rho$ on logistic regression, cf. $r_{\mathrm{coco}}, r_{\mathrm{sc}}$).
(165) naive, uncontrolled term	$\mathbb{E}_{i(t)}[Z_{t+1} + Y_{t+1}] \le \frac{Z_t}{1+\eta\sigma} + (1-\theta_1)Y_t + \frac{\nu}{\theta_1}D_F(w_t, x_t)$	$v(t) = \dfrac{\frac{\nu}{\theta_1}D_F(w_t, x_t)}{\frac{\eta\sigma}{1+\eta\sigma}Z_t + \theta_1 Y_t}$, the uncontrolled term over the certified decrease of $Z + Y$: $v(t) \ge 1$ exactly when the one-step inequality certifies no decrease of $Z_t + Y_t$ at step t.
(166) naive, κ-cost	$\frac{\nu}{\theta_1}D_F \le 2\kappa\nu[\dots]$	ratio of the two sides; expected $\ll 1$ (the bound is loose by $\approx \kappa$), confirming that the obstruction is the proof, not the quantity.

Table 5: The cancellation, the contraction of the potential, the global rate, and the two inequalities of the naive scheme, with the ratio to record along a trajectory.

Method	Gradient evaluations to accuracy ε	Source and constants
Gradient descent, $\gamma = 1/R^2$	$n\,\kappa \log \frac{1}{\varepsilon}$	Notes, Section 5.4.4 ("running time $dn\frac{R^2}{\mu}\log\frac{1}{\varepsilon}$"); with $\gamma = 1/L_F$, $n\frac{L_F}{\mu}\log\frac{1}{\varepsilon}$ where $L_F \le R^2$.
Accelerated GD (5.16)–(5.17)	$n\sqrt{\kappa}\log\frac{1}{\varepsilon}$	Notes, Section 5.2.5: rate $(1-\sqrt{\mu/L})^t$; Proposition 5.17: $(1-\frac{1}{2}\sqrt{\mu/L})^t$.
SGD, $\gamma_t = 1/(\mu t)$	$\mathcal{O}(\kappa/\varepsilon)$ (no logarithm, sublinear)	Notes, Section 5.4.4 ("the convergence rate has been shown to be $\mathcal{O}(\kappa/t)$ in Section 5.4.1") and the table of Section 5.5 (smooth case, rate $L/(t\mu)$); Prop. 5.8 is the Lipschitz case, $2B^2(1+\log t)/(\mu t)$.
SAGA, $\gamma = 1/(4R^2)$	$\big(3n + \frac{16}{3}\kappa\big)\log\frac{2n}{\varepsilon}$	Prop. 5.9 of the notes: contraction $1-\min\{\frac{1}{3n}, \frac{3\mu}{16R^2}\}$, one gradient per iteration; [7].
SVRG (epoch length m)	$\mathcal{O}\big((n+\kappa)\log\frac{1}{\varepsilon}\big)$ with $m \asymp \kappa$	[20]; as quoted in [21, Section 1]: "the iteration complexity of SVRG is $\mathcal{O}((n+L/\mu)\log 1/\varepsilon)$".
L-SVRG, $\eta = 1/(6L)$, $p = 1/n$	$3\max\{6\kappa, 2n\}\log\frac{\Phi_0}{\varepsilon}$ without storage, $2\max\{6\kappa, 2n\}\log\frac{\Phi_0}{\varepsilon}$ with	[21, Theorem 3.5]: $\mathbb{E}\Phi^k \le \max\{1-\frac{\mu}{6L}, 1-\frac{1}{2n}\}^k\Phi^0$, $\Phi^k = \|x^k-x^*\|^2 + \frac{4\eta^2}{pn}\sum_i \|\nabla f_i(w^k) - \nabla f_i(x^*)\|^2$; $2+pn = 3$ gradients per iteration without storage, $1+pn = 2$ with.
Katyusha, $m = 2n$	$\mathcal{O}\big((n+\sqrt{n\kappa})\log\frac{F(\theta_0)-F_*}{\varepsilon}\big)$	[?, Theorem 2.1, Remark 2.2 and footnote 11]: 2.5 stochastic gradients per iteration without storage, 1.5 with (quoted below).
L-Katyusha, (144)	$n + \big(12n + 15\sqrt{n\kappa}\big)\log\frac{3L(n+3)\|\theta_0-\eta_*\|^2}{8\varepsilon}$ without storage, $n + \big(8n + 10\sqrt{n\kappa}\big)\log\frac{3L(n+3)\|\theta_0-\eta_*\|^2}{8\varepsilon}$ with	Corollary 5.12; [21, Theorem 4.6]: $\rho = \min\{\frac{\sigma}{6\theta_1}, \frac{\theta_1}{2n}\}$; $2+pn = 3$ gradients per iteration without storage, $1+pn = 2$ with (quoted below); the leading n is the initial $F'(w_0)$.
Catalyst on SAGA/SVRG	$\tilde{\mathcal{O}}\big(\sqrt{n\kappa}\log\frac{1}{\varepsilon}\big)$ for $n \le \kappa$ (extra $\log\kappa$ factor)	[23, Table 1].
Naive AGD+SVRG, $b = 1$	no bound better than $\min\{n+\kappa, n\sqrt{\kappa}\}\log\frac{1}{\varepsilon}$: Nitanda $\tilde{\mathcal{O}}(n + \frac{n-b}{n-1}\kappa + b\sqrt{\kappa})$; Driggs et al. $\mathcal{O}(n\sqrt{\kappa}\log\frac{1}{\varepsilon})$	[?, Remark 5.7] on [27]; [9, Theorem 14] with $b = 1$, $p = n$ (Section 5.4).
Lower bound, randomized incremental gradient methods (any sampling distribution)	$\Omega\big((n+\sqrt{n\kappa})\log\frac{\|\theta_0-\eta_*\|^2}{\varepsilon}\big)$, for a large enough dimension	[22, Corollary 3] (quoted below; their m is our n, and in their sum convention $L = \sum_i L_i$ is our L after rescaling $f_i \to f_i/n$).
Lower bound, any randomized algorithm, $\kappa > 161\,n$	$\Omega\big(n + \sqrt{n\kappa}\log\big(\frac{\varepsilon_0}{\varepsilon}\sqrt{\frac{n}{\kappa}}\big)\big)$, with no $n\log\frac{1}{\varepsilon}$ term	[32, Theorem 8] (quoted below); see hypothesis (A1) and [15] for the $n\log\frac{1}{\varepsilon}$ term and the role of the span assumption.

Table 6: Gradient complexity of the methods discussed in the notes and in this note, in terms of n, $\kappa = L/\mu$ and ε.

Quotations used in Table 6. Gradient counts. [**?**, footnote 11]: "The claim 'SVRG or Katyusha computes 1.5 stochastic gradients' requires one to store $\nabla f_i(\tilde{x})$ in the memory for each $i \in [n]$, and this costs $O(dn)$ space in the most general setting. If one does not store $\nabla f_i(\tilde{x})$ in the memory, then each iteration of SVRG or Katyusha computes 2.5 stochastic gradients for the choice $m = 2n$." [21, Section 4]: "Algorithm 2 calls stochastic gradient oracle $\mathcal{O}(1 + pn)$ times per iteration in expectation." Lower bounds. [22, Corollary 3]: "The number of gradient evaluations performed by any randomized incremental gradient methods for finding a solution $\bar{x} \in X$ of problem (1.1) such that $\mathbb{E}[\|\bar{x} - x^*\|_2^2] \leq \epsilon$ cannot be smaller than $\Omega\big((\sqrt{mC} + m)\log \frac{\|x^0 - x^*\|_2^2}{\epsilon}\big)$ if n is sufficiently large, where $C = L/\mu$ and $L = \sum_{i=1}^m L_i$" (their n is the dimension), the worst-case instance being analyzed "for any arbitrary distribution used to choose f_i at each iteration". [32, Theorem 8], for m components that are γ-smooth and λ-strongly convex with $\gamma/\lambda > 161m$ and initial gap ϵ_0: any randomized algorithm $\mathcal{A}$ "must make $\Omega\big(m + \sqrt{m\frac{\gamma}{\lambda}}\log\big(\frac{\epsilon_0}{\epsilon}\sqrt{\frac{m\lambda}{\gamma}}\big)\big)$ queries to h_F".

In the regime $\kappa \gg n$ of hypothesis (A4), the rows read $n\kappa$ (GD), $n\sqrt{\kappa}$ (AGD), κ (SAGA, L-SVRG), $\sqrt{n\kappa}$ (L-Katyusha, optimal): among the incremental methods ($\mathcal{O}(1)$ gradients per iteration) the exponent of κ drops from 1 to $\frac{1}{2}$ only for the one whose proof contains the cancellation (160); accelerated gradient descent also has the exponent $\frac{1}{2}$ but pays the factor n per iteration.

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